

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 01

Algebraic Proofs

5 Concepts | 444 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 01 | Algebraic Proofs

CLASSROOM ROUTE

Concept 1 | Binomial and Multinomial Theorems**70 questions**

Official lessons: 1-1, 1-2, 1-3

Concept 2 | Polynomial Division**60 questions**

Official lessons: 1-4, 1-5

Concept 3 | Rational Expressions**110 questions**

Official lessons: 1-6, 1-7, 1-8

Concept 4 | Identities and Proofs**106 questions**

Official lessons: 1-9, 1-10, 1-11, 1-12

Concept 5 | Inequalities**98 questions**

Official lessons: 1-13, 1-14, 1-15, 1-16

The Formula Bank changes at each Concept boundary.

Concept 1 | Binomial and Multinomial Theorems

OFFICIAL LESSONS 1-1, 1-2, 1-3

Binomial and Multinomial Theorems

Equation-first recall | use the same rail beside every question in this Concept

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Binomial and Multinomial Theorems

C01-T01-COMBINED | V01

Q001 | CHALLENGE MCQ

C01-T01-C01

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

A Coefficient = 720; sum = -1

B Coefficient = 720; sum = 1

C Coefficient = 540; sum = -1

D Coefficient = 240; sum = 32

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-COMBINED | V02

Q002 | CHALLENGE MCQ

C01-T01-C02

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

A First = 60; second = -120

B First = 60; second = 120

C First = 120; second = -60

D First = -60; second = -120

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-COMBINED | V03

Q003 | CHALLENGE MCQ

C01-T01-C03

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

A Greatest coefficient = 19

B Greatest coefficient = 16

C Greatest coefficient = 20

D Greatest coefficient = 24

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCrn - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q004 | WRITTEN

C01-T01-S03

Expand $(x + 2)^5$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q005 | MCQ

C01-T01-S05

Expand $(3x - 2)^4$ using the Binomial Theorem:

A $81x^4 - 216x^3 - 96x + 16$

B $81x^4 - 216x^3 + 432x^2 - 96x + 16$

C $-(3x - 2)^4$

D $81x^4 - 216x^3 + 216x^2 - 96x + 16$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q006 | WRITTEN

C01-T01-S09

Expand $(x + 2)^6$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q007 | MCQ

C01-T01-S2a

Expand $(x + 1)^4$ using the Binomial Theorem:

A $x^4 + 4x^3 + 4x + 1$

B $x^4 + 4x^3 + 6x^2 + 4x + 1$

C $x^4 + 4x^3 + 12x^2 + 4x + 1$

D $-(x + 1)^4$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q008 | MCQ

C01-T01-S2b

Expand $(a + b)^4$ using the Binomial Theorem:

A $a^4 + 4a^3b + 4ab^3 + b^4$

B $a^4 + 4a^3b + 12a^2b^2 + 4ab^3 + b^4$

C $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

D $-(a + b)^4$

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q009 | WRITTEN

C01-T01-S2c

Expand $(a - 2b)^4$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q010 | MCQ

C01-T01-S2d

Expand $(x - 2)^5$ using the Binomial Theorem:

A $x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

B $x^5 - 10x^4 + 40x^3 - 80x^2 - 32$

C $x^5 - 10x^4 + 40x^3 - 80x^2 + 160x - 32$

D $-(x - 2)^5$

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q011 | WRITTEN

C01-T01-S2e

Expand $(3x + 2y)^5$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q012 | WRITTEN

C01-T01-S2f

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q013 | WRITTEN

C01-T01-S2g

Expand $(x - 3)^6$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q014 | WRITTEN

C01-T01-S2h

Expand $(2x - 3y)^6$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q015 | WRITTEN

C01-T01-S2i

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q016 | MCQ

C01-T01-P0101

Expand $(x + 3)^3$ using the Binomial Theorem:

A $x^3 + 9x^2 + 27$

B $x^3 + 9x^2 + 27x + 27$

C $x^3 + 9x^2 + 54x + 27$

D $-(x + 3)^3$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q017 | MCQ

C01-T01-P0103

Expand $(a + 2b)^5$ using the Binomial Theorem:

A $a^5 + 10a^4b + 40a^3b^2 + 80ab^4 + 32b^5$

B $a^5 + 10a^4b + 40a^3b^2 + 160a^2b^3 + 80ab^4 + 32b^5$

C $-(a + 2b)^5$

D $a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q018 | MCQ

C01-T01-P0107

Expand $(x + 2)^2$ using the Binomial Theorem:

A $x^2 + 4$

B $x^2 + 8x + 4$

C $-(x + 2)^2$

D $x^2 + 4x + 4$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q019 | MCQ

C01-T01-P0108

Expand $(a - b)^3$ using the Binomial Theorem:

A $a^3 - 3a^2b + 3ab^2 - b^3$

B $a^3 - 3a^2b - b^3$

C $a^3 - 3a^2b + 6ab^2 - b^3$

D $-(a - b)^3$

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V01

Q020 | MCQ

C01-T01-P0109

Expand $(2x + 1)^5$ using the Binomial Theorem:

A $32x^5 + 80x^3 + 40x^2 + 10x + 1$

B $32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$

C $32x^5 + 160x^4 + 80x^3 + 40x^2 + 10x + 1$

D $-(2x + 1)^5$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V02

Q021 | WRITTEN

C01-T01-S10

Expand $(3a - 2b)^5$ using the Binomial Theorem.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V02

Q022 | MCQ

C01-T01-P0102

Expand $(2x - 1)^4$ using the Binomial Theorem:

A $16x^4 - 32x^3 - 8x + 1$

B $16x^4 - 32x^3 + 48x^2 - 8x + 1$

C $16x^4 - 32x^3 + 24x^2 - 8x + 1$

D $-(2x - 1)^4$

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V02

Q023 | MCQ

C01-T01-P0110

Expand $(x - 3)^4$ using the Binomial Theorem:

A $x^4 - 12x^3 - 108x + 81$

B $x^4 - 12x^3 + 54x^2 - 108x + 81$

C $x^4 - 12x^3 + 108x^2 - 108x + 81$

D $-(x - 3)^4$

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V02

Q024 | MCQ

C01-T01-P0111

Expand $(2a + b)^3$ using the Binomial Theorem:

A $8a^3 + 6ab^2 + b^3$

B $8a^3 + 24a^2b + 6ab^2 + b^3$

C $8a^3 + 12a^2b + 6ab^2 + b^3$

D $-(2a + b)^3$

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V03

Q025 | MCQ

C01-T01-P0106

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:

A $16a^8 + 96a^6 + 216a^2 + 81$

B $16a^8 + 96a^6 + 432a^4 + 216a^2 + 81$

C $16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$

D $-(2a^2 + 3)^4$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F01 | V03

Q026 | MCQ

C01-T01-P0114

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:

A $27a^6 - 54a^4 - 8$

B $27a^6 - 54a^4 + 36a^2 - 8$

C $27a^6 - 54a^4 + 72a^2 - 8$

D $-(3a^2 - 2)^3$

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q027 | WRITTEN

C01-T01-S01

Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q028 | WRITTEN

C01-T01-S02

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q029 | WRITTEN

C01-T01-S04

Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r {}^{n-r} C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q030 | WRITTEN

C01-T01-S06

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q031 | MCQ

C01-T01-S07

Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:

A 720

B -720

C -1440

D -719

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q032 | WRITTEN

C01-T01-S08

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q033 | MCQ

C01-T01-S1a

Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:

A 1120

B -1120

C 2240

D 1121

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q034 | MCQ

C01-T01-S1b

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:

A -12

B 12

C 24

D 13

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q035 | MCQ

C01-T01-S1c

Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:

A -160

B 320

C 160

D 161

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q036 | MCQ

C01-T01-S1d

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:

A -405

B 810

C 406

D 405

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q037 | WRITTEN

C01-T01-S1e

Find the coefficient of a in the expansion of $(2a - 3)^4$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q038 | WRITTEN

C01-T01-S1f

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q039 | WRITTEN

C01-T01-S1g

Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r {}^{n-r} C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q040 | MCQ

C01-T01-S1h

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:

A 90

B -180

C -89

D -90

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^s C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F02 | V01

Q041 | WRITTEN

C01-T01-S1i

Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r {}^{n-r} C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F03 | V01

Q042 | WRITTEN

C01-T01-S30

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F04 | V01

Q043 | MCQ

C01-T01-P0401

Find the coefficient of xyz in the expansion of $(x + y + z)^3$ ∴

A 6

B -6

C 3

D 12

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q044 | MCQ

C01-T01-S40

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:

A -280

B 560

C 280

D 281

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q045 | MCQ

C01-T01-S42

Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:

A 60

B -60

C 120

D 61

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q046 | MCQ

C01-T01-S43

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:

A -90

B 90

C 180

D 91

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q047 | MCQ

C01-T01-S5a

Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:

A 42

B -42

C 84

D 43

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q048 | WRITTEN

C01-T01-S5b

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q049 | WRITTEN

C01-T01-S5c

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q050 | WRITTEN

C01-T01-S5d

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q051 | WRITTEN

C01-T01-S5e

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V01

Q052 | WRITTEN

C01-T01-S5f

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q053 | MCQ

C01-T01-S41

Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:

A 6720

B -13440

C -6719

D -6720

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q054 | MCQ

C01-T01-S44

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:

A -3240

B 6480

C 3240

D 3241

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q055 | MCQ

C01-T01-S45

Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:

A 4320

B -8640

C -4319

D -4320

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q056 | WRITTEN

C01-T01-S5g

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q057 | WRITTEN

C01-T01-S5h

Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q058 | WRITTEN

C01-T01-S5i

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q059 | WRITTEN

C01-T01-S5j

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q060 | WRITTEN

C01-T01-S5k

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F05 | V02

Q061 | WRITTEN

C01-T01-S5I

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

F06 | V02

Q062 | WRITTEN

C01-T01-S60

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-MCQ

Quiz M1 | QUIZ MCQ

C01-T01-Q01

The coefficient of x^4 in $(2x - 3)^5$ is:

A -240

B 240

C -120

D 120

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-MCQ

Quiz M2 | QUIZ MCQ

C01-T01-Q04

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

A 30

B 60

C 90

D 120

WORKING AREA

FORMULA BANK

Binomial

$$nCr a^r b^{n-r}$$

Multinomial / Trinomial

$$nCr n - rCs a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-MCQ

Quiz M3 | QUIZ MCQ

C01-T01-Q07

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

A -4320

B -2160

C 2160

D 4320

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-MCQ

Quiz M4 | QUIZ MCQ

C01-T01-Q10

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

A 0

B 8

C 16

D 32

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-WRITTEN

Quiz W1 | QUIZ WRITTEN

C01-T01-Q11

Expand $(2x - 1)^4$ fully.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-WRITTEN

Quiz W2 | QUIZ WRITTEN

C01-T01-Q12

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-WRITTEN

Quiz W3 | QUIZ WRITTEN

C01-T01-Q14

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

WORKING AREA

FORMULA BANK

Binomial

$${}^nC_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^nC_r n - r {}^C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 1 | Binomial and Multinomial Theorems

C01-T01-QUIZ | C01-T01-QUIZ-WRITTEN

Quiz W4 | QUIZ WRITTEN

C01-T01-Q15

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

WORKING AREA

FORMULA BANK

Binomial

$${}^n C_r a^r b^{n-r}$$

Multinomial / Trinomial

$${}^n C_r n - r {}^n C_s a^r b^s c^{n-r-s}$$

Expansion check

$$r + (n - r) = n$$

Concept 2 | Polynomial Division

OFFICIAL LESSONS 1-4, 1-5

Polynomial Division

Equation-first recall | use the same rail beside every question in this Concept

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q001 | CHALLENGE MCQ

C01-T02-C07

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$.
Then verify the result using $A = B \times Q + R$:

A $Q = 2x - 2, \quad R = -2x + 2$

B $Q = 2x - 1, \quad R = -2x + 1$

C $Q = 2x - 1, \quad R = 2x - 1$

D $Q = 2x, \quad R = -2x + 1$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q002 | CHALLENGE MCQ

C01-T02-C08

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

A $3x^3 - 7x^2 + 12x - 1$

B $(3x - 1) \times (x^2 - 2x + 3)$

C $3x^3 - 7x^2 + 10x - 5$

D $x^2 + 2x + 4$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q003 | CHALLENGE MCQ

C01-T02-C09

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

A $\frac{x^3 + 5x^2 + 9x - 11}{x + 3}$

B $-x^2 - 2x + 1$

C $\frac{x^3 + 5x^2 + 5x - 3}{x + 4}$

D $x^2 + 2x - 1$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q004 | CHALLENGE MCQ

C01-T02-C10

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

A $2x^3 + 10x^2 + 6x - 18$

B $2x + 6$

C $2x + 5$

D $\frac{1}{2x + 5}$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q005 | CHALLENGE MCQ

C01-T02-C11

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$.
Decide whether the result is exact and justify from R:

A $Q = 3x^2 + 5x + 5, \quad R = -2x - 10$

B $Q = 3x^2 + 5x + 6, \quad R = -2x - 11$

C $Q = 3x^2 + 5x + 6, \quad R = 2x + 11$

D $Q = 3x^2 + 5x + 7, \quad R = -2x - 11$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-COMBINED | V01

Q006 | CHALLENGE MCQ

C01-T02-C12

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

A $Q = x \times (x^2 - 2x + 1), \quad R = 6$

B $Q = x \times (x^2 - 2x + 1), \quad R = -6$

C $Q = x^3 - 2x^2 + x + 1, \quad R = 6$

D $Q = x^3 - 2x^2 + x - 1, \quad R = 7$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V01

Q007 | WRITTEN

C01-T02-S05

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V01

Q008 | WRITTEN

C01-T02-S06

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V01

Q009 | MCQ

C01-T02-O01

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:

A $Q = 3x^2 + 8x + 9, \quad R = -23$

B $Q = 3x^2 + 8x + 10, \quad R = 23$

C $Q = 3x^2 + 8x + 8, \quad R = 24$

D $Q = 3x^2 + 8x + 9, \quad R = 23$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V01

Q010 | MCQ

C01-T02-O03

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

A $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{37}{8}, R = \frac{111}{8}$

B $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = \frac{103}{8}$

C $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = -\frac{103}{8}$

D $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{53}{8}, R = \frac{103}{8}$

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

WORKING AREA

Concept 2 | Polynomial Division

F01 | V01

Q011 | MCQ

C01-T02-O51

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:

A $Q = x \times (3x - 1), \quad R = 9$

B $Q = 3x^2 - x + 1, \quad R = 8$

C $Q = 3x^2 - x + 1, \quad R = -8$

D $Q = 3x^2 - x + 2, \quad R = 8$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V01

Q012 | MCQ

C01-T02-O52

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:

A $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$

B $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = -38$

C $Q = 2x^3 - 3x^2 + 6x - 16, \quad R = 38$

D $Q = 2x^3 - 3x^2 + 6x - 18, \quad R = 39$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V02

Q013 | MCQ

C01-T02-O05

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:

A $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = -36$

B $Q = 4x^3 + 5x^2 + 10x + 22, \quad R = 36$

C $Q = 4x^3 + 5x^2 + 10x + 20, \quad R = 37$

D $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F01 | V02

Q014 | MCQ

C01-T02-O07

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:

$$\text{A } Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{100}{27}, \quad R = -\frac{92}{27}$$

$$\text{B } Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

$$\text{C } Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = \frac{119}{27}$$

$$\text{D } Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{46}{27}, \quad R = -\frac{119}{27}$$

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

WORKING AREA

Concept 2 | Polynomial Division

F02 | V01

Q015 | WRITTEN

C01-T02-S01

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q016 | WRITTEN

C01-T02-S03

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q017 | WRITTEN

C01-T02-S07

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q018 | WRITTEN

C01-T02-S09

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q019 | MCQ

C01-T02-O09

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:

A $Q = x + 1, \quad R = 4x - 6$

B $Q = x + 2, \quad R = -4x + 6$

C $Q = x, \quad R = -4x + 7$

D $Q = x + 1, \quad R = -4x + 6$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q020 | MCQ

C01-T02-O11

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:

A $Q = x^2 - 5x + 10, \quad R = -25x + 7$

B $Q = x^2 - 5x + 11, \quad R = -25x + 6$

C $Q = x^2 - 5x + 11, \quad R = 25x - 6$

D $Q = x^2 - 5x + 12, \quad R = -25x + 6$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q021 | MCQ

C01-T02-O53

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:

A $Q = 2x + 5, \quad R = -2x + 5$

B $Q = 2x + 6, \quad R = 2x - 5$

C $Q = 2x + 4, \quad R = 2x - 4$

D $Q = 2x + 5, \quad R = 2x - 5$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q022 | MCQ

C01-T02-O54

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:

A $Q = x^2 - 5x + 22, \quad R = -69x + 24$

B $Q = x^2 - 5x + 20, \quad R = -69x + 25$

C $Q = x^2 - 5x + 21, \quad R = -69x + 24$

D $Q = x^2 - 5x + 21, \quad R = 69x - 24$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q023 | MCQ

C01-T02-O55

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:

A $Q = 3x + 2, \quad R = x + 6$

B $Q = 3x + 3, \quad R = x + 5$

C $Q = 3x + 3, \quad R = -x - 5$

D $Q = 3x + 4, \quad R = x + 5$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V01

Q024 | MCQ

C01-T02-O56

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:

A $Q = 2x^2 - 3x + 1, \quad R = 8x - 10$

B $Q = 2x^2 - 3x + 1, \quad R = -8x + 10$

C $Q = 2x^2 - 3x + 2, \quad R = 8x - 10$

D $Q = x \times (2x - 3), \quad R = 8x - 9$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V02

Q025 | WRITTEN

C01-T02-S08

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V02

Q026 | WRITTEN

C01-T02-S10

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V02

Q027 | MCQ

C01-T02-O13

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:

A $Q = 2x - \frac{3}{2}, \quad R = -\frac{35x}{2} + 9$

B $Q = 2x - \frac{1}{2}, \quad R = \frac{35x}{2} - 9$

C $Q = 2x - \frac{5}{2}, \quad R = \frac{35x}{2} - 8$

D $Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V02

Q028 | MCQ

C01-T02-O15

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:

A $Q = x - 1, \quad R = -4x + 3$

B $Q = x, \quad R = -4x + 2$

C $Q = x, \quad R = 4x - 2$

D $Q = x + 1, \quad R = -4x + 2$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F02 | V02

Q029 | MCQ

C01-T02-O57

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

A $Q = \frac{5x}{2} - \frac{3}{4}, \quad R = \frac{9x}{4} + \frac{21}{4}$

B $Q = \frac{5x}{2} + \frac{1}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$

C $Q = \frac{5x}{2} - \frac{7}{4}, \quad R = -\frac{9x}{4} - \frac{17}{4}$

D $Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

WORKING AREA

Concept 2 | Polynomial Division

F02 | V02

Q030 | MCQ

C01-T02-O58

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

A $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{67}{64}, R = \frac{37x}{64} + \frac{163}{32}$

B $Q = \frac{3x^2}{4} + \frac{5x}{16} - \frac{61}{64}, R = \frac{37x}{64} + \frac{195}{32}$

C $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = \frac{37x}{64} + \frac{163}{32}$

D $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = -\frac{37x}{64} - \frac{163}{32}$

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

WORKING AREA

Concept 2 | Polynomial Division

F02 | V03

Q031 | MCQ

C01-T02-O17

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:

A $Q = \frac{3x}{2} + \frac{11}{4}, \quad R = \frac{11x}{4} - \frac{3}{4}$

B $Q = \frac{3x}{2} + \frac{15}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$

C $Q = \frac{3x}{2} + \frac{7}{4}, \quad R = -\frac{11x}{4} + \frac{7}{4}$

D $Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

WORKING AREA

Concept 2 | Polynomial Division

F02 | V03

Q032 | MCQ

C01-T02-O19

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

A $Q = 3x - \frac{11}{2}, \quad R = \frac{29x}{2} - 7$

B $Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$

C $Q = 3x - \frac{9}{2}, \quad R = -\frac{29x}{2} + 8$

D $Q = 3x - \frac{7}{2}, \quad R = \frac{29x}{2} - 8$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q033 | WRITTEN

C01-T02-S02

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q034 | WRITTEN

C01-T02-S04

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$.
Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q035 | WRITTEN

C01-T02-S11

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q036 | WRITTEN

C01-T02-S12

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$.
Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q037 | WRITTEN

C01-T02-S13

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q038 | MCQ

C01-T02-O21

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

A $Q = 4a^2 - 3ax + x^2, \quad R = -2a^3$

B $Q = 4a^2 - 3ax + x^2 + 1, \quad R = 2a^3$

C $Q = 4a^2 - 3ax + x^2 - 1, \quad R = 2a^3 + 1$

D $Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V01

Q039 | MCQ

C01-T02-O23

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

A $Q = x + 2y - 1, \quad R = -6y^2 + 1$

B $Q = x + 2y, \quad R = -6y^2$

C $Q = x + 2y, \quad R = 6y^2$

D $Q = x + 2y + 1, \quad R = -6y^2$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F03 | V02

Q040 | WRITTEN

C01-T02-S14

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F04 | V01

Q041 | WRITTEN

C01-T02-S16

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F04 | V01

Q042 | WRITTEN

C01-T02-S20

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F04 | V01

Q043 | WRITTEN

C01-T02-S21

Find the polynomial A when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F04 | V01

Q044 | WRITTEN

C01-T02-S22

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F04 | V02

Q045 | WRITTEN

C01-T02-S18

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V01

Q046 | WRITTEN

C01-T02-S17

When $4x^3 + 8x - 9$ is divided by B , the quotient is $2x - 3$ and the remainder is $x + 15$. Find B .

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V01

Q047 | WRITTEN

C01-T02-S19

When $x^3 - x^2 + 3x + 1$ is divided by B, the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V02

Q048 | WRITTEN

C01-T02-S23

When $2x^3 + x^2 - 3x + 5$ is divided by B , the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B .

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V02

Q049 | WRITTEN

C01-T02-S24

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V02

Q050 | WRITTEN

C01-T02-S25

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F05 | V03

Q051 | WRITTEN

C01-T02-S26

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

F06 | V01

Q052 | WRITTEN

C01-T02-S15

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-MCQ

Quiz M1 | QUIZ MCQ

C01-T02-Q01

For $x^2 - 5x + 1$ divided by $x - 2$, what is R?:

A -5

B -1

C 5

D 1

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-MCQ

Quiz M2 | QUIZ MCQ

C01-T02-Q04

In a polynomial in x , a is treated as... Which of the following is correct?:

A *another variable to divide by*

B a constant coefficient

C zero

D the divisor

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-MCQ

Quiz M3 | QUIZ MCQ

C01-T02-Q07

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

A $B = A \times Q + R$

B $B = \frac{A + R}{Q}$

C $B = \frac{A - R}{Q}$

D $B = \frac{Q}{A - R}$

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-MCQ

Quiz M4 | QUIZ MCQ

C01-T02-Q10

If B has degree 2, a valid remainder can have degree...:

A 3

B 2

C 1

D 4

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-WRITTEN

Quiz W1 | QUIZ WRITTEN

C01-T02-Q11

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q , R , and verify $A = B \times Q + R$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-WRITTEN

Quiz W2 | QUIZ WRITTEN

C01-T02-Q12

Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-WRITTEN

Quiz W3 | QUIZ WRITTEN

C01-T02-Q14

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 2 | Polynomial Division

C01-T02-QUIZ | C01-T02-QUIZ-WRITTEN

Quiz W4 | QUIZ WRITTEN

C01-T02-Q15

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

WORKING AREA

FORMULA BANK

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder theorem

$$R = P(a) \quad \text{when divisor} = x - a$$

Recovery

$$Q = \frac{A - R}{B} \quad B = \frac{A - R}{Q}$$

Concept 3 | Rational Expressions

OFFICIAL LESSONS 1-6, 1-7, 1-8

Rational Expressions

Equation-first recall | use the same rail beside every question in this Concept

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q001 | CHALLENGE MCQ

C01-T03-C01-MCQ

After simplifying this transfer chain before substitution, which result is correct for 2 ()?:

A $\frac{2x(x+3)}{(x-3)(x-1)}$

B $\frac{-2x^2 - 6x}{x^2 - 4x + 3}$

C $\frac{4x^2 + 12x}{x^2 - 4x + 3}$

D $\frac{x^2 + 3x}{x^2 - 4x + 3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q002 | CHALLENGE MCQ

C01-T03-C02-MCQ

After simplifying this transfer chain before substitution, which result is correct for 3 ()?:

A $\frac{-3x^2 + 3x + 18}{x^2 + 3x}$

B $\frac{3(x-3)(x+2)}{x(x+3)}$

C $\frac{6x^2 - 6x - 36}{x^2 + 3x}$

D $\frac{3x^2 - 3x - 18}{2x^2 + 6x}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q003 | CHALLENGE MCQ

C01-T03-C03-MCQ

After simplifying this transfer chain before substitution, which result is correct for 4():?

A $\frac{-8x^2 + 24x + 32}{x^2 - 5x + 6}$

B $\frac{16x^2 - 48x - 64}{x^2 - 5x + 6}$

C $\frac{8(x-4)(x+1)}{(x-3)(x-2)}$

D $\frac{4x^2 - 12x - 16}{x^2 - 5x + 6}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q004 | CHALLENGE MCQ

C01-T03-C04-MCQ

After simplifying this transfer chain before substitution, which result is correct for 8 ()?:

A $-\frac{8}{x}$

B $\frac{16}{x}$

C $\frac{4}{x}$

D $\frac{8}{x}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q005 | CHALLENGE MCQ

C01-T03-C05-MCQ

After simplifying this transfer chain before substitution, which result is correct for 9 ()?:

A $\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$

B $-\frac{9x^5}{2} + \frac{9}{2}$

C $9x^5 - 9$

D $\frac{9x^5}{4} - \frac{9}{4}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-COMBINED | V01

Q006 | CHALLENGE MCQ

C01-T03-C06-MCQ

After simplifying this transfer chain before substitution, which result is correct for 10():

A $\frac{-10x^3 + 10x^2 + 160x - 160}{x^3 - 5x^2 + 6x}$

B $\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$

C $\frac{20x^3 - 20x^2 - 320x + 320}{x^3 - 5x^2 + 6x}$

D $\frac{5x^3 - 5x^2 - 80x + 80}{x^3 - 5x^2 + 6x}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q007 | WRITTEN

C01-T03-S-WU1A

Simplify $\frac{12ab^5}{9a^3b^2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q008 | WRITTEN

C01-T03-S-TR1A

Simplify $\frac{6x^2y^3}{20x^3y}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q009 | WRITTEN

C01-T03-S-EX1A

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q010 | WRITTEN

C01-T03-S-EX1B

Simplify $\frac{2a^2b}{8a^3b^3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q011 | MCQ

C01-T03-S-WU1A-MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

A $-\frac{4b^3}{a^2}$

B $\frac{4b^3}{a^2}$

C $\frac{8b^3}{a^2}$

D $\frac{2b^3}{a^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q012 | MCQ

C01-T03-S-TR1A-MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

A $\frac{9y^2}{10x}$

B $-\frac{9y^2}{10x}$

C $\frac{9y^2}{5x}$

D $\frac{9y^2}{20x}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q013 | MCQ

C01-T03-S-EX1A-MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

A $\frac{3a}{b^3}$

B $-\frac{3a}{b^3}$

C $\frac{6a}{b^3}$

D $\frac{3a}{2b^3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q014 | MCQ

C01-T03-S-EX1B-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

A $-\frac{1}{2ab^2}$

B $\frac{1}{2ab^2}$

C $\frac{1}{ab^2}$

D $\frac{1}{4ab^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q015 | MCQ

C01-T03-O01-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

A $\frac{3a^2}{2b^3}$

B $-\frac{3a^2}{2b^3}$

C $\frac{3a^2}{b^3}$

D $\frac{3a^2}{4b^3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q016 | MCQ

C01-T03-O02-MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

A $\frac{9x^2y^2}{4}$

B $-\frac{9x^2y^2}{4}$

C $-\frac{9x^2y^2}{2}$

D $-\frac{9x^2y^2}{8}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q017 | MCQ

C01-T03-O03-MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

A $-\frac{3b^3}{a^3c}$

B $\frac{6b^3}{a^3c}$

C $\frac{3b^3}{a^3c}$

D $\frac{3b^3}{2a^3c}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q018 | MCQ

C01-T03-O04-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

A $-\frac{4x^3}{3y^2}$

B $\frac{8x^3}{3y^2}$

C $\frac{2x^3}{3y^2}$

D $\frac{4x^3}{3y^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q019 | MCQ

C01-T03-O05-MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

A $\frac{2a^3}{b^2}$

B $-\frac{2a^3}{b^2}$

C $\frac{4a^3}{b^2}$

D $\frac{a^3}{b^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q020 | MCQ

C01-T03-O06-MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)?$:

A $-3x^2y^3$

B $3x^2y^3$

C $6x^2y^3$

D $\frac{3x^2y^3}{2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q021 | MCQ

C01-T03-O07-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

A $\frac{3c^2}{2a^3}$

B $-\frac{3c^2}{a^3}$

C $-\frac{3c^2}{2a^3}$

D $-\frac{3c^2}{4a^3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q022 | WRITTEN

C01-T03-S-EX2A

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q023 | WRITTEN

C01-T03-S-EX2B

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q024 | MCQ

C01-T03-S-EX2A-MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

A $-\frac{12c^2}{a}$

B $\frac{24c^2}{a}$

C $\frac{12c^2}{a}$

D $\frac{6c^2}{a}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q025 | MCQ

C01-T03-S-EX2B-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

A $-\frac{8a^2c^3}{b^2}$

B $\frac{16a^2c^3}{b^2}$

C $\frac{4a^2c^3}{b^2}$

D $\frac{8a^2c^3}{b^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q026 | MCQ

C01-T03-O08-MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

A $-\frac{27ac}{2b^2}$

B $\frac{27ac}{b^2}$

C $\frac{27ac}{4b^2}$

D $\frac{27ac}{2b^2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q027 | MCQ

C01-T03-O09-MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

A $\frac{9ax^2}{y}$

B $-\frac{9ax^2}{y}$

C $\frac{18ax^2}{y}$

D $\frac{9ax^2}{2y}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q028 | MCQ

C01-T03-O10-MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

A $-\frac{64bc}{5a}$

B $\frac{64bc}{5a}$

C $\frac{128bc}{5a}$

D $\frac{32bc}{5a}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q029 | MCQ

C01-T03-O11-MCQ

After reducing coefficients and powers, which result is correct for $3 \left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy} \right)$?:

A $-\frac{18abxy^2}{5}$

B $\frac{36abxy^2}{5}$

C $\frac{18abxy^2}{5}$

D $\frac{9abxy^2}{5}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q030 | WRITTEN

C01-T03-S-WU1B

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q031 | WRITTEN

C01-T03-S-TR1B

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q032 | WRITTEN

C01-T03-S-EX1C

Simplify $\frac{x^2+3x}{x^2-9}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q033 | WRITTEN

C01-T03-S-EX1D

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q034 | WRITTEN

C01-T03-S-EX1E

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V03

Q035 | WRITTEN

C01-T03-S-EX1F

Simplify $\frac{x^3-1}{x^2+2x-3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V01

Q036 | WRITTEN

C01-T03-S-TR2A

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V01

Q037 | WRITTEN

C01-T03-S-EX2D

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V01

Q038 | WRITTEN

C01-T03-S-EX2E

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V02

Q039 | WRITTEN

C01-T03-S-EX2F

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V01

Q040 | WRITTEN

C01-T03-S-WU2

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V01

Q041 | WRITTEN

C01-T03-S-TR2B

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V02

Q042 | WRITTEN

C01-T03-S-EX2C

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V03

Q043 | WRITTEN

C01-T03-S-EX2G

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V03

Q044 | WRITTEN

C01-T03-S-EX2H

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V03

Q045 | WRITTEN

C01-T03-S-EX2I

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F05 | V01

Q046 | WRITTEN

C01-T03-S-CH1

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ for $x = \sqrt{123}$.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T04-COMBINED | V01

Q047 | CHALLENGE WRITTEN

C01-T04-C01

Simplify

$$\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right); \text{ state every excluded value.}$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q048 | WRITTEN

C01-T04-S-EX1

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q049 | MCQ

C01-T04-O01

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1} :$$

A $\frac{x^2 + 5x + 3}{x + 1}$

B $\frac{x^2 + 3x + 1}{x + 1}$

C $\frac{x^2 + 4x + 2}{x + 1}$

D $-\frac{x^2 + 4x + 2}{x + 1}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q050 | MCQ

C01-T04-O02

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4} :$$

A $\frac{3x^2 + x - 3}{(x - 2)(x + 2)}$

B $\frac{x^2 + x + 5}{(x - 2)(x + 2)}$

C $-\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

D $\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F01 | V01

Q051 | MCQ

C01-T04-O03

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)} :$$

A

$$\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

B

$$\frac{4x^2 + 4x - 7}{(x - 2)(x + 1)}$$

C

$$\frac{2x^2 + 6x - 3}{(x - 2)(x + 1)}$$

D

$$-\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F01 | V01

Q052 | MCQ

C01-T04-O04

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2} :$$

A $\frac{x^3 + 2x^2 + 2x + 5}{x + 2}$

B $\frac{x^3 + 2x^2 + x + 3}{x + 2}$

C $\frac{x^3 + 2x^2 + 1}{x + 2}$

D $-\frac{x^3 + 2x^2 + x + 3}{x + 2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q053 | WRITTEN

C01-T04-S-EX2

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q054 | MCQ

C01-T04-O07

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2} :$$

A $\frac{x(x+3)}{x+2}$

B $\frac{x^2 + 4x + 2}{x + 2}$

C $\frac{x^2 + 2x - 2}{x + 2}$

D $-\frac{x(x+3)}{x+2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q055 | MCQ

C01-T04-O08

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1} :$$

A

$$\frac{4x^2 - 4x + 5}{(x - 1)(x + 1)}$$

C

$$\frac{2x^2 - 4x + 7}{(x - 1)(x + 1)}$$

B

$$\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

D

$$-\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F01 | V02

Q056 | MCQ

C01-T04-O09

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)} :$$

A $\frac{x^2 - 2x + 4}{(x - 1)(x + 2)}$

B $-\frac{x^2 + 4x - 8}{(x - 1)(x + 2)}$

C $-\frac{3(x - 2)}{(x - 1)(x + 2)}$

D $\frac{3(x - 2)}{(x - 1)(x + 2)}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q057 | MCQ

C01-T04-O10

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3} :$$

A $\frac{2x^3 - x^2 + 2x - 10}{x - 3}$

B $\frac{2x^3 - x^2 - 4}{x - 3}$

C $\frac{2x^3 - x^2 + x - 7}{x - 3}$

D $-\frac{2x^3 - x^2 + x - 7}{x - 3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q058 | WRITTEN

C01-T04-S-WU1

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q059 | WRITTEN

C01-T04-S-TR1

Simplify

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q060 | MCQ

C01-T04-O13

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)} :$$

A

$$\frac{x^3 + 12x^2 + 30x + 11}{(x+1)(x+3)(x+5)}$$

C

$$-\frac{x^3 + 6x^2 + 16x + 19}{(x+1)(x+3)(x+5)}$$

B

$$\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

D

$$-\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F02 | V01

Q061 | MCQ

C01-T04-O14

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)} :$$

A

$$\frac{x^3 + 14x^2 + 53x + 62}{(x+2)(x+4)(x+6)}$$

B

$$-\frac{x^3 + 10x^2 + 35x + 34}{(x+2)(x+4)(x+6)}$$

C

$$\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

D

$$-\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q062 | MCQ

C01-T04-O15

Simplify

$$\frac{3x+2}{(x+1)(x+4)} + \frac{x+5}{(x+4)(x+7)} :$$

A

$$\frac{x^3 + 16x^2 + 68x + 47}{(x+1)(x+4)(x+7)}$$

B

$$-\frac{x^3 + 8x^2 + 10x + 9}{(x+1)(x+4)(x+7)}$$

C

$$-\frac{4x^2 + 29x + 19}{(x+1)(x+4)(x+7)}$$

D

$$\frac{4x^2 + 29x + 19}{(x+1)(x+4)(x+7)}$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q063 | MCQ

C01-T04-O16

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)} :$$

A

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

B

$$\frac{x^3 + 20x^2 + 80x + 82}{(x+2)(x+5)(x+8)}$$

C

$$\frac{x^3 + 10x^2 + 52x + 78}{(x+2)(x+5)(x+8)}$$

D

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q064 | MCQ

C01-T04-O17

Simplify

$$\frac{2x+1}{(x+3)(x+6)} + \frac{x+4}{(x+6)(x+9)} :$$

A

$$\frac{x^3 + 21x^2 + 125x + 183}{(x+3)(x+6)(x+9)}$$

B

$$\frac{3x^2 + 26x + 21}{(x+3)(x+6)(x+9)}$$

C

$$-\frac{x^3 + 15x^2 + 73x + 141}{(x+3)(x+6)(x+9)}$$

D

$$-\frac{3x^2 + 26x + 21}{(x+3)(x+6)(x+9)}$$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F02 | V01

Q065 | MCQ

C01-T04-O18

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)} :$$

A $\frac{x^3 + 18x^2 + 63x + 25}{(x+1)(x+5)(x+8)}$

B $-\frac{x^3 + 10x^2 + 43x + 55}{(x+1)(x+5)(x+8)}$

C $\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$

D $-\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q066 | WRITTEN

C01-T04-S-TR2

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q067 | WRITTEN

C01-T04-S-EX3

Simplify

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q068 | WRITTEN

C01-T04-S-EX5

Simplify

$$\frac{2}{x-1} + \frac{3}{x^2 + 5x - 6}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q069 | MCQ

C01-T04-O23

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1} :$$

A

$$\frac{x^2 + 3x - 1}{(x+1)(x+3)}$$

B

$$-\frac{x^2 + 5x + 7}{(x+1)(x+3)}$$

C

$$-\frac{x+4}{(x+1)(x+3)}$$

D

$$\frac{x+4}{(x+1)(x+3)}$$

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

WORKING AREA

Concept 3 | Rational Expressions

F02 | V03

Q070 | WRITTEN

C01-T04-S-EX4

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V03

Q071 | WRITTEN

C01-T04-S-EX10

Simplify

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V03

Q072 | WRITTEN

C01-T04-S-EX11

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V01

Q073 | WRITTEN

C01-T04-S-EX6

Simplify

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V02

Q074 | WRITTEN

C01-T04-S-EX7

Simplify

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V02

Q075 | WRITTEN

C01-T04-S-EX8

Simplify

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V02

Q076 | WRITTEN

C01-T04-S-EX9

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V01

Q077 | MCQ

C01-T04-O53

A learner combines $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

A

$$\frac{1}{(x-1)(x+2)}$$

B

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

C

$$-\frac{2x^2+2x-5}{(x-1)(x+2)}$$

D

$$\frac{x^2+x-3}{(x-1)(x+2)}$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F05 | V01

Q078 | WRITTEN

C01-T04-S-CH1

Simplify

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T05-COMBINED | V01

Q079 | CHALLENGE WRITTEN

C01-T05-C01

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V01

Q080 | WRITTEN

C01-T05-S-WU1

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q081 | WRITTEN

C01-T05-S-TR1

Simplify

$$\frac{1}{1 - \frac{x}{x+1}}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q082 | MCQ

C01-T05-O01

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}} :$$

A $-x$ B $-x - 2$ C $x + 1$ D $-x - 1$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q083 | MCQ

C01-T05-O02

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}} :$$

A $-x - 2$

B $-x - 1$

C $-x - 3$

D $x + 2$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q084 | MCQ

C01-T05-O03

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}} :$$

A $-\frac{1}{x}$

B $-\frac{x+1}{x}$

C $-\frac{2x+1}{x}$

D $\frac{x+1}{x}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q085 | MCQ

C01-T05-O04

Simplify

$$\frac{1}{1 - \frac{2x+3}{x+1}} :$$

A $\frac{1}{x+2}$

B $-\frac{2x+3}{x+2}$

C $-\frac{x+1}{x+2}$

D $\frac{x+1}{x+2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q086 | MCQ

C01-T05-O05

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}} :$$

A $\frac{x-3}{2x-1}$

B $-\frac{3x+1}{2x-1}$

C $\frac{x+2}{2x-1}$

D $-\frac{x+2}{2x-1}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q087 | MCQ

C01-T05-O06

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}} :$$

A $-x - 3$

B $-x - 2$

C $-x - 4$

D $x + 3$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F01 | V02

Q088 | MCQ

C01-T05-O07

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}} :$$

A $\frac{3}{x+4}$

B $-\frac{x+1}{x+4}$

C $-\frac{2x+5}{x+4}$

D $\frac{x+1}{x+4}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q089 | WRITTEN

C01-T05-S-WU2

Simplify

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q090 | MCQ

C01-T05-O15

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

A $\frac{2x+5}{3}$

B $\frac{2(x+4)}{3}$

C $\frac{2(x+1)}{3}$

D $-\frac{2x+5}{3}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V01

Q091 | MCQ

C01-T05-O19

Simplify

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

A $\frac{x+3}{2}$

B $\frac{x+5}{2}$

C $\frac{x+1}{2}$

D $-\frac{x+3}{2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q092 | WRITTEN

C01-T05-S-TR2

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q093 | MCQ

C01-T05-O13

Simplify

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

A 2

B 0

C 1

D -1

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q094 | MCQ

C01-T05-O14

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

A 2

B 0

C -1

D 1

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V02

Q095 | MCQ

C01-T05-O16

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

A 2

B 1

C 0

D -1

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F02 | V03

Q096 | WRITTEN

C01-T05-S-EX3

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V01

Q097 | WRITTEN

C01-T05-S-TR3

Simplify

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F03 | V02

Q098 | WRITTEN

C01-T05-S-EX5

Simplify

$$\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F04 | V01

Q099 | WRITTEN

C01-T05-S-EX2

Simplify

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F05 | V01

Q100 | WRITTEN

C01-T05-S-EX1

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F05 | V02

Q101 | WRITTEN

C01-T05-S-EX4

Simplify

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

F06 | V01

Q102 | WRITTEN

C01-T05-S-CH1

Simplify

$$\frac{\frac{1}{1 + \frac{1}{1 + \frac{1}{1+x}}}}{1 + \frac{1}{1 + \frac{1}{1+x}}}.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-QUIZ | C01-T03-QUIZ-MCQ

Quiz M1 | QUIZ MCQ

C01-T03-Q01

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

A $\frac{x-2}{3x+1}$

B $\frac{x+2}{3x+1}$

C $\frac{x-2}{3x+2}$

D $\frac{x^2-4}{3x+1}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T04-QUIZ | C01-T04-QUIZ-MCQ

Quiz M2 | QUIZ MCQ

C01-T04-Q01

Simplify
 $\frac{x+1}{x-2} + \frac{2}{x-2} :$

A $\frac{x+3}{x-2}$

B $\frac{x-1}{x-2}$

C $\frac{x+1}{2x-4}$

D $\frac{x+3}{x+2}$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T04-QUIZ | C01-T04-QUIZ-MCQ

Quiz M3 | QUIZ MCQ

C01-T04-Q10

In $\frac{3}{x^2 + x} - \frac{3 - x}{x^2 + x}$, the numerator after the minus sign is... Which of the following is correct?:

A $3 - (3 - x)$

B $3 - 3 - x$

C $3 + 3 - x$

D $3 - x$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T05-QUIZ | C01-T05-QUIZ-MCQ

Quiz M4 | QUIZ MCQ

C01-T05-Q10

Which statement is safe when simplifying a compound fraction?:

A Multiply top and bottom by the same inner denominator

B Add all denominators

C Cancel terms across +

D Drop all restrictions

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T03-QUIZ | C01-T03-QUIZ-WRITTEN

Quiz W1 | QUIZ WRITTEN

C01-T03-Q11

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T04-QUIZ | C01-T04-QUIZ-WRITTEN

Quiz W2 | QUIZ WRITTEN

C01-T04-Q11

Simplify

 $\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T04-QUIZ | C01-T04-QUIZ-WRITTEN

Quiz W3 | QUIZ WRITTEN

C01-T04-Q15

Simplify

$$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y} \text{ in } x \text{ and } y.$$

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 3 | Rational Expressions

C01-T05-QUIZ | C01-T05-QUIZ-WRITTEN

Quiz W4 | QUIZ WRITTEN

C01-T05-Q15

Explain why $(x^2-16)/(x-4)=x+4$ is not an identity for every real x .

WORKING AREA

FORMULA BANK

Restrictions

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Concept 4 | Identities and Proofs

OFFICIAL LESSONS 1-9, 1-10, 1-11, 1-12

Identities and Proofs

Equation-first recall | use the same rail beside every question in this Concept

Proof route

LHS $\longrightarrow$ RHS

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Identities and Proofs

C01-T06-COMBINED | V01

Q001 | CHALLENGE WRITTEN

C01-T06-C01

$2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find a,b,c.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q002 | WRITTEN

C01-T06-S-WU1

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q003 | WRITTEN

C01-T06-S-TR1

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q004 | MCQ

C01-T06-O01

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

A (2, 2, -3)

B (0, 2, -3)

C (1, 2, -3)

D (1, 3, -3)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q005 | MCQ

C01-T06-O02

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

A (3, -3, -1)

B (1, -3, -1)

C (2, -3, -1)

D (2, -2, -1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q006 | MCQ

C01-T06-O03

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

A (4, 4, 1)

B (2, 4, 1)

C (3, 5, 1)

D (3, 4, 1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q007 | MCQ

C01-T06-O04

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

A (1, -5, 3)

B (2, -5, 3)

C (0, -5, 3)

D (1, -4, 3)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q008 | WRITTEN

C01-T06-S-EX1

$$a(x + 1) - b(x + 5) = 8x$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q009 | MCQ

C01-T06-O07

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:

A (2, 1)

B (3, 1)

C (1, 1)

D (2, 2)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q010 | MCQ

C01-T06-O08

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

A (4, -2)

B (3, -2)

C (2, -2)

D (3, -1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q011 | MCQ

C01-T06-O09

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

A (5, 3)

B (3, 3)

C (4, 4)

D (4, 3)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q012 | MCQ

C01-T06-O10

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

A (6, -4)

B (4, -4)

C (5, -3)

D (5, -4)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q013 | WRITTEN

C01-T06-S-EX2

$$(x + a)(b + 3) = 4x + 12$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q014 | MCQ

C01-T06-O13

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

A (3, -1)

B (2, -1)

C (1, -1)

D (2, 0)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q015 | MCQ

C01-T06-O14

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

A (4, 0)

B (2, 0)

C (3, 0)

D (3, 1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q016 | MCQ

C01-T06-O15

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

A (5, 1)

B (3, 1)

C (4, 1)

D (4, 2)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q017 | WRITTEN

C01-T06-S-EX3

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q018 | WRITTEN

C01-T06-S-EX4

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q019 | WRITTEN

C01-T06-S-EX5

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q020 | MCQ

C01-T06-O18

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

A (2, -2)

B (1, -2)

C (0, -2)

D (1, -1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q021 | MCQ

C01-T06-O19

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

A (3, -1)

B (1, -1)

C (2, -1)

D (2, 0)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q022 | MCQ

C01-T06-O20

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

A (4, 0)

B (3, 0)

C (2, 0)

D (3, 1)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Q023 | WRITTEN

C01-T06-S-WU2

$$\frac{3x+1}{(x+2)(x-3)} = \frac{a}{x+2} + \frac{b}{x-3}$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Q024 | WRITTEN

C01-T06-S-EX7

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V02

Q025 | WRITTEN

C01-T06-S-TR2

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V03

Q026 | WRITTEN

C01-T06-S-EX6

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Q027 | WRITTEN

C01-T06-S-EX8

$$\frac{3x-8}{3x^2+5x-2} = \frac{a}{x+2} + \frac{b}{3x-1}$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V02

Q028 | WRITTEN

C01-T06-S-CH1

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, t > 0$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q029 | WRITTEN

C01-T07-S-WU1

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q030 | WRITTEN

C01-T07-S-EX6

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q031 | MCQ

C01-T07-P-F01-V01-01

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, the value of the simplified left-hand side is:

A $8xy + 8y$

B $8xy + 8y + 1$

C $8xy + 8y - 1$

D $-8xy - 8y$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q032 | MCQ

C01-T07-P-F01-V01-02

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

A $12xy + 24y + 1$

B $12xy + 24y - 1$

C $-12xy - 24y$

D $12xy + 24y$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q033 | MCQ

C01-T07-P-F01-V01-03

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

A $4xy + 12y + 1$

B $4xy + 12y - 1$

C $-4xy - 12y$

D $4xy + 12y$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q034 | MCQ

C01-T07-P-F01-V01-04

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

A $8xy + 32y$

B $8xy + 32y + 1$

C $8xy + 32y - 1$

D $-8xy - 32y$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q035 | WRITTEN

C01-T07-S-TR1

Prove that $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q036 | MCQ

C01-T07-P-F01-V02-01

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

A $2x^2 + 4x + 8y^2 + 3$

B $2x^2 + 4x + 8y^2 + 1$

C $2x^2 + 4x + 8y^2 + 2$

D $-2x^2 - 4x - 8y^2 - 2$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q037 | MCQ

C01-T07-P-F01-V02-02

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

A $2x^2 + 8x + 18y^2 + 9$

B $2x^2 + 8x + 18y^2 + 7$

C $2x^2 + 8x + 18y^2 + 8$

D $-2x^2 - 8x - 18y^2 - 8$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q038 | MCQ

C01-T07-P-F01-V02-03

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

A $2x^2 + 12x + 2y^2 + 19$

B $2x^2 + 12x + 2y^2 + 17$

C $2x^2 + 12x + 2y^2 + 18$

D $-2x^2 - 12x - 2y^2 - 18$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q039 | MCQ

C01-T07-P-F01-V02-04

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

A $2x^2 + 16x + 8y^2 + 33$

B $2x^2 + 16x + 8y^2 + 31$

C $-2x^2 - 16x - 8y^2 - 32$

D $2x^2 + 16x + 8y^2 + 32$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q040 | WRITTEN

C01-T07-S-EX4

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q041 | MCQ

C01-T07-P-F01-V03-01

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, the value of the simplified left-hand side is:

A $8xy - 8x + 8y - 7$

B $8xy - 8x + 8y - 9$

C $-8xy + 8x - 8y + 8$

D $8xy - 8x + 8y - 8$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q042 | MCQ

C01-T07-P-F01-V03-02

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

A $24xy - 24x + 24y - 23$

B $24xy - 24x + 24y - 25$

C $24xy - 24x + 24y - 24$

D $-24xy + 24x - 24y + 24$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Q043 | WRITTEN

C01-T07-S-EX1

Prove that $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V02

Q044 | WRITTEN

C01-T07-S-CH1

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Q045 | WRITTEN

C01-T07-S-WU2

Prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Q046 | WRITTEN

C01-T07-S-EX5

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V02

Q047 | WRITTEN

C01-T07-S-TR2

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F04 | V01

Q048 | WRITTEN

C01-T07-S-EX2

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F04 | V02

Q049 | WRITTEN

C01-T07-S-EX3

Prove that $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F99 | V01

Q050 | CHALLENGE MCQ

C01-T07-C-01

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

A $8x^3y + 8x^2y + 8xy + 8y + 1$

B $8x^3y + 8x^2y + 8xy + 8y$

C $8x^3y + 8x^2y + 8xy + 8y - 1$

D $-8x^3y - 8x^2y - 8xy - 8y$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q051 | WRITTEN

C01-T08-S-TR1

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q052 | WRITTEN

C01-T08-S-EX1

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q053 | MCQ

C01-T08-P-F01-V01-01

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

A $-2xy + 5$

B $-2xy + 3$

C $-2(xy - 2)$

D $2(xy - 2)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q054 | MCQ

C01-T08-P-F01-V01-02

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

A $-2(xy - 5)$

B $-2(xy - 4)$

C $2xy - 9$

D $-2xy + 9$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q055 | MCQ

C01-T08-P-F01-V01-03

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

A $-2(xy - 1)$

B $-2xy$

C $-2xy + 1$

D $2xy - 1$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q056 | MCQ

C01-T08-P-F01-V01-04

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

A $-2(xy - 8)$

B $-2xy + 17$

C $-2xy + 15$

D $2(xy - 8)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q057 | WRITTEN

C01-T08-S-EX2

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q058 | MCQ

C01-T08-P-F01-V02-01

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

A $2(xy + 1)$

B $2xy + 1$

C $2xy$

D $-2xy - 1$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q059 | MCQ

C01-T08-P-F01-V02-02

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

A $2xy + 5$

B $2(xy + 2)$

C $2xy + 3$

D $-2(xy + 2)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q060 | MCQ

C01-T08-P-F01-V02-03

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

A $2xy + 5$

B $2xy + 3$

C $2(xy + 2)$

D $-2(xy + 2)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q061 | MCQ

C01-T08-P-F01-V02-04

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

A $2(xy + 5)$

B $2(xy + 4)$

C $-2xy - 9$

D $2xy + 9$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q062 | WRITTEN

C01-T08-S-EX3

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Q063 | WRITTEN

C01-T08-S-WU1

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V02

Q064 | WRITTEN

C01-T08-S-TR2

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Q065 | WRITTEN

C01-T08-S-EX4

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V02

Q066 | WRITTEN

C01-T08-S-EX5

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F04 | V01

Q067 | WRITTEN

C01-T08-S-CH1

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F99 | V01

Q068 | CHALLENGE MCQ

C01-T08-C-01

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

A $-2xy + 11$

B $-2xy + 9$

C $-2(xy - 5)$

D $2(xy - 5)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q069 | WRITTEN

C01-T09-S-WU1

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q070 | MCQ

C01-T09-P-001

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q071 | MCQ

C01-T09-P-002

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\frac{cd}{c^2 + d^2}$

D $\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q072 | MCQ

C01-T09-P-003

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q073 | MCQ

C01-T09-P-004

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\frac{cd}{c^2 + d^2}$

D $\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q074 | CHALLENGE MCQ

C01-T09-C-01

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\frac{cd}{c^2 + d^2}$

D $\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V01

Q075 | CHALLENGE MCQ

C01-T09-C-02

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q076 | WRITTEN

C01-T09-S-TR1

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q077 | MCQ

C01-T09-P-008

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q078 | MCQ

C01-T09-P-009

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2+c^2}{(a-c)(a+c)}$

D $\frac{a^2+c^2}{(a-c)(a+c)}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q079 | MCQ

C01-T09-P-010

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2+c^2}{(a-c)(a+c)}$

D $\frac{a^2+c^2}{(a-c)(a+c)}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V02

Q080 | MCQ

C01-T09-P-011

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q081 | WRITTEN

C01-T09-S-EX1

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q082 | MCQ

C01-T09-P-015

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{2c}{c-d}$

B $\frac{c+d}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q083 | MCQ

C01-T09-P-016

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{2c}{c-d}$

B $\frac{c+d}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q084 | MCQ

C01-T09-P-017

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V03

Q085 | MCQ

C01-T09-P-018

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q086 | WRITTEN

C01-T09-S-EX2

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V04

Q087 | MCQ

C01-T09-P-022

Under the condition $a/b = c/d$, the required expression is equal to:

A $\frac{2a - c}{2b - d}$

B $\frac{2a + 2b - c - d}{2b - d}$

C $\frac{2a - 2b - c + d}{2b - d}$

D $-\frac{2a - c}{2b - d}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V05

Q088 | WRITTEN

C01-T09-S-EX3

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}$, $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F01 | V06

Q089 | WRITTEN

C01-T09-S-CH1

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Q090 | WRITTEN

C01-T09-S-TR2

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} =$

$$\frac{z}{c}, \quad \frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V02

Q091 | WRITTEN

C01-T09-S-EX4

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} =$

$$\frac{z}{c}, \quad \frac{x+y+z}{a+b+c} = \frac{x}{a}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V03

Q092 | WRITTEN

C01-T09-S-EX5

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} =$

$$\frac{z}{c}, \quad \frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Q093 | WRITTEN

C01-T09-S-WU2

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression:
 $x : y : z = 2 : 3 : 5, \frac{2xy+yz+3zx}{x^2+y^2+z^2}.$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V02

Q094 | WRITTEN

C01-T09-S-TR3

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression:
 $a : b : c = 2 : 3 : 4, \frac{ab+bc+ca}{a^2+b^2+c^2}.$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V03

Q095 | WRITTEN

C01-T09-S-EX7

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression:

$$a : b : c = 3 : 5 : 2, \quad \frac{a^2 + b^2 - c^2}{a^2 - b^2 + c^2}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V04

Q096 | WRITTEN

C01-T09-S-EX8

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression:

$$x : y : z = 2 : -3 : 4, \quad \frac{x^2 + y^2 + z^2}{xy + yz + zx}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V05

Q097 | WRITTEN

C01-T09-S-EX9

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq$

$$0, \frac{x^2+5y^2}{x^2+xy+y^2}.$$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F04 | V01

Q098 | WRITTEN

C01-T09-S-EX6

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

C01-T06-QUIZ | C01-T06-QUIZ-MCQ

Quiz M1 | QUIZ MCQ

C01-T06-Q01

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

A (1, 3, -2)

B (1, 2, -1)

C (2, 3, -2)

D (1, -3, 2)

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Quiz M2 | QUIZ MCQ

C01-T07-Q-M04

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

A $x^4 + 4$

B $x^4 + 5$

C $x^4 + 3$

D $-x^4 - 4$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Quiz M3 | QUIZ MCQ

C01-T08-Q-M07

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

A $3b(b + c)$

B $3b^2 + 3bc + 1$

C $3b^2 + 3bc - 1$

D $-3b(b + c)$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Quiz M4 | QUIZ MCQ

C01-T09-Q-M10

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

A $\frac{5}{2}$

B $\frac{3}{2}$

C $\frac{1}{2}$

D $-\frac{3}{2}$

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ RHS

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

C01-T06-QUIZ | C01-T06-QUIZ-WRITTEN

Quiz W1 | QUIZ WRITTEN

C01-T06-Q11

Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F02 | V01

Quiz W2 | QUIZ WRITTEN

C01-T07-Q-W02

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F03 | V01

Quiz W3 | QUIZ WRITTEN

C01-T08-Q-W04

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 4 | Identities and Proofs

F04 | V01

Quiz W4 | QUIZ WRITTEN

C01-T09-Q-W05

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

WORKING AREA

FORMULA BANK

Proof route

LHS $\longrightarrow$ **RHS**

Useful identities

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Rational identity

$$\frac{A}{B} = \frac{AC}{BC} \quad (C \neq 0)$$

Concept 5 | Inequalities

OFFICIAL LESSONS 1-13, 1-14, 1-15, 1-16

Inequalities

Equation-first recall | use the same rail beside every question in this Concept

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Start with the family cue, write the first move, then use the working grid.

Concept 5 | Inequalities

F01 | V01

Q001 | WRITTEN

C01-T10-S-WU1

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q002 | MCQ

C01-T10-G-001

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q003 | MCQ

C01-T10-G-002

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

A $x - y + 1$

B $x - y - 1$

C $x - y$

D $-x + y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q004 | MCQ

C01-T10-G-003

If $x > y$, prove that $3(x + y) > 2(x + 2y)$.

If $x > y$, the positive difference between the two sides is:

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q005 | MCQ

C01-T10-G-004

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q006 | MCQ

C01-T10-G-005

If $x > y$, prove that $3x + 5y > 2(x + 3y)$.

If $x > y$, the positive difference between the two sides is:

A $x - y + 1$

B $x - y$

C $x - y - 1$

D $-x + y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q007 | WRITTEN

C01-T10-S-TR1

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q008 | WRITTEN

C01-T10-S-EX2

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q009 | MCQ

C01-T10-G-006

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

A $ac - ad - bc + bd + 1$

B $ac - ad - bc + bd - 1$

C $-(a - b)(c - d)$

D $(a - b)(c - d)$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q010 | MCQ

C01-T10-G-007

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

A $ac - ad - 2bc + 2bd + 1$

B $ac - ad - 2bc + 2bd - 1$

C $(a - 2b)(c - d)$

D $-(a - 2b)(c - d)$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q011 | MCQ

C01-T10-G-008

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

A $(a - 3b)(c - d)$

B $ac - ad - 3bc + 3bd + 1$

C $ac - ad - 3bc + 3bd - 1$

D $-(a - 3b)(c - d)$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V03

Q012 | WRITTEN

C01-T10-S-EX1

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V01

Q013 | WRITTEN

C01-T10-S-WU2

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q014 | WRITTEN

C01-T10-S-TR2

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V03

Q015 | WRITTEN

C01-T10-S-EX3

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F03 | V01

Q016 | WRITTEN

C01-T10-S-EX4

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F03 | V02

Q017 | WRITTEN

C01-T10-S-CH1

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F99 | V01

Q018 | CHALLENGE MCQ

C01-T10-C-001

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

A $2ax - 2ay - 2bx + 2by + 1$

B $2ax - 2ay - 2bx + 2by - 1$

C $-2(a - b)(x - y)$

D $2(a - b)(x - y)$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q019 | WRITTEN

C01-T11-S-WU1

Prove that $a^2 + 4b^2 \geq 4ab$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q020 | MCQ

C01-T11-G-001

Under a, b real, prove that $a^2 + 4b^2 \geq 4ab$ select the correct factorisation of LHS – RHS:

A $a^2 - 4ab + 4b^2 + 1$

B $(a - 2b - 1)(a - 2b + 1)$

C $-(a - 2b)^2$

D $(a - 2b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q021 | MCQ

C01-T11-G-002

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS – RHS:

A $a^2 - 6ab + 9b^2 + 1$

B $(a - 3b - 1)(a - 3b + 1)$

C $(a - 3b)^2$

D $-(a - 3b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q022 | MCQ

C01-T11-G-003

Under a, b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS – RHS:

A $a^2 - 8ab + 16b^2 + 1$

B $(a - 4b)^2$

C $(a - 4b - 1)(a - 4b + 1)$

D $-(a - 4b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q023 | MCQ

C01-T11-G-004

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q024 | MCQ

C01-T11-G-005

Under a, b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS – RHS:

A $a^2 - 12ab + 36b^2 + 1$

B $(a - 6b - 1)(a - 6b + 1)$

C $(a - 6b)^2$

D $-(a - 6b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q025 | MCQ

C01-T11-G-006

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS – RHS:

A $a^2 - 14ab + 49b^2 + 1$

B $(a - 7b)^2$

C $(a - 7b - 1)(a - 7b + 1)$

D $-(a - 7b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q026 | WRITTEN

C01-T11-S-TR2

Prove that $a^2 + 5b^2 \geq 4ab$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q027 | WRITTEN

C01-T11-S-EX1

Prove that $9x^2 \geq y(6x - y)$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q028 | WRITTEN

C01-T11-S-EX2

Prove that $2(x^2 + y^2) \geq (x + y)^2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q029 | WRITTEN

C01-T11-S-EX3

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q030 | WRITTEN

C01-T11-S-EX4

Prove that $(2a + 3b)^2 \geq 12ab$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q031 | MCQ

C01-T11-G-007

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS – RHS:

A $(2x - y)^2$

B $4x^2 - 4xy + y^2 + 1$

C $(2x - y - 1)(2x - y + 1)$

D $-(2x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q032 | MCQ

C01-T11-G-008

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS – RHS:

A $9x^2 - 6xy + y^2 + 1$

B $(3x - y - 1)(3x - y + 1)$

C $(3x - y)^2$

D $-(3x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q033 | MCQ

C01-T11-G-009

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS – RHS:

A $16x^2 - 8xy + y^2 + 1$

B $(4x - y - 1)(4x - y + 1)$

C $-(4x - y)^2$

D $(4x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q034 | MCQ

C01-T11-G-010

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS – RHS:

A $25x^2 - 10xy + y^2 + 1$

B $(5x - y - 1)(5x - y + 1)$

C $-(5x - y)^2$

D $(5x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q035 | MCQ

C01-T11-G-011

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS – RHS:

A $(6x - y)^2$

B $36x^2 - 12xy + y^2 + 1$

C $(6x - y - 1)(6x - y + 1)$

D $-(6x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V03

Q036 | MCQ

C01-T11-G-012

Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS – RHS:

A $x^2 - 2xy + y^2 + 1$

B $(x - y - 1)(x - y + 1)$

C $(x - y)^2$

D $-(x - y)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V01

Q037 | WRITTEN

C01-T11-S-WU2

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q038 | WRITTEN

C01-T11-S-EX5

Prove that $x^2 + 2 > 2x$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q039 | WRITTEN

C01-T11-S-EX6

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V03

Q040 | WRITTEN

C01-T11-S-CH1

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F03 | V01

Q041 | WRITTEN

C01-T11-S-TR1

Prove that $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F99 | V01

Q042 | CHALLENGE WRITTEN

C01-T11-C-01

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q043 | WRITTEN

C01-T12-S-WU1

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q044 | MCQ

C01-T12-G-001

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-b)^2}{4}$

B $\frac{a^2 - 2ab + b^2 + 4}{4}$

C $\frac{(a-b-2)(a-b+2)}{4}$

D $-\frac{(a-b)^2}{4}$

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

WORKING AREA

Concept 5 | Inequalities

F01 | V01

Q045 | MCQ

C01-T12-G-002

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-2b)^2}{4}$

B $\frac{a^2 - 4ab + 4b^2 + 4}{4}$

C $\frac{(a-2b-2)(a-2b+2)}{4}$

D $-\frac{(a-2b)^2}{4}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q046 | MCQ

C01-T12-G-003

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{a^2 - 6ab + 9b^2 + 4}{4}$

B $\frac{(a - 3b - 2)(a - 3b + 2)}{4}$

C $\frac{(a - 3b)^2}{4}$

D $-\frac{(a - 3b)^2}{4}$

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

WORKING AREA

Concept 5 | Inequalities

F01 | V01

Q047 | MCQ

C01-T12-G-004

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-4b)^2}{4}$

B $\frac{a^2 - 8ab + 16b^2 + 4}{4}$

C $\frac{(a-4b-2)(a-4b+2)}{4}$

D $-\frac{(a-4b)^2}{4}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q048 | MCQ

C01-T12-G-005

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-5b)^2}{4}$

B $\frac{a^2 - 10ab + 25b^2 + 4}{4}$

C $\frac{(a-5b-2)(a-5b+2)}{4}$

D $-\frac{(a-5b)^2}{4}$

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

WORKING AREA

Concept 5 | Inequalities

F01 | V01

Q049 | MCQ

C01-T12-G-006

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{a^2 - 12ab + 36b^2 + 4}{4}$

B $\frac{(a - 6b)^2}{4}$

C $\frac{(a - 6b - 2)(a - 6b + 2)}{4}$

D $-\frac{(a - 6b)^2}{4}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q050 | MCQ

C01-T12-G-007

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-7b)^2}{4}$

B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$

C $\frac{(a-7b-2)(a-7b+2)}{4}$

D $-\frac{(a-7b)^2}{4}$

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

WORKING AREA

Concept 5 | Inequalities

F01 | V01

Q051 | MCQ

C01-T12-G-008

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-8b)^2}{4}$

B $\frac{a^2 - 16ab + 64b^2 + 4}{4}$

C $\frac{(a-8b-2)(a-8b+2)}{4}$

D $-\frac{(a-8b)^2}{4}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q052 | WRITTEN

C01-T12-S-TR1

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q053 | WRITTEN

C01-T12-S-EX1

Prove that $a + 2 \geq \sqrt{3 + 6a}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q054 | WRITTEN

C01-T12-S-EX2

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q055 | WRITTEN

C01-T12-S-EX3

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q056 | MCQ

C01-T12-G-009

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $2\sqrt{a}\sqrt{b} + 1$

B $2\sqrt{a}\sqrt{b}$

C $2\sqrt{a}\sqrt{b} - 1$

D $-2\sqrt{a}\sqrt{b}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q057 | MCQ

C01-T12-G-010

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $4\sqrt{a}\sqrt{b}$

B $4\sqrt{a}\sqrt{b} + 1$

C $4\sqrt{a}\sqrt{b} - 1$

D $-4\sqrt{a}\sqrt{b}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q058 | MCQ

C01-T12-G-011

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $6\sqrt{a}\sqrt{b} + 1$

B $6\sqrt{a}\sqrt{b} - 1$

C $6\sqrt{a}\sqrt{b}$

D $-6\sqrt{a}\sqrt{b}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q059 | MCQ

C01-T12-G-012

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $8\sqrt{a}\sqrt{b} + 1$

B $8\sqrt{a}\sqrt{b}$

C $8\sqrt{a}\sqrt{b} - 1$

D $-8\sqrt{a}\sqrt{b}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V01

Q060 | WRITTEN

C01-T12-S-WU2

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q061 | WRITTEN

C01-T12-S-TR2

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q062 | WRITTEN

C01-T12-S-EX4

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q063 | WRITTEN

C01-T12-S-EX5

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q064 | WRITTEN

C01-T12-S-EX6

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F03 | V01

Q065 | WRITTEN

C01-T12-S-CH1

If $a > b > 0$, prove that $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F99 | V01

Q066 | CHALLENGE WRITTEN

C01-T12-C-01

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q067 | WRITTEN

C01-T13-S-WU1

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q068 | WRITTEN

C01-T13-S-TR1

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q069 | MCQ

C01-T13-G-001

Under $a > 0$ and $b > 0$, prove that $\frac{a^2 + b^2}{ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{(a-b)^2}{ab}$

B $\frac{a^2 - ab + b^2}{ab}$

C $\frac{a^2 - 3ab + b^2}{ab}$

D $-\frac{(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q070 | MCQ

C01-T13-G-002

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{4a^2 - 2ab + b^2}{2ab}$

B $\frac{4a^2 - 6ab + b^2}{2ab}$

C $\frac{(2a - b)^2}{2ab}$

D $-\frac{(2a - b)^2}{2ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q071 | MCQ

C01-T13-G-003

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{9a^2 - 3ab + b^2}{3ab}$

B $\frac{(3a - b)^2}{3ab}$

C $\frac{9a^2 - 9ab + b^2}{3ab}$

D $-\frac{(3a - b)^2}{3ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q072 | MCQ

C01-T13-G-004

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{(4a - b)^2}{4ab}$

B $\frac{16a^2 - 4ab + b^2}{4ab}$

C $\frac{16a^2 - 12ab + b^2}{4ab}$

D $-\frac{(4a - b)^2}{4ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q073 | MCQ

C01-T13-G-005

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{25a^2 - 5ab + b^2}{5ab}$

B $\frac{25a^2 - 15ab + b^2}{5ab}$

C $-\frac{(5a - b)^2}{5ab}$

D $\frac{(5a - b)^2}{5ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q074 | MCQ

C01-T13-G-006

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{36a^2 - 6ab + b^2}{6ab}$

B $\frac{36a^2 - 18ab + b^2}{6ab}$

C $-\frac{(6a - b)^2}{6ab}$

D $\frac{(6a - b)^2}{6ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q075 | MCQ

C01-T13-G-007

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{49a^2 - 7ab + b^2}{7ab}$

B $\frac{49a^2 - 21ab + b^2}{7ab}$

C $\frac{(7a - b)^2}{7ab}$

D $-\frac{(7a - b)^2}{7ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Q076 | MCQ

C01-T13-G-008

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

A $\frac{64a^2 - 8ab + b^2}{8ab}$

B $\frac{64a^2 - 24ab + b^2}{8ab}$

C $-\frac{(8a - b)^2}{8ab}$

D $\frac{(8a - b)^2}{8ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q077 | WRITTEN

C01-T13-S-WU2

Prove that $(a + b) \left(\frac{1}{a} + \frac{1}{b} \right) \geq 4$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q078 | WRITTEN

C01-T13-S-EX4

Prove that $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q079 | WRITTEN

C01-T13-S-EX5

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q080 | WRITTEN

C01-T13-S-EX6

Prove that $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q081 | MCQ

C01-T13-G-009

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

A $\frac{a^2 - ab + b^2}{ab}$

B $\frac{a^2 - 3ab + b^2}{ab}$

C $-\frac{(a-b)^2}{ab}$

D $\frac{(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q082 | MCQ

C01-T13-G-010

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q083 | MCQ

C01-T13-G-011

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS – RHS:

A $\frac{9a^2 - 17ab + 9b^2}{ab}$

B $\frac{9a^2 - 19ab + 9b^2}{ab}$

C $\frac{9(a-b)^2}{ab}$

D $-\frac{9(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Q084 | MCQ

C01-T13-G-012

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

A $\frac{16a^2 - 31ab + 16b^2}{ab}$

B $\frac{16a^2 - 33ab + 16b^2}{ab}$

C $-\frac{16(a-b)^2}{ab}$

D $\frac{16(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V01

Q085 | WRITTEN

C01-T13-S-TR2

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q086 | WRITTEN

C01-T13-S-EX1

Prove that $a + \frac{4}{a} \geq 4$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q087 | WRITTEN

C01-T13-S-EX2

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V02

Q088 | WRITTEN

C01-T13-S-EX3

Prove that $a + b + \frac{9}{a+b} \geq 6$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F03 | V01

Q089 | WRITTEN

C01-T13-S-CH1

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F99 | V01

Q090 | CHALLENGE WRITTEN

C01-T13-C-01

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Quiz M1 | QUIZ MCQ

C01-T10-Q-M01

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Quiz M2 | QUIZ MCQ

C01-T11-Q-M04

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Quiz M3 | QUIZ MCQ

C01-T12-Q-M07

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

A $\frac{(a-7b)^2}{4}$

B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$

C $\frac{(a-7b-2)(a-7b+2)}{4}$

D $-\frac{(a-7b)^2}{4}$

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

WORKING AREA

Concept 5 | Inequalities

F01 | V02

Quiz M4 | QUIZ MCQ

C01-T13-Q-M10

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V01

Quiz W1 | QUIZ WRITTEN

C01-T10-Q-W01

If $x > y$, prove that $3x + 4y > 2x + 5y$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V03

Quiz W2 | QUIZ WRITTEN

C01-T11-Q-W02

Under x, y real, prove that $x^2 + y^2 \geq 2xy$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F01 | V02

Quiz W3 | QUIZ WRITTEN

C01-T12-Q-W04

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a + b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

Concept 5 | Inequalities

F02 | V01

Quiz W4 | QUIZ WRITTEN

C01-T13-Q-W05

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

WORKING AREA

FORMULA BANK

Order rule

$$0 < a < b \Rightarrow a^2 < b^2$$

Square rule

$$(x - y)^2 \geq 0$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab} \quad (a, b > 0)$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 02

Sequences

5 Concepts | 988 synchronized questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 02 | Sequences

CLASSROOM ROUTE

Concept 1 | Arithmetic Sequences**288 questions**

Official lessons: 2-1, 2-2, 2-3, 2-4, 2-5, 2-6

Concept 2 | Geometric Sequences**194 questions**

Official lessons: 2-7, 2-8, 2-9, 2-10

Concept 3 | Summation**282 questions**

Official lessons: 2-11, 2-12, 2-13, 2-14, 2-15, 2-16, 2-17

Concept 4 | Advanced Sequences**54 questions**

Official lessons: 2-18, 2-19

Concept 5 | Recurrence and Induction**170 questions**

Official lessons: 2-20, 2-21, 2-22, 2-23, 2-24, 2-25, 2-26

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 1 | Arithmetic Sequences

OFFICIAL LESSONS 2-1, 2-2, 2-3, 2-4, 2-5, 2-6

Arithmetic Sequences

Equation-first recall | use the same rail beside every question in this Concept

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Arithmetic Sequences

F01 | V01

Q001 | WRITTEN

C02-T01-S-TRY1A

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q002 | WRITTEN

C02-T01-S-TRY1C

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q003 | WRITTEN

C02-T01-S-EX1A

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q004 | WRITTEN

C02-T01-S-EX1B

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q005 | WRITTEN

C02-T01-S-EX1C

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q006 | WRITTEN

C02-T01-S-EX1G

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q007 | WRITTEN

C02-T01-S-EX1H

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q008 | WRITTEN

C02-T01-S-EX11

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q009 | MCQ

C02-T01-G-M01

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

A $(-2, 1, 5)$ B $(-2, 1, 4)$ C $(-2, 0, 4)$ D $(2, -1, -4)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q010 | MCQ

C02-T01-G-M02

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

A (11, 15, 20)

B (11, 14, 19)

C (11, 15, 19)

D (-11, -15, -19)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q011 | MCQ

C02-T01-G-M03

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

A (3, 2, 4)

B (3, 1, 3)

C (-3, -2, -3)

D (3, 2, 3)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q012 | MCQ

C02-T01-G-M04

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

A (5, 11, 21)

B (5, 11, 22)

C (5, 10, 21)

D (-5, -11, -21)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q013 | MCQ

C02-T01-G-M05

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

A (4, 10, 17)

B (4, 10, 16)

C (4, 9, 16)

D (-4, -10, -16)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q014 | MCQ

C02-T01-G-M06

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

A (3, 1, 0)

B (3, 0, -1)

C (3, 1, -1)

D (-3, -1, 1)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q015 | MCQ

C02-T01-G-M07

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

A (4, 3, 7)

B (4, 2, 6)

C (-4, -3, -6)

D (4, 3, 6)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q016 | MCQ

C02-T01-G-M08

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

A $(-3, -3, -1)$ B $(-3, -3, 0)$ C $(-3, -4, -1)$ D $(3, 3, 1)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q017 | MCQ

C02-T01-G-M09

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

A (1, 10, 26)

B (1, 10, 25)

C (1, 9, 25)

D (-1, -10, -25)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q018 | MCQ

C02-T01-G-M10

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

A (3, 6, 12)

B (3, 5, 11)

C (3, 6, 11)

D (-3, -6, -11)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q019 | MCQ

C02-T01-G-M11

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

A $(-3, 4, 12)$ B $(-3, 3, 11)$ C $(3, -4, -11)$ D $(-3, 4, 11)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q020 | MCQ

C02-T01-G-M12

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

A (2, 9, 20)

B (2, 9, 21)

C (2, 8, 20)

D (-2, -9, -20)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q021 | WRITTEN

C02-T01-S-TRY1B

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q022 | WRITTEN

C02-T01-S-EX1D

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q023 | WRITTEN

C02-T01-S-EX1E

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q024 | WRITTEN

C02-T01-S-EX1F

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q025 | MCQ

C02-T01-G-M16

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

A (2, 4, 8)

B (2, 4, 9)

C (2, 3, 8)

D (-2, -4, -8)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q026 | MCQ

C02-T01-G-M17

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

A $(-2, 4, -7)$ B $(-2, 4, -8)$ C $(-2, 3, -8)$ D $(2, -4, 8)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q027 | MCQ

C02-T01-G-M18

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

A (6, 18, 55)

B (6, 17, 54)

C (6, 18, 54)

D (-6, -18, -54)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q028 | MCQ

C02-T01-G-M19

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

A $(-15, 45, -134)$ B $(-15, 44, -135)$ C $(15, -45, 135)$ D $(-15, 45, -135)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q029 | MCQ

C02-T01-G-M20

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

A (3, 15, 63)

B (3, 15, 64)

C (3, 14, 63)

D (-3, -15, -63)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q030 | MCQ

C02-T01-G-M21

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

A (2, 6, 19)

B (2, 6, 18)

C (2, 5, 18)

D (-2, -6, -18)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q031 | MCQ

C02-T01-G-M22

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

A (5, -25, 126)

B (5, -26, 125)

C (5, -25, 125)

D (-5, 25, -125)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q032 | MCQ

C02-T01-G-M23

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

A $(-1, 11, -24)$ B $(-1, 10, -25)$ C $(1, -11, 25)$ D $(-1, 11, -25)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q033 | WRITTEN

C02-T01-S-WU1

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q034 | WRITTEN

C02-T01-S-TRY1D

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q035 | WRITTEN

C02-T01-S-EX1J

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q036 | WRITTEN

C02-T01-S-EX1K

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q037 | WRITTEN

C02-T01-S-EX1L

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q038 | MCQ

C02-T01-G-M26

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

A $(\frac{2}{3}, \frac{3}{4}, \frac{4}{6})$

B $(\frac{2}{3}, \frac{3}{3}, \frac{4}{5})$

C $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5})$

D $(-\frac{2}{3}, -\frac{3}{4}, -\frac{4}{5})$

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

WORKING AREA

Concept 1 | Arithmetic Sequences

F01 | V03

Q039 | MCQ

C02-T01-G-M27

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

A $(\frac{1}{4}, \frac{3}{7}, \frac{1}{3})$

B $(\frac{1}{4}, \frac{3}{6}, \frac{1}{2})$

C $(-\frac{1}{4}, -\frac{3}{7}, -\frac{1}{2})$

D $(\frac{1}{4}, \frac{3}{7}, \frac{1}{2})$

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

WORKING AREA

Concept 1 | Arithmetic Sequences

F01 | V03

Q040 | MCQ

C02-T01-G-M28

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

A $(1, \frac{5}{3}, \frac{5}{2})$

B $(1, \frac{5}{3}, \frac{5}{3})$

C $(1, \frac{5}{2}, \frac{5}{2})$

D $(-1, -\frac{5}{3}, -\frac{5}{2})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q041 | MCQ

C02-T01-G-M29

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

A $(3, 2, \frac{9}{6})$

B $(3, 2, \frac{9}{5})$

C $(3, 1, \frac{9}{5})$

D $(-3, -2, -\frac{9}{5})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q042 | MCQ

C02-T01-G-M30

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

A $(1, \frac{7}{6}, \frac{9}{8})$

B $(1, \frac{7}{5}, \frac{9}{7})$

C $(1, \frac{7}{6}, \frac{9}{7})$

D $(-1, -\frac{7}{6}, -\frac{9}{7})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q043 | MCQ

C02-T01-G-M31

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

A $(1, \frac{1}{4}, \frac{3}{8})$

B $(1, \frac{1}{3}, \frac{3}{7})$

C $(-1, -\frac{1}{4}, -\frac{3}{7})$

D $(1, \frac{1}{4}, \frac{3}{7})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q044 | WRITTEN

C02-T01-S-WU2A

For the sequence $\{a_n\} : 0, 1, 4, 9, 16, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q045 | WRITTEN

C02-T01-S-TRY2A

For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q046 | WRITTEN

C02-T01-S-EX2B

For the sequence $\{a_n\} : 1, 4, 9, 16, 25, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q047 | WRITTEN

C02-T01-S-EX2G

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q048 | MCQ

C02-T01-G-M33

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

A n^2

B n^3

C n^5

D n^6

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q049 | MCQ

C02-T01-G-M34

Which formula gives the general term of $\{a_n\} : 9, 16, 25, 36, \dots$?:

A $(n+2)^3$

B $(n+2)^4$

C $(n+2)^2$

D $(n+2)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q050 | MCQ

C02-T01-G-M35

Which formula gives the general term of $\{a_n\} : -1, 0, 1, 8, \dots$?:

A $(n - 2)^2$

B $(n - 2)^5$

C $(n - 2)^6$

D $(n - 2)^3$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q051 | MCQ

C02-T01-G-M36

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:

A $(n + 5)^2$

B $(n + 5)^3$

C $(n + 5)^4$

D $(n + 5)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q052 | MCQ

C02-T01-G-M37

Which formula gives the general term of $\{a_n\} : 1, 9, 25, 49, \dots$?:

A $(2n + 1)^2$

B $(2n - 1)^2$

C $(2n - 1)^4$

D $(2n - 1)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q053 | MCQ

C02-T01-G-M38

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:

A $(n+1)^4$

B $(n-1)^6$

C $(n-1)^4$

D $(n-1)^7$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q054 | WRITTEN

C02-T01-S-TRY2C

For the sequence $\{a_n\} : 2, 5, 8, 11, 14, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q055 | WRITTEN

C02-T01-S-EX2A

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q056 | WRITTEN

C02-T01-S-EX2C

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q057 | WRITTEN

C02-T01-S-EX2H

For the sequence $\{a_n\} : 4, 9, 14, 19, 24, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q058 | WRITTEN

C02-T01-S-EX2I

For the sequence $\{a_n\} : 6, 13, 20, 27, 34, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q059 | MCQ

C02-T01-G-M40

Which formula gives the general term of $\{a_n\} : 7, 12, 17, 22, \dots$?:

A $5n + 2$

B $(5n + 2) + 1$

C $(5n + 2) - 1$

D $-(5n + 2)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q060 | MCQ

C02-T01-G-M41

Which formula gives the general term of $\{a_n\} : -4, -1, 2, 5, \dots$?:

A $(3n - 7) + 1$

B $3n - 7$

C $(3n - 7) - 1$

D $-(3n - 7)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q061 | MCQ

C02-T01-G-M42

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2, ...?:

A $(14 - 4n) + 1$

B $(14 - 4n) - 1$

C $14 - 4n$

D $-(14 - 4n)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q062 | MCQ

C02-T01-G-M43

Which formula gives the general term of $\{a_n\} : \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$?:

A $\left(\frac{n}{2}\right) + 1$

B $\left(\frac{n}{2}\right) - 1$

C $-\left(\frac{n}{2}\right)$

D $\frac{n}{2}$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q063 | WRITTEN

C02-T01-S-WU2B

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q064 | WRITTEN

C02-T01-S-TRY2B

For the sequence $\{a_n\} : -3, 6, -9, 12, -15, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q065 | WRITTEN

C02-T01-S-EX2D

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q066 | WRITTEN

C02-T01-S-EX2F

For the sequence $\{a_n\} : -1, 4, -7, 10, -13, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q067 | WRITTEN

C02-T01-S-WU2C

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q068 | WRITTEN

C02-T01-S-TRY2D

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q069 | WRITTEN

C02-T01-S-EX2E

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q070 | WRITTEN

C02-T01-S-EX2J

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V05

Q071 | WRITTEN

C02-T01-S-EX2K

For the sequence $\{a_n\} : 1 \times 3, 2 \times 9, 3 \times 27, 4 \times 81, 5 \times 243, \dots$, deduce the general term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F99 | V01

Q072 | WRITTEN

C02-T01-S-CH1

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q073 | WRITTEN

C02-T02-S-WU1

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q074 | WRITTEN

C02-T02-S-TR1

Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q075 | MCQ

C02-T02-G-001

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A (4, 4)

B (4, 5)

C (5, 4)

D (4, -5)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q076 | MCQ

C02-T02-G-002

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A $(-7, 4)$ B $(-7, -4)$ C $(7, 4)$ D $(-3, 4)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q077 | MCQ

C02-T02-G-003

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A (18, 5)

B (13, -5)

C (18, -5)

D (-18, -5)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q078 | MCQ

C02-T02-G-006

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A (12, 4)

B (8, -4)

C (-12, -4)

D (12, -4)

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q079 | WRITTEN

C02-T02-S-EX1

Find the first term and common difference of: (a) $3, 5, 7, 9, 11, \dots$; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q080 | MCQ

C02-T02-G-004

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A $(\frac{1}{2}, \frac{2}{3})$

B $(\frac{1}{2}, \frac{1}{3})$

C $(\frac{1}{3}, \frac{1}{2})$

D $(-\frac{1}{2}, \frac{1}{3})$

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

WORKING AREA

Concept 1 | Arithmetic Sequences

F01 | V03

Q081 | MCQ

C02-T02-G-005

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A $(-\frac{3}{4}, \frac{1}{2})$

B $(-\frac{3}{4}, -\frac{1}{2})$

C $(\frac{3}{4}, \frac{1}{2})$

D $(-\frac{1}{4}, \frac{1}{2})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q082 | MCQ

C02-T02-G-007

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A $(\frac{5}{3}, \frac{2}{3})$

B $(2, \frac{1}{3})$

C $(\frac{5}{3}, \frac{1}{3})$

D $(-\frac{5}{3}, \frac{1}{3})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q083 | MCQ

C02-T02-G-008

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

A $(-\frac{5}{2}, -\frac{3}{2})$

B $(-\frac{5}{2}, \frac{3}{2})$

C $(\frac{5}{2}, \frac{3}{2})$

D $(-1, \frac{3}{2})$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q084 | WRITTEN

C02-T02-S-WU2

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q085 | WRITTEN

C02-T02-S-TR2

The sequences are arithmetic. Find the common difference and the missing numbers: (a) 2, 9, A , B , ...; (b) A , 17, B , 5,

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q086 | MCQ

C02-T02-G-014

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

A $d = 4, A = 10, B = 18$

B $d = 8, A = 14, B = 22$

C $d = 4, A = 12, B = 18$

D $d = -4, A = 2, B = 10$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q087 | MCQ

C02-T02-G-015

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

A $d = 4, A = 9, B = 13$

B $d = -4, A = 9, B = 1$

C $d = 4, A = 1, B = 13$

D $d = 2, A = 3, B = 11$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q088 | MCQ

C02-T02-G-016

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

A $d = 3, A = 14, B = 17$

B $d = 4, A = 15, B = 19$

C $d = 4, A = 14, B = 17$

D $d = -4, A = 7, B = 3$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q089 | WRITTEN

C02-T02-S-EX2

Find the common difference and missing terms: (a) 8, 15, A , B , ...; (b) 7, 13, A , B , ...; (c) A , 9, 4, B , ...; (d) A , 15, B , 9, ...; (e) 11, A , B , 20, ...; (f) 17, A , B , 5, C ,

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q090 | WRITTEN

C02-T02-S-TR3

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q091 | WRITTEN

C02-T02-S-EX3

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V04

Q092 | WRITTEN

C02-T02-S-WU3

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V04

Q093 | WRITTEN

C02-T02-S-EX4

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F99 | V01

Q094 | WRITTEN

C02-T02-S-CH1

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q095 | WRITTEN

C02-T03-S-TR1A

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q096 | WRITTEN

C02-T03-S-EX1A

The first term is 6, and the 4th term is 18. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q097 | WRITTEN

C02-T03-S-EX1B

The first term is 20, and the 8th term is -1 . Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q098 | WRITTEN

C02-T03-S-EX1C

The first term is -8 , and the 8th term is 55 . Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q099 | WRITTEN

C02-T03-S-EX1D

The first term is -18 , and the 10th term is -63 . Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q100 | MCQ

C02-T03-G-M01

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

A $(3n - 1, 3)$

B $(4n, 4)$

C $(3n + 1, 4)$

D $(a_n, d) = (3n + 1, 3)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q101 | MCQ

C02-T03-G-M02

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

A $(4n - 3, 4)$

B $(a_n, d) = (4n - 7, 4)$

C $(5n - 8, 5)$

D $(4n - 7, -4)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q102 | MCQ

C02-T03-G-M03

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

A $(a_n, d) = (16 - 4n, -4)$

B $(12 - 4n, -4)$

C $(16 + 4n, 4)$

D $(16 - 3n, -3)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q103 | MCQ

C02-T03-G-M04

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

A $(5n, 5)$ B $(a_n, d) = (4n + 1, 4)$ C $(4n - 1, 4)$ D $(4n + 1, -4)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q104 | WRITTEN

C02-T03-S-TR1B

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q105 | WRITTEN

C02-T03-S-EX1E

The common difference is 2, and the 8th term is 4. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q106 | WRITTEN

C02-T03-S-EX1F

The common difference is -4 , and the 5th term is -7 . Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q107 | WRITTEN

C02-T03-S-EX1G

The common difference is -7 , and the 6th term is -24 . Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q108 | MCQ

C02-T03-G-M05

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

A $(a_n, a) = (9 - 2n, 7)$

B $(7 - 2n, 9)$

C $(1 - 2n, 3)$

D $(9 + 2n, 7)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q109 | MCQ

C02-T03-G-M06

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

A $(5n - 3, -2)$

B $(17 + 5n, 2)$

C $(13 - 5n, 8)$

D $(a_n, a) = (5n - 13, -8)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q110 | MCQ

C02-T03-G-M07

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

A $(4n + 2, -2)$

B $(2n + 4, 6)$

C $(a_n, a) = (4n - 2, 2)$

D $(4n - 2, 4)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q111 | MCQ

C02-T03-G-M08

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

A $(3 - 3n, 6)$

B $(6 + 3n, 3)$

C $(6 - 2n, 2)$

D $(a_n, a) = (6 - 3n, 3)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q112 | WRITTEN

C02-T03-S-WU1

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q113 | WRITTEN

C02-T03-S-TR1C

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q114 | WRITTEN

C02-T03-S-EX11

The 4th term is 14, and the 10th term is 62. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q115 | WRITTEN

C02-T03-S-EX1J

The 3rd term is 30, and the 10th term is 2. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q116 | WRITTEN

C02-T03-S-EX1K

The 10th term is 49, and the 23rd term is 10. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q117 | WRITTEN

C02-T03-S-EX1L

The 3rd term is 6, and the 10th term is 62. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q118 | MCQ

C02-T03-G-M09

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

A $(2n + 1, -1)$

B $(n + 3, 2)$

C $(a_n, a) = (2n - 1, 1)$

D $(2n - 1, 5)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q119 | MCQ

C02-T03-G-M10

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

A $(13 - 2n, 15)$

B $(15 + 2n, 5)$

C $(a_n, a) = (15 - 2n, 13)$

D $(2n - 15, -13)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q120 | MCQ

C02-T03-G-M11

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

A $(5n + 4, -1)$

B $(a_n, a) = (5n - 4, 1)$

C $(4n - 5, 2)$

D $(5n - 4, 31)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q121 | MCQ

C02-T03-G-M12

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

A $(a_n, a) = (3n - 10, -7)$

B $(3n - 4, -1)$

C $(2n - 7, -5)$

D $(10 - 3n, 7)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q122 | MCQ

C02-T03-G-M13

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

A $\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$

B $\left(a_n = \frac{n+2}{2}, -\frac{5}{2}\right)$

C $\left(a_n = \frac{n-2}{2}, \frac{1}{2}\right)$

D $\left(a_n = \frac{2n-1}{3}, -\frac{1}{3}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Q123 | MCQ

C02-T03-G-M14

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

A $(a_n = 2n + 4, 2)$

B $(a_n = 2n + 2, 4)$

C $(a_n = n + 4, 5)$

D $(a_n = 2n - 2, 8)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V04

Q124 | WRITTEN

C02-T03-S-EX1H

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V04

Q125 | MCQ

C02-T03-G-M15

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

A $\left(a_n = -\frac{3(n+1)}{2}, -3\right)$

B $\left(a_n = -\frac{3(n-1)}{2}, -6\right)$

C $\left(a_n = \frac{3(n+1)}{2}, 3\right)$

D $\left(a_n = -\frac{n+1}{2}, -3\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V04

Q126 | MCQ

C02-T03-G-M16

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

A $\left(a_n = \frac{2n+4}{3}, 2\right)$

B $\left(a_n = \frac{2n-6}{3}, -\frac{8}{3}\right)$

C $\left(a_n = \frac{n+6}{3}, \frac{8}{3}\right)$

D $\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q127 | WRITTEN

C02-T03-S-WU2A

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q128 | WRITTEN

C02-T03-S-TR2A

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q129 | WRITTEN

C02-T03-S-EX2A

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q130 | WRITTEN

C02-T03-S-EX3A

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q131 | MCQ

C02-T03-G-M17

For $a = 3, d = 4$, which term of the arithmetic sequence is 59?:

A $n = 14$ B $n = 16$ C $n = 15$ D $n = 13$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q132 | MCQ

C02-T03-G-M18

For $a = -5$, $d = 6$, which term of the arithmetic sequence is 55?:

A $n = 10$ B $n = 12$ C $n = 9$ D $n = 11$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q133 | MCQ

C02-T03-G-M19

For $a = 20$, $d = -3$, which term of the arithmetic sequence is -16 ?:

A $n = 13$ B $n = 12$ C $n = 14$ D $n = 11$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q134 | MCQ

C02-T03-G-M20

For $a = 1, d = 5$, which term of the arithmetic sequence is 86?:

A $n = 17$ B $n = 18$ C $n = 19$ D $n = 16$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q135 | WRITTEN

C02-T03-S-WU2B

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q136 | WRITTEN

C02-T03-S-TR2B

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q137 | MCQ

C02-T03-G-M21

For $a = 2, d = 7$, find the first term greater than 100:

A $n = 16$ B $n = 15$ C $n = 17$ D $n = 14$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q138 | MCQ

C02-T03-G-M22

For $a = -9$, $d = 4$, find the first term greater than 50:

A $n = 15$ B $n = 17$ C $n = 14$ D $n = 16$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q139 | MCQ

C02-T03-G-M23

For $a = 6$, $d = 3$, find the first term greater than 200:

A $n = 65$

B $n = 67$

C $n = 64$

D $n = 66$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q140 | MCQ

C02-T03-G-M24

For $a = -1$, $d = 9$, find the first term greater than 300:

A $n = 34$ B $n = 35$ C $n = 36$ D $n = 33$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q141 | WRITTEN

C02-T03-S-EX2B

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q142 | WRITTEN

C02-T03-S-EX3B

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q143 | MCQ

C02-T03-G-M25

For $a = 10$, $d = -4$, find the first term smaller than -100 :

A $n = 28$

B $n = 29$

C $n = 30$

D $n = 27$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q144 | MCQ

C02-T03-G-M26

For $a = 25$, $d = -6$, find the first term smaller than -250 :

A $n = 46$

B $n = 47$

C $n = 48$

D $n = 45$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V03

Q145 | MCQ

C02-T03-G-M27

For $a = -3$, $d = -5$, find the first term smaller than -200 :

A $n = 40$

B $n = 42$

C $n = 41$

D $n = 39$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q146 | MCQ

C02-T03-G-M28

For $a = 4, d = 6$, find the first term that is at least 100:

A $n = 17$

B $n = 16$

C $n = 18$

D $n = 15$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V04

Q147 | MCQ

C02-T03-G-M29

For the expression For $a = 5, d = 4$, does the sequence contain the term 42?, the required value is:

A $n = 9$

B No term

C $n = 10$ D $n = 11$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q148 | WRITTEN

C02-T03-S-EX4

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q149 | WRITTEN

C02-T03-S-WU3

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q150 | WRITTEN

C02-T03-S-TR3

When $x^2, x, -8$ form an arithmetic sequence in this order, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q151 | WRITTEN

C02-T03-S-EX5

When $-3, x, x^2$ form an arithmetic sequence in this order, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F99 | V01

Q152 | CHALLENGE WRITTEN

C02-T03-S-CH1

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q153 | WRITTEN

C02-T04-S-WU1A

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q154 | WRITTEN

C02-T04-S-TR1A

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q155 | WRITTEN

C02-T04-S-EX1A

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q156 | WRITTEN

C02-T04-S-EX1B

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q157 | WRITTEN

C02-T04-S-EX1C

Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q158 | MCQ

C02-T04-G-M01

A sequence has first term 6, last term 30, and 7 terms. Find its sum:

A 105

B 126

C 132

D 168

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q159 | MCQ

C02-T04-G-M02

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

A 64

B 72

C 81

D 144

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q160 | MCQ

C02-T04-G-M03

Find the sum of 10 terms whose first term is 15 and last term is 3:

A 45

B 90

C 150

D 180

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q161 | MCQ

C02-T04-G-M04

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

A 168

B 196

C 224

D 392

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q162 | MCQ

C02-T04-G-M05

Find the sum of 6 terms with first term -12 and last term 8 :

A -120 B -12 C 12 D -2

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q163 | MCQ

C02-T04-G-M06

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

A 12

B 16

C 20

D 24

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q164 | WRITTEN

C02-T04-S-WU1C

Find (S_{13}) when the first term is (20) , the common difference is (-5) , and the number of terms is (13) .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q165 | WRITTEN

C02-T04-S-TR1C

Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q166 | WRITTEN

C02-T04-S-EX1G

Find (S_{15}) when the first term is (10), the common difference is (-3) , and there are (15) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q167 | WRITTEN

C02-T04-S-EX1H

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q168 | WRITTEN

C02-T04-S-EX11

Find (S_{18}) when the first term is (3), the common difference is (2), and there are (18) terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q169 | MCQ

C02-T04-G-M07

For $a = 4, d = 3, n = 12$, find S_{12} :

A 198

B 228

C 246

D 264

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q170 | MCQ

C02-T04-G-M08

For $a = 9, d = -2, n = 10$, find S_{10} :

A -90

B 0

C 9

D 90

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q171 | MCQ

C02-T04-G-M09

For $a = -5, d = 4, n = 9$, find S_9 :

A -88

B 44

C 88

D 99

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q172 | MCQ

C02-T04-G-M10

For $a = 2, d = -3, n = 15$, find S_{15} :

A -315

B -285

C -270

D 285

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q173 | MCQ

C02-T04-G-M11

For $a = 11, d = 0, n = 8$, find S_8 :

A 11

B 44

C 88

D 96

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q174 | MCQ

C02-T04-G-M12

For $a = -3, d = -2, n = 7$, find S_7 :

A -42

B -54

C 54

D -63

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q175 | WRITTEN

C02-T04-S-WU1B

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q176 | WRITTEN

C02-T04-S-TR1B

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q177 | WRITTEN

C02-T04-S-EX1D

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q178 | WRITTEN

C02-T04-S-EX1E

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q179 | WRITTEN

C02-T04-S-EX1F

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q180 | MCQ

C02-T04-G-M13

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

A 205

B 225

C 230

D 246

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q181 | MCQ

C02-T04-G-M15

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

A 228

B 240

C 246

D 264

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q182 | MCQ

C02-T04-G-M16

A sequence has $a = 12, d = 2, l = 50$. Find its sum:

A 580

B 600

C 620

D 640

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q183 | MCQ

C02-T04-G-M18

A sequence has $a = -10, d = 6, l = 50$. Find its sum:

A 198

B 210

C 220

D 240

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q184 | MCQ

C02-T04-G-M14

A sequence has $a = 18, d = -3, l = -9$. Find its sum:

A -45

B 36

C 45

D 90

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q185 | MCQ

C02-T04-G-M17

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

A 168

B 196

C 224

D 280

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q186 | WRITTEN

C02-T04-S-TR2

For the arithmetic sequence with first term (48) and common difference (-2), find the sum from the 20th term to the 30th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q187 | WRITTEN

C02-T04-S-EX3

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q188 | MCQ

C02-T04-G-M19

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

A 252

B 264

C 276

D 288

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q189 | MCQ

C02-T04-G-M20

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :

A -14

B 7

C 14

D 28

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q190 | MCQ

C02-T04-G-M22

For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :

A 216

B 234

C 243

D 270

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q191 | MCQ

C02-T04-G-M23

For $a = 1, d = 7$, find the sum from a_3 to a_9 :

A 224

B 245

C 252

D 266

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q192 | MCQ

C02-T04-G-M25

For $a = 60$, $d = -3$, find the sum from a_9 to a_{17} :

A 198

B 216

C 225

D 234

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q193 | MCQ

C02-T04-G-M26

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

A 196

B 210

C 224

D 238

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q194 | WRITTEN

C02-T04-S-WU2

For the arithmetic sequence (90, 84, 78, ...), find the sum from the 10th term to the 20th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q195 | WRITTEN

C02-T04-S-EX2

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q196 | WRITTEN

C02-T04-S-EX4

For the arithmetic sequence (74, 69, 64, ...), find the sum from the 8th term to the 18th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q197 | MCQ

C02-T04-G-M21

For $a = 4, d = 5$, find the sum from a_6 to a_{13} :

A 248

B 336

C 372

D 384

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q198 | MCQ

C02-T04-G-M24

For $a = -25$, $d = 6$, find the sum from a_5 to a_{11} :

A 105

B 112

C 119

D 126

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F05 | V01

Q199 | WRITTEN

C02-T04-S-WU1D

Find (S_n) for a sequence with first term (1) and common difference (4).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F05 | V01

Q200 | WRITTEN

C02-T04-S-TR1D

Find (S_n) for a sequence with first term (2) and common difference (-3) .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F05 | V01

Q201 | WRITTEN

C02-T04-S-EX1J

Find (S_n) for a sequence with first term (0) and common difference (5).

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F05 | V01

Q202 | WRITTEN

C02-T04-S-EX1K

Find (S_n) for a sequence with first term (3) and common difference (-4) .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F05 | V01

Q203 | WRITTEN

C02-T04-S-EX1L

Find (S_n) for a sequence with first term (-4) and common difference (-2) .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F06 | V01

Q204 | CHALLENGE WRITTEN

C02-T04-S-CH1

For the arithmetic sequence (94, 89, 84, ...), find the sum of the odd-numbered terms from the first term to the 11th term.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q205 | WRITTEN

C02-T05-S-WU1

Find the sum of the natural numbers from 1 to 100.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q206 | WRITTEN

C02-T05-S-T1

Find the sum of the natural numbers from 1 to 200.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q207 | WRITTEN

C02-T05-S-E1

Find the sum of the natural numbers from 1 to 150.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q208 | WRITTEN

C02-T05-S-E2

Find the sum of the natural numbers from 1 to 70.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q209 | WRITTEN

C02-T05-S-E3

Find the sum of the natural numbers from 1 to 100.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q210 | WRITTEN

C02-T05-S-E4

Find the sum of the natural numbers from 1 to 400.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q211 | WRITTEN

C02-T05-S-E5

Find the sum of the natural numbers from 1 to 250.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q212 | WRITTEN

C02-T05-S-E6

Find the sum of the natural numbers from 1 to 81.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q213 | WRITTEN

C02-T05-S-E7

Find the sum of the natural numbers from 1 to 301.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q214 | WRITTEN

C02-T05-S-E8

Find the sum of the natural numbers from 1 to 99.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q215 | MCQ

C02-T05-G-M01

Find the sum of the natural numbers from 1 to 30:

A 435

B 465

C 495

D 525

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q216 | MCQ

C02-T05-G-M02

Find the sum of the natural numbers from 1 to 45:

A 990

B 1035

C 1080

D 1125

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q217 | MCQ

C02-T05-G-M03

Find the sum of the natural numbers from 1 to 60:

A 1770

B 1830

C 1890

D 1950

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q218 | MCQ

C02-T05-G-M04

Find the sum of the natural numbers from 1 to 75:

A 2775

B 2850

C 2925

D 3000

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q219 | MCQ

C02-T05-G-M05

Find the sum of the natural numbers from 1 to 90:

A 4005

B 4095

C 4185

D 4275

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q220 | MCQ

C02-T05-G-M06

Find the sum of the natural numbers from 1 to 125:

A 7750

B 7875

C 8000

D 8125

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q221 | MCQ

C02-T05-G-M07

Find the sum of the natural numbers from 1 to 180:

A 16110

B 16290

C 16470

D 16650

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q222 | MCQ

C02-T05-G-M08

Find the sum of the natural numbers from 1 to 220:

A 24090

B 24310

C 24530

D 24750

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q223 | MCQ

C02-T05-G-M09

Find the sum of the natural numbers from 1 to 275:

A 37675

B 37950

C 38225

D 38500

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q224 | MCQ

C02-T05-G-M10

Find the sum of the natural numbers from 1 to 350:

A 61075

B 61425

C 61775

D 62125

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q225 | WRITTEN

C02-T05-S-WU2

Find the sum of the natural numbers from 101 to 200.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q226 | WRITTEN

C02-T05-S-T2

Find the sum of the natural numbers from 201 to 300.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q227 | WRITTEN

C02-T05-S-EX4

Find the sum of all two-digit natural numbers.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q228 | MCQ

C02-T05-G-M11

A sequence has first term 5, last term 35, and 7 terms. Select its sum:

A 133

B 140

C 147

D 154

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q229 | MCQ

C02-T05-G-M12

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

A 351

B 360

C 369

D 378

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q230 | MCQ

C02-T05-G-M13

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

A 63

B 72

C 81

D 90

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q231 | MCQ

C02-T05-G-M14

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

A 188

B 196

C 204

D 212

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q232 | MCQ

C02-T05-G-M15

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

A 80

B 90

C 100

D 110

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q233 | MCQ

C02-T05-G-M16

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

A 507

B 520

C 533

D 546

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q234 | MCQ

C02-T05-G-M17

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

A 342

B 354

C 366

D 378

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q235 | MCQ

C02-T05-G-M18

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

A 984

B 1000

C 1016

D 1032

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q236 | MCQ

C02-T05-G-M19

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

A 14950

B 15050

C 15150

D 15250

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q237 | MCQ

C02-T05-G-M20

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

A 960

B 976

C 992

D 1008

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q238 | WRITTEN

C02-T05-S-WU3

Find the sum of the multiples of 3 among the two-digit natural numbers.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q239 | WRITTEN

C02-T05-S-T3

Find the sum of all two-digit natural numbers that are multiples of 5.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q240 | WRITTEN

C02-T05-S-EX6

Find the sum of the multiples of 4 from 1 to 100.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q241 | WRITTEN

C02-T05-S-EX8

Find the sum of all multiples of 6 from 10 to 100.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q242 | MCQ

C02-T05-G-M21

Select the sum of the multiples of 3 from 3 to 99:

A 1680

B 1683

C 1686

D 1689

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V02

Q243 | MCQ

C02-T05-G-M22

Select the sum of the multiples of 4 from 4 to 100:

A 1296

B 1300

C 1304

D 1308

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V01

Q244 | WRITTEN

C02-T05-S-EX7

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q245 | WRITTEN

C02-T05-S-T4

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q246 | WRITTEN

C02-T05-S-EX5

Find the sum of all two-digit odd natural numbers.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V02

Q247 | WRITTEN

C02-T05-S-EX9

Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F99 | V01

Q248 | CHALLENGE WRITTEN

C02-T05-S-CH1

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q249 | WRITTEN

C02-T06-S-WU1

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q250 | WRITTEN

C02-T06-S-E1

Find n when $a = 3$, $d = 2$ and $S_n = 63$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q251 | MCQ

C02-T06-G-M01

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

A 5

B 6

C 7

D 8

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q252 | MCQ

C02-T06-G-M02

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

A 7

B 8

C 9

D 10

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q253 | MCQ

C02-T06-G-M03

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:

A 9

B 10

C 11

D 12

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q254 | MCQ

C02-T06-G-M04

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:

A 11

B 12

C 13

D 14

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q255 | MCQ

C02-T06-G-M05

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:

A 14

B 15

C 16

D 17

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q256 | MCQ

C02-T06-G-M06

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:

A 17

B 18

C 19

D 20

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q257 | MCQ

C02-T06-G-M07

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:

A 19

B 20

C 21

D 22

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q258 | MCQ

C02-T06-G-M08

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:

A 23

B 24

C 25

D 26

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q259 | MCQ

C02-T06-G-M09

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:

A 27

B 28

C 29

D 30

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q260 | MCQ

C02-T06-G-M10

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:

A 31

B 32

C 33

D 34

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q261 | MCQ

C02-T06-G-M11

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:

A 35

B 36

C 37

D 38

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Q262 | MCQ

C02-T06-G-M12

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:

A 39

B 40

C 41

D 42

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q263 | WRITTEN

C02-T06-S-T1

For first term 17 and common difference -2 , find how many terms have sum -63 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q264 | WRITTEN

C02-T06-S-E2

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q265 | MCQ

C02-T06-G-M13

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:

A 3

B 4

C 5

D 6

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q266 | MCQ

C02-T06-G-M14

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:

A 5

B 6

C 7

D 8

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q267 | MCQ

C02-T06-G-M15

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:

A 7

B 8

C 9

D 10

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V02

Q268 | MCQ

C02-T06-G-M16

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:

A 9

B 10

C 11

D 12

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q269 | WRITTEN

C02-T06-S-WU2

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q270 | WRITTEN

C02-T06-S-T2

Find the general term with first term 8 and sum of the first 13 terms -130 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q271 | WRITTEN

C02-T06-S-E3

Find the general term with first term 4 and $S_6 = 114$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F02 | V01

Q272 | WRITTEN

C02-T06-S-E6

Find the general term when $a_3 = 10$ and $S_{10} = 175$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q273 | WRITTEN

C02-T06-S-WU3

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q274 | WRITTEN

C02-T06-S-T3

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q275 | WRITTEN

C02-T06-S-E4

For $a = 20$, $d = -6$, find n giving the maximum sum and the sum.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F03 | V02

Q276 | WRITTEN

C02-T06-S-E5

For $a = 126$, $d = -13$, find n giving the maximum sum and the sum.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q277 | WRITTEN

C02-T06-S-T4

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q278 | WRITTEN

C02-T06-S-E7

Find the general term where $S_6 = 3$ and $S_{10} = 65$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Q279 | WRITTEN

C02-T06-S-E8

Find the general term where $S_{20} = 400$ and $S_{30} = 900$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F99 | V01

Q280 | CHALLENGE WRITTEN

C02-T06-S-CH1

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Quiz M1 | QUIZ MCQ

C02-T01-Q-M01

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

A $\frac{11}{7}$

B $\frac{10}{7}$

C $\frac{7}{11}$

D $\frac{9}{7}$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Quiz M2 | QUIZ MCQ

C02-T03-Q-M01

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

A $a_n = 4n + 2$

B $a_n = 4n - 2$

C $a_n = 5n - 5$

D $a_n = 2n + 4$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F06 | V01

Quiz M3 | QUIZ MCQ

C02-T04-Q-M04

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

A 16

B 18

C 20

D 24

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Quiz M4 | QUIZ MCQ

C02-T06-Q-M4

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

A $\frac{1}{10}$

B $\frac{1}{5}$

C $\frac{2}{5}$

D $\frac{1}{2}$

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V01

Quiz W1 | QUIZ WRITTEN

C02-T01-Q-W02

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F01 | V03

Quiz W2 | QUIZ WRITTEN

C02-T03-Q-W01

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F06 | V01

Quiz W3 | QUIZ WRITTEN

C02-T04-Q-W04

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 1 | Arithmetic Sequences

F04 | V01

Quiz W4 | QUIZ WRITTEN

C02-T06-Q-W4

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

WORKING AREA

FORMULA BANK

General term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic test

$$a_{n+1} - a_n = d$$

Concept 2 | Geometric Sequences

OFFICIAL LESSONS 2-7, 2-8, 2-9, 2-10

Geometric Sequences

Equation-first recall | use the same rail beside every question in this Concept

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Geometric Sequences

F01 | V01

Q001 | WRITTEN

C02-T07-S-WU1a

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q002 | WRITTEN

C02-T07-S-T1a

Find the general term when the first term is 3 and the common ratio is 4.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q003 | WRITTEN

C02-T07-S-E1a

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q004 | WRITTEN

C02-T07-S-E1b

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q005 | WRITTEN

C02-T07-S-E1c

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q006 | MCQ

C02-T07-G-M01

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

A $a_n = 2(3)^4$

B $a_n = 2(3)^5$

C $a_n = 2(3)^3$

D $a_n = 2 + (3)^4$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q007 | MCQ

C02-T07-G-M02

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

A $a_n = 3(2)^5$

B $a_n = 3(2)^6$

C $a_n = 3(2)^4$

D $a_n = 3 + (2)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q008 | MCQ

C02-T07-G-M03

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

A $a_n = 5(4)^3$

B $a_n = 5(4)^4$

C $a_n = 5(4)^2$

D $a_n = 5 + (4)^3$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q009 | MCQ

C02-T07-G-M04

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

A $a_n = 4(5)^2$

B $a_n = 4(5)^3$

C $a_n = 4(5)^1$

D $a_n = 4 + (5)^2$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q010 | MCQ

C02-T07-G-M14

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

A $a_n = 5(2)^6$

B $a_n = 5(2)^7$

C $a_n = 5(2)^5$

D $a_n = 5 + (2)^6$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q011 | MCQ

C02-T07-G-M15

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

A $a_n = 7(4)^2$

B $a_n = 7(4)^3$

C $a_n = 7(4)^1$

D $a_n = 7 + (4)^2$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q012 | WRITTEN

C02-T07-S-T1b

Find the general term when the first term is 5 and the common ratio is -2 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q013 | WRITTEN

C02-T07-S-E1d

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q014 | WRITTEN

C02-T07-S-E1e

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q015 | WRITTEN

C02-T07-S-E1f

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q016 | MCQ

C02-T07-G-M05

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

A $a_n = 2(-3)^4$

B $a_n = 2(-3)^5$

C $a_n = 2(-3)^3$

D $a_n = 2 + (-3)^4$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q017 | MCQ

C02-T07-G-M06

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?:

A $a_n = 3(-2)^5$

B $a_n = 3(-2)^6$

C $a_n = 3(-2)^4$

D $a_n = 3 + (-2)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q018 | MCQ

C02-T07-G-M07

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

A $a_n = 6(-1)^3$

B $a_n = 6(-1)^4$

C $a_n = 6(-1)^2$

D $a_n = 6 + (-1)^3$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q019 | MCQ

C02-T07-G-M08

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?:

A $a_n = 8(-2)^2$

B $a_n = 8(-2)^3$

C $a_n = 8(-2)^1$

D $a_n = 8 + (-2)^2$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q020 | MCQ

C02-T07-G-M16

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

A $a_n = 9(-2)^4$

B $a_n = 9(-2)^5$

C $a_n = 9(-2)^3$

D $a_n = 9 + (-2)^4$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q021 | WRITTEN

C02-T07-S-T1c

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q022 | WRITTEN

C02-T07-S-E1g

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q023 | WRITTEN

C02-T07-S-E1h

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q024 | WRITTEN

C02-T07-S-E1i

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q025 | MCQ

C02-T07-G-M09

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

A $a_n = 4\left(\frac{1}{3}\right)^4$

B $a_n = 4\left(\frac{1}{3}\right)^5$

C $a_n = 4\left(\frac{1}{3}\right)^3$

D $a_n = 4 + \left(\frac{1}{3}\right)^4$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q026 | MCQ

C02-T07-G-M10

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

A $a_n = 3\left(-\frac{1}{2}\right)^5$

B $a_n = 3\left(-\frac{1}{2}\right)^6$

C $a_n = 3\left(-\frac{1}{2}\right)^4$

D $a_n = 3 + \left(-\frac{1}{2}\right)^5$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q027 | MCQ

C02-T07-G-M11

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?:

A $a_n = 6\left(\frac{1}{2}\right)^3$

B $a_n = 6\left(\frac{1}{2}\right)^4$

C $a_n = 6\left(\frac{1}{2}\right)^2$

D $a_n = 6 + \left(\frac{1}{2}\right)^3$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q028 | MCQ

C02-T07-G-M12

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?:

A $a_n = 2\left(\frac{3}{2}\right)^4$

B $a_n = 2\left(\frac{3}{2}\right)^5$

C $a_n = 2\left(\frac{3}{2}\right)^3$

D $a_n = 2 + \left(\frac{3}{2}\right)^4$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q029 | MCQ

C02-T07-G-M13

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

A $a_n = 8\left(-\frac{3}{2}\right)^3$

B $a_n = 8\left(-\frac{3}{2}\right)^4$

C $a_n = 8\left(-\frac{3}{2}\right)^2$

D $a_n = 8 + \left(-\frac{3}{2}\right)^3$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q030 | WRITTEN

C02-T07-S-WU1b

Find the general term of 5, 10, 20, 40,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q031 | WRITTEN

C02-T07-S-T1d

Find the general term of 2, 6, 18, 54,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q032 | WRITTEN

C02-T07-S-E1j

Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q033 | MCQ

C02-T07-G-M17

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:

A $3 \times 2^{n-1}$

B 3×2^n

C $3 \times 2^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q034 | MCQ

C02-T07-G-M18

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:

A $2 \times 5^{n-1}$

B 2×5^n

C $2 \times 5^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q035 | MCQ

C02-T07-G-M26

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:

A $4 \times 2^{n-1}$

B 4×2^n

C $4 \times 2^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q036 | MCQ

C02-T07-G-M27

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:

A $7 \times 3^{n-1}$

B 7×3^n

C $7 \times 3^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q037 | MCQ

C02-T07-G-M31

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:

A 4^{n-1}

B 4^n

C 4^{n-2}

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q038 | WRITTEN

C02-T07-S-E1k

Find the general term of the geometric sequence in part k. The sequence is 2, -6, 18, -54,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q039 | WRITTEN

C02-T07-S-E11

Find the general term of the geometric sequence in part 1. The sequence is 5, -10, 20, -40,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q040 | MCQ

C02-T07-G-M19

Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \dots$:

A $5(-2)^{n-1}$

B $5(-2)^n$

C $5(-2)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q041 | MCQ

C02-T07-G-M25

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:

A $2(-3)^{n-1}$

B $2(-3)^n$

C $2(-3)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q042 | MCQ

C02-T07-G-M30

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:

A $-4(-2)^{n-1}$

B $-4(-2)^n$

C $-4(-2)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q043 | MCQ

C02-T07-G-M32

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \dots$:

A $6(-3)^{n-1}$

B $6(-3)^n$

C $6(-3)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q044 | WRITTEN

C02-T07-S-WU1c

Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q045 | WRITTEN

C02-T07-S-T1e

Find the general term of $-8, 12, -18, 27, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q046 | WRITTEN

C02-T07-S-T1f

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q047 | WRITTEN

C02-T07-S-E1m

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q048 | WRITTEN

C02-T07-S-E1n

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q049 | WRITTEN

C02-T07-S-E1o

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q050 | WRITTEN

C02-T07-S-E1p

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q051 | WRITTEN

C02-T07-S-E1q

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q052 | WRITTEN

C02-T07-S-E1r

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q053 | MCQ

C02-T07-G-M20

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:

A $-8\left(-\frac{3}{2}\right)^{n-1}$

B $-8\left(-\frac{3}{2}\right)^n$

C $-8\left(-\frac{3}{2}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q054 | MCQ

C02-T07-G-M21

Select the correct option: Find the general term of the geometric sequence 48, 24, 12, 6, ...:

A $48\left(\frac{1}{2}\right)^{n-1}$

B $48\left(\frac{1}{2}\right)^n$

C $48\left(\frac{1}{2}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q055 | MCQ

C02-T07-G-M22

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

A $\frac{1}{6} \left(\frac{3}{2}\right)^{n-1}$

B $\frac{1}{6} \left(\frac{3}{2}\right)^n$

C $\frac{1}{6} \left(\frac{3}{2}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q056 | MCQ

C02-T07-G-M23

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

A $8(2\sqrt{2})^{n-1}$

B $8(2\sqrt{2})^n$

C $8(2\sqrt{2})^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q057 | MCQ

C02-T07-G-M24

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

A $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

B $9\left(-\frac{\sqrt{3}}{3}\right)^n$

C $9\left(-\frac{\sqrt{3}}{3}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q058 | MCQ

C02-T07-G-M28

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

A $3\left(-\frac{1}{2}\right)^{n-1}$

B $3\left(-\frac{1}{2}\right)^n$

C $3\left(-\frac{1}{2}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q059 | MCQ

C02-T07-G-M29

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

A $10\left(\frac{1}{2}\right)^{n-1}$

B $10\left(\frac{1}{2}\right)^n$

C $10\left(\frac{1}{2}\right)^{n-2}$

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q060 | WRITTEN

C02-T07-S-T2

Find the common ratio and missing values in $A, -18, 6, B, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q061 | MCQ

C02-T07-G-M33

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

A $r = \frac{1}{2}, A = 24, B = 3$

B $r = 2, A = 1, B = 2$

C $r = -1, A = 0, B = 0$

D $r = 3, A = 3, B = 9$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V02

Q062 | WRITTEN

C02-T07-S-WU2

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V02

Q063 | WRITTEN

C02-T07-S-E2a

Find the common ratio and missing values in $A, -3, 9, B$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V02

Q064 | WRITTEN

C02-T07-S-E2b

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V03

Q065 | WRITTEN

C02-T07-S-E2c

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F99 | V01

Q066 | CHALLENGE WRITTEN

C02-T07-S-CH1

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q067 | WRITTEN

C02-T08-S-WU1a

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q068 | WRITTEN

C02-T08-S-T1a

Find the general term when the 3rd term is 18 and the 5th term is 162.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q069 | WRITTEN

C02-T08-S-E1a

Find the requested values. The 2nd term is 6 and the 4th term is 54.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q070 | WRITTEN

C02-T08-S-E1b

Find the requested values. The 3rd term is 27 and the 5th term is 243.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q071 | WRITTEN

C02-T08-S-E1d

Find the requested values. The 2nd term is 14 and the 5th term is 112.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q072 | WRITTEN

C02-T08-S-WU1b

Find the general term when the 3rd term is 12 and the 6th term is -96 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q073 | WRITTEN

C02-T08-S-T1b

Find the general term when the 2nd term is -6 and the 5th term is 48 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q074 | WRITTEN

C02-T08-S-T1c

Find the general term when the 3rd term is 12 and the 7th term is 192.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q075 | WRITTEN

C02-T08-S-E1c

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q076 | WRITTEN

C02-T08-S-E1e

Find the requested values. The 4th term is 16 and the 7th term is -128 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q077 | WRITTEN

C02-T08-S-E1f

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q078 | WRITTEN

C02-T08-S-E1g

Find the requested values. The 2nd term is 3 and the 6th term is 768.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q079 | WRITTEN

C02-T08-S-E1h

Find the requested values. The 3rd term is 9 and the 7th term is 729/16.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q080 | WRITTEN

C02-T08-S-E1i

Find the requested values. The 4th term is $\frac{2}{9}$ and the 8th term is 18.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q081 | MCQ

C02-T08-G-M01

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

A $a_n = 2(3)^{n-1}$

B $a_n = 2(3)^n$

C $a_n = 2(3)^{n+1}$

D $a_n = 2 + 3^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q082 | MCQ

C02-T08-G-M02

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

A $a_n = 3(2)^{n-1}$

B $a_n = 3(2)^n$

C $a_n = 3(2)^{n+1}$

D $a_n = 3 + 2^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q083 | MCQ

C02-T08-G-M03

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

A $a_n = 5(4)^{n-1}$

B $a_n = 5(4)^n$

C $a_n = 5(4)^{n+1}$

D $a_n = 5 + 4^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q084 | MCQ

C02-T08-G-M04

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

A $a_n = 4(5)^{n-1}$

B $a_n = 4(5)^n$

C $a_n = 4(5)^{n+1}$

D $a_n = 4 + 5^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q085 | MCQ

C02-T08-G-M05

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

A $a_n = 2(-3)^{n-1}$

B $a_n = 2(-3)^n$

C $a_n = 2(-3)^{n+1}$

D $a_n = 2 - 3^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q086 | MCQ

C02-T08-G-M06

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

A $a_n = 3(-2)^{n-1}$

B $a_n = 3(-2)^n$

C $a_n = 3(-2)^{n+1}$

D $a_n = 3 - 2^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q087 | MCQ

C02-T08-G-M07

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

A $a_n = 6(-1)^{n-1}$

B $a_n = 6(-1)^n$

C $a_n = 6(-1)^{n+1}$

D $a_n = 6 - 1^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q088 | MCQ

C02-T08-G-M08

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

A $a_n = 8(-2)^{n-1}$

B $a_n = 8(-2)^n$

C $a_n = 8(-2)^{n+1}$

D $a_n = 8 - 2^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q089 | MCQ

C02-T08-G-M09

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

A $a_n = 4\left(\frac{1}{3}\right)^{n-1}$

B $a_n = 4\left(\frac{1}{3}\right)^n$

C $a_n = 4\left(\frac{1}{3}\right)^{n+1}$

D $a_n = 4 + \frac{1}{3}^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q090 | MCQ

C02-T08-G-M10

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

A $a_n = 3\left(-\frac{1}{2}\right)^{n-1}$

B $a_n = 3\left(-\frac{1}{2}\right)^n$

C $a_n = 3\left(-\frac{1}{2}\right)^{n+1}$

D $a_n = 3 - \frac{1}{2}^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q091 | MCQ

C02-T08-G-M11

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

A $a_n = 6\left(\frac{1}{2}\right)^{n-1}$

B $a_n = 6\left(\frac{1}{2}\right)^n$

C $a_n = 6\left(\frac{1}{2}\right)^{n+1}$

D $a_n = 6 + \frac{1}{2}^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q092 | MCQ

C02-T08-G-M12

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

A $a_n = 2\left(\frac{3}{2}\right)^{n-1}$

B $a_n = 2\left(\frac{3}{2}\right)^n$

C $a_n = 2\left(\frac{3}{2}\right)^{n+1}$

D $a_n = 2 + \frac{3}{2}^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q093 | MCQ

C02-T08-G-M13

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

A $a_n = 8\left(-\frac{3}{2}\right)^{n-1}$

B $a_n = 8\left(-\frac{3}{2}\right)^n$

C $a_n = 8\left(-\frac{3}{2}\right)^{n+1}$

D $a_n = 8 - \frac{3}{2}^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q094 | MCQ

C02-T08-G-M14

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

A $a_n = 5(2)^{n-1}$

B $a_n = 5(2)^n$

C $a_n = 5(2)^{n+1}$

D $a_n = 5 + 2^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q095 | MCQ

C02-T08-G-M15

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n :

A $a_n = 7(4)^{n-1}$

B $a_n = 7(4)^n$

C $a_n = 7(4)^{n+1}$

D $a_n = 7 + 4^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q096 | MCQ

C02-T08-G-M16

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n :

A $a_n = 9(-2)^{n-1}$

B $a_n = 9(-2)^n$

C $a_n = 9(-2)^{n+1}$

D $a_n = 9 - 2^{n-1}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q097 | WRITTEN

C02-T08-S-WU2

When 2, x , 8 form a geometric sequence in this order, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q098 | WRITTEN

C02-T08-S-E2

Find the requested values. When 1, x , 25 form a geometric sequence, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q099 | WRITTEN

C02-T08-S-E3

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q100 | WRITTEN

C02-T08-S-T2

When 1, x , 12 form a geometric sequence in this order, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q101 | MCQ

C02-T08-G-M17

Select the correct option: If x is the middle term of the geometric triple 2, x , 8, find x :

A ± 4 B $+4$ C -4

D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q102 | MCQ

C02-T08-G-M18

Select the correct option: If x is the middle term of the geometric triple 1, x , 25, find x :

A ± 5 B $+5$ C -5 D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q103 | MCQ

C02-T08-G-M19

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

A ± 2 B $+2$ C -2 D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q104 | MCQ

C02-T08-G-M20

Select the correct option: If x is the middle term of the geometric triple 3, x , 12, find x :

A ± 6 B $+6$ C -6 D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q105 | MCQ

C02-T08-G-M21

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

A ± 6 B $+6$ C -6 D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q106 | MCQ

C02-T08-G-M22

Select the correct option: If x is the middle term of the geometric triple 2, x , 18, find x :

A ± 6 B $+6$ C -6 D 0

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q107 | WRITTEN

C02-T08-S-WU3

The numbers 6, a , b form an arithmetic sequence and a , b , 16 form a geometric sequence. Find a , b .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q108 | WRITTEN

C02-T08-S-E4

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V02

Q109 | WRITTEN

C02-T08-S-T3

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F99 | V01

Q110 | CHALLENGE WRITTEN

C02-T08-S-CH1

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q111 | WRITTEN

C02-T09-S-WU1a

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q112 | WRITTEN

C02-T09-S-TRY1a

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q113 | WRITTEN

C02-T09-S-EX1a

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q114 | WRITTEN

C02-T09-S-EX1d

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q115 | MCQ

C02-T09-G-M01

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

A 161

B 159

C 162

D 160

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q116 | MCQ

C02-T09-G-M02

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

A 316

B 314

C 315

D 317

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q117 | MCQ

C02-T09-G-M03

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

A 683

B 682

C 681

D 684

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q118 | MCQ

C02-T09-G-M04

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :

A 468

B 469

C 467

D 470

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q119 | MCQ

C02-T09-G-M05

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

A 218

B 216

C 219

D 217

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q120 | MCQ

C02-T09-G-M06

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :

A 79

B 77

C 78

D 80

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q121 | WRITTEN

C02-T09-S-WU1b

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q122 | WRITTEN

C02-T09-S-TRY1c

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q123 | WRITTEN

C02-T09-S-EX1c

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q124 | WRITTEN

C02-T09-S-EX1f

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q125 | MCQ

C02-T09-G-M07

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

A 16

B 15

C 14

D 17

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q126 | MCQ

C02-T09-G-M08

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

A 13

B 14

C 12

D 15

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q127 | MCQ

C02-T09-G-M09

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

A $\frac{47}{2}$

B 23

C $\frac{95}{4}$

D $\frac{93}{4}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q128 | MCQ

C02-T09-G-M10

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

A $\frac{53}{8}$

B $\frac{13}{2}$

C $\frac{105}{16}$

D $\frac{107}{16}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q129 | MCQ

C02-T09-G-M11

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

A $\frac{79}{4}$

B $\frac{315}{16}$

C $\frac{157}{8}$

D $\frac{317}{16}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q130 | MCQ

C02-T09-G-M12

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

A $\frac{160}{27}$

B $\frac{161}{27}$

C $\frac{53}{9}$

D 6

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q131 | WRITTEN

C02-T09-S-TRY1b

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q132 | WRITTEN

C02-T09-S-EX1b

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q133 | WRITTEN

C02-T09-S-EX1e

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q134 | MCQ

C02-T09-G-M13

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :

A 45

B 43

C 46

D 44

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q135 | MCQ

C02-T09-G-M14

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

A -59

B -61

C -60

D -58

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q136 | MCQ

C02-T09-G-M15

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

A 1

B 0

C -1

D 2

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q137 | MCQ

C02-T09-G-M16

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

A -105

B -104

C -106

D -103

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q138 | MCQ

C02-T09-G-M17

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

A 8

B 6

C 9

D 7

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q139 | MCQ

C02-T09-G-M18

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

A 43

B 41

C 42

D 44

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q140 | WRITTEN

C02-T09-S-TRY2a

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q141 | WRITTEN

C02-T09-S-EX2a

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q142 | WRITTEN

C02-T09-S-EX2b

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q143 | WRITTEN

C02-T09-S-EX2e

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q144 | MCQ

C02-T09-G-M19

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

A $3(2^{n-1} - 1)$

B $6(2^n - 1)$

C $3(2^n - 1)$

D $3(1 - 2^n)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q145 | MCQ

C02-T09-G-M20

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

A $5\{1 - (\frac{1}{2})^n\}$

B $10\{1 - (\frac{1}{2})^{n-1}\}$

C $10\{1 - (2)^n\}$

D $10\{1 - (\frac{1}{2})^n\}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q146 | MCQ

C02-T09-G-M21

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

A $\frac{4\{1 - (-2)^n\}}{3}$

B $\frac{4\{1 - (-2)^{n-1}\}}{3}$

C $\frac{4\{1 + (-2)^n\}}{3}$

D $\frac{4\{1 - 2^n\}}{3}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q147 | WRITTEN

C02-T09-S-EX2c

Find S_n from the first term to the n -th term of 2, 6, 18, 54,

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q148 | WRITTEN

C02-T09-S-WU2

Find S_n from the first term to the n -th term of $2, -6, 18, -54, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q149 | WRITTEN

C02-T09-S-TRY2b

Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q150 | WRITTEN

C02-T09-S-EX2d

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q151 | WRITTEN

C02-T09-S-EX2f

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Q152 | WRITTEN

C02-T09-S-EX2g

Find S_n from the first term to the n -th term of $1, -2, 4, -8, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q153 | MCQ

C02-T09-G-M38

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

A 120

B 121

C 119

D 122

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F99 | V01

Q154 | CHALLENGE WRITTEN

C02-T09-S-CH1

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q155 | MCQ

C02-T10-G-M01

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

A $(a, r) = (2, 2)$

B $(a, r) = (1, 3)$

C $(a, r) = \left(\frac{10}{3}, -2\right)$

D $(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q156 | MCQ

C02-T10-G-M02

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

D $(a, r) = \left(\frac{33}{7}, -3\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q157 | MCQ

C02-T10-G-M03

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

A $(a, r) = (4, 2)$

B $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

C $(a, r) = (3, 3)$

D $(a, r) = (8, -2)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q158 | MCQ

C02-T10-G-M04

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

A $(a, r) = (4, 2)$ or $(a, r) = \left(\frac{28}{3}, -2\right)$

B $(a, r) = (5, 2)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(\frac{31}{3}, -2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q159 | MCQ

C02-T10-G-M05

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = \left(\frac{20}{7}, -3\right)$

D $(a, r) = (1, 3)$ or $(a, r) = \left(\frac{13}{7}, -3\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q160 | MCQ

C02-T10-G-M06

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

A $(a, r) = (6, 2)$

B $(a, r) = (5, 3)$

C $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

D $(a, r) = \left(\frac{38}{3}, -2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Q161 | MCQ

C02-T10-G-M09

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

A $(a, r) = \left(3, \frac{3}{2}\right)$

B $(a, r) = \left(2, \frac{5}{2}\right)$

C $(a, r) = \left(\frac{45}{7}, -\frac{3}{2}\right)$

D $(a, r) = \left(2, \frac{3}{2}\right)$ or $(a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q162 | WRITTEN

C02-T10-S-E1

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Q163 | MCQ

C02-T10-G-M10

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

A $(a, r) = \left(5, \frac{1}{2}\right)$

B $(a, r) = \left(4, \frac{3}{2}\right)$

C $(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

D $(a, r) = \left(\frac{31}{3}, -\frac{1}{2}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q164 | WRITTEN

C02-T10-S-WU1

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q165 | WRITTEN

C02-T10-S-T1

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q166 | WRITTEN

C02-T10-S-E2

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q167 | WRITTEN

C02-T10-S-E3

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q168 | MCQ

C02-T10-G-M07

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

A $(a, r) = (3, -2)$

B $(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

C $(a, r) = (2, -1)$

D $(a, r) = \left(\frac{13}{7}, 2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V03

Q169 | MCQ

C02-T10-G-M08

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

A $(a, r) = (3, -2)$ or $(a, r) = \left(\frac{9}{7}, 2\right)$

B $(a, r) = (4, -2)$

C $(a, r) = (3, -1)$

D $(a, r) = \left(\frac{16}{7}, 2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q170 | WRITTEN

C02-T10-S-T3

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q171 | WRITTEN

C02-T10-S-E7

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q172 | MCQ

C02-T10-G-M11

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :

A $(a, r) = (4, 2)$

B $(a, r) = (3, 3)$

C $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

D $(a, r) = \left(13, \frac{1}{2}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q173 | MCQ

C02-T10-G-M12

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = \left(19, \frac{1}{3}\right)$

D $(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q174 | MCQ

C02-T10-G-M14

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :

A $(a, r) = (5, 2)$

B $(a, r) = (4, 2)$ or $(a, r) = \left(16, \frac{1}{2}\right)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(17, \frac{1}{2}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q175 | MCQ

C02-T10-G-M15

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

D $(a, r) = \left(10, \frac{1}{3}\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V01

Q176 | MCQ

C02-T10-G-M16

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

A $(a, r) = \left(6, \frac{1}{2}\right)$

B $(a, r) = \left(5, \frac{3}{2}\right)$

C $(a, r) = \left(\frac{9}{4}, 2\right)$

D $(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q177 | WRITTEN

C02-T10-S-WU3

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q178 | WRITTEN

C02-T10-S-E8

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Q179 | WRITTEN

C02-T10-S-E9

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q180 | WRITTEN

C02-T10-S-WU2

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q181 | WRITTEN

C02-T10-S-T2

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Q182 | WRITTEN

C02-T10-S-E5

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F04 | V03

Q183 | WRITTEN

C02-T10-S-E4

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F04 | V03

Q184 | WRITTEN

C02-T10-S-E6

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F05 | V01

Q185 | MCQ

C02-T10-G-M40

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

A 537

B 487

C 562

D 512

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F99 | V01

Q186 | CHALLENGE WRITTEN

C02-T10-S-CH1

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V01

Quiz M1 | QUIZ MCQ

C02-T07-Q-M1

Select the correct option: For $a = 3, r = 2$, find a_6 :

A 48

B 96

C 192

D 64

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Quiz M2 | QUIZ MCQ

C02-T08-Q-M2

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

A 4

B -4

C ± 4

D 16

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V02

Quiz M3 | QUIZ MCQ

C02-T09-Q-M3

For 1, 3, 9, ..., find S_n :

A $\frac{3^{n-1} - 1}{2}$

B $\frac{3^n - 1}{2}$

C $\frac{3^n + 1}{2}$

D $\frac{3^n - 1}{3}$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F05 | V01

Quiz M4 | QUIZ MCQ

C02-T10-Q-M4

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

A 528

B 496

C 544

D 512

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F01 | V02

Quiz W1 | QUIZ WRITTEN

C02-T07-Q-W1

Find the general term of $-4, 8, -16, 32, \dots$

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F02 | V03

Quiz W2 | QUIZ WRITTEN

C02-T08-Q-W2

If 1, x , 25 are consecutive geometric terms, find x .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F03 | V01

Quiz W3 | QUIZ WRITTEN

C02-T09-Q-W3

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_{\infty} = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 2 | Geometric Sequences

F05 | V01

Quiz W4 | QUIZ WRITTEN

C02-T10-Q-W4

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

WORKING AREA

FORMULA BANK

General term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1} \quad (r \neq 1)$$

Infinite sum

$$S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Concept 3 | Summation

OFFICIAL LESSONS 2-11, 2-12, 2-13, 2-14, 2-15, 2-16, 2-17

Summation

Equation-first recall | use the same rail beside every question in this Concept

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Summation

F01 | V01

Q001 | WRITTEN

C02-T11-S-WU1a

Write $\sum_{k=1}^5 2k$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q002 | WRITTEN

C02-T11-S-T1a

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q003 | WRITTEN

C02-T11-S-E1a

Write $\sum_{k=1}^5 3k$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q004 | WRITTEN

C02-T11-S-E1b

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q005 | WRITTEN

C02-T11-S-E1c

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q006 | MCQ

C02-T11-G-M01

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

A $4 + 8 + 12$

B $4 + 8 + 16$

C $8 + 12 + 16$

D $4 + 12 + 20$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q007 | MCQ

C02-T11-G-M04

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

A $2 + 4 + 8 + 16$

B $1 + 2 + 4 + 8$

C $2 + 6 + 18 + 54$

D $4 + 8 + 16 + 32$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q008 | MCQ

C02-T11-G-M07

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

A $4 + 24 + 124$

B $5 + 25 + 125$

C $4 + 25 + 124$

D $1 + 4 + 24$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q009 | MCQ

C02-T11-G-M10

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

A $5 + 6 + 7 + 8 + 9$

B $4 + 5 + 6 + 7 + 8$

C $1 + 2 + 3 + 4 + 5$

D $6 + 7 + 8 + 9 + 10$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q010 | MCQ

C02-T11-G-M13

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

A $7 + 9 + 11 + 13$

B $5 + 7 + 9 + 11$

C $4 + 7 + 10 + 13$

D $8 + 10 + 12 + 14$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q011 | MCQ

C02-T11-G-M16

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

A $3 + 5 + 9$ B $2 + 4 + 8$ C $1 + 3 + 5$ D $4 + 6 + 10$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q012 | MCQ

C02-T11-G-M19

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

A $4 + 10 + 18$ B $3 + 8 + 15$ C $1 + 4 + 9$ D $6 + 12 + 18$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q013 | MCQ

C02-T11-G-M22

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

A $5 + 10 + 17 + 26$

B $2 + 5 + 10 + 17$

C $4 + 9 + 16 + 25$

D $6 + 11 + 18 + 27$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q014 | WRITTEN

C02-T11-S-WU1b

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q015 | WRITTEN

C02-T11-S-T1b

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q016 | WRITTEN

C02-T11-S-E1d

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q017 | WRITTEN

C02-T11-S-E1e

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q018 | WRITTEN

C02-T11-S-E1f

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q019 | MCQ

C02-T11-G-M02

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

A $3 + 5 + 7 + 9$

B $2 + 4 + 6 + 8$

C $3 + 6 + 9 + 12$

D $1 + 3 + 5 + 7$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q020 | MCQ

C02-T11-G-M05

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

A $5 + 8 + 11 + 14$

B $2 + 5 + 8 + 11$

C $5 + 9 + 13 + 17$

D $4 + 7 + 10 + 13$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q021 | MCQ

C02-T11-G-M08

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

A $-1 + 5 + 15 + 29$

B $-1 + 1 + 5 + 13$

C $2 + 5 + 10 + 17$

D $-3 + 5 + 13 + 29$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q022 | MCQ

C02-T11-G-M11

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

A $4 + 13 + 28 + 49$

B $3 + 12 + 27 + 48$

C $4 + 10 + 19 + 28$

D $1 + 4 + 9 + 16$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q023 | MCQ

C02-T11-G-M14

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

A $0 + 2 + 6 + 12$ B $1 + 4 + 9 + 16$ C $0 + 1 + 4 + 9$ D $2 + 6 + 12 + 20$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q024 | MCQ

C02-T11-G-M17

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

A $10 + 15 + 20$ B $5 + 10 + 15$ C $2 + 4 + 6$ D $15 + 20 + 25$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q025 | MCQ

C02-T11-G-M20

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

A $3 + 3 + 3 + 3$

B $3 + 6 + 9 + 12$

C $1 + 1 + 1 + 1$

D $4 + 4 + 4 + 4$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q026 | MCQ

C02-T11-G-M23

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

A $6 + 11 + 16 + 21$

B $5 + 10 + 15 + 20$

C $1 + 6 + 11 + 16$

D $7 + 12 + 17 + 22$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q027 | WRITTEN

C02-T11-S-WU1c

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q028 | WRITTEN

C02-T11-S-WU1d

Write $\sum_{k=1}^3 2$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q029 | WRITTEN

C02-T11-S-T1c

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q030 | WRITTEN

C02-T11-S-T1d

Write $\sum_{k=1}^6 3$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q031 | WRITTEN

C02-T11-S-E1g

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q032 | WRITTEN

C02-T11-S-E1h

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q033 | WRITTEN

C02-T11-S-E1i

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q034 | WRITTEN

C02-T11-S-E1j

Write $\sum_{k=1}^4 5$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q035 | WRITTEN

C02-T11-S-E1k

Write $\sum_{k=1}^3 3$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q036 | WRITTEN

C02-T11-S-E11

Write $\sum_{k=1}^5 4$ as a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q037 | MCQ

C02-T11-G-M03

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

A $3 + 6 + 11$ B $2 + 5 + 10$ C $3 + 5 + 9$ D $1 + 4 + 9$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q038 | MCQ

C02-T11-G-M06

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

A $7 + 7 + 7 + 7 + 7$

B $7 + 14 + 21 + 28 + 35$

C $7 + 7 + 7 + 7$

D $5 + 5 + 5 + 5 + 5$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q039 | MCQ

C02-T11-G-M09

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

A $9 + 16 + 25 + 36$

B $1 + 4 + 9 + 16$

C $9 + 25 + 49 + 81$

D $3 + 4 + 5 + 6$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q040 | MCQ

C02-T11-G-M12

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

A $6 + 36 + 216$

B $1 + 6 + 36$

C $6 + 12 + 18$

D $36 + 72 + 108$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q041 | MCQ

C02-T11-G-M15

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

A $1 + 5 + 9 + 13 + 17$

B $4 + 8 + 12 + 16 + 20$

C $1 + 4 + 7 + 10 + 13$

D $-1 + 3 + 7 + 11 + 15$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q042 | MCQ

C02-T11-G-M18

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

A $-2 + 1 + 4 + 7$

B $-2 + 0 + 2 + 4$

C $3 + 6 + 9 + 12$

D $-5 - 2 + 1 + 4$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q043 | MCQ

C02-T11-G-M21

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

A $2 + 14 + 62$

B $4 + 16 + 64$

C $2 + 12 + 60$

D $-2 + 2 + 14$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q044 | MCQ

C02-T11-G-M24

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

A $1 + 6 + 15$ B $2 + 7 + 18$ C $1 + 3 + 5$ D $0 + 6 + 12$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q045 | WRITTEN

C02-T11-S-WU2a

Express $3 + 7 + 11 + \cdots + (4n - 1)$ using the summation symbol Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q046 | WRITTEN

C02-T11-S-T2a

Express $5 + 8 + 11 + \cdots + (3n + 2)$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q047 | WRITTEN

C02-T11-S-T2b

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q048 | WRITTEN

C02-T11-S-E2a

Express $1 + 3 + 5 + \cdots + (2n - 1)$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q049 | WRITTEN

C02-T11-S-E2d

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q050 | WRITTEN

C02-T11-S-E2e

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q051 | WRITTEN

C02-T11-S-E2f

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q052 | MCQ

C02-T11-G-M25

Select the correct option: Express $2 + 5 + 8 + \cdots + (3n - 1)$ using the summation symbol Σ :

A $\sum_{k=1}^n (3k - 1)$

B $\sum_{k=0}^n (3k - 1)$

C $\sum_{k=1}^n (3k + 1)$

D $\sum_{k=1}^n (2k - 1)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q053 | MCQ

C02-T11-G-M28

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :

A $\sum_{k=1}^{10} (4k + 3)$

B $\sum_{k=1}^9 (4k + 3)$

C $\sum_{k=0}^{10} (4k + 3)$

D $\sum_{k=1}^{10} (4k - 3)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q054 | MCQ

C02-T11-G-M31

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :

A $\sum_{k=1}^n (4k + 2)$

B $\sum_{k=0}^n (4k + 2)$

C $\sum_{k=1}^n (4k - 2)$

D $\sum_{k=1}^n (3k + 2)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q055 | MCQ

C02-T11-G-M34

Select the correct option: Express $8 + 13 + 18 + \cdots + 103$ using the summation symbol Σ :

A $\sum_{k=1}^{20} (5k + 3)$

B $\sum_{k=1}^{19} (5k + 3)$

C $\sum_{k=0}^{20} (5k + 3)$

D $\sum_{k=1}^{20} (5k - 3)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q056 | MCQ

C02-T11-G-M37

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :

A $\sum_{k=11}^{19} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=11}^{19} k$

D $\sum_{k=12}^{19} k^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q057 | MCQ

C02-T11-G-M40

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :

A $\sum_{k=2}^{10} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=2}^{10} k$

D $\sum_{k=3}^{10} k^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q058 | MCQ

C02-T11-G-M43

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :

A $\sum_{k=0}^{n-1} 3^k$

B $\sum_{k=1}^n 3^k$

C $\sum_{k=1}^{n-1} 3^k$

D $\sum_{k=0}^n 3^k$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q059 | MCQ

C02-T11-G-M46

Select the correct option: Express $2 + 7 + 12 + \cdots + (5n - 3)$ using the summation symbol Σ :

A $\sum_{k=1}^n (5k - 3)$

B $\sum_{k=0}^n (5k - 3)$

C $\sum_{k=1}^n (5k + 3)$

D $\sum_{k=1}^n (4k - 3)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q060 | WRITTEN

C02-T11-S-WU2b

Express $6^2 + 7^2 + 8^2 + \cdots + 15^2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q061 | WRITTEN

C02-T11-S-T2c

Express $4^2 + 5^2 + 6^2 + \cdots + 13^2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q062 | WRITTEN

C02-T11-S-E2b

Express $1 + 7 + 25 + 79 + \cdots + (3^n - 2)$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q063 | WRITTEN

C02-T11-S-E2c

Express $1 + 2 + 4 + 8 + \cdots + 2^{n-1}$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q064 | WRITTEN

C02-T11-S-E2g

Express $7^2 + 8^2 + \cdots + 21^2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q065 | WRITTEN

C02-T11-S-E2h

Express $12^2 + 13^2 + \cdots + 18^2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q066 | WRITTEN

C02-T11-S-E2i

Express $10^2 + 11^2 + \cdots + 20^2$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q067 | MCQ

C02-T11-G-M26

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :

A $\sum_{k=4}^{12} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=4}^{12} k$

D $\sum_{k=5}^{12} k^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q068 | MCQ

C02-T11-G-M29

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :

A $\sum_{k=1}^n 3^k$

B $\sum_{k=0}^n 3^k$

C $\sum_{k=1}^n 3^{k-1}$

D $\sum_{k=1}^n k^3$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q069 | MCQ

C02-T11-G-M32

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :

A $\sum_{k=1}^n 2^k$

B $\sum_{k=0}^n 2^k$

C $\sum_{k=1}^n 2^{k-1}$

D $\sum_{k=1}^n k^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q070 | MCQ

C02-T11-G-M35

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :

A $\sum_{k=0}^{n-1} 2^k$

B $\sum_{k=1}^n 2^k$

C $\sum_{k=1}^{n-1} 2^k$

D $\sum_{k=0}^n 2^k$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q071 | MCQ

C02-T11-G-M38

Select the correct option: Express $7 + 14 + 21 + \cdots + 7n$ using the summation symbol Σ :

A $\sum_{k=1}^n 7k$

B $\sum_{k=0}^n 7k$

C $\sum_{k=1}^n (7k + 1)$

D $\sum_{k=1}^n k^7$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q072 | MCQ

C02-T11-G-M41

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :

A $\sum_{k=2}^n 5k$

B $\sum_{k=1}^n 5k$

C $\sum_{k=2}^n (5k + 1)$

D $\sum_{k=0}^n 5k$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q073 | MCQ

C02-T11-G-M44

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

A $\sum_{k=6}^{16} k^2$

B $\sum_{k=1}^{11} k^2$

C $\sum_{k=6}^{16} k$

D $\sum_{k=7}^{16} k^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q074 | WRITTEN

C02-T11-S-T2d

Express $2 + 5 + 8 + \cdots + 29$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q075 | WRITTEN

C02-T11-S-E2j

Express $8 + 16 + 24 + \cdots + 56$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q076 | WRITTEN

C02-T11-S-E2k

Express $7 + 13 + 19 + \cdots + 73$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q077 | WRITTEN

C02-T11-S-E2I

Express $4 + 9 + 14 + \cdots + 119$ using Σ .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q078 | CHALLENGE WRITTEN

C02-T11-S-CH1

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

WORKING AREA

FORMULA BANK

Sigma linearity

$$\Sigma (au_n + bv_n) = a \Sigma u_n + b \Sigma v_n$$

Standard sums

$$\Sigma k = \frac{n(n+1)}{2} \quad \Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q079 | WRITTEN

C02-T12-S-WU-A

Find $\sum_{k=1}^n (-3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q080 | WRITTEN

C02-T12-S-T-A

Find $\sum_{k=1}^n 3$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q081 | WRITTEN

C02-T12-S-E-A

Find $\sum_{k=1}^n 6$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q082 | WRITTEN

C02-T12-S-E-B

Find $\sum_{k=1}^n (-4)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q083 | WRITTEN

C02-T12-S-E-C

Find $\sum_{k=1}^n 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q084 | MCQ

C02-T12-G-M01

Select the correct option: Find $\sum_{k=1}^n 9$:

A $9n$

B $n + 9$

C $9n^2$

D $\frac{9n(n+1)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q085 | MCQ

C02-T12-G-M02

Select the correct option: Find $\sum_{k=1}^n (-2)$:

A $-2n$ B $2n$ C $-n - 2$ D $-n^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q086 | MCQ

C02-T12-G-M09

Select the correct option: Find $\sum_{k=1}^7 4$:

A 28

B 11

C 16

D 49

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q087 | MCQ

C02-T12-G-M12

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

A -60

B 60

C -17

D -144

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q088 | MCQ

C02-T12-G-M03

Select the correct option: Find $\sum_{k=1}^n k$:

A $\frac{n(n+1)}{2}$

B $n(n+1)$

C $\frac{n(n-1)}{2}$

D n^2

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q089 | MCQ

C02-T12-G-M05

Select the correct option: Find $\sum_{k=1}^n 4k$:

A $2n(n+1)$

B $4n(n+1)$

C $2n(n-1)$

D $4n$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q090 | MCQ

C02-T12-G-M07

Select the correct option: Find $\sum_{k=1}^{10} k$:

A 55

B 45

C 100

D 110

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q091 | MCQ

C02-T12-G-M10

Select the correct option: Find $\sum_{k=1}^8 3k$:

A 108

B 72

C 96

D 120

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q092 | MCQ

C02-T12-G-M13

Select the correct option: Find $\sum_{k=1}^6 5k$:

A 105

B 75

C 90

D 120

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q093 | MCQ

C02-T12-G-M15

Select the correct option: Find $\sum_{k=1}^{20} k$:

A 210

B 190

C 200

D 420

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q094 | MCQ

C02-T12-G-M04

Select the correct option: Find $\sum_{k=1}^n k^2$:

A $\frac{n(n+1)(2n+1)}{6}$

B $\frac{n(n+1)}{2}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n^2(n+1)^2}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q095 | MCQ

C02-T12-G-M06

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

A $n(n+1)(2n+1)$

B $3n(n+1)$

C $n(n+1)(2n-1)$

D $6n^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q096 | MCQ

C02-T12-G-M08

Select the correct option: Find $\sum_{k=1}^5 k^2$:

A 55

B 25

C 30

D 50

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q097 | MCQ

C02-T12-G-M11

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

A 60

B 30

C 40

D 80

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q098 | MCQ

C02-T12-G-M14

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

A 56

B 36

C 48

D 72

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q099 | MCQ

C02-T12-G-M16

Select the correct option: Find $\sum_{k=1}^7 k^2$:

A 140

B 49

C 91

D 196

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q100 | WRITTEN

C02-T12-S-WU-B

Find $\sum_{k=1}^n (5k + 4)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q101 | WRITTEN

C02-T12-S-T-B

Find $\sum_{k=1}^n (2k - 3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q102 | WRITTEN

C02-T12-S-E-D

Find $\sum_{k=1}^n (4k - 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q103 | WRITTEN

C02-T12-S-E-E

Find $\sum_{k=1}^n (3k + 2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q104 | WRITTEN

C02-T12-S-E-F

Find $\sum_{k=1}^n (6k - 3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q105 | MCQ

C02-T12-G-M17

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

A $\frac{n(3n+1)}{2}$

B $\frac{n(3n-1)}{2}$

C $n(3n+1)$

D $\frac{3n(n+1)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q106 | MCQ

C02-T12-G-M18

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

A $n(n + 6)$

B $n(n + 5)$

C $n(n + 1) + 5$

D $2n(n + 6)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q107 | MCQ

C02-T12-G-M19

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

A $4n^2$

B $4n(n + 1)$

C $8n^2$

D $4n(n - 1)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q108 | MCQ

C02-T12-G-M26

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

A $\frac{n(5n + 11)}{2}$

B $\frac{n(5n + 3)}{2}$

C $\frac{n(5n + 5)}{2}$

D $n(5n + 11)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q109 | WRITTEN

C02-T12-S-E-I

Find $\sum_{k=1}^n (k-1)^2$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q110 | MCQ

C02-T12-G-M20

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

A $\frac{n(n+1)(n+2)}{3}$

B $\frac{n(n+1)(2n+1)}{6}$

C $\frac{n(n+1)(n+2)}{6}$

D $n(n+1)(n+2)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q111 | MCQ

C02-T12-G-M21

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

A $\frac{n(n+1)(4n-1)}{6}$

B $\frac{n(n+1)(4n+1)}{6}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n(n-1)(4n-1)}{6}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q112 | MCQ

C02-T12-G-M22

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

A $\frac{n(2n^2 + 3n - 5)}{6}$

B $\frac{n(2n^2 + 3n - 1)}{6}$

C $\frac{n(2n^2 - 3n + 1)}{6}$

D $\frac{n(n+1)(2n+1)}{6}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q113 | MCQ

C02-T12-G-M27

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

A $\frac{n(n+1)(2n+3)}{2}$

B $\frac{n(n+1)(2n+1)}{2}$

C $n(n+1)(2n+3)$

D $\frac{n(n-1)(2n+3)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q114 | MCQ

C02-T12-G-M28

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

A $\frac{n(8n^2 + 3n + 7)}{6}$

B $\frac{n(8n^2 + 9n + 7)}{6}$

C $\frac{n(4n^2 + 3n + 2)}{6}$

D $\frac{n(8n^2 - 3n + 7)}{6}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q115 | WRITTEN

C02-T12-S-WU-C

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q116 | WRITTEN

C02-T12-S-T-C

Find $\sum_{k=1}^n (k-1)(3k-1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q117 | WRITTEN

C02-T12-S-T-D

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q118 | WRITTEN

C02-T12-S-E-G

Find $\sum_{k=1}^n k(k-2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q119 | WRITTEN

C02-T12-S-E-H

Find $\sum_{k=1}^n k(2k-2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q120 | WRITTEN

C02-T12-S-E-J

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q121 | WRITTEN

C02-T12-S-E-K

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q122 | WRITTEN

C02-T12-S-E-L

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n+1)(n+4)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q123 | WRITTEN

C02-T12-S-WU-D

Find $\sum_{k=1}^4 (2k - 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q124 | WRITTEN

C02-T12-S-T-E

Find $\sum_{k=1}^5 (3k - 2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q125 | WRITTEN

C02-T12-S-E-M

Find $\sum_{k=1}^{20} (2k - 7)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q126 | WRITTEN

C02-T12-S-E-N

Find $\sum_{k=1}^8 (4k + 5)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q127 | WRITTEN

C02-T12-S-E-O

Find $\sum_{k=1}^{10} (2k - 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q128 | CHALLENGE WRITTEN

C02-T12-S-CH

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q129 | WRITTEN

C02-T13-S-WU-a

Find the sum $\sum_{k=1}^n k^2(k+1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q130 | WRITTEN

C02-T13-S-Try-a

Find the sum $\sum_{k=1}^n k^2(k-2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q131 | WRITTEN

C02-T13-S-Ex-b

Find the sum $\sum_{k=1}^n k^2(k+2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q132 | WRITTEN

C02-T13-S-Ex-d

Find the sum $\sum_{k=1}^n k^2(k-3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q133 | MCQ

C02-T13-G-M01

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

A $\frac{n(n+1)(n+2)(3n+1)}{12}$

B $\frac{n(n+1)(n+2)(3n-1)}{12}$

C $\frac{n(n+1)(3n+1)}{12}$

D $\frac{n(n+1)(n+2)(3n+2)}{6}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q134 | MCQ

C02-T13-G-M02

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

A $\frac{n(n+1)(3n^2-5n-4)}{12}$

B $\frac{n(n+1)(3n^2+5n-4)}{12}$

C $\frac{n(n+1)(3n^2-5n+4)}{12}$

D $\frac{n(n+1)(3n^2-4n-5)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q135 | MCQ

C02-T13-G-M04

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

A $\frac{n(n+1)(3n^2+11n+4)}{12}$

B $\frac{n(n+1)(3n^2+11n-4)}{12}$

C $\frac{n(n+1)(3n^2+9n+4)}{12}$

D $\frac{n(n+1)(3n^2+4n+11)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q136 | MCQ

C02-T13-G-M06

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

A $\frac{n(n+1)(n^2-3n-2)}{4}$

B $\frac{n(n+1)(n^2+3n-2)}{4}$

C $\frac{n(n+1)(n^2-3n+2)}{4}$

D $\frac{n(n+1)(n^2-2n-3)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q137 | MCQ

C02-T13-G-M07

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

A $\frac{n(n+1)(n^2+5n+2)}{4}$

B $\frac{n(n+1)(n^2+5n-2)}{4}$

C $\frac{n(n+1)(n^2+3n+2)}{4}$

D $\frac{n(n+1)(n^2+2n+5)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q138 | MCQ

C02-T13-G-M08

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

A $\frac{n(n+1)(3n^2-13n-8)}{12}$

B $\frac{n(n+1)(3n^2-13n+8)}{12}$

C $\frac{n(n+1)(3n^2-11n-8)}{12}$

D $\frac{n(n+1)(3n^2-8n-13)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q139 | MCQ

C02-T13-G-M11

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

A $\frac{n(n+1)(3n^2+19n+8)}{12}$

B $\frac{n(n+1)(3n^2+19n-8)}{12}$

C $\frac{n(n+1)(3n^2+17n+8)}{12}$

D $\frac{n(n+1)(3n^2+8n+19)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q140 | MCQ

C02-T13-G-M18

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

A $\frac{n(n+1)(3n^2-5n-4)}{12}$

B $\frac{n(n+1)(3n^2-5n+4)}{12}$

C $\frac{n(n+1)(3n^2-4n-5)}{12}$

D $\frac{n(n+1)(3n^2-n-4)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q141 | WRITTEN

C02-T13-S-Ex-a

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q142 | WRITTEN

C02-T13-S-Ex-c

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q143 | MCQ

C02-T13-G-M03

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

A $\frac{n(n+1)(n-1)(n+2)}{4}$

B $\frac{n(n+1)(n+1)(n+2)}{4}$

C $\frac{n(n-1)(n+2)}{4}$

D $\frac{n(n+1)(n-2)(n+3)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q144 | MCQ

C02-T13-G-M05

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

A $\frac{n(n+1)(n^2+n+4)}{4}$

B $\frac{n(n+1)(n^2+n-4)}{4}$

C $\frac{n(n+1)(n^2+2n+4)}{4}$

D $\frac{n(n+1)(n^2+n+2)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q145 | MCQ

C02-T13-G-M09

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

A $\frac{n(n+1)(n^2+n+6)}{4}$

B $\frac{n(n+1)(n^2+n-6)}{4}$

C $\frac{n(n+1)(n^2+3n+6)}{4}$

D $\frac{n(n+1)(n^2+n+3)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q146 | MCQ

C02-T13-G-M10

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

A $\frac{n(n+1)(n^2+n-4)}{4}$

B $\frac{n(n+1)(n^2+n+4)}{4}$

C $\frac{n(n+1)(n^2-2n-4)}{4}$

D $\frac{n(n+1)(n^2+n-2)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q147 | MCQ

C02-T13-G-M12

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

A $\frac{n(n+1)(n^2+n+8)}{4}$

B $\frac{n(n+1)(n^2+n-8)}{4}$

C $\frac{n(n+1)(n^2+4n+8)}{4}$

D $\frac{n(n+1)(n^2+n+4)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q148 | MCQ

C02-T13-G-M13

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

A $\frac{n(n+1)(n^2+n-6)}{4}$

B $\frac{n(n+1)(n^2+n+6)}{4}$

C $\frac{n(n+1)(n^2-3n-6)}{4}$

D $\frac{n(n+1)(n^2+n-3)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q149 | MCQ

C02-T13-G-M14

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

A $\frac{n(n+1)(3n^2+23n+10)}{12}$

B $\frac{n(n+1)(3n^2+23n-10)}{12}$

C $\frac{n(n+1)(3n^2+21n+10)}{12}$

D $\frac{n(n+1)(3n^2+10n+23)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q150 | MCQ

C02-T13-G-M15

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

A $\frac{n(n+1)(3n^2-17n-10)}{12}$

B $\frac{n(n+1)(3n^2-17n+10)}{12}$

C $\frac{n(n+1)(3n^2-15n-10)}{12}$

D $\frac{n(n+1)(3n^2-10n-17)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q151 | MCQ

C02-T13-G-M16

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

A $\frac{n(n+1)(n^2+n+10)}{4}$

B $\frac{n(n+1)(n^2+n-10)}{4}$

C $\frac{n(n+1)(n^2+5n+10)}{4}$

D $\frac{n(n+1)(n^2+n+5)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q152 | MCQ

C02-T13-G-M17

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

A $\frac{n(n+1)(n^2+n-10)}{4}$

B $\frac{n(n+1)(n^2+n+10)}{4}$

C $\frac{n(n+1)(n^2-5n-10)}{4}$

D $\frac{n(n+1)(n^2+n-5)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q153 | MCQ

C02-T13-G-M19

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

A $\frac{n(n+1)(3n^2+27n+12)}{12}$

B $\frac{n(n+1)(3n^2+27n-12)}{12}$

C $\frac{n(n+1)(3n^2+25n+12)}{12}$

D $\frac{n(n+1)(3n^2+12n+27)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q154 | MCQ

C02-T13-G-M20

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

A $\frac{n(n+1)(n^2+n+12)}{4}$

B $\frac{n(n+1)(n^2+n-12)}{4}$

C $\frac{n(n+1)(n^2+6n+12)}{4}$

D $\frac{n(n+1)(n^2+n+6)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q155 | MCQ

C02-T13-G-M21

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-6)$:

A $\frac{n(n+1)(3n^2-25n-12)}{12}$

B $\frac{n(n+1)(3n^2-25n+12)}{12}$

C $\frac{n(n+1)(3n^2-23n-12)}{12}$

D $\frac{n(n+1)(3n^2-12n-25)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q156 | MCQ

C02-T13-G-M22

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:

A $\frac{n(n+1)(n^2+n-12)}{4}$

B $\frac{n(n+1)(n^2+n+12)}{4}$

C $\frac{n(n+1)(n^2-6n-12)}{4}$

D $\frac{n(n+1)(n^2+n-6)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q157 | MCQ

C02-T13-G-M23

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

A $\frac{n(n+1)(3n^2+31n+14)}{12}$

B $\frac{n(n+1)(3n^2+31n-14)}{12}$

C $\frac{n(n+1)(3n^2+29n+14)}{12}$

D $\frac{n(n+1)(3n^2+14n+31)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q158 | MCQ

C02-T13-G-M24

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

A $\frac{n(n+1)(3n^2+35n+16)}{12}$

B $\frac{n(n+1)(3n^2+35n-16)}{12}$

C $\frac{n(n+1)(3n^2+33n+16)}{12}$

D $\frac{n(n+1)(3n^2+16n+35)}{12}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q159 | WRITTEN

C02-T13-S-Ex-e

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q160 | WRITTEN

C02-T13-S-Ex-g

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q161 | WRITTEN

C02-T13-S-Ex-o

Find the sum $\sum_{k=1}^n 2^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q162 | WRITTEN

C02-T13-S-WU-b

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q163 | WRITTEN

C02-T13-S-Try-b

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q164 | WRITTEN

C02-T13-S-Try-d

Find the sum $\sum_{k=1}^n (-5)^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q165 | WRITTEN

C02-T13-S-Ex-f

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q166 | WRITTEN

C02-T13-S-Ex-h

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q167 | WRITTEN

C02-T13-S-Ex-k

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q168 | WRITTEN

C02-T13-S-Ex-m

Find the sum $\sum_{k=1}^n (-3)^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q169 | WRITTEN

C02-T13-S-WU-c

Find the sum $\sum_{k=1}^n 4^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q170 | WRITTEN

C02-T13-S-Try-c

Find the sum $\sum_{k=1}^n 4^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q171 | WRITTEN

C02-T13-S-Ex-i

Find the sum $\sum_{k=1}^n 3^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q172 | WRITTEN

C02-T13-S-Ex-j

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q173 | WRITTEN

C02-T13-S-Ex-I

Find the sum $\sum_{k=1}^3 3^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q174 | WRITTEN

C02-T13-S-Ex-n

Find the sum $\sum_{k=1}^5 2^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q175 | WRITTEN

C02-T13-S-Ex-p

Find the sum $\sum_{k=1}^4 (-3)^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q176 | CHALLENGE WRITTEN

C02-T13-S-CH1

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q177 | WRITTEN

C02-T14-S-T1a

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q178 | WRITTEN

C02-T14-S-E1a

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q179 | WRITTEN

C02-T14-S-E2a

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q180 | MCQ

C02-T14-G-M01

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

A $\frac{n}{n+1}$

B $\frac{1}{n+1}$

C $\frac{n}{2(n+1)}$

D $\frac{n+1}{n}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q181 | MCQ

C02-T14-G-M02

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \dots + \frac{1}{(n+1)(n+2)}$:

A $\frac{n}{2(n+2)}$

B $\frac{1}{2(n+2)}$

C $\frac{n}{n+2}$

D $\frac{n+1}{2(n+2)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q182 | MCQ

C02-T14-G-M03

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \dots + \frac{1}{(n+2)(n+3)}$:

A $\frac{n}{3(n+3)}$

B $\frac{n}{n+3}$

C $\frac{1}{3(n+3)}$

D $\frac{n+1}{3(n+3)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q183 | MCQ

C02-T14-G-M09

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

A $\frac{n}{4(n+4)}$

B $\frac{n}{3(n+4)}$

C $\frac{n+1}{4(n+4)}$

D $\frac{1}{4(n+4)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q184 | MCQ

C02-T14-G-M12

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

A $\frac{n}{6(n+6)}$

B $\frac{n}{5(n+6)}$

C $\frac{n+1}{6(n+6)}$

D $\frac{1}{6(n+6)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q185 | MCQ

C02-T14-G-M15

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

A $\frac{n}{7(n+7)}$

B $\frac{n}{6(n+7)}$

C $\frac{n+1}{7(n+7)}$

D $\frac{1}{7(n+7)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q186 | MCQ

C02-T14-G-M18

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \dots + \frac{1}{(n+8)(n+9)}$:

A $\frac{n}{9(n+9)}$

B $\frac{n}{8(n+9)}$

C $\frac{n+1}{9(n+9)}$

D $\frac{1}{9(n+9)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q187 | MCQ

C02-T14-G-M21

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \dots + \frac{1}{(n+9)(n+10)}$:

A $\frac{n}{10(n+10)}$

B $\frac{n}{9(n+10)}$

C $\frac{n+1}{10(n+10)}$

D $\frac{1}{10(n+10)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q188 | MCQ

C02-T14-G-M24

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \dots + \frac{1}{(n+10)(n+11)}$:

A $\frac{n}{11(n+11)}$

B $\frac{n}{10(n+11)}$

C $\frac{n+1}{11(n+11)}$

D $\frac{1}{11(n+11)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q189 | WRITTEN

C02-T14-S-WU1a

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q190 | WRITTEN

C02-T14-S-T1b

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q191 | WRITTEN

C02-T14-S-E1c

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q192 | WRITTEN

C02-T14-S-E2b

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q193 | WRITTEN

C02-T14-S-E2c

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q194 | MCQ

C02-T14-G-M04

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

A $\frac{n}{2n+1}$

B $\frac{n}{2(2n+1)}$

C $\frac{1}{2n+1}$

D $\frac{n+1}{2n+1}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q195 | MCQ

C02-T14-G-M05

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \dots + \frac{1}{(3n-2)(3n+1)}$:

A $\frac{n}{3n+1}$

B $\frac{n}{3(3n+1)}$

C $\frac{1}{3n+1}$

D $\frac{n+1}{3n+1}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q196 | MCQ

C02-T14-G-M06

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \dots + \frac{1}{(3n-2)(3n+4)}$:

A $\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$

B $\frac{1}{6}(1 - \frac{1}{3n+4})$

C $\frac{1}{2(3n+4)}$

D $\frac{n}{2(3n+4)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q197 | MCQ

C02-T14-G-M10

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \dots + \frac{1}{(4n-2)(4n+2)}$:

A $\frac{n}{2(4n+2)}$

B $\frac{n}{4(4n+2)}$

C $\frac{1}{4n+2}$

D $\frac{n+1}{2(4n+2)}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q198 | MCQ

C02-T14-G-M13

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

A $\frac{n}{3(5n+3)}$

B $\frac{n}{5(5n+3)}$

C $\frac{n+1}{3(5n+3)}$

D $\frac{1}{5n+3}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q199 | MCQ

C02-T14-G-M16

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \dots + \frac{1}{(4n-3)(4n+1)}$:

A $\frac{n}{4n+1}$

B $\frac{n}{4(4n+1)}$

C $\frac{1}{4n+1}$

D $\frac{n+1}{4n+1}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q200 | MCQ

C02-T14-G-M19

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \dots + \frac{1}{(5n-3)(5n+2)}$:

A $\frac{n}{2(5n+2)}$

B $\frac{n}{5(5n+2)}$

C $\frac{n+1}{2(5n+2)}$

D $\frac{1}{5n+2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q201 | MCQ

C02-T14-G-M22

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \dots + \frac{1}{(6n-2)(6n+4)}$:

A $\frac{n}{4(6n+4)}$

B $\frac{n}{6(6n+4)}$

C $\frac{n+1}{4(6n+4)}$

D $\frac{1}{6n+4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q202 | WRITTEN

C02-T14-S-T1c

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q203 | WRITTEN

C02-T14-S-E1b

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q204 | WRITTEN

C02-T14-S-WU1b

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q205 | WRITTEN

C02-T14-S-T1d

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q206 | WRITTEN

C02-T14-S-E1d

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q207 | WRITTEN

C02-T14-S-E2e

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V02

Q208 | WRITTEN

C02-T14-S-E2d

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q209 | WRITTEN

C02-T14-S-E1e

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q210 | CHALLENGE WRITTEN

C02-T14-S-CH1

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}.$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q211 | WRITTEN

C02-T15-S-WU

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q212 | WRITTEN

C02-T15-G-W01

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q213 | WRITTEN

C02-T15-G-W21

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q214 | WRITTEN

C02-T15-G-W09

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q215 | WRITTEN

C02-T15-G-W29

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q216 | MCQ

C02-T15-G-M01

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

A $3^n n$

B $-(3^n n)$

C $(3^n n) + 1$

D $(3^n n) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q217 | MCQ

C02-T15-G-M25

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

A $\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$

B $-\left(\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}\right)$

C $\left(\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}\right) + 1$

D $\left(\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q218 | MCQ

C02-T15-G-M37

Select the correct option: Find the sum $\sum_{k=1}^n (k+4) \times (2)^{k-1}$:

A $2^n n + 3 \cdot 2^n - 3$

B $-(2^n n + 3 \cdot 2^n - 3)$

C $(2^n n + 3 \cdot 2^n - 3) + 1$

D $(2^n n + 3 \cdot 2^n - 3) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q219 | WRITTEN

C02-T15-S-TRY

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \cdots + (n+1) \times 3^{n-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q220 | MCQ

C02-T15-G-M04

Select the correct option: Find the sum $\sum_{k=1}^n (k+1) \times (3)^{k-1}$:

A $\frac{2 \cdot 3^n n + 3^n - 1}{4}$

B $-\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right)$

C $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q221 | MCQ

C02-T15-G-M29

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

A $\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$

B $-\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right)$

C $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q222 | MCQ

C02-T15-G-M41

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 3) \times (3)^{k-1}$:

A $3^n n - 2 \cdot 3^n + 2$

B $-(3^n n - 2 \cdot 3^n + 2)$

C $(3^n n - 2 \cdot 3^n + 2) + 1$

D $(3^n n - 2 \cdot 3^n + 2) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q223 | MCQ

C02-T15-G-M13

Select the correct option: Find the sum $\sum_{k=1}^n (2k+1) \times (-2)^{k-1}$:

A $-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$

B $-\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right)$

C $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) + 1$

D $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q224 | MCQ

C02-T15-G-M19

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

A $-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$

B $-\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right)$

C $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) + 1$

D $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q225 | WRITTEN

C02-T15-S-EX1

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q226 | WRITTEN

C02-T15-S-EX2

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n - 1) \times 3^n$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V03

Q227 | MCQ

C02-T15-G-M02

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

A $\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$

B $-\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right)$

C $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) + 1$

D $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q228 | CHALLENGE WRITTEN

C02-T15-S-CH

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q229 | WRITTEN

C02-T16-S-WU-A

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the k -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q230 | WRITTEN

C02-T16-S-TRY-1

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the k -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q231 | WRITTEN

C02-T16-S-EX2-A

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the k -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q232 | WRITTEN

C02-T16-S-EX1-A

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the k -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q233 | MCQ

C02-T16-G-M01

Select the correct option: For the cumulative sequence 2, 2+4, 2+4+6, 2+4+6+8, ..., what is a_k :

A $k(k+1)$

B $k(k+3)$

C $2k^2$

D $k(k+2)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q234 | MCQ

C02-T16-G-M02

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

A $\frac{3k(k+1)}{2}$

B $\frac{3k(k+3)}{2}$

C $3k^2$

D $\frac{3k(k+2)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q235 | MCQ

C02-T16-G-M03

Select the correct option: For the cumulative sequence 4, 4+6, 4+6+8, 4+6+8+10, ..., what is a_k :

A $k(k+3)$

B $k(k+5)$

C $2k(k+1)$

D $k(k+4)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q236 | MCQ

C02-T16-G-M04

Select the correct option: For the cumulative sequence 2, $2 + 6$, $2 + 6 + 10$, $2 + 6 + 10 + 14$, ..., what is a_k :

A $2k^2$

B $2k(k + 2)$

C $2k(2k - 1)$

D $2k(k + 1)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q237 | MCQ

C02-T16-G-M05

Select the correct option: For the cumulative sequence 5, $5 + 8$, $5 + 8 + 11$, $5 + 8 + 11 + 14$, ..., what is a_k :

A $\frac{k(3k+7)}{2}$

B $\frac{k(3k+13)}{2}$

C $k(3k+2)$

D $\frac{k(3k+10)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q238 | MCQ

C02-T16-G-M06

Select the correct option: For the cumulative sequence 3, $3 + 8$, $3 + 8 + 13$, $3 + 8 + 13 + 18$, ..., what is a_k :

A $\frac{k(5k+1)}{2}$

B $\frac{k(5k+11)}{2}$

C $k(5k-2)$

D $\frac{k(5k+6)}{2}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q239 | MCQ

C02-T16-G-M07

Select the correct option: For the cumulative sequence 6, $6 + 8$, $6 + 8 + 10$, $6 + 8 + 10 + 12$, ..., what is a_k :

A $k(k + 5)$

B $k(k + 7)$

C $2k(k + 2)$

D $k(k + 6)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q240 | MCQ

C02-T16-G-M08

Select the correct option: For the cumulative sequence 4, $4 + 8$, $4 + 8 + 12$, $4 + 8 + 12 + 16$, ..., what is a_k :

A $2k(k + 1)$

B $2k(k + 3)$

C $4k^2$

D $2k(k + 2)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q241 | WRITTEN

C02-T16-S-WU-B

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the sum from the first term to the n -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q242 | WRITTEN

C02-T16-S-TRY-2

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the sum from the first term to the n -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q243 | WRITTEN

C02-T16-S-EX1-B

For the sequence 1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, ..., find the sum from the first term to the n -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q244 | WRITTEN

C02-T16-S-EX2-B

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the sum from the first term to the n -th term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q245 | MCQ

C02-T16-G-M25

Select the correct option: For the cumulative sequence 2, 2+5, 2+5+8, 2+5+8+11, ..., find a_5 :

A 40

B 46

C 34

D 52

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q246 | CHALLENGE WRITTEN

C02-T16-S-CH

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q247 | WRITTEN

C02-T17-S-WU-A

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q248 | WRITTEN

C02-T17-S-TRY-1

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q249 | WRITTEN

C02-T17-S-EX-1

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q250 | WRITTEN

C02-T17-S-EX-2

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q251 | MCQ

C02-T17-G-M01

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

A $2n + 3$

B $2n - 1$

C $2(n + 2)$

D $a_n = 2n + 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q252 | MCQ

C02-T17-G-M02

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

A $4n + 1$

B $2(2n + 3)$

C $a_n = 2(2n + 1)$

D $4n + 5$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q253 | MCQ

C02-T17-G-M03

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

A $6n$

B $a_n = 6(n - 1)$

C $6n - 5$

D $2(3n - 4)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q254 | MCQ

C02-T17-G-M04

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

A $a_n = 2(4n - 3)$

B $4(2n - 1)$

C $8n - 7$

D $2(4n + 1)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q255 | MCQ

C02-T17-G-M05

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

A $10n + 3$

B $2(5n - 1)$

C $10(n + 1)$

D $a_n = 10n$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q256 | MCQ

C02-T17-G-M19

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

A $4n - 1$

B $4n + 5$

C $a_n = 4n + 1$

D $4n + 3$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Q257 | MCQ

C02-T17-G-M20

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

A $3(2n - 1)$

B $2(3n + 2)$

C $a_n = 2(3n - 1)$

D $6n + 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q258 | MCQ

C02-T17-G-M06

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

A $3n^2 + n + 1$

B $3n^2 + 7n + 6$

C $a_n = 3n^2 + n + 2$

D $3n^2 + n + 3$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q259 | MCQ

C02-T17-G-M07

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

A $2(n+1)(3n+1)$

B $a_n = 2n(3n-2)$

C $2(3n^2 - 2n + 1)$

D $2(n-1)(3n+1)$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q260 | MCQ

C02-T17-G-M08

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

A $a_n = 9n^2 - 13n + 9$

B $9n^2 - 13n + 12$

C $9n^2 - 13n + 8$

D $9n^2 + 5n + 5$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q261 | MCQ

C02-T17-G-M09

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

A $3n^2 - 5n + 5$

B $(n - 1)(3n - 2)$

C $3n^2 + n + 2$

D $a_n = 3n^2 - 5n + 4$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q262 | MCQ

C02-T17-G-M10

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

A $2(3n^2 + 1)$

B $6n^2 - 1$

C $6(n+1)^2$

D $a_n = 6n^2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Q263 | MCQ

C02-T17-G-M21

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

A $3(n+1)(n+2)$

B $a_n = 3n(n+1)$

C $3n^2 + 3n + 1$

D $3n^2 + 3n - 2$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q264 | WRITTEN

C02-T17-S-TRY-3

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q265 | WRITTEN

C02-T17-S-EX-3

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Q266 | WRITTEN

C02-T17-S-EX-7

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q267 | WRITTEN

C02-T17-S-WU-B

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q268 | WRITTEN

C02-T17-S-TRY-2

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q269 | WRITTEN

C02-T17-S-EX-4

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Q270 | WRITTEN

C02-T17-S-EX-6

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q271 | WRITTEN

C02-T17-S-EX-5

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Q272 | WRITTEN

C02-T17-S-EX-8

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V01

Q273 | MCQ

C02-T17-G-M25

Select the correct option: If $S_n = n(n + 4)$, find a_5 :

A -13

B 13

C 14

D 12

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Q274 | CHALLENGE WRITTEN

C02-T17-S-CH

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V01

Quiz M1 | QUIZ MCQ

C02-T11-Q-M1

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

A $2 + 5 + 8 + 11$ B $3 + 6 + 9 + 12$ C $1 + 4 + 7 + 10$ D $2 + 6 + 10 + 14$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Quiz M2 | QUIZ MCQ

C02-T13-Q-M4

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

A $\frac{n(n+1)(n-1)(n+2)}{4}$

B $\frac{n(n+1)(n+1)(n+2)}{4}$

C $\frac{n(n-1)(n+2)}{4}$

D $\frac{n(n+1)(n-2)(n+3)}{4}$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V01

Quiz M3 | QUIZ MCQ

C02-T15-Q-M2

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

A $(n-1) \times 2^n + 1$

B $-((n-1) \times 2^n + 1)$

C $((n-1) \times 2^n + 1) + 1$

D $((n-1) \times 2^n + 1) - 1$

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F03 | V03

Quiz M4 | QUIZ MCQ

C02-T17-Q-M4

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

A 30

B -32

C 32

D 34

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V02

Quiz W1 | QUIZ WRITTEN

C02-T11-Q-W1

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F01 | V03

Quiz W2 | QUIZ WRITTEN

C02-T13-Q-W1

Find the sum $\sum_{k=1}^n k^2(k+5)$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F02 | V03

Quiz W3 | QUIZ WRITTEN

C02-T15-Q-W3

Find the sum $\sum_{k=1}^n (4k - 1) \times 3^k$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 3 | Summation

F99 | V01

Quiz W4 | QUIZ WRITTEN

C02-T17-Q-W4

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

WORKING AREA

FORMULA BANK

Sigma linearity

$$\sum (au_n + bv_n) = a \sum u_n + b \sum v_n$$

Standard sums

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Concept 4 | Advanced Sequences

OFFICIAL LESSONS 2-18, 2-19

Advanced Sequences

Equation-first recall | use the same rail beside every question in this Concept

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

group index $\longrightarrow$ term formula

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Advanced Sequences

F01 | V01

Q001 | WRITTEN

C02-T18-S-W1

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q002 | MCQ

C02-T18-G-M01

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

A $n^2 - 3$

B $(n + 1)^2$

C $a_n = n^2$

D $n^2 + 3$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q003 | MCQ

C02-T18-G-M02

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

A $\frac{3n^2 - n + 4}{2}$

B $a_n = \frac{3n^2 - 7n + 8}{2}$

C $\frac{3n^2 - 7n + 16}{2}$

D $\frac{3n^2 - 7n + 6}{2}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q004 | MCQ

C02-T18-G-M07

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

A $a_n = 3n^2 - 8n + 12$

B $3n^2 - 8n + 13$

C $3n^2 - 8n + 9$

D $3n^2 - 2n + 7$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q005 | MCQ

C02-T18-G-M33

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :

A -25

B 25

C 30

D 23

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q006 | MCQ

C02-T18-G-M34

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :

A 37

B 38

C 34

D -37

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q007 | MCQ

C02-T18-G-M39

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :

A 73

B 70

C -72

D 72

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q008 | WRITTEN

C02-T18-S-T1A

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q009 | WRITTEN

C02-T18-S-E1D

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q010 | MCQ

C02-T18-G-M03

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

A $a_n = \frac{(n+1)(5n-2)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{5n^2 + 3n - 6}{2}$

D $\frac{(n+2)(5n+3)}{2}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q011 | MCQ

C02-T18-G-M04

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

A $\frac{7n^2 - 13n + 18}{2}$

B $\frac{7n^2 - 13n + 8}{2}$

C $\frac{7n^2 + n + 8}{2}$

D $a_n = \frac{7n^2 - 13n + 14}{2}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q012 | MCQ

C02-T18-G-M06

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

A $2(n + 1)(n + 3)$

B $a_n = 2n(n + 2)$

C $2(n^2 + 2n + 2)$

D $2(n^2 + 2n - 1)$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V03

Q013 | WRITTEN

C02-T18-S-E1B

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V01

Q014 | WRITTEN

C02-T18-S-W2

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V01

Q015 | WRITTEN

C02-T18-S-E1A

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V03

Q016 | WRITTEN

C02-T18-S-T1B

Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\longrightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Q017 | WRITTEN

C02-T18-S-E1C

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V02

Q018 | WRITTEN

C02-T18-S-E1E

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F04 | V01

Q019 | MCQ

C02-T18-G-M25

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

A $a_n = \frac{(n+1)(2n^2 - 5n + 6)}{6}$

B $\frac{2n^3 - 3n^2 + n + 12}{6}$

C $\frac{2n^3 - 3n^2 + n - 12}{6}$

D $\frac{(n+2)(2n^2 - n + 3)}{6}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F99 | V01

Q020 | CHALLENGE WRITTEN

C02-T18-S-CH1

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q021 | WRITTEN

C02-T19-S-W1

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q022 | WRITTEN

C02-T19-S-T1

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q023 | WRITTEN

C02-T19-S-E1A

For groups with $a_j = 1 + (j-1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q024 | MCQ

C02-T19-G-M01

Select the correct option: find the first term of group 3:

A 6

B 4

C 5

D 3

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q025 | MCQ

C02-T19-G-M04

Select the correct option: find the first term of group 6:

A 77

B 108

C 78

D 79

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q026 | MCQ

C02-T19-G-M07

Select the correct option: find the first term of group 9:

A 38

B 36

C 45

D 37

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q027 | MCQ

C02-T19-G-M10

Select the correct option: find the first term of group 4:

A 48

B 30

C 31

D 29

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q028 | MCQ

C02-T19-G-M13

Select the correct option: find the first term of group 7:

A 21

B 28

C 22

D 23

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Q029 | MCQ

C02-T19-G-M16

Select the correct option: find the first term of group 10:

A 247

B 245

C 300

D 246

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q030 | MCQ

C02-T19-G-M02

Select the correct option: find the first term of group 4:

A 20

B 21

C 19

D 32

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q031 | MCQ

C02-T19-G-M05

Select the correct option: find the first term of group 7:

A 138

B 110

C 111

D 109

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q032 | MCQ

C02-T19-G-M08

Select the correct option: find the first term of group 10:

A 163

B 200

C 164

D 165

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q033 | MCQ

C02-T19-G-M11

Select the correct option: find the first term of group 5:

A 58

B 59

C 57

D 78

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Q034 | MCQ

C02-T19-G-M14

Select the correct option: find the first term of group 8:

A 128

B 100

C 101

D 99

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V03

Q035 | WRITTEN

C02-T19-S-E2A

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V03

Q036 | MCQ

C02-T19-G-M03

Select the correct option: find the first term of group 5:

A 30

B 28

C 39

D 29

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V03

Q037 | MCQ

C02-T19-G-M06

Select the correct option: find the first term of group 8:

A 89

B 90

C 88

D 110

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V01

Q038 | WRITTEN

C02-T19-S-W2

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V02

Q039 | WRITTEN

C02-T19-S-T2

For the same grouped sequence, find the sum of the terms in group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V02

Q040 | WRITTEN

C02-T19-S-E1B

For the same grouped sequence, find the sum of the terms in group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F02 | V03

Q041 | WRITTEN

C02-T19-S-E2B

For the same grouped sequence, find the sum of the terms in group 10.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Q042 | WRITTEN

C02-T19-S-W3

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Q043 | WRITTEN

C02-T19-S-T3

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V02

Q044 | WRITTEN

C02-T19-S-E1C

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V03

Q045 | WRITTEN

C02-T19-S-E2C

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F99 | V01

Q046 | CHALLENGE WRITTEN

C02-T19-S-CH1

For the grouped sequence 2, 6 | 10, 14, 18 | 22, 26, 30, 34 | ..., solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Quiz M1 | QUIZ MCQ

C02-T18-Q-M1

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

A $a_n = \frac{3n^2 - n + 2}{2}$

B $\frac{3n^2 - n + 4}{2}$

C $\frac{n(3n - 1)}{2}$

D $\frac{3n^2 + 5n + 4}{2}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Quiz M2 | QUIZ MCQ

C02-T18-Q-M3

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

A $\frac{(-2)^n + 2}{6}$

B $-\frac{(-2)^n - 10}{3}$

C $a_n = \frac{(-2)^n + 20}{6}$

D $\frac{(-2)^n + 38}{6}$

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Quiz M3 | QUIZ MCQ

C02-T19-Q-M4

Select the correct option: find the first term of group 9:

A 195

B 196

C 194

D 243

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Quiz M4 | QUIZ MCQ

C02-T19-Q-M3

Select the correct option: the value 29 is in:

A Group 5, position 2

B Group 5, position 1

C Group 6, position 1

D Group 4, position 2

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V01

Quiz W1 | QUIZ WRITTEN

C02-T18-Q-W1

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V01

Quiz W2 | QUIZ WRITTEN

C02-T18-Q-W3

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F01 | V02

Quiz W3 | QUIZ WRITTEN

C02-T19-Q-W4

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 4 | Advanced Sequences

F03 | V02

Quiz W4 | QUIZ WRITTEN

C02-T19-Q-W3

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

WORKING AREA

FORMULA BANK

Difference sequence

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping
group index $\rightarrow$ term formula

Concept 5 | Recurrence and Induction

OFFICIAL LESSONS 2-20, 2-21, 2-22, 2-23, 2-24, 2-25, 2-26

Recurrence and Induction

Equation-first recall | use the same rail beside every question in this Concept

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Start with the family cue, write the first move, then use the working grid.

Concept 5 | Recurrence and Induction

F01 | V01

Q001 | WRITTEN

C02-T20-S-W1

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q002 | WRITTEN

C02-T20-S-T1A

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q003 | WRITTEN

C02-T20-S-E1B

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q004 | WRITTEN

C02-T20-S-E1D

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q005 | WRITTEN

C02-T20-S-E1E

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q006 | WRITTEN

C02-T20-S-E1F

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q007 | MCQ

C02-T20-G-M27

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

A -81

B 81

C 82

D 80

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q008 | MCQ

C02-T20-G-M30

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

A 133

B 127

C -128

D 128

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q009 | MCQ

C02-T20-G-M33

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

A 34

B 37

C 33

D -34

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q010 | MCQ

C02-T20-G-M36

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

A -171

B 171

C 172

D 170

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q011 | MCQ

C02-T20-G-M25

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

A 17

B 11

C -13

D 13

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q012 | MCQ

C02-T20-G-M28

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

A -17

B -15

C -19

D 17

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q013 | MCQ

C02-T20-G-M31

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

A -15

B 13

C -13

D -12

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q014 | MCQ

C02-T20-G-M34

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

A 22

B 16

C -18

D 18

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q015 | MCQ

C02-T20-G-M26

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

A 20

B -23

C 23

D 28

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q016 | MCQ

C02-T20-G-M29

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

A -1

B -7

C 4

D -4

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q017 | MCQ

C02-T20-G-M32

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

A -9

B 9

C 11

D 6

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q018 | MCQ

C02-T20-G-M35

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

A -6

B 3

C -3

D 2

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q019 | WRITTEN

C02-T20-S-W2A

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q020 | WRITTEN

C02-T20-S-T2A

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q021 | WRITTEN

C02-T20-S-T2B

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q022 | WRITTEN

C02-T20-S-E2A

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q023 | WRITTEN

C02-T20-S-E2B

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q024 | WRITTEN

C02-T20-S-E2E

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q025 | WRITTEN

C02-T20-S-E2F

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q026 | MCQ

C02-T20-G-M03

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

A $3^n + 2$

B $3^n - 1$

C $3^n + 3$

D $a_n = 3^n$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q027 | MCQ

C02-T20-G-M06

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

A $a_n = 2^{2n-1}$

B $\frac{2^{2n} + 2}{2}$

C $\frac{2^{2n} - 2}{2}$

D $\frac{2^{2n} + 4}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q028 | MCQ

C02-T20-G-M15

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

A $a_n = 4 \cdot 5^{n-1}$

B $\frac{4 \cdot 5^n + 5}{5}$

C $\frac{4 \cdot 5^n - 5}{5}$

D $\frac{2(2 \cdot 5^n + 5)}{5}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q029 | MCQ

C02-T20-G-M24

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

A $a_n = 3 \cdot 2^n$

B $3 \cdot 2^n + 1$

C $3 \cdot 2^n - 1$

D $3 \cdot 2^n + 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q030 | WRITTEN

C02-T20-S-W2B

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q031 | WRITTEN

C02-T20-S-T2C

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q032 | WRITTEN

C02-T20-S-E2C

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q033 | WRITTEN

C02-T20-S-E2G

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q034 | MCQ

C02-T20-G-M01

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

A $4n - 1$

B $a_n = 4n - 3$

C $4n + 1$

D $4n - 5$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q035 | MCQ

C02-T20-G-M10

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

A $\frac{2n^3 - 3n^2 + n + 36}{6}$

B $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$

C $\frac{2n^3 - 3n^2 + n + 48}{6}$

D $\frac{2n^3 - 3n^2 + n + 12}{6}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q036 | MCQ

C02-T20-G-M16

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

A $-\frac{(-4)^n - 8}{4}$

B $-\frac{(-4)^n + 8}{4}$

C $-\frac{(-4)^n - 12}{4}$

D $a_n = (-4)^{n-1}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q037 | MCQ

C02-T20-G-M19

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

A $a_n = \frac{3^n + 7}{2}$

B $\frac{3^n + 9}{2}$

C $\frac{3^n + 3}{2}$

D $\frac{3^n + 11}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q038 | WRITTEN

C02-T20-S-T2D

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q039 | WRITTEN

C02-T20-S-E2D

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q040 | WRITTEN

C02-T20-S-E2H

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q041 | MCQ

C02-T20-G-M05

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

A $10 - 3n$

B $a_n = 8 - 3n$

C $-3(n - 4)$

D $5 - 3n$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q042 | MCQ

C02-T20-G-M08

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

A $\frac{n(n-1)}{2}$

B $\frac{n^2 - n + 10}{2}$

C $a_n = \frac{n^2 - n + 6}{2}$

D $\frac{n^2 - n + 12}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q043 | MCQ

C02-T20-G-M14

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

A $-5(n - 2)$

B $a_n = 8 - 5n$

C $12 - 5n$

D $-5(n - 1)$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q044 | MCQ

C02-T20-G-M17

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

A $6n - 7$

B $2(3n - 1)$

C $a_n = 2(3n - 2)$

D $6n - 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q045 | WRITTEN

C02-T20-S-T1B

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q046 | WRITTEN

C02-T20-S-E1A

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q047 | WRITTEN

C02-T20-S-E1C

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q048 | CHALLENGE WRITTEN

C02-T20-S-CH1

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q049 | WRITTEN

C02-T21-S-W1

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q050 | WRITTEN

C02-T21-S-T1B

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q051 | WRITTEN

C02-T21-S-E1A

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q052 | WRITTEN

C02-T21-S-E1B

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q053 | WRITTEN

C02-T21-S-E1D

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n+1)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q054 | WRITTEN

C02-T21-S-E1E

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q055 | MCQ

C02-T21-G-M03

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

A $\frac{(n+1)(2n^2 - 5n + 6)}{6}$

B $\frac{(n+2)(2n^2 - 7n + 15)}{6}$

C $a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$

D $\frac{2n^3 - 3n^2 + n + 36}{6}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q056 | MCQ

C02-T21-G-M09

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

A $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

B $\frac{(n+1)(2n^2 - 5n + 6)}{2}$

C $\frac{n(n-1)(2n-1)}{2}$

D $\frac{2n^3 - 3n^2 + n + 8}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q057 | MCQ

C02-T21-G-M12

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

A $\frac{(n-1)(2n^2 - n + 6)}{6}$

B $\frac{2n^3 - 3n^2 + 7n + 18}{6}$

C $a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$

D $\frac{(2n+3)(n^2 - 3n + 8)}{6}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q058 | MCQ

C02-T21-G-M15

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

A $\frac{8n^3 - 15n^2 + 7n + 42}{6}$

B $\frac{8n^3 - 15n^2 + 7n + 18}{6}$

C $\frac{8n^3 - 15n^2 + 7n + 48}{6}$

D $a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q059 | MCQ

C02-T21-G-M21

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

A $4n(n-1)$

B $4(n^2 - n + 1)$

C $a_n = 2(2n^2 - 2n + 1)$

D $4n^2 - 4n + 5$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q060 | MCQ

C02-T21-G-M27

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

A 17

B 19

C 15

D -17

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q061 | MCQ

C02-T21-G-M33

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

A 96

B 90

C -92

D 92

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q062 | MCQ

C02-T21-G-M36

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

A 9

B 11

C 7

D -9

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q063 | MCQ

C02-T21-G-M39

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

A -210

B 210

C 215

D 208

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q064 | MCQ

C02-T21-G-M45

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

A 10

B 12

C 8

D -10

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q065 | WRITTEN

C02-T21-S-T1A

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q066 | MCQ

C02-T21-G-M01

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

A $a_n = n^2$

B $n^2 + 1$

C $n^2 - 3$

D $n^2 + 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q067 | WRITTEN

C02-T21-S-E1C

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q068 | WRITTEN

C02-T21-S-E1F

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q069 | WRITTEN

C02-T21-S-E1G

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q070 | CHALLENGE WRITTEN

C02-T21-S-CH1

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q071 | WRITTEN

C02-T22-S-W1

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q072 | WRITTEN

C02-T22-S-T1A

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q073 | WRITTEN

C02-T22-S-E1A

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q074 | WRITTEN

C02-T22-S-E1B

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q075 | WRITTEN

C02-T22-S-E1C

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q076 | WRITTEN

C02-T22-S-E1D

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q077 | WRITTEN

C02-T22-S-E1E

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q078 | WRITTEN

C02-T22-S-E1F

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q079 | WRITTEN

C02-T22-S-T1B

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q080 | MCQ

C02-T22-G-M03

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

A $-\frac{2(-2)^n - 11}{3}$

B $a_n = -\frac{2(-2)^n - 5}{3}$

C $-\frac{2(-2)^n - 17}{3}$

D $-\frac{2((-2)^n + 2)}{3}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q081 | MCQ

C02-T22-G-M06

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

A $\frac{3 \cdot 3^{-n} (7 \cdot 2^n - 6 \cdot 3^n)}{2}$

B $\frac{3^{-n} (21 \cdot 2^n - 8 \cdot 3^n)}{2}$

C $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 4 \cdot 3^n)}{2}$

D $\frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q082 | MCQ

C02-T22-G-M09

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

A $\frac{4(7^n + 14)}{21}$

B $\frac{4 \cdot 7^n - 49}{21}$

C $\frac{4 \cdot 7^n + 77}{21}$

D $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q083 | MCQ

C02-T22-G-M12

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

A $\frac{2^n + 14}{2}$

B $a_n = \frac{2^n + 10}{2}$

C $\frac{2^n + 18}{2}$

D $\frac{2^n + 4}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q084 | MCQ

C02-T22-G-M15

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

A $-\frac{2(2(-2)^n + 1)}{3}$

B $-\frac{4(-2)^n - 13}{3}$

C $a_n = -\frac{4(-2)^n - 7}{3}$

D $-\frac{4((-2)^n - 4)}{3}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q085 | MCQ

C02-T22-G-M18

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

A $-\frac{13(-5)^n - 25}{15}$

B $-\frac{13(-5)^n + 50}{15}$

C $-\frac{13(-5)^n - 40}{15}$

D $a_n = -\frac{13(-5)^n + 5}{15}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q086 | MCQ

C02-T22-G-M21

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

A $-\frac{(-3)^n - 13}{4}$

B $a_n = -\frac{(-3)^n - 5}{4}$

C $-\frac{(-3)^n - 21}{4}$

D $-\frac{(-3)^n + 7}{4}$

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

WORKING AREA

Concept 5 | Recurrence and Induction

F02 | V01

Q087 | MCQ

C02-T22-G-M24

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

A $2^{-n}(11 \cdot 2^n - 26)$

B $2 \cdot 2^{-n}(8 \cdot 2^n - 13)$

C $a_n = 2 \cdot 2^{-n}(7 \cdot 2^n - 13)$

D $2^{-n}(17 \cdot 2^n - 26)$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q088 | MCQ

C02-T22-G-M01

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

A 3

B 0

C 4

D $a_n = 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q089 | MCQ

C02-T22-G-M27

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

A $-\frac{2(-2)^n - 11}{3}$

B $-\frac{2((-2)^n + 2)}{3}$

C $-\frac{2((-2)^n - 7)}{3}$

D $a_n = -\frac{2(-2)^n - 5}{3}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q090 | CHALLENGE WRITTEN

C02-T22-G-CH1

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q091 | WRITTEN

C02-T23-S-W1

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q092 | WRITTEN

C02-T23-S-E1

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q093 | WRITTEN

C02-T23-G-W03

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q094 | MCQ

C02-T23-G-M03

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

A $\frac{n(n+1)(n+2)}{3}$

B $\frac{n^3 + 3n^2 + 2n + 3}{3}$

C $\frac{n^3 + 3n^2 + 2n - 3}{3}$

D $\frac{(n+3)(n^2+2)}{3}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q095 | MCQ

C02-T23-G-M06

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

A $4^n + 1$

B $4^n - 1$

C $4^n + 3$

D $4^n - 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q096 | MCQ

C02-T23-G-M09

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

A $n^2 + 4n - 1$

B $n^2 + 4n + 2$

C $n(n + 4)$

D $(n + 1)(n + 3)$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q097 | MCQ

C02-T23-G-M15

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

A $\frac{4n^3 + 15n^2 + 11n + 12}{6}$

B $\frac{n(n+1)(4n+11)}{6}$

C $\frac{4n^3 + 15n^2 + 11n + 24}{6}$

D $\frac{(n+2)(4n^2 + 7n - 3)}{6}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q098 | MCQ

C02-T23-G-M18

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

A $\frac{2n^3 + 5n^2 + 3n - 2}{2}$

B $\frac{2n^3 + 5n^2 + 3n + 4}{2}$

C $\frac{n(n+1)(2n+3)}{2}$

D $\frac{2n^3 + 5n^2 + 3n + 6}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q099 | MCQ

C02-T23-G-M21

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

A $7(2^n - 1)$

B $72^n - 6$

C $72^n - 8$

D $72^n - 5$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q100 | MCQ

C02-T23-G-M27

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

A $\frac{n^3 + 3n^2 + 2n - 3}{3}$

B $\frac{(n+3)(n^2+2)}{3}$

C $\frac{n(n+1)(n+2)}{3}$

D $\frac{n^3 + 3n^2 + 2n + 9}{3}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q101 | MCQ

C02-T23-G-M30

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

A $4^n - 1$

B 4^n

C $4^n - 2$

D $4^n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q102 | MCQ

C02-T23-G-M33

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

A $n^2 + 4n + 2$

B $n(n + 4)$

C $(n + 2)^2$

D $n^2 + 4n - 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q103 | WRITTEN

C02-T23-S-T1

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q104 | WRITTEN

C02-T23-S-T2

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q105 | WRITTEN

C02-T23-S-E3

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q106 | WRITTEN

C02-T23-S-E2

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V02

Q107 | WRITTEN

C02-T23-S-E4

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10^{n+1}-10}{9}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q108 | CHALLENGE WRITTEN

C02-T23-G-CH1

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2+3n+5)}{3}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q109 | WRITTEN

C02-T24-S-W1

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q110 | WRITTEN

C02-T24-S-T1

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q111 | WRITTEN

C02-T24-S-E1

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q112 | WRITTEN

C02-T24-G-W03

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q113 | WRITTEN

C02-T24-G-W09

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q114 | WRITTEN

C02-T24-G-W15

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q115 | WRITTEN

C02-T24-G-W21

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q116 | MCQ

C02-T24-G-M03

Select the correct option: which inequality is proved for $n \geq 3$?:

A $2^n \geq 1n + 2$

B $2^n > 2n + 2$

C $2^n + 1 > 1n + 2$

D $2^n > 1n + 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q117 | MCQ

C02-T24-G-M06

Select the correct option: which inequality is proved for $n \geq 2$?:

A $3^n > 2n$

B $3^n \geq 2n + 0$

C $3^n > 3n + 0$

D $3^n + 1 > 2n + 0$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q118 | MCQ

C02-T24-G-M09

Select the correct option: which inequality is proved for $n \geq 4$?:

A $3^n + 1 > 5n + 1$

B $3^n > 5n + 1$

C $3^n \geq 5n + 1$

D $3^n > 6n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q119 | MCQ

C02-T24-G-M12

Select the correct option: which inequality is proved for $n \geq 3$?:

A $3^n \geq 3n + 0$

B $3^n > 4n + 0$

C $3^n + 1 > 3n + 0$

D $3^n > 3n$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q120 | MCQ

C02-T24-G-M15

Select the correct option: which inequality is proved for $n \geq 4$?:

A $3^n > 5n + 2$

B $3^n \geq 5n + 2$

C $3^n > 6n + 2$

D $3^n + 1 > 5n + 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q121 | MCQ

C02-T24-G-M18

Select the correct option: which inequality is proved for $n \geq 3$?:

A $4^n + 1 > 3n + 1$

B $4^n > 3n + 1$

C $4^n \geq 3n + 1$

D $4^n > 4n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q122 | MCQ

C02-T24-G-M21

Select the correct option: which inequality is proved for $n \geq 4$?:

A $4^n \geq 2n + 3$

B $4^n > 3n + 3$

C $4^n + 1 > 2n + 3$

D $4^n > 2n + 3$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q123 | MCQ

C02-T24-G-M24

Select the correct option: which inequality is proved for $n \geq 4$?:

A $4^n > 4n + 1$

B $4^n \geq 4n + 1$

C $4^n > 5n + 1$

D $4^n + 1 > 4n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q124 | WRITTEN

C02-T24-S-E2

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q125 | MCQ

C02-T24-G-M27

Select the correct option: which inequality is proved for $n \geq 3$?:

A $4^n + 1 > 3n + 0$

B $4^n > 3n$

C $4^n \geq 3n + 0$

D $4^n > 4n + 0$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q126 | CHALLENGE WRITTEN

C02-T24-S-CH1

When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q127 | WRITTEN

C02-T25-S-W1

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q128 | WRITTEN

C02-T25-S-T1

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q129 | WRITTEN

C02-T25-G-W01

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q130 | WRITTEN

C02-T25-G-W06

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q131 | WRITTEN

C02-T25-G-W07

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q132 | WRITTEN

C02-T25-G-W12

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q133 | MCQ

C02-T25-G-M01

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

A 2

B 3

C 4

D 5

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q134 | MCQ

C02-T25-G-M06

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

A 3

B 4

C 5

D 2

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q135 | MCQ

C02-T25-G-M07

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

A 4

B 5

C 2

D 3

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q136 | MCQ

C02-T25-G-M12

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?

A 4

B 5

C 2

D 3

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q137 | MCQ

C02-T25-G-M13

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?

A 9

B 6

C 7

D 8

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q138 | MCQ

C02-T25-G-M18

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?

A 6

B 7

C 8

D 9

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q139 | MCQ

C02-T25-G-M19

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

A 7

B 8

C 9

D 6

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q140 | MCQ

C02-T25-G-M24

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

A 7

B 8

C 9

D 6

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q141 | MCQ

C02-T25-G-M02

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

A 3

B 4

C 5

D 2

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q142 | WRITTEN

C02-T25-S-E1

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V02

Q143 | WRITTEN

C02-T25-S-E2

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q144 | CHALLENGE WRITTEN

C02-T25-S-CH1

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \quad \text{and} \quad P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V01

Q145 | WRITTEN

C02-T26-S-W1

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V02

Q146 | WRITTEN

C02-T26-S-T1

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q147 | WRITTEN

C02-T26-S-E1A

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F01 | V03

Q148 | WRITTEN

C02-T26-S-E2A

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q149 | WRITTEN

C02-T26-S-W2

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q150 | MCQ

C02-T26-G-M01

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

A $a_n = 2n$

B $a_n = 2(n - 1)$

C $a_n = 1 - 2n$

D $a_n = 2n - 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q151 | MCQ

C02-T26-G-M02

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

A $a_n = 2n - 1$

B $a_n = -2n$

C $a_n = 2n$

D $a_n = 2n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q152 | MCQ

C02-T26-G-M03

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

A $a_n = -2n - 1$

B $a_n = 2n + 1$

C $a_n = 2(n + 1)$

D $a_n = 2n$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q153 | MCQ

C02-T26-G-M04

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

A $a_n = 2(n + 1)$

B $a_n = 2n + 3$

C $a_n = 2n + 1$

D $a_n = -2(n + 1)$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q154 | MCQ

C02-T26-G-M05

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

A $a_n = 3n - 1$

B $a_n = 3(n - 1)$

C $a_n = 2 - 3n$

D $a_n = 3n - 2$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q155 | MCQ

C02-T26-G-M06

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

A $a_n = 3n - 2$

B $a_n = 1 - 3n$

C $a_n = 3n - 1$

D $a_n = 3n$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q156 | MCQ

C02-T26-G-M07

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

A $a_n = -3n$

B $a_n = 3n$

C $a_n = 3n + 1$

D $a_n = 3n - 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Q157 | MCQ

C02-T26-G-M08

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

A $a_n = 3n + 1$

B $a_n = 3n + 2$

C $a_n = 3n$

D $a_n = -3n - 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Q158 | WRITTEN

C02-T26-S-E1B

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Q159 | WRITTEN

C02-T26-S-T2

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V01

Q160 | MCQ

C02-T26-G-M33

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

A $a_n = \frac{2^n - 2}{2}$

B $a_n = -2^{n-1}$

C $a_n = 2^{n-1}$

D $a_n = \frac{2^n + 2}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F04 | V01

Q161 | WRITTEN

C02-T26-S-E2B

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F99 | V01

Q162 | CHALLENGE WRITTEN

C02-T26-S-CH1

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Quiz M1 | QUIZ MCQ

C02-T20-Q-M1

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

A $4n$

B $4n - 5$

C $4n + 1$

D $a_n = 2(2n - 1)$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Quiz M2 | QUIZ MCQ

C02-T22-Q-M2

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

A $a_n = -\frac{(-2)^n - 2}{2}$

B $-\frac{(-2)^n - 4}{2}$

C $-\frac{(-2)^n + 4}{2}$

D $-\frac{(-2)^n - 6}{2}$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Quiz M3 | QUIZ MCQ

C02-T24-Q-M3

Select the correct option: which inequality is proved for $n \geq 4$?:

A $2^n + 1 > 3n + 1$

B $2^n > 3n + 1$

C $2^n \geq 3n + 1$

D $2^n > 4n + 1$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V01

Quiz M4 | QUIZ MCQ

C02-T26-Q-M4

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

A $a_n = 2n + 1$

B $a_n = -2(n + 1)$

C $a_n = 2(n + 1)$

D $a_n = 2n + 3$

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V02

Quiz W1 | QUIZ WRITTEN

C02-T20-Q-W2

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V02

Quiz W2 | QUIZ WRITTEN

C02-T22-Q-W2

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F03 | V02

Quiz W3 | QUIZ WRITTEN

C02-T24-G-Q-W3

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = m(q')$$

Concept 5 | Recurrence and Induction

F02 | V03

Quiz W4 | QUIZ WRITTEN

C02-T26-Q-W2

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

WORKING AREA

FORMULA BANK

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1) \text{ and } P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = m(q')$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 03

Complex Numbers and Equations

4 Concepts | 702 synchronized questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 03 | Complex Numbers and Equations

CLASSROOM ROUTE

Concept 1 | Complex Numbers**210 questions**

Official lessons: 3-1, 3-2, 3-3, 3-4

Concept 2 | Quadratic Equations**248 questions**

Official lessons: 3-5, 3-6, 3-7, 3-8, 3-9, 3-10, 3-11

Concept 3 | Polynomial Theorems**126 questions**

Official lessons: 3-12, 3-13, 3-14, 3-15, 3-16

Concept 4 | Higher-Degree Equations**118 questions**

Official lessons: 3-17, 3-18, 3-19, 3-20

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 1 | Complex Numbers

OFFICIAL LESSONS 3-1, 3-2, 3-3, 3-4

Complex Numbers

Equation-first recall | use the same rail beside every question in this Concept

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Complex Numbers

F01 | V01

Q001 | WRITTEN

C03-T01-31-WU1a

State the real and imaginary parts of $3 - 2i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V02

Q002 | WRITTEN

C03-T01-31-WU1b

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V03

Q003 | WRITTEN

C03-T01-31-WU1c

State the real and imaginary parts of -8 .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V04

Q004 | WRITTEN

C03-T01-31-WU1d

State the real and imaginary parts of $2i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q005 | WRITTEN

C03-T01-31-WU2

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q006 | WRITTEN

C03-T01-31-TR1a

State the real and imaginary parts of $-3 + 5i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V02

Q007 | WRITTEN

C03-T01-31-TR1b

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V03

Q008 | WRITTEN

C03-T01-31-TR1c

State the real and imaginary parts of 1.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V04

Q009 | WRITTEN

C03-T01-31-TR1d

State the real and imaginary parts of $3i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V03

Q010 | WRITTEN

C03-T01-31-TR2a

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0, x, y \text{ real}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q011 | WRITTEN

C03-T01-31-TR2b

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q012 | WRITTEN

C03-T01-31-EX1a

State the real and imaginary parts of $2 + 5i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q013 | WRITTEN

C03-T01-31-EX1b

State the real and imaginary parts of $-25 + 13i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V02

Q014 | WRITTEN

C03-T01-31-EX1c

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V02

Q015 | WRITTEN

C03-T01-31-EX1d

State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V03

Q016 | WRITTEN

C03-T01-31-EX1e

State the real and imaginary parts of -4 .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V03

Q017 | WRITTEN

C03-T01-31-EX1f

State the real and imaginary parts of $1/4$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V04

Q018 | WRITTEN

C03-T01-31-EX1g

State the real and imaginary parts of $\sqrt{3}i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V04

Q019 | WRITTEN

C03-T01-31-EX1h

State the real and imaginary parts of $-i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V01

Q020 | WRITTEN

C03-T01-31-EX2a

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V01

Q021 | WRITTEN

C03-T01-31-EX2b

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V01

Q022 | WRITTEN

C03-T01-31-EX2c

Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V01

Q023 | WRITTEN

C03-T01-31-EX2d

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V01

Q024 | WRITTEN

C03-T01-31-EX2e

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V01

Q025 | WRITTEN

C03-T01-31-CH

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q026 | WRITTEN

C03-T01-32-WUa

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q027 | WRITTEN

C03-T01-32-WUb

Calculate $(i - 2)^2$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V04

Q028 | WRITTEN

C03-T01-32-WUc

Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q029 | WRITTEN

C03-T01-32-TRa

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q030 | WRITTEN

C03-T01-32-TRb

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V03

Q031 | WRITTEN

C03-T01-32-TRc

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q032 | WRITTEN

C03-T01-32-EXa

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q033 | WRITTEN

C03-T01-32-EXb

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q034 | WRITTEN

C03-T01-32-EXc

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q035 | WRITTEN

C03-T01-32-EXd

Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q036 | WRITTEN

C03-T01-32-EXe

Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V01

Q037 | WRITTEN

C03-T01-32-EXf

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q038 | WRITTEN

C03-T01-32-EXg

Calculate $(2i - 1)^2$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q039 | WRITTEN

C03-T01-32-EXh

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q040 | WRITTEN

C03-T01-32-EXi

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V03

Q041 | WRITTEN

C03-T01-32-EXj

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V03

Q042 | WRITTEN

C03-T01-32-EXk

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V03

Q043 | WRITTEN

C03-T01-32-EXI

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V02

Q044 | WRITTEN

C03-T01-32-CH

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q045 | WRITTEN

C03-T01-33-WU1a

State the complex conjugate of $2 + 3i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q046 | WRITTEN

C03-T01-33-WU1b

State the complex conjugate of $7 - i\sqrt{11}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q047 | WRITTEN

C03-T01-33-WU1c

State the complex conjugate of 6.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q048 | WRITTEN

C03-T01-33-WU1d

State the complex conjugate of $-5i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q049 | WRITTEN

C03-T01-33-WU2a

Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V03

Q050 | WRITTEN

C03-T01-33-WU2b

Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q051 | WRITTEN

C03-T01-33-TR1a

State the complex conjugate of $1 - 4i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q052 | WRITTEN

C03-T01-33-TR1b

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q053 | WRITTEN

C03-T01-33-TR1c

State the complex conjugate of 2.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q054 | WRITTEN

C03-T01-33-TR1d

State the complex conjugate of $-8i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q055 | WRITTEN

C03-T01-33-TR2a

Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V03

Q056 | WRITTEN

C03-T01-33-TR2b

Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q057 | WRITTEN

C03-T01-33-EX1a

State the complex conjugate of $3 + 4i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q058 | WRITTEN

C03-T01-33-EX1b

State the complex conjugate of $9 + 8i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q059 | WRITTEN

C03-T01-33-EX1c

State the complex conjugate of $8 + i\sqrt{3}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q060 | WRITTEN

C03-T01-33-EX1d

State the complex conjugate of 5.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q061 | WRITTEN

C03-T01-33-EX1e

State the complex conjugate of -15 .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q062 | WRITTEN

C03-T01-33-EX1f

State the complex conjugate of $\sqrt{7}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q063 | WRITTEN

C03-T01-33-EX1g

State the complex conjugate of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q064 | WRITTEN

C03-T01-33-EX1h

State the complex conjugate of $-50i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q065 | WRITTEN

C03-T01-33-EX1i

State the complex conjugate of $i\sqrt{6}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q066 | WRITTEN

C03-T01-33-EX2a

Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q067 | WRITTEN

C03-T01-33-EX2b

Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q068 | WRITTEN

C03-T01-33-EX2c

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q069 | WRITTEN

C03-T01-33-EX2d

Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q070 | WRITTEN

C03-T01-33-EX2e

Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q071 | WRITTEN

C03-T01-33-EX2f

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V03

Q072 | WRITTEN

C03-T01-33-EX2g

Calculate $1/i$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V03

Q073 | WRITTEN

C03-T01-33-EX2h

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V03

Q074 | WRITTEN

C03-T01-33-CH

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)} \right)^{2027}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q075 | WRITTEN

C03-T01-34-WU1

Express the square roots of -6 in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q076 | WRITTEN

C03-T01-34-WU2a

Express $\sqrt{-12}\sqrt{-6}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q077 | WRITTEN

C03-T01-34-WU2b

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q078 | WRITTEN

C03-T01-34-TR1a

Express $\sqrt{-7}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q079 | WRITTEN

C03-T01-34-TR1b

Express $\sqrt{-27}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q080 | WRITTEN

C03-T01-34-TR1c

Express $-\sqrt{-12}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q081 | WRITTEN

C03-T01-34-TR1d

Express the square roots of -12 in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q082 | WRITTEN

C03-T01-34-TR2a

Express $\sqrt{-3}\sqrt{-6}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q083 | WRITTEN

C03-T01-34-TR2b

Express $\sqrt{27}/\sqrt{-3}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V03

Q084 | WRITTEN

C03-T01-34-TR2c

Express $(3 + \sqrt{-2})^2$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q085 | WRITTEN

C03-T01-34-EX1a

Express $\sqrt{-2}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q086 | WRITTEN

C03-T01-34-EX1b

Express $\sqrt{-9}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q087 | WRITTEN

C03-T01-34-EX1c

Express $\sqrt{-45}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q088 | WRITTEN

C03-T01-34-EX1d

Express $-\sqrt{-24}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q089 | WRITTEN

C03-T01-34-EX1e

Express $-\sqrt{-36}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q090 | WRITTEN

C03-T01-34-EX1f

Express the square roots of -2 in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q091 | WRITTEN

C03-T01-34-EX1g

Express the square roots of -18 in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q092 | WRITTEN

C03-T01-34-EX1h

Express the square roots of -16 in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q093 | WRITTEN

C03-T01-34-EX2a

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q094 | WRITTEN

C03-T01-34-EX2b

Express $\sqrt{-28}\sqrt{-35}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q095 | WRITTEN

C03-T01-34-EX2c

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q096 | WRITTEN

C03-T01-34-EX2d

Express $\sqrt{27}/\sqrt{-12}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q097 | WRITTEN

C03-T01-34-EX2e

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V02

Q098 | WRITTEN

C03-T01-34-EX2f

Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V03

Q099 | WRITTEN

C03-T01-34-EX2g

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V03

Q100 | WRITTEN

C03-T01-34-EX2h

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V04

Q101 | WRITTEN

C03-T01-34-CH

Solve over the complex numbers: $(2x - 8)^2 = -10$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q102 | MCQ

C03-K01-G-MCQ-001

For $2 + 3i$, select the ordered pair (real part, imaginary part):

A (2, 4)

B (3, 2)

C (2, 3)

D (3, 3)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q103 | MCQ

C03-K01-G-MCQ-002

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

A $(-3, 5)$ B $(-4, 6)$ C $(5, -4)$ D $(-4, 5)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q104 | MCQ

C03-K01-G-MCQ-003

For $7 - 2i$, select the ordered pair (real part, imaginary part):

A (8, -2)

B (7, -1)

C (-2, 7)

D (7, -2)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q105 | MCQ

C03-K01-G-MCQ-004

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

A $\left(-\frac{\sqrt{2}}{3}, \frac{1}{3}\right)$

B $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$

C $\left(\frac{4}{3}, -\frac{\sqrt{2}}{3}\right)$

D $\left(\frac{1}{3}, 1 - \frac{\sqrt{2}}{3}\right)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q106 | MCQ

C03-K01-G-MCQ-005

For -6 , select the ordered pair (real part, imaginary part):

A $(-5, 0)$ B $(-6, 1)$ C $(0, -6)$ D $(-6, 0)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q107 | MCQ

C03-K01-G-MCQ-006

For $4i$, select the ordered pair (real part, imaginary part):

A (0, 4)

B (1, 4)

C (0, 5)

D (4, 0)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q108 | MCQ

C03-K01-G-MCQ-007

For $-i$, select the ordered pair (real part, imaginary part):A $(-1, 0)$ B $(0, -1)$ C $(1, -1)$ D $(0, 0)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q109 | MCQ

C03-K01-G-MCQ-008

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

A $(0, 1 + \sqrt{5})$

B $(\sqrt{5}, 0)$

C $(0, \sqrt{5})$

D $(1, \sqrt{5})$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q110 | MCQ

C03-K01-G-MCQ-009

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

A $(\sqrt{3}, \frac{3}{2})$

B $(\frac{3}{2}, \sqrt{3})$

C $(\frac{5}{2}, \sqrt{3})$

D $(\frac{3}{2}, 1 + \sqrt{3})$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q111 | MCQ

C03-K01-G-MCQ-010

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):

A $\left(\frac{3}{4}, -2\right)$

B $\left(-\frac{1}{4}, -1\right)$

C $\left(-2, -\frac{1}{4}\right)$

D $\left(-\frac{1}{4}, -2\right)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q112 | MCQ

C03-K01-G-MCQ-011

For $5/3$, select the ordered pair (real part, imaginary part):

A $\left(\frac{8}{3}, 0\right)$

B $\left(\frac{5}{3}, 1\right)$

C $\left(0, \frac{5}{3}\right)$

D $\left(\frac{5}{3}, 0\right)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q113 | MCQ

C03-K01-G-MCQ-012

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

A $(0, 1 - \sqrt{7})$

B $(-\sqrt{7}, 0)$

C $(0, -\sqrt{7})$

D $(1, -\sqrt{7})$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q114 | MCQ

C03-K01-G-MCQ-013

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

A $(-8, \sqrt{2})$

B $(-9, 1 + \sqrt{2})$

C $(\sqrt{2}, -9)$

D $(-9, \sqrt{2})$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q115 | MCQ

C03-K01-G-MCQ-014

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

A $\left(-\frac{3}{4}, \frac{2}{5}\right)$

B $\left(\frac{2}{5}, -\frac{3}{4}\right)$

C $\left(\frac{7}{5}, -\frac{3}{4}\right)$

D $\left(\frac{2}{5}, \frac{1}{4}\right)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q116 | MCQ

C03-K01-G-MCQ-015

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

A $(0, -\sqrt{11})$

B $(-\sqrt{11}, 0)$

C $(1 - \sqrt{11}, 0)$

D $(-\sqrt{11}, 1)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | V01

Q117 | MCQ

C03-K01-G-MCQ-016

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

A $(0, 1 + \sqrt{13})$

B $(\sqrt{13}, 0)$

C $(0, \sqrt{13})$

D $(1, \sqrt{13})$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q118 | MCQ

C03-K01-G-MCQ-017

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y):

A (3, -1)

B (2, -2)

C (-1, 2)

D (2, -1)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q119 | MCQ

C03-K01-G-MCQ-018

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

A (-1, 2)

B (0, 2)

C (-1, 1)

D (2, -1)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q120 | MCQ

C03-K01-G-MCQ-019

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

A (3, 1)

B (1, 3)

C (2, 3)

D (1, 2)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q121 | MCQ

C03-K01-G-MCQ-020

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x,y):

A (1, -2)

B (-2, 1)

C (-1, 1)

D (-2, 0)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q122 | MCQ

C03-K01-G-MCQ-021

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x,y):

A (3, -2)

B (-1, 3)

C (3, -1)

D (4, -1)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q123 | MCQ

C03-K01-G-MCQ-022

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x,y):

A (2, -2)

B (1, -3)

C (-2, 1)

D (1, -2)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q124 | MCQ

C03-K01-G-MCQ-023

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x,y):

A (0, -2)

B (-1, -3)

C (-2, -1)

D (-1, -2)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | V02

Q125 | MCQ

C03-K01-G-MCQ-024

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

A (2, 1)

B 3

C (2, 2)

D (3, 2)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q126 | MCQ

C03-K01-G-MCQ-025

For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:

A $1 + 6i$

B $-2 - 6i$

C $2 + 6i$

D $3 + 6i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q127 | MCQ

C03-K01-G-MCQ-026

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:

A $2 - 2i$

B $-3 + 2i$

C $3 - 2i$

D $4 - 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q128 | MCQ

C03-K01-G-MCQ-027

For $(1 + 2i)(3 - i)$, select the simplified standard form:A $6 + 5i$ B $4 + 5i$ C $-5 - 5i$ D $5 + 5i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q129 | MCQ

C03-K01-G-MCQ-028

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

A 4

B 5

C 3

D -4

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q130 | MCQ

C03-K01-G-MCQ-029

For $(2 - i)^2$, select the simplified standard form:

A $2 - 4i$

B $-3 + 4i$

C $(2 - i)^2$

D $4 - 4i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q131 | MCQ

C03-K01-G-MCQ-030

For $(1 + 3i)^2$, select the simplified standard form:

A $-8 + 6i$

B $-7 + 6i$

C $-9 + 6i$

D $8 - 6i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q132 | MCQ

C03-K01-G-MCQ-031

For i^{37} , select the simplified standard form:A $-i$ B i C $1 + i$ D $-1 + i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q133 | MCQ

C03-K01-G-MCQ-032

For $i^{22} + i^9$, select the simplified standard form:

A $1 - i$ B $-1 + i$ C i D $-2 + i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q134 | MCQ

C03-K01-G-MCQ-033

For $i^{15} - i^6$, select the simplified standard form:

A $-1 + i$ B $1 - i$ C $2 - i$ D $-i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q135 | MCQ

C03-K01-G-MCQ-034

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:

A $4 + 3i$

B $-5 - 3i$

C $5 + 3i$

D $6 + 3i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q136 | MCQ

C03-K01-G-MCQ-035

For $(2 + i)(2 - i)$, select the simplified standard form:

A 5

B 6

C 4

D -5

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q137 | MCQ

C03-K01-G-MCQ-036

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

A $-3 + 4\sqrt{2}i$

B $2 - 4\sqrt{2}i$

C $(\sqrt{2} + 2i)^2$

D $-1 + 4\sqrt{2}i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q138 | MCQ

C03-K01-G-MCQ-037

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

A $-1 - i$ B i C $-i$ D $1 - i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q139 | MCQ

C03-K01-G-MCQ-038

For $(3 - i)^3$, select the simplified standard form:

A $19 - 26i$

B $17 - 26i$

C $-18 + 26i$

D $(3 - i)^3$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q140 | MCQ

C03-K01-G-MCQ-039

For $(1 - 2i)(-2 + i)$, select the simplified standard form:

A $1 + 5i$ B $-1 + 5i$ C $-5i$ D $5i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q141 | MCQ

C03-K01-G-MCQ-040

For $i^5 + i^{10}$, select the simplified standard form:

A i B $-2 + i$ C $1 - i$ D $-1 + i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q142 | MCQ

C03-K01-G-MCQ-041

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

A $3 + 6i$

B $1 + 6i$

C $-2 - 6i$

D $2 + 6i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | V02

Q143 | MCQ

C03-K01-G-MCQ-042

For $(2 - 3i)^2$, select the simplified standard form:

A $-6 - 12i$

B $5 + 12i$

C $(2 - 3i)^2$

D $-4 - 12i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q144 | MCQ

C03-K01-G-MCQ-043

For the expression State the complex conjugate of $3 - 2i$; the required value is:

A $4 + 2i$

B $3 - 2i$

C $3 + 2i$

D $-3 - 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q145 | MCQ

C03-K01-G-MCQ-044

For the expression State the complex conjugate of $-4 + \sqrt{3}i$, the required value is:

A $-4 - \sqrt{3}i$

B $4 + \sqrt{3}i$

C $-3 - \sqrt{3}i$

D $-4 + \sqrt{3}i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q146 | MCQ

C03-K01-G-MCQ-045

For the expression State the complex conjugate of $7i$., the required value is:

A $1 - 7i$ B 3 C $-7i$ D $7i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q147 | MCQ

C03-K01-G-MCQ-046

For the expression State the complex conjugate of $-\sqrt{5}$;, the required value is:

A $1 - \sqrt{5}$

B 3

C $-\sqrt{5}$

D $\sqrt{5}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q148 | MCQ

C03-K01-G-MCQ-047

For the expression State the complex conjugate of $1/2 - 3i$, the required value is:

A $\frac{1}{2} - 3i$

B $\frac{1}{2} + 3i$

C $-\frac{1}{2} - 3i$

D $\frac{3}{2} + 3i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q149 | MCQ

C03-K01-G-MCQ-048

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:

A $\frac{1}{2} + \frac{3i}{2}$

B $\frac{3}{2} + \frac{3i}{2}$

C $-\frac{1}{2} + \frac{3i}{2}$

D $-\frac{1}{2} - \frac{3i}{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q150 | MCQ

C03-K01-G-MCQ-049

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:A $1 - 2i$ B $2 - 2i$ C $-2i$ D $-1 + 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q151 | MCQ

C03-K01-G-MCQ-050

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:

A $-\frac{1}{10} - \frac{7i}{10}$

B $\frac{1}{10} + \frac{7i}{10}$

C $\frac{11}{10} + \frac{7i}{10}$

D $-\frac{9}{10} + \frac{7i}{10}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q152 | MCQ

C03-K01-G-MCQ-051

For $5i/(2 - i)$, select the value in the form $a+bi$:A $-2 + 2i$ B $1 - 2i$ C $i(2 + i)$ D $2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q153 | MCQ

C03-K01-G-MCQ-052

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:A $1 - i$ B $-1 - i$ C i D $-i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q154 | MCQ

C03-K01-G-MCQ-053

For $1/i$, select the value in the form $a+bi$:A $1 - i$ B $-1 - i$ C i D $-i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q155 | MCQ

C03-K01-G-MCQ-054

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:

A $\frac{5}{13} - \frac{12i}{13}$

B $-\frac{5}{13} + \frac{12i}{13}$

C $\frac{8}{13} + \frac{12i}{13}$

D $-\frac{18}{13} + \frac{12i}{13}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q156 | MCQ

C03-K01-G-MCQ-055

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:

A $\frac{-\sqrt{2} - i}{\sqrt{2} - i}$

B $\frac{\sqrt{2} + i}{\sqrt{2} - i}$

C $\frac{2\sqrt{2}}{\sqrt{2} - i}$

D $\frac{2i}{\sqrt{2} - i}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q157 | MCQ

C03-K01-G-MCQ-056

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:

A $\frac{8}{13} - \frac{14i}{13}$

B $-\frac{18}{13} - \frac{14i}{13}$

C $\frac{5}{13} + \frac{14i}{13}$

D $-\frac{5}{13} - \frac{14i}{13}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q158 | MCQ

C03-K01-G-MCQ-057

For $7i/(1-i)$, select the value in the form $a+bi$:

A $-\frac{9}{2} + \frac{7i}{2}$

B $\frac{7}{2} - \frac{7i}{2}$

C $-\frac{7}{2} + \frac{7i}{2}$

D $-\frac{5}{2} + \frac{7i}{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q159 | MCQ

C03-K01-G-MCQ-058

For the expression State the complex conjugate of $-2 + 5i$, the required value is:

A $-2 + 5i$

B $-2 - 5i$

C $2 + 5i$

D $-1 - 5i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q160 | MCQ

C03-K01-G-MCQ-059

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

A $-6 - \sqrt{7}i$

B $7 + \sqrt{7}i$

C $6 - \sqrt{7}i$

D $6 + \sqrt{7}i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q161 | MCQ

C03-K01-G-MCQ-060

For the expression State the complex conjugate of $-i\sqrt{2}$., the required value is:

A 3

B $\sqrt{2}i$ C $-\sqrt{2}i$ D $1 + \sqrt{2}i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q162 | MCQ

C03-K01-G-MCQ-061

For the expression State the complex conjugate of 9:, the required value is:

A 10

B 3

C 9

D -9

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q163 | MCQ

C03-K01-G-MCQ-062

For the expression State the complex conjugate of $-3/4 + 2i$, the required value is:

A $-\frac{3}{4} - 2i$

B $\frac{3}{4} + 2i$

C $\frac{1}{4} - 2i$

D $-\frac{3}{4} + 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q164 | MCQ

C03-K01-G-MCQ-063

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

A $\frac{6}{5} + \frac{12i}{5}$

B $-\frac{4}{5} + \frac{12i}{5}$

C $-\frac{1}{5} - \frac{12i}{5}$

D $\frac{1}{5} + \frac{12i}{5}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q165 | MCQ

C03-K01-G-MCQ-064

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:

A $\frac{7}{17} - \frac{6i}{17}$

B $\frac{24}{17} - \frac{6i}{17}$

C $-\frac{10}{17} - \frac{6i}{17}$

D $-\frac{7}{17} + \frac{6i}{17}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q166 | MCQ

C03-K01-G-MCQ-065

For $3i/(2 + 3i)$, select the value in the form $a+bi$:

A $\frac{9}{13} + \frac{6i}{13}$

B $\frac{22}{13} + \frac{6i}{13}$

C $-\frac{4}{13} + \frac{6i}{13}$

D $-\frac{9}{13} - \frac{6i}{13}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q167 | MCQ

C03-K01-G-MCQ-066

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

A $-\frac{2i}{\sqrt{3} + i}$

B $\frac{-\sqrt{3} + i}{\sqrt{3} + i}$

C $\frac{\sqrt{3} - i}{\sqrt{3} + i}$

D $\frac{2\sqrt{3}}{\sqrt{3} + i}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q168 | MCQ

C03-K01-G-MCQ-067

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

A $\frac{1}{2} - \frac{i}{2}$

B $\frac{3}{2} - \frac{i}{2}$

C $-\frac{1}{2} - \frac{i}{2}$

D $-\frac{1}{2} + \frac{i}{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q169 | MCQ

C03-K01-G-MCQ-068

For the expression State the complex conjugate of $4 - 7i$; the required value is:

A $-4 - 7i$

B $5 + 7i$

C $4 - 7i$

D $4 + 7i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q170 | MCQ

C03-K01-G-MCQ-069

For the expression State the complex conjugate of $\sqrt{10} + 2i$, the required value is:

A $\sqrt{10} - 2i$

B $-\sqrt{10} + 2i$

C $1 + \sqrt{10} - 2i$

D $\sqrt{10} + 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V01

Q171 | MCQ

C03-K01-G-MCQ-070

For the expression State the complex conjugate of $-8i$; the required value is:

A 3

B $8i$ C $-8i$ D $1 + 8i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | V02

Q172 | MCQ

C03-K01-G-MCQ-071

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

A $-\frac{5}{13} - \frac{12i}{13}$

B $\frac{5}{13} + \frac{12i}{13}$

C $\frac{18}{13} + \frac{12i}{13}$

D $-\frac{8}{13} + \frac{12i}{13}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q173 | MCQ

C03-K01-G-MCQ-072

For $\sqrt{-5}$, select the simplified value in terms of i :

A $5i$ B $i\sqrt{5}$ C $-i\sqrt{5}$ D $\sqrt{5}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q174 | MCQ

C03-K01-G-MCQ-073

For $\sqrt{-20}$, select the simplified value in terms of i :

A $2i\sqrt{5}$

B $-2i\sqrt{5}$

C $4i\sqrt{5}$

D $i\sqrt{20}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q175 | MCQ

C03-K01-G-MCQ-074

For $-\sqrt{-18}$, select the simplified value in terms of i :

A $-3i\sqrt{2}$

B $3i\sqrt{2}$

C $-6i$

D $-i\sqrt{18}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q176 | MCQ

C03-K01-G-MCQ-075

For *squarerootsof* -8 , select the simplified value in terms of i :

A $2i\sqrt{2}$

B $\pm 8i$

C $\pm 2i\sqrt{2}$

D $\pm i\sqrt{8}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q177 | MCQ

C03-K01-G-MCQ-076

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i :

A 3

B -3 C $3i$ D $\sqrt{9}i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q178 | MCQ

C03-K01-G-MCQ-077

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i :

A 2

B -4

C 4

D $-4i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q179 | MCQ

C03-K01-G-MCQ-078

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

A -2 B $i\sqrt{4}$ C $-2i$ D $2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q180 | MCQ

C03-K01-G-MCQ-079

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

A $1 - 4i\sqrt{3}$

B $7 + 4i$

C $4 + 2i\sqrt{3}$

D $1 + 4i\sqrt{3}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q181 | MCQ

C03-K01-G-MCQ-080

For $\sqrt{-72}$, select the simplified value in terms of i :

A $-6i\sqrt{2}$

B $12i$

C $i\sqrt{72}$

D $6i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q182 | MCQ

C03-K01-G-MCQ-081

For *squarerootsof* -50 , select the simplified value in terms of i :

A $\pm i\sqrt{50}$

B $5i\sqrt{2}$

C $\pm 10i$

D $\pm 5i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q183 | MCQ

C03-K01-G-MCQ-082

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

A -6 B 6 C $-6i$ D 3

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q184 | MCQ

C03-K01-G-MCQ-083

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

A $2i$ B -2 C $-i\sqrt{2}$ D $-2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q185 | MCQ

C03-K01-G-MCQ-084

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

A $6\sqrt{2}$

B $6i\sqrt{2}$

C $2i\sqrt{2}$

D $-6i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q186 | MCQ

C03-K01-G-MCQ-085

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

A $4\sqrt{2}$

B $4i\sqrt{2}$

C $8i\sqrt{2}$

D $-4i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q187 | MCQ

C03-K01-G-MCQ-086

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i :

A $3 + 2i\sqrt{10}$

B $7 + 2i\sqrt{10}$

C $3 - 2i\sqrt{10}$

D $5 + 2i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q188 | MCQ

C03-K01-G-MCQ-087

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i :

A 4

B $14i$ C $9 - 5i$

D 14

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q189 | MCQ

C03-K01-G-MCQ-088

For $\sqrt{-7}$, select the simplified value in terms of i :

A $7i$ B $\sqrt{7}$ C $i\sqrt{7}$ D $-i\sqrt{7}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q190 | MCQ

C03-K01-G-MCQ-089

For $\sqrt{-28}$, select the simplified value in terms of i :

A $2i\sqrt{7}$

B $-2i\sqrt{7}$

C $4i\sqrt{7}$

D $i\sqrt{28}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q191 | MCQ

C03-K01-G-MCQ-090

For *squarerootsof* -12 , select the simplified value in terms of i :

A $\pm 2i\sqrt{3}$

B $\pm i\sqrt{12}$

C $2i\sqrt{3}$

D $\pm 4i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q192 | MCQ

C03-K01-G-MCQ-091

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

A $-3i$ B $3i$ C -3 D $i\sqrt{3}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q193 | MCQ

C03-K01-G-MCQ-092

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

A 6

B $-6i$ C -3 D -6

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q194 | MCQ

C03-K01-G-MCQ-093

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

A -4 B $-i\sqrt{4}$ C $-4i$ D $4i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q195 | MCQ

C03-K01-G-MCQ-094

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

A $-5i\sqrt{2}$

B $5\sqrt{2}$

C $5i\sqrt{2}$

D $i\sqrt{26}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \implies a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q196 | MCQ

C03-K01-G-MCQ-095

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

A $-1 + 2i\sqrt{2}$

B $3 + 2i\sqrt{2}$

C $-1 - 2i\sqrt{2}$

D $1 + 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q197 | MCQ

C03-K01-G-MCQ-096

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

A $16 - 3i$ B 19 C 13 D $19i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | V01

Q198 | MCQ

C03-K01-G-MCQ-097

For $\sqrt{-75}$, select the simplified value in terms of i :

A $-5i\sqrt{3}$

B $15i$

C $i\sqrt{75}$

D $5i\sqrt{3}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V01

Q199 | MCQ

C03-K01-G-MCQ-098

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:

A $\left(-\frac{1}{2}, -7\right)$

B $\left(-\frac{1}{2}, 7\right)$

C $\left(-\frac{1}{2}, 6\right)$

D $\left(\frac{1}{2}, 7\right)$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V02

Q200 | MCQ

C03-K01-G-MCQ-099

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:

A $\{0, 1, 2\}$

B $\{2, \frac{1}{2}\}$

C $\{-2, 2, \frac{1}{2}\}$

D $\{0, 2, \frac{1}{2}\}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V03

Q201 | MCQ

C03-K01-G-MCQ-100

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:

A $((1 + 7i)/5)^{2027}$

B $((7 + i)/5)^{2027}$

C $((7 - i)/5)^{2027}$

D $(7 + i)^{2027}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F99 | V04

Q202 | MCQ

C03-K01-G-MCQ-101

Solve over the complex numbers: $(3x - 6)^2 = -8$:

A $x = 2 \pm 2i\sqrt{2}$

B $x = 2 \pm (2/3)i\sqrt{2}$

C $x = 2 \pm i\sqrt{2}$

D $x = 3 \pm i\sqrt{2}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | C03-T01-F01-VQ

Quiz M1 | QUIZ MCQ

C03-K01-G-MCQ-151

For $4-3i$, select the ordered pair (real part, imaginary part):

A (4, -3)

B (4, 3)

C (-4, -3)

D (-3, 4)

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | C03-T01-F03-VQ

Quiz M2 | QUIZ MCQ

C03-K01-G-MCQ-152

For $(1+2i)(2-i)$, select the simplified value:A $4 + 3i$ B 1 C 2 D $-4 - 3i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F04 | C03-T01-F04-VQ

Quiz M3 | QUIZ MCQ

C03-K01-G-MCQ-153

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

A 1

B 2

C $-1 - 2i$ D $1 + 2i$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | C03-T01-F05-VQ

Quiz M4 | QUIZ MCQ

C03-K01-G-MCQ-154

For $\sqrt{-27}$, select the expression in terms of i :

A $3i\sqrt{3}$

B $-3i\sqrt{3}$

C $9i$

D $i\sqrt{27}$

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F01 | C03-T01-F01-VQ

Quiz W1 | QUIZ WRITTEN

C03-K01-QUIZ-WRITTEN-1

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F02 | C03-T01-F02-VQ

Quiz W2 | QUIZ WRITTEN

C03-K01-QUIZ-WRITTEN-2

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F03 | C03-T01-F03-VQ

Quiz W3 | QUIZ WRITTEN

C03-K01-QUIZ-WRITTEN-3

Calculate $i^{18} + i^7 - i^3$ in standard form.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 1 | Complex Numbers

F05 | C03-T01-F05-VQ

Quiz W4 | QUIZ WRITTEN

C03-K01-QUIZ-WRITTEN-4

Solve over the complex numbers: $(x - 3)^2 = -12$.

WORKING AREA

FORMULA BANK

Standard form

$$z = a + bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers of i

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Conjugate and division

$$\overline{a + bi} = a - bi \quad \frac{z}{w} = \frac{z \overline{w}}{w \overline{w}}$$

Negative roots

$$\sqrt{-a} = i\sqrt{a} \quad (a > 0)$$

Concept 2 | Quadratic Equations

OFFICIAL LESSONS 3-5, 3-6, 3-7, 3-8, 3-9, 3-10, 3-11

Quadratic Equations

Equation-first recall | use the same rail beside every question in this Concept

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

$$D = b^2 - 4ac$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P \quad \alpha^3 + \beta^3 = S^3 - 3PS$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign} \quad \alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Quadratic Equations

F01 | V01

Q001 | WRITTEN

C03-T02-35-WU1a

Solve $x^2 + 8 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q002 | WRITTEN

C03-T02-35-TR1a

Solve $x^2 + 4 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q003 | WRITTEN

C03-T02-35-EX1a

Solve $x^2 + 49 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q004 | WRITTEN

C03-T02-35-EX1b

Solve $9x^2 - 4 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q005 | WRITTEN

C03-T02-35-EX2c

Solve $x^2 - 12 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q006 | WRITTEN

C03-T02-35-EX2f

Solve $4x^2 + 3 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q007 | MCQ

C03-K02-G-MCQ-001

Solve $x^2 + 8 = 0$ Select the correct answer:

A $x = \pm 2i\sqrt{2}$

B $x = 2i\sqrt{2}$

C $x = -2i\sqrt{2}$

D $x = +2i\sqrt{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q008 | MCQ

C03-K02-G-MCQ-004

Solve $x^2 + 4 = 0$ Select the correct answer:

A $x = \pm 2i$

B $x = 2i$

C $x = -2i$

D $x = +2i$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q009 | MCQ

C03-K02-G-MCQ-008

Solve $x^2 + 49 = 0$ Select the correct answer:

A $x = -7i$

B $x = +7i$

C $x = \pm 7i$

D $x = 7i$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q010 | MCQ

C03-K02-G-MCQ-009

Solve $9x^2 - 4 = 0$ Select the correct answer:

A $x = -\frac{2}{3}$

B $x = +\frac{2}{3}$

C $x = \pm\frac{2}{3}$

D $x = \frac{2}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q011 | MCQ

C03-K02-G-MCQ-020

Solve $x^2 - 12 = 0$ Select the correct answer:

A $x = +2\sqrt{3}$

B $x = \pm 2\sqrt{3}$

C $x = 2\sqrt{3}$

D $x = -2\sqrt{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V01

Q012 | MCQ

C03-K02-G-MCQ-023

Solve $4x^2 + 3 = 0$ Select the correct answer:

A $x = +\frac{i\sqrt{3}}{2}$

B $x = \pm\frac{i\sqrt{3}}{2}$

C $x = \frac{i\sqrt{3}}{2}$

D $x = -\frac{i\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q013 | WRITTEN

C03-T02-35-TR1b

Solve $(x - 3)^2 = -8$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q014 | WRITTEN

C03-T02-35-EX1c

Solve $(x - 2)^2 = 5$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q015 | WRITTEN

C03-T02-35-EX1d

Solve $(x + 6)^2 = -6$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q016 | WRITTEN

C03-T02-35-EX2b

Solve $(x + 4)^2 = -18$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q017 | WRITTEN

C03-T02-35-EX2i

Solve $(2x - 3)^2 = 7$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q018 | MCQ

C03-K02-G-MCQ-005

Solve $(x - 3)^2 = -8$ Select the correct answer:

A $x = 3 \pm 2i\sqrt{2}$

B $x = 32i\sqrt{2}$

C $x = 3 - 2i\sqrt{2}$

D $x = 3 + 2i\sqrt{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q019 | MCQ

C03-K02-G-MCQ-010

Solve $(x - 2)^2 = 5$ Select the correct answer:

A $x = 2 + \sqrt{5}$

B $x = 2 \pm \sqrt{5}$

C $x = 2\sqrt{5}$

D $x = 2 - \sqrt{5}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q020 | MCQ

C03-K02-G-MCQ-011

Solve $(x + 6)^2 = -6$ Select the correct answer:

A $x = -6 - i\sqrt{6}$

B $x = -6 + i\sqrt{6}$

C $x = -6 \pm i\sqrt{6}$

D $x = -6i\sqrt{6}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q021 | MCQ

C03-K02-G-MCQ-019

Solve $(x + 4)^2 = -18$ Select the correct answer:

A $x = -4 - 3i\sqrt{2}$

B $x = -4 + 3i\sqrt{2}$

C $x = -4 \pm 3i\sqrt{2}$

D $x = -43i\sqrt{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F01 | V02

Q022 | MCQ

C03-K02-G-MCQ-026

Solve $(2x - 3)^2 = 7$ Select the correct answer:

A $x = \frac{3\sqrt{7}}{2}$

B $x = \frac{3 - \sqrt{7}}{2}$

C $x = \frac{3 + \sqrt{7}}{2}$

D $x = \frac{3 \pm \sqrt{7}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q023 | WRITTEN

C03-T02-35-WU1b

Solve $2x^2 - 7x - 9 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q024 | WRITTEN

C03-T02-35-TR1c

Solve $2x^2 + 5x - 3 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q025 | WRITTEN

C03-T02-35-EX1e

Solve $2x^2 + 7x = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q026 | WRITTEN

C03-T02-35-EX1f

Solve $2x^2 + 5x - 12 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q027 | WRITTEN

C03-T02-35-EX2e

Solve $x^2 - 8x + 15 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q028 | MCQ

C03-K02-G-MCQ-002

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:

A $x = -1, \frac{9}{2}$

B $-(x = -1, \frac{9}{2})$

C $x = -1, \frac{9}{2} + 1$

D $x = -1, \frac{9}{2} - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q029 | MCQ

C03-K02-G-MCQ-006

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:

A $-(x = \frac{1}{2}, -3)$

B $x = \frac{1}{2}, -3 + 1$

C $x = \frac{1}{2}, -3 - 1$

D $x = \frac{1}{2}, -3$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q030 | MCQ

C03-K02-G-MCQ-012

Solve $2x^2 + 7x = 0$ Select the correct answer:

A $-(x = 0, -\frac{7}{2})$

B $x = 0, -\frac{7}{2} + 1$

C $x = 0, -\frac{7}{2} - 1$

D $x = 0, -\frac{7}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q031 | MCQ

C03-K02-G-MCQ-013

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:

A $x = \frac{3}{2}, -4$

B $-(x = \frac{3}{2}, -4)$

C $x = \frac{3}{2}, -4 + 1$

D $x = \frac{3}{2}, -4 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | V01

Q032 | MCQ

C03-K02-G-MCQ-022

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:

A $x = 3, 5$

B $x = 3, 6$

C $x = 3, 5 - 1$

D $x = 3, 5$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V01

Q033 | WRITTEN

C03-T02-35-EX2d

Solve $3x^2 - 4x - 1 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V01

Q034 | MCQ

C03-K02-G-MCQ-021

Solve $3x^2 - 4x - 1 = 0$ Select the correct answer:

A $x = \frac{2 + \sqrt{7}}{3}$

B $x = \frac{2 \pm \sqrt{7}}{3}$

C $x = \frac{2\sqrt{7}}{3}$

D $x = \frac{2 - \sqrt{7}}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q035 | WRITTEN

C03-T02-35-WU1c

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q036 | WRITTEN

C03-T02-35-TR1d

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q037 | WRITTEN

C03-T02-35-EX1g

Solve $x^2 + x + 1 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q038 | WRITTEN

C03-T02-35-EX1h

Solve $3x^2 - 5x + 3 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q039 | WRITTEN

C03-T02-35-EX1i

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q040 | WRITTEN

C03-T02-35-EX1j

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q041 | WRITTEN

C03-T02-35-EX2a

Solve $x^2 - x + 2 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q042 | WRITTEN

C03-T02-35-EX2g

Solve $3x^2 - \sqrt{5}x + 1 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q043 | WRITTEN

C03-T02-35-EX2h

Solve $6x^2 - 12x + 15 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q044 | MCQ

C03-K02-G-MCQ-003

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$ Select the correct answer:

A $x = \frac{-\sqrt{6} + 4i}{2}$

B $x = \frac{-\sqrt{6} \pm 4i}{2}$

C $x = \frac{-\sqrt{64}i}{2}$

D $x = \frac{-\sqrt{6} - 4i}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q045 | MCQ

C03-K02-G-MCQ-007

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ Select the correct answer:

A $x = \frac{-\sqrt{3} - i\sqrt{7}}{2}$

B $x = \frac{-\sqrt{3} + i\sqrt{7}}{2}$

C $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

D $x = \frac{-\sqrt{3}i\sqrt{7}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q046 | MCQ

C03-K02-G-MCQ-014

Solve $x^2 + x + 1 = 0$ Select the correct answer:

A $x = \frac{-1+i\sqrt{3}}{2}$

B $x = \frac{-1-i\sqrt{3}}{2}$

C $x = \frac{-1+i\sqrt{3}}{2}$

D $x = \frac{-1 \pm i\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q047 | MCQ

C03-K02-G-MCQ-015

Solve $3x^2 - 5x + 3 = 0$ Select the correct answer:

A $x = \frac{5 + i\sqrt{11}}{6}$

B $x = \frac{5 \pm i\sqrt{11}}{6}$

C $x = \frac{5i\sqrt{11}}{6}$

D $x = \frac{5 - i\sqrt{11}}{6}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q048 | MCQ

C03-K02-G-MCQ-016

Solve $x^2 - 2\sqrt{3}x + 5 = 0$ Select the correct answer:

A $x = \sqrt{3}i\sqrt{2}$

B $x = \sqrt{3} - i\sqrt{2}$

C $x = \sqrt{3} + i\sqrt{2}$

D $x = \sqrt{3} \pm i\sqrt{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q049 | MCQ

C03-K02-G-MCQ-017

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ Select the correct answer:

A $x = \frac{-\sqrt{2} + i}{3}$

B $x = \frac{-\sqrt{2} \pm i}{3}$

C $x = \frac{-\sqrt{2}i}{3}$

D $x = \frac{-\sqrt{2} - i}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q050 | MCQ

C03-K02-G-MCQ-018

Solve $x^2 - x + 2 = 0$ Select the correct answer:

A $x = \frac{1 - i\sqrt{7}}{2}$

B $x = \frac{1 + i\sqrt{7}}{2}$

C $x = \frac{1 \pm i\sqrt{7}}{2}$

D $x = \frac{1i\sqrt{7}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q051 | MCQ

C03-K02-G-MCQ-024

Solve $3x^2 - \sqrt{5}x + 1 = 0$ Select the correct answer:

A $x = \frac{\sqrt{5} - i\sqrt{7}}{6}$

B $x = \frac{\sqrt{5} + i\sqrt{7}}{6}$

C $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

D $x = \frac{\sqrt{5}i\sqrt{7}}{6}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | V02

Q052 | MCQ

C03-K02-G-MCQ-025

Solve $6x^2 - 12x + 15 = 0$ Select the correct answer:

A $x = 1 + \frac{i\sqrt{6}}{2}$

B $x = 1 \pm \frac{i\sqrt{6}}{2}$

C $x = 1 - \frac{i\sqrt{6}}{2}$

D $x = 1 - \frac{i\sqrt{6}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q053 | WRITTEN

C03-T02-36-WU11

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q054 | WRITTEN

C03-T02-36-WU12

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q055 | WRITTEN

C03-T02-36-WU13

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q056 | WRITTEN

C03-T02-36-TR1a

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q057 | WRITTEN

C03-T02-36-TR1b

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q058 | WRITTEN

C03-T02-36-TR1c

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q059 | WRITTEN

C03-T02-36-EX1a

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q060 | WRITTEN

C03-T02-36-EX1b

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q061 | WRITTEN

C03-T02-36-EX1c

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q062 | WRITTEN

C03-T02-36-EX1d

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q063 | WRITTEN

C03-T02-36-EX1e

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q064 | WRITTEN

C03-T02-36-EX1f

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q065 | MCQ

C03-K02-G-MCQ-028

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:

A Two distinct imaginary (non-real complex) solutions

B Two repeated imaginary (non-real complex) solutions

C Two distinct imaginary (non-imaginary complex) solutions

D Two distinct real (non-real complex) solutions

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q066 | MCQ

C03-K02-G-MCQ-029

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions
Select the correct answer:

A $-2 \geq m \geq 6$

B $+2 \leq m \leq 6$

C $-2 \leq m \leq 7$

D $-2 \leq m \leq 6$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q067 | MCQ

C03-K02-G-MCQ-030

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

A $k < -6$ or $k > 2$: distinct imaginary; $k = -6, 2$: repeated imaginary; $-6 < k < 2$: distinct imaginary

C $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

B $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct real

D $k < -6$ or $k > 2$: repeated real; $k = -6, 2$: repeated real; $-6 < k < 2$: repeated imaginary

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q068 | MCQ

C03-K02-G-MCQ-031

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions. Select the correct answer:

A Two distinct imaginary (non-imaginary complex) solutions

B Two distinct real (non-real complex) solutions

C Two distinct imaginary (non-real complex) solutions

D Two repeated imaginary (non-real complex) solutions

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q069 | MCQ

C03-K02-G-MCQ-032

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions. Select the correct answer:

A Two distinct real solutions

B Two repeated real solutions

C Two distinct imaginary solutions

D Two distinct real solutions;(sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q070 | MCQ

C03-K02-G-MCQ-033

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions. Select the correct answer:

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q071 | MCQ

C03-K02-G-MCQ-036

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:

A Two distinct imaginary solutions

B Two distinct real solutions;(sign error)

C Two distinct real solutions

D Two repeated real solutions

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q072 | MCQ

C03-K02-G-MCQ-037

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:

A 0

B 1

C One repeated real solution

D One repeated real solution;(sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q073 | MCQ

C03-K02-G-MCQ-038

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:

A Two distinct real (non-real complex) solutions

B Two distinct imaginary (non-real complex) solutions

C Two repeated imaginary (non-real complex) solutions

D Two distinct imaginary (non-imaginary complex) solutions

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q074 | MCQ

C03-K02-G-MCQ-039

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q075 | MCQ

C03-K02-G-MCQ-040

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:

A Two distinct imaginary (non-real complex) solutions

B Two repeated imaginary (non-real complex) solutions

C Two distinct imaginary (non-imaginary complex) solutions

D Two distinct real (non-real complex) solutions

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V01

Q076 | MCQ

C03-K02-G-MCQ-041

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

A Two distinct real solutions

B Two repeated real solutions

C Two distinct imaginary solutions

D Two distinct real solutions;(sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q077 | WRITTEN

C03-T02-36-TR22

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q078 | WRITTEN

C03-T02-36-TR33

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q079 | WRITTEN

C03-T02-36-EX2a

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q080 | WRITTEN

C03-T02-36-EX2b

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q081 | WRITTEN

C03-T02-36-EX2c

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q082 | MCQ

C03-K02-G-MCQ-034

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions. Select the correct answer:

A $+1 < m < 2$

B $-1 < m < 3$

C $-1 < m < 2$

D $-1 > m > 2$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q083 | MCQ

C03-K02-G-MCQ-035

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

A $a < 2$ or $a > 6$: distinct imaginary; $a = 2, 6$: repeated imaginary; $2 < a < 6$: distinct imaginary

B $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct real

C $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

D $a < 2$ or $a > 6$: repeated real; $a = 2, 6$: repeated real; $2 < a < 6$: repeated imaginary

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q084 | MCQ

C03-K02-G-MCQ-042

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution. Select the correct answer:

A $a = 3$ or $a = -5$

B $a = 3$ or $a = -4$

C $a = 3$ or $a = -5 - 1$

D $a = 3$ or $a = -5$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q085 | MCQ

C03-K02-G-MCQ-043

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions. Select the correct answer:

A $-18 > m > -2$

B $+18 < m < -2$

C $-18 < m < -1$

D $-18 < m < -2$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V02

Q086 | MCQ

C03-K02-G-MCQ-044

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions. Select the correct answer:

A $0 \leq m \leq 4$

B $0 \leq m \leq 3 - 1$

C $0 \leq m \leq 3$

D $0 \geq m \geq 3$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q087 | WRITTEN

C03-T02-37-WU11

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q088 | WRITTEN

C03-T02-37-TR11

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q089 | WRITTEN

C03-T02-37-EX1a

Find the sum and product of the roots of $x^2 + x + 2 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q090 | WRITTEN

C03-T02-37-EX1b

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q091 | MCQ

C03-K02-G-MCQ-081

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:

A $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} + 1$

B $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} - 1$

C $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

D $-(\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q092 | MCQ

C03-K02-G-MCQ-086

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:

A $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

B $-(\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4})$

C $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} + 1$

D $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q093 | MCQ

C03-K02-G-MCQ-091

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

A $S = -1, \quad P = 2$ (sign error)

B $S = -1, \quad P = 2$

C $S = -1, \quad P = 3$

D $S = -1, \quad P = 2 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V01

Q094 | MCQ

C03-K02-G-MCQ-092

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

A $S = \frac{7}{3}, P = -2$

B $-(S = \frac{7}{3}, P = -2)$

C $S = \frac{7}{3}, P = -2 + 1$

D $S = \frac{7}{3}, P = -2 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q095 | WRITTEN

C03-T02-37-WU2a

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q096 | WRITTEN

C03-T02-37-WU2b

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q097 | WRITTEN

C03-T02-37-TR2a

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q098 | WRITTEN

C03-T02-37-TR2b

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q099 | WRITTEN

C03-T02-37-EX2a

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q100 | WRITTEN

C03-T02-37-EX2b

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q101 | WRITTEN

C03-T02-37-EX3a

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q102 | WRITTEN

C03-T02-37-EX3b

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q103 | MCQ

C03-K02-G-MCQ-082

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$. Select the correct answer:

A 12

B 13

C $12 - 1$

D 12 (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q104 | MCQ

C03-K02-G-MCQ-083

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

A $\frac{81}{2} - 1$

B $\frac{81}{2}$

C $-(\frac{81}{2})$

D $\frac{81}{2} + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q105 | MCQ

C03-K02-G-MCQ-087

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

A 1

B 2

C $1 - 1$

D 1 (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q106 | MCQ

C03-K02-G-MCQ-088

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

A -1 (sign error)

B -1

C 0

D -1 - 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q107 | MCQ

C03-K02-G-MCQ-094

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

A -1

B -2 - 1

C -2 (sign error)

D -2

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q108 | MCQ

C03-K02-G-MCQ-095

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

A -10 (sign error)

B -10

C -9

D -10 - 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q109 | MCQ

C03-K02-G-MCQ-098

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$. Select the correct answer:

A $\frac{14}{3} - 1$

B $\frac{14}{3}$

C $-(\frac{14}{3})$

D $\frac{14}{3} + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V02

Q110 | MCQ

C03-K02-G-MCQ-099

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

A -9 B $-10 - 1$ C -10 (sign error)D -10

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q111 | WRITTEN

C03-T02-37-WU2c

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q112 | WRITTEN

C03-T02-37-TR2c

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q113 | WRITTEN

C03-T02-37-EX1c

Find the sum and product of the roots of $5x^2 - 3 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q114 | WRITTEN

C03-T02-37-EX2c

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q115 | WRITTEN

C03-T02-37-EX3c

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q116 | MCQ

C03-K02-G-MCQ-084

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

A $-4 - 1$ B -4 (sign error)C -4 D -3

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q117 | MCQ

C03-K02-G-MCQ-089

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$. Select the correct answer:

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q118 | MCQ

C03-K02-G-MCQ-093

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:

A $-(S = 0, \quad P = -\frac{3}{5})$

B $S = 0, \quad P = -\frac{3}{5} + 1$

C $S = 0, \quad P = -\frac{3}{5} - 1$

D $S = 0, \quad P = -\frac{3}{5}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q119 | MCQ

C03-K02-G-MCQ-096

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$. Select the correct answer:

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V03

Q120 | MCQ

C03-K02-G-MCQ-100

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

A 7

B $6 - 1$

C 6 (sign error)

D 6

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q121 | WRITTEN

C03-T02-37-WU2d

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q122 | WRITTEN

C03-T02-37-TR2d

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q123 | WRITTEN

C03-T02-37-EX2d

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q124 | WRITTEN

C03-T02-37-EX3d

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q125 | MCQ

C03-K02-G-MCQ-085

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

A $\frac{5}{2}$

B $-(\frac{5}{2})$

C $\frac{5}{2} + 1$

D $\frac{5}{2} - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q126 | MCQ

C03-K02-G-MCQ-090

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

A $\frac{1}{2} + 1$

B $\frac{1}{2} - 1$

C $\frac{1}{2}$

D $-(\frac{1}{2})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q127 | MCQ

C03-K02-G-MCQ-097

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

A 12

B $11 - 1$

C 11 (sign error)

D 11

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V04

Q128 | MCQ

C03-K02-G-MCQ-101

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

A $-(\frac{23}{3})$

B $\frac{23}{3} + 1$

C $\frac{23}{3} - 1$

D $\frac{23}{3}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q129 | WRITTEN

C03-T02-38-TR11

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q130 | WRITTEN

C03-T02-38-EX11

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q131 | WRITTEN

C03-T02-38-EX12

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q132 | MCQ

C03-K02-G-MCQ-049

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots. Select the correct answer:

A $k = 6, \quad x = 2, 6$

B $k = 6, \quad x = 2, 7$

C $k = 6, \quad x = 2, 6 - 1$

D $k = 6, \quad x = 2, 6$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q133 | MCQ

C03-K02-G-MCQ-051

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots. Select the correct answer:

A $m = 8, \quad x = -2, -4$

B $m = 8, \quad x = -2, -3$

C $m = 8, \quad x = -2, -4 - 1$

D $m = 8, \quad x = -2, -4$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V01

Q134 | MCQ

C03-K02-G-MCQ-052

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

A $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{3})$

B $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2}) - 1$

C $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$ (sign error) =

D $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q135 | WRITTEN

C03-T02-38-WU11

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q136 | WRITTEN

C03-T02-38-TR12

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q137 | WRITTEN

C03-T02-38-EX13

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q138 | WRITTEN

C03-T02-38-EX14

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q139 | MCQ

C03-K02-G-MCQ-048

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots
Select the correct answer:

A $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 3)$

B $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2) - 1$

C $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$ (sign error)

D $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q140 | MCQ

C03-K02-G-MCQ-050

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots
Select the correct answer:

A $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

B $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, 0)$

C $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1) - 1$

D (m, x_1, x_2)
 $(2, 1, 2)$ or $(-4, -2, -1)$ (sign error) =

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q141 | MCQ

C03-K02-G-MCQ-053

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots
Select the correct answer:

A $m = 5$, $x = -5, -1$ (sign error)B $m = 5$, $x = -5, -1$ C $m = 5$, $x = -5, 0$ D $m = 5$, $x = -5, -1 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V02

Q142 | MCQ

C03-K02-G-MCQ-054

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots. Select the correct answer:

A (m, x_1, x_2)
 $(-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$ (sign error) =

B $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

C $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

D $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2}) - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V03

Q143 | WRITTEN

C03-T02-38-EX15

If the ratio of the roots of $x^2 - 2x + m = 0$ is 2 : 3, find m and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V03

Q144 | MCQ

C03-K02-G-MCQ-055

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots. Select the correct answer:

A $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

B $-(m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5})$

C $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} + 1$

D $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, \quad v = \beta + t \Rightarrow u + v = S + 2t, \quad uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, \quad v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V04

Q145 | WRITTEN

C03-T02-38-EX16

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V04

Q146 | MCQ

C03-K02-G-MCQ-056

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

A $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$ (sign error)

B $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

C $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 10)$

D $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9) - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q147 | WRITTEN

C03-T02-39-WU1a

Factorise $2x^2 - 3x + 2$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q148 | WRITTEN

C03-T02-39-TR1a

Factorise $x^2 + 2x + 5$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q149 | WRITTEN

C03-T02-39-TR1b

Factorise $2x^2 - 6x + 1$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q150 | WRITTEN

C03-T02-39-EX1a

Factorise $x^2 + 4x + 5$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q151 | WRITTEN

C03-T02-39-EX1b

Factorise $x^2 + 4x + 2$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q152 | WRITTEN

C03-T02-39-EX1c

Factorise $x^2 - 4x + 7$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q153 | WRITTEN

C03-T02-39-EX1d

Factorise $2x^2 - x + 1$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q154 | WRITTEN

C03-T02-39-EX1e

Factorise $3x^2 - 5x + 3$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q155 | WRITTEN

C03-T02-39-EX1f

Factorise $2x^2 - 2x - 1$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q156 | MCQ

C03-K02-G-MCQ-058

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:

A $2\left(x - \frac{3-i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

B $-(2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right))$

C $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

D $2\left(x - \frac{3+\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q157 | MCQ

C03-K02-G-MCQ-060

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:

A $(x + 1 - 2i)(x + 1 + 2i)$

B $(x + 1 - 2)(x + 1 + 2i)$

C $(x + 1 - 2 - i)(x + 1 + 2i)$

D $(x + 1 - 2i)(x + 13i)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q158 | MCQ

C03-K02-G-MCQ-061

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:

A $-2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

B $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right) + 1$

C $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right) - 1$

D $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q159 | MCQ

C03-K02-G-MCQ-063

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:

A $(x + 2 - i)(x + 2 + i)$

B $(x + 2 + i)(x + 2 + i)$

C $(x + 2 - i)(x + 3 + i)$

D $(x + 2 - i)(x + 2 + i)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q160 | MCQ

C03-K02-G-MCQ-064

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

A $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$ (sign error)

B $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

C $(x + 2 - \sqrt{2})(x + 2 + \sqrt{3})$

D $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2}) - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q161 | MCQ

C03-K02-G-MCQ-065

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:

A $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

B $(x - 2 - \sqrt{3})(x - 2 + i\sqrt{3})$

C $(x - 2 + i\sqrt{3})(x - 2 + i\sqrt{3})$

D $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{4})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q162 | MCQ

C03-K02-G-MCQ-066

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:

A $2\left(x - \frac{1-i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

B $-(2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right))$

C $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

D $2\left(x - \frac{1+\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q163 | MCQ

C03-K02-G-MCQ-067

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:

A $3 \left(x - \frac{5 + i\sqrt{11}}{6} \right) \left(x - \frac{5 - i\sqrt{11}}{6} \right)$

B $3 \left(x - \frac{5 + \sqrt{11}}{6} \right) \left(x - \frac{5 - i\sqrt{11}}{6} \right)$

C $3 \left(x - \frac{5 - i\sqrt{11}}{6} \right) \left(x - \frac{5 - i\sqrt{11}}{6} \right)$

D $-3 \left(x - \frac{5 + i\sqrt{11}}{6} \right) \left(x - \frac{5 - i\sqrt{11}}{6} \right)$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q164 | MCQ

C03-K02-G-MCQ-068

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:

A $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right) - 1$

B $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

C $-2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

D $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right) + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q165 | WRITTEN

C03-T02-39-WU1b

Factorise $x^4 + 2x^2 - 3$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q166 | WRITTEN

C03-T02-39-TR1c

Factorise $x^4 - x^2 - 6$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q167 | WRITTEN

C03-T02-39-EX1g

Factorise $x^4 - 9$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q168 | WRITTEN

C03-T02-39-EX1h

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q169 | MCQ

C03-K02-G-MCQ-059

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:

A $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{4})$

B $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

C $(x - 1)(x + 1)(x - \sqrt{3})(x + i\sqrt{3})$

D $(x - 1)(x + 1)(x + i\sqrt{3})(x + i\sqrt{3})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q170 | MCQ

C03-K02-G-MCQ-062

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

A $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{3})$

B $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

C $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{2})(x + i\sqrt{2})$

D $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{2})(x + i\sqrt{2})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q171 | MCQ

C03-K02-G-MCQ-069

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:

A $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{3})(x + i\sqrt{3})$

B $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{3})(x + i\sqrt{3})$

C $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{4})$

D $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V02

Q172 | MCQ

C03-K02-G-MCQ-070

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

A $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2}) - 1$

B $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$ (sign error)

C $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

D $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{3})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q173 | WRITTEN

C03-T02-310-WU11

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q174 | WRITTEN

C03-T02-310-WU12

Find two numbers whose sum is -1 and product is 2 .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q175 | WRITTEN

C03-T02-310-TR11

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q176 | WRITTEN

C03-T02-310-TR12

Find two numbers whose sum and product are both 1.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q177 | WRITTEN

C03-T02-310-EX1a

Create a quadratic equation with roots -1 and 2 .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q178 | WRITTEN

C03-T02-310-EX1b

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q179 | WRITTEN

C03-T02-310-EX1c

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q180 | WRITTEN

C03-T02-310-EX2a

Find two numbers whose sum is 2 and product is 5.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q181 | WRITTEN

C03-T02-310-EX2b

Find two numbers whose sum is 4 and product is 2.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q182 | MCQ

C03-K02-G-MCQ-106

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots. Select the correct answer:

A $2x^2 - 2x + 1 = 0$ (sign error)

B $2x^2 - 2x + 1 = 0$

C $2x^2 - 2x + 1 = 1$

D $2x^2 - 2x + 1 = 0 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q183 | MCQ

C03-K02-G-MCQ-107

Find two numbers whose sum is -1 and product is 2 Select the correct answer:

A $-\left(\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}\right)$

B $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

C $\frac{-1+\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

D $\frac{-1-i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q184 | MCQ

C03-K02-G-MCQ-109

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$. Select the correct answer:

A $x^2 - 6x + 13 = 0$ (sign error)

B $x^2 - 6x + 13 = 0$

C $x^2 - 6x + 13 = 1$

D $x^2 - 6x + 13 = 0 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q185 | MCQ

C03-K02-G-MCQ-110

Find two numbers whose sum and product are both 1 Select the correct answer:

A $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

B $\frac{1+\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

C $\frac{1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

D $-(\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2})$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q186 | MCQ

C03-K02-G-MCQ-112

Create a quadratic equation with roots -1 and 2 Select the correct answer:

A $x^2 - x - 2 = 0$

B $x^2 - x - 2 = 1$

C $x^2 - x - 2 = 0 - 1$

D $x^2 - x - 2 = 0$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q187 | MCQ

C03-K02-G-MCQ-113

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$. Select the correct answer:

A $x^2 - 2x + 17 = 1$

B $x^2 - 2x + 17 = 0 - 1$

C $x^2 - 2x + 17 = 0$ (sign error)

D $x^2 - 2x + 17 = 0$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q188 | MCQ

C03-K02-G-MCQ-114

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$. Select the correct answer:

A $3x^2 - 10x + 6 = 0$

B $3x^2 - 10x + 6 = 1$

C $3x^2 - 10x + 6 = 0 - 1$

D $3x^2 - 10x + 6 = 0$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q189 | MCQ

C03-K02-G-MCQ-115

Find two numbers whose sum is 2 and product is 5 Select the correct answer:

A $1 + 2i, 1 - i$

B $1 + 2i, 1 - 2i$

C $1 + 2, 1 - 2i$

D $1 + 2 - i, 1 - 2i$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V01

Q190 | MCQ

C03-K02-G-MCQ-116

Find two numbers whose sum is 4 and product is 2 Select the correct answer:

A $2 + \sqrt{2}, 2 - \sqrt{2}$ (sign error)

B $2 + \sqrt{2}, 2 - \sqrt{2}$

C $2 + \sqrt{2}, 2 - \sqrt{3}$

D $2 + \sqrt{2}, 2 - \sqrt{2} - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V02

Q191 | WRITTEN

C03-T02-310-WU13

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V02

Q192 | MCQ

C03-K02-G-MCQ-108

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$. Select the correct answer:

A $x^2 - 2x - 4 = 0 - 1$

B $x^2 - 2x - 4 = 0$ (sign error)

C $x^2 - 2x - 4 = 0$

D $x^2 - 2x - 4 = 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V03

Q193 | WRITTEN

C03-T02-310-TR13

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V03

Q194 | WRITTEN

C03-T02-310-EX44

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V03

Q195 | MCQ

C03-K02-G-MCQ-111

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

A $x^2 - 7x + 9 = 1$

B $x^2 - 7x + 9 = 0 - 1$

C $x^2 - 7x + 9 = 0$ (sign error)

D $x^2 - 7x + 9 = 0$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V03

Q196 | MCQ

C03-K02-G-MCQ-118

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

A $x^2 + 10x + 49 = 0 - 1$

B $x^2 + 10x + 49 = 0$ (sign error)

C $x^2 + 10x + 49 = 0$

D $x^2 + 10x + 49 = 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V04

Q197 | WRITTEN

C03-T02-310-EX55

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V04

Q198 | MCQ

C03-K02-G-MCQ-119

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$

Select the correct answer:

A $4x^2 + 3x + 2 = 0$

B $4x^2 + 3x + 2 = 1$

C $4x^2 + 3x + 2 = 0 - 1$

D $4x^2 + 3x + 2 = 0$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V05

Q199 | WRITTEN

C03-T02-310-EX33

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V05

Q200 | MCQ

C03-K02-G-MCQ-117

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.
Select the correct answer:

A $x^2 + 8x + 24 = 1$

B $x^2 + 8x + 24 = 0 - 1$

C $x^2 + 8x + 24 = 0$ (sign error)

D $x^2 + 8x + 24 = 0$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q201 | WRITTEN

C03-T02-311-TR11

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q202 | WRITTEN

C03-T02-311-EX11

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q203 | MCQ

C03-K02-G-MCQ-073

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

A $-2 < m < -1$

B $-2 > m > -1$

C $+2 < m < -1$

D $-2 < m < 0$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q204 | MCQ

C03-K02-G-MCQ-075

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m . Select the correct answer:

A $-1 < m < -\frac{1}{2} - 1$

B $-1 < m < -\frac{1}{2}$

C $-(-1 < m < -\frac{1}{2})$

D $-1 < m < -\frac{1}{2} + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q205 | WRITTEN

C03-T02-311-WU11

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q206 | WRITTEN

C03-T02-311-EX13

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q207 | MCQ

C03-K02-G-MCQ-072

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m
Select the correct answer:

A $+6 < m < -2$

B $-6 < m < -1$

C $-6 < m < -2$

D $-6 > m > -2$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q208 | MCQ

C03-K02-G-MCQ-077

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

A $k > 10$ (sign error)B $k > 10$ C $k > 11$ D $k > 10 - 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q209 | WRITTEN

C03-T02-311-TR12

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q210 | WRITTEN

C03-T02-311-EX12

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q211 | MCQ

C03-K02-G-MCQ-074

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m
Select the correct answer:

A $m > -4$

B $m > -3$

C $m > -4 - 1$

D $m > -4$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q212 | MCQ

C03-K02-G-MCQ-076

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

A $k > -2$

B $k > -3 - 1$

C $k > -3$ (sign error)

D $k > -3$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q213 | CHALLENGE WRITTEN

C03-T02-36-CHa

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q214 | CHALLENGE WRITTEN

C03-T02-36-CHb

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q215 | CHALLENGE WRITTEN

C03-T02-36-CHc

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q216 | CHALLENGE WRITTEN

C03-T02-37-CHa

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q217 | CHALLENGE WRITTEN

C03-T02-37-CHb

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q218 | CHALLENGE WRITTEN

C03-T02-37-CHc

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q219 | CHALLENGE WRITTEN

C03-T02-37-CHd

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V05

Q220 | CHALLENGE WRITTEN

C03-T02-38-CH

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q221 | CHALLENGE WRITTEN

C03-T02-39-CH

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

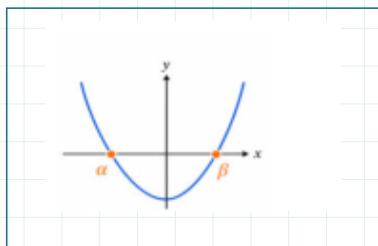
F08 | V04

Q222 | CHALLENGE WRITTEN

C03-T02-310-CH

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.

WORKING AREA



Official source figure

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q223 | CHALLENGE WRITTEN

C03-T02-311-CHa

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q224 | CHALLENGE WRITTEN

C03-T02-311-CHb

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q225 | CHALLENGE WRITTEN

C03-T02-311-CHc

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F99 | V01

Q226 | CHALLENGE WRITTEN

C03-T02-35-CH

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q227 | CHALLENGE MCQ

C03-K02-G-MCQ-045

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

A $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct real

B $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

C $a < -4$ or $a > 0$: repeated real; $a = -4, 0$: repeated real; $-4 < a < 0$: repeated imaginary

D $a < -4$ or $a > 0$: distinct imaginary; $a = -4, 0$: repeated imaginary; $-4 < a < 0$: distinct imaginary

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q228 | CHALLENGE MCQ

C03-K02-G-MCQ-046

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

A $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary - 1

C $-(\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary)

B $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

D $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary + 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | V03

Q229 | CHALLENGE MCQ

C03-K02-G-MCQ-047

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies
Select the correct answer:

A $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct real

B $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

C $a > 3$: repeated real; $a = 3$: repeated real; $a < 3$: repeated imaginary

D $a > 3$: distinct imaginary; $a = 3$: repeated imaginary; $a < 3$: distinct imaginary

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q230 | CHALLENGE MCQ

C03-K02-G-MCQ-102

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

A $\frac{9}{2} - 1$

B $\frac{9}{2}$

C $-(\frac{9}{2})$

D $\frac{9}{2} + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q231 | CHALLENGE MCQ

C03-K02-G-MCQ-103

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

A 4

B $3 - 1$

C 3 (sign error)

D 3

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q232 | CHALLENGE MCQ

C03-K02-G-MCQ-104

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

A $\frac{27}{2} - 1$

B $\frac{27}{2}$

C $-(\frac{27}{2})$

D $\frac{27}{2} + 1$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | V05

Q233 | CHALLENGE MCQ

C03-K02-G-MCQ-105

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

A 9

B 10

C $9 - 1$

D 9 (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F06 | V05

Q234 | CHALLENGE MCQ

C03-K02-G-MCQ-057

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$. Select the correct answer:

A Common root c ; other roots a, b ; equation $x^2 + cx + ab = 0$ (equation = -1).

B Common root c ; other roots a, b ; equation $x^2 + cx + ab = 0$ (sign error).

C Common root c ; other roots a, b ; equation $x^2 + cx + ab = 0$.

D Common root c ; other roots a, b ; equation $x^2 + cx + ab = 0$ (equation = 1).

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F07 | V01

Q235 | CHALLENGE MCQ

C03-K02-G-MCQ-071

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

A $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$

B $x = \frac{\sqrt{6}}{2}$, $x = \frac{i\sqrt{3}}{3}$; real: $\sqrt{6}/2$

C $x = -\frac{\sqrt{6}}{2}$, $x = -\frac{i\sqrt{3}}{3}$; real: $-\sqrt{6}/2$

D $x = +\frac{\sqrt{6}}{2}$, $x = +\frac{i\sqrt{3}}{3}$; real: $+\sqrt{6}/2$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | V04

Q236 | CHALLENGE MCQ

C03-K02-G-MCQ-120

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:

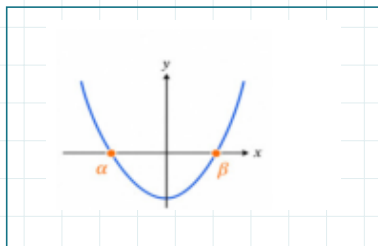
A If $S=\alpha+\beta$ and $P=\alpha\beta \neq 0$, then $Px^2-Sx+1=0$.

B If $S=\alpha+\beta$ and $P=\alpha\beta \neq 0$, then $Px^2-Sx+1=1$.

C If $S=\alpha+\beta$ and $P=\alpha\beta \neq 0$, then $Px^2-Sx+1=0, -1$

D If $S=\alpha+\beta$ and $P=\alpha\beta \neq 0$, then $Px^2-Sx+1=0$; (sign error)

WORKING AREA



Official source figure

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V01

Q237 | CHALLENGE MCQ

C03-K02-G-MCQ-078

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

A $+4 < m < -3$

B $-4 < m < -2$

C $-4 < m < -3$

D $-4 > m > -3$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V02

Q238 | CHALLENGE MCQ

C03-K02-G-MCQ-079

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

A $m > 6$

B $m > 5 - 1$

C $m > 5$ (sign error)

D $m > 5$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F09 | V03

Q239 | CHALLENGE MCQ

C03-K02-G-MCQ-080

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

A $m > -4$

B $m < +4$

C $m < -3$

D $m < -4$

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F99 | V01

Q240 | CHALLENGE MCQ

C03-K02-G-MCQ-027

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

A $x = \frac{-\sqrt{3}i\sqrt{3}}{3}$; no real equilibrium

B $x = \frac{-\sqrt{3} - i\sqrt{3}}{3}$; no real equilibrium

C $x = \frac{-\sqrt{3} + i\sqrt{3}}{3}$; no real equilibrium

D $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F02 | C03-K02-F02-VQ

Quiz M1 | QUIZ MCQ

C03-K02-QUIZ-MCQ-1

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

A 2, 3 (sign error)

B 2, 3

C 2, 4

D 2, 3 - 1

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | C03-K02-F04-VQ

Quiz M2 | QUIZ MCQ

C03-K02-QUIZ-MCQ-2

For $x^2 - 4x + 7 = 0$, determine the nature of the roots. Select the correct answer:

A Two distinct non-real complex roots

B Two repeated non-real complex roots

C Two distinct non-imaginary complex roots

D Two distinct non-real complex roots;(sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | C03-K02-F05-VQ

Quiz M3 | QUIZ MCQ

C03-K02-QUIZ-MCQ-3

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$. Select the correct answer:

A $7 - 1$

B 7 (sign error)

C 7

D 8

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | C03-K02-F08-VQ

Quiz M4 | QUIZ MCQ

C03-K02-QUIZ-MCQ-4

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

A $x^2 - 4x + 5 = 0$

B $x^2 - 4x + 5 = 1$

C $x^2 - 4x + 5 = 0 - 1$

D $x^2 - 4x + 5 = 0$ (sign error)

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$
$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F03 | C03-K02-F03-VQ

Quiz W1 | QUIZ WRITTEN

C03-K02-QUIZ-WRITTEN-1

Solve $2x^2 + 3x - 2 = 0$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$
$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F04 | C03-K02-F04-VQ

Quiz W2 | QUIZ WRITTEN

C03-K02-QUIZ-WRITTEN-2

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F05 | C03-K02-F05-VQ

Quiz W3 | QUIZ WRITTEN

C03-K02-QUIZ-WRITTEN-3

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 2 | Quadratic Equations

F08 | C03-K02-F08-VQ

Quiz W4 | QUIZ WRITTEN

C03-K02-QUIZ-WRITTEN-4

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

WORKING AREA

FORMULA BANK

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Transformed roots

$$u = \alpha + t, v = \beta + t \Rightarrow u + v = S + 2t, uv = P + tS + t^2$$

Discriminant

$$D = b^2 - 4ac$$

Reciprocal roots

$$u = \frac{1}{\alpha}, v = \frac{1}{\beta} \Rightarrow Px^2 - Sx + 1 = 0$$

Vieta sum and product

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Complex factorisation

$$ax^2 + bx + c = a(x - \alpha)(x - \beta)$$

Root expressions

$$\alpha^2 + \beta^2 = S^2 - 2P$$

$$\alpha^3 + \beta^3 = S^3 - 3PS$$

Square-root route

$$(x - h)^2 = k \Rightarrow x = h \pm \sqrt{k}$$

Root signs

$$\alpha\beta > 0 \Rightarrow \text{same sign}$$

$$\alpha\beta < 0 \Rightarrow \text{opposite signs}$$

Concept 3 | Polynomial Theorems

OFFICIAL LESSONS 3-12, 3-13, 3-14, 3-15, 3-16

Polynomial Theorems

Equation-first recall | use the same rail beside every question in this Concept

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \iff P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Polynomial Theorems

F01 | V01

Q001 | WRITTEN

C03-T03-312-EX1

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V01

Q002 | WRITTEN

C03-T03-312-EX2

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V01

Q003 | MCQ

C03-K03-G-MCQ-005

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

A $2 - 1$

B 2 (sign error)

C 2

D 3

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V01

Q004 | MCQ

C03-K03-G-MCQ-006

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

A $6 - 1$

B 6 (sign error)

C 6

D 7

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q005 | WRITTEN

C03-T03-312-WU1

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q006 | WRITTEN

C03-T03-312-TR1

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q007 | WRITTEN

C03-T03-312-EX3

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q008 | WRITTEN

C03-T03-312-EX4

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q009 | MCQ

C03-K03-G-MCQ-001

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

A 9

B $8 - 1$

C 8 (sign error)

D 8

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q010 | MCQ

C03-K03-G-MCQ-003

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:

A $-\frac{1}{8}$

B $-(\frac{1}{8})$

C $-\frac{1}{8} + 1$

D $-\frac{1}{8} - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q011 | MCQ

C03-K03-G-MCQ-007

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

A 1

B $0 - 1$

C 0 (sign error)

D 0

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V02

Q012 | MCQ

C03-K03-G-MCQ-008

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

A 4

B 5

C $4 - 1$

D 4 (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q013 | WRITTEN

C03-T03-312-WU2

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q014 | WRITTEN

C03-T03-312-TR2

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q015 | WRITTEN

C03-T03-312-EX5

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q016 | WRITTEN

C03-T03-312-EX6

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q017 | MCQ

C03-K03-G-MCQ-002

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

A $a = 13/3$ (sign error)B $a = 13/3$ C $a = 13/3 + 1$ D $a = 13/3 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q018 | MCQ

C03-K03-G-MCQ-004

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

A $a = -4$ (sign error)B $a = -4$ C $a = -3$ D $a = -4 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q019 | MCQ

C03-K03-G-MCQ-009

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

A $a = -4 - 1$

B $a = -4$ (sign error)

C $a = -4$

D $a = -3$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V01

Q020 | MCQ

C03-K03-G-MCQ-010

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

A $a = -16$

B $a = -15$

C $a = -16 - 1$

D $a = -16$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V02

Q021 | WRITTEN

C03-T03-312-EX7

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V02

Q022 | MCQ

C03-K03-G-MCQ-011

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

A $a = -4$

B $a = -3$

C $a = -4 - 1$

D $a = -4$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q023 | WRITTEN

C03-T03-313-WU1

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q024 | WRITTEN

C03-T03-313-TR1

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q025 | WRITTEN

C03-T03-313-EX1

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q026 | WRITTEN

C03-T03-313-EX2

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q027 | MCQ

C03-K03-G-MCQ-013

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

A $2x + 1$ B $2x^2$ C $2x + 1 - 1$ D $2x + 1$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q028 | MCQ

C03-K03-G-MCQ-014

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$. Select the correct answer:

A $-10x + 13$ B $-10x14$ C $-10x + 13 - 1$ D $-10x + 13$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q029 | MCQ

C03-K03-G-MCQ-015

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

A $2x + 3$ B $2x^4$ C $2x + 3 - 1$ D $2x + 3$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V01

Q030 | MCQ

C03-K03-G-MCQ-016

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$. Select the correct answer:

A $-15x + 19$ B $-15x20$ C $-15x + 19 - 1$ D $-15x + 19$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q031 | WRITTEN

C03-T03-314-WU1

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q032 | WRITTEN

C03-T03-314-TR1

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q033 | WRITTEN

C03-T03-314-EX1

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q034 | WRITTEN

C03-T03-314-EX2

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q035 | MCQ

C03-K03-G-MCQ-018

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$ Select the correct answer:

A $3x - 2$ (sign error)B $3x - 2$ C $3x - 1$ D $3x - 2 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q036 | MCQ

C03-K03-G-MCQ-019

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$. Select the correct answer:

A 1

B x C x (sign error)

D 0

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q037 | MCQ

C03-K03-G-MCQ-020

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$. Select the correct answer:

A $3x - 1$ B $3x0$ C $3x - 1 - 1$ D $3x - 1$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V01

Q038 | MCQ

C03-K03-G-MCQ-021

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$. Select the correct answer:

A $7x - 11$ B $7x - 12 - 1$ C $7x - 12$ (sign error)D $7x - 12$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V02

Q039 | WRITTEN

C03-T03-314-EX3

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F04 | V02

Q040 | MCQ

C03-K03-G-MCQ-022

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$. Select the correct answer:

A $x^2 + 3x - 4$

B $x^2 + 3x - 3$

C $x^2 + 3x - 4 - 1$

D $x^2 + 3x - 4$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V01

Q041 | WRITTEN

C03-T03-315-WU1

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V01

Q042 | MCQ

C03-K03-G-MCQ-024

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$. Select the correct answer:

A $Q = x^2 - 2x + 9, R = -7$

B $Q = x^2 - 2x + 9, R = -6$

C $Q = x^2 - 2x + 9, R = -7 - 1$

D $Q = x^2 - 2x + 9, R = -7$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q043 | WRITTEN

C03-T03-315-TR1a

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q044 | WRITTEN

C03-T03-315-TR1b

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q045 | WRITTEN

C03-T03-315-EX1a

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q046 | WRITTEN

C03-T03-315-EX1b

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q047 | WRITTEN

C03-T03-315-EX1c

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q048 | MCQ

C03-K03-G-MCQ-025

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$. Select the correct answer:

A $Q = x^2 + x + 6, R = 0$ (sign error)

B $Q = x^2 + x + 6, R = 0$

C $Q = x^2 + x + 6, R = 1$

D $Q = x^2 + x + 6, R = 0 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q049 | MCQ

C03-K03-G-MCQ-026

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$. Select the correct answer:

A $Q = 3x^2 + 4x + 9, R = 27$

B $Q = 3x^2 + 4x + 9, R = 26 - 1$

C $Q = 3x^2 + 4x + 9, R = 26$ (sign error)

D $Q = 3x^2 + 4x + 9, R = 26$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q050 | MCQ

C03-K03-G-MCQ-027

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$. Select the correct answer:

A $Q = x^2 + 2x - 1, R = 5$

B $Q = x^2 + 2x - 1, R = 4 - 1$

C $Q = x^2 + 2x - 1, R = 4$ (sign error)

D $Q = x^2 + 2x - 1, R = 4$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q051 | MCQ

C03-K03-G-MCQ-028

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

A $Q = x^2 + 3x + 2, R = -1$

B $Q = x^2 + 3x + 2, R = 0$

C $Q = x^2 + 3x + 2, R = -1 - 1$

D $Q = x^2 + 3x + 2, R = -1$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V02

Q052 | MCQ

C03-K03-G-MCQ-029

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$. Select the correct answer:

A $Q = 4x + 2, R = 4$ (sign error)

B $Q = 4x + 2, R = 4$

C $Q = 4x + 2, R = 5$

D $Q = 4x + 2, R = 4 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V03

Q053 | WRITTEN

C03-T03-315-EX1d

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V03

Q054 | MCQ

C03-K03-G-MCQ-030

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$. Select the correct answer:

A $Q = 3x^3 - x^2 + 2x - 6, R = 1$

B $Q = 3x^3 - x^2 + 2x - 6, R = 0 - 1$

C $Q = 3x^3 - x^2 + 2x - 6, R = 0$ (sign error)

D $Q = 3x^3 - x^2 + 2x - 6, R = 0$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q055 | WRITTEN

C03-T03-316-WU1a

Factorise $x^3 + 2x^2 - x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q056 | WRITTEN

C03-T03-316-TR1a

Factorise $x^3 - 2x^2 - 5x + 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q057 | WRITTEN

C03-T03-316-TR1b

Factorise $x^3 - x^2 - 16x - 20$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q058 | WRITTEN

C03-T03-316-TR1c

Factorise $2x^3 - x^2 - 5x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q059 | WRITTEN

C03-T03-316-EX1a

Factorise $x^3 - 7x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q060 | WRITTEN

C03-T03-316-EX1b

Factorise $x^3 + 4x^2 + x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q061 | WRITTEN

C03-T03-316-EX1c

Factorise $x^3 + 6x^2 + 11x + 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q062 | WRITTEN

C03-T03-316-EX1d

Factorise $x^3 - 3x + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q063 | WRITTEN

C03-T03-316-EX1e

Factorise $x^3 + x^2 - 8x - 12$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q064 | WRITTEN

C03-T03-316-EX1f

Factorise $x^3 - 3x^2 - 9x - 5$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q065 | WRITTEN

C03-T03-316-EX1g

Factorise $2x^3 - 3x^2 - 3x + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q066 | WRITTEN

C03-T03-316-EX1h

Factorise $3x^3 + x^2 - 8x + 4$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q067 | WRITTEN

C03-T03-316-EX1i

Factorise $2x^3 + 7x^2 + 8x + 3$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q068 | MCQ

C03-K03-G-MCQ-032

Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:

A $(x - 1)(x + 1)(x3)$

B $(x - 1)(x + 1)(x + 2) - 1$

C $(x - 1)(x + 1)(x + 2)$ (sign error)

D $(x - 1)(x + 1)(x + 2)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q069 | MCQ

C03-K03-G-MCQ-034

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:

A $(x - 1)(x - 3)(x3)$

B $(x - 1)(x - 3)(x + 2) - 1$

C $(x - 1)(x - 3)(x + 2)$ (sign error)

D $(x - 1)(x - 3)(x + 2)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q070 | MCQ

C03-K03-G-MCQ-035

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:

A $(x + 2)^2(x - 5) - 1$

B $(x + 2)^2(x - 5)$ (sign error)

C $(x + 2)^2(x - 5)$

D $(x + 2)^2(x - 4)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q071 | MCQ

C03-K03-G-MCQ-036

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:A $(x - 2)(2x + 1)(x + 1)$ (sign error)B $(x - 2)(2x + 1)(x + 1)$ C $(x - 2)(2x + 1)(x2)$ D $(x - 2)(2x + 1)(x + 1) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q072 | MCQ

C03-K03-G-MCQ-040

Factorise $x^3 - 7x - 6$ Select the correct answer:A $(x - 3)(x + 1)(x + 2)$ (sign error)B $(x - 3)(x + 1)(x + 2)$ C $(x - 3)(x + 1)(x3)$ D $(x - 3)(x + 1)(x + 2) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q073 | MCQ

C03-K03-G-MCQ-041

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:

A $(x - 1)(x + 2)(x + 3) - 1$

B $(x - 1)(x + 2)(x + 3)$ (sign error)

C $(x - 1)(x + 2)(x + 3)$

D $(x - 1)(x + 2)(x + 4)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q074 | MCQ

C03-K03-G-MCQ-042

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:A $(x + 1)(x + 2)(x + 3)$ (sign error)B $(x + 1)(x + 2)(x + 3)$ C $(x + 1)(x + 2)(x + 4)$ D $(x + 1)(x + 2)(x + 3) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q075 | MCQ

C03-K03-G-MCQ-043

Factorise $x^3 - 3x + 2$ Select the correct answer:

A $(x - 1)^2(x + 3)$

B $(x - 1)^2(x + 2) - 1$

C $(x - 1)^2(x + 2)$ (sign error)

D $(x - 1)^2(x + 2)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q076 | MCQ

C03-K03-G-MCQ-044

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:

A $(x - 3)(x + 2)^3$

B $(x - 3)(x + 2)^2 - 1$

C $(x - 3)(x + 2)^2$ (sign error)

D $(x - 3)(x + 2)^2$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q077 | MCQ

C03-K03-G-MCQ-045

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:

A $(x - 5)(x + 1)^3$

B $(x - 5)(x + 1)^2 - 1$

C $(x - 5)(x + 1)^2$ (sign error)

D $(x - 5)(x + 1)^2$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q078 | MCQ

C03-K03-G-MCQ-046

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:

A $(x - 2)(x + 1)(2x - 1)$

B $(x - 2)(x + 1)(2x0)$

C $(x - 2)(x + 1)(2x - 1) - 1$

D $(x - 2)(x + 1)(2x - 1)$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q079 | MCQ

C03-K03-G-MCQ-047

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:

A $(x - 1)(x + 2)(3x - 2) - 1$

B $(x - 1)(x + 2)(3x - 2)$ (sign error)

C $(x - 1)(x + 2)(3x - 2)$

D $(x - 1)(x + 2)(3x - 1)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V01

Q080 | MCQ

C03-K03-G-MCQ-048

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:

A $(x + 1)^2(2x + 3)$

B $(x + 1)^2(2x + 3) - 1$

C $(x + 1)^2(2x + 3)$ (sign error)

D $(x + 1)^2(2x + 3)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q081 | WRITTEN

C03-T03-316-WU1b

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q082 | WRITTEN

C03-T03-316-TR1e

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q083 | WRITTEN

C03-T03-316-TR1f

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q084 | WRITTEN

C03-T03-316-EX1m

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q085 | WRITTEN

C03-T03-316-EX1n

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q086 | WRITTEN

C03-T03-316-EX1o

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q087 | WRITTEN

C03-T03-316-EX1p

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q088 | WRITTEN

C03-T03-316-EX1q

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q089 | WRITTEN

C03-T03-316-EX1r

Factorise $x^4 - x^3 + x^2 + x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q090 | MCQ

C03-K03-G-MCQ-033

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:

A $(x - 2)^2(x^2 - 2x - 3)$

B $(x - 2)^2(x^2 - 2x - 4) - 1$

C $(x - 2)^2(x^2 - 2x - 4)$ (sign error)

D $(x - 2)^2(x^2 - 2x - 4)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q091 | MCQ

C03-K03-G-MCQ-038

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:

A $(x - 3)(x - 2)(x + 1)(x + 7)$

B $(x - 3)(x - 2)(x + 1)(x + 8)$

C $(x - 3)(x - 2)(x + 1)(x + 7) - 1$

D $(x - 3)(x - 2)(x + 1)(x + 7)$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q092 | MCQ

C03-K03-G-MCQ-039

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:

A $(x - 1)^2(x^2 - 2x + 3)$

B $(x - 1)^2(x^2 - 2x + 3) - 1$

C $(x - 1)^2(x^2 - 2x + 3)$ (sign error)

D $(x - 1)^2(x^2 - 2x + 3)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q093 | MCQ

C03-K03-G-MCQ-052

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ Select the correct answer:

A $(x - 4)(x - 3)(x - 2)(x - 0)$

B $(x - 4)(x - 3)(x - 2)(x - 1) - 1$

C $(x - 4)(x - 3)(x - 2)(x - 1)$ (sign error)

D $(x - 4)(x - 3)(x - 2)(x - 1)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q094 | MCQ

C03-K03-G-MCQ-053

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ Select the correct answer:

A $(x - 5)(x - 1)^2(x + 2) - 1$

B $(x - 5)(x - 1)^2(x + 2)$ (sign error)

C $(x - 5)(x - 1)^2(x + 2)$

D $(x - 5)(x - 1)^2(x3)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q095 | MCQ

C03-K03-G-MCQ-054

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ Select the correct answer:

A $(x - 4)(x - 2)(x + 2)(2x + 4)$

B $(x - 4)(x - 2)(x + 2)(2x + 3) - 1$

C $(x - 4)(x - 2)(x + 2)(2x + 3)$ (sign error)

D $(x - 4)(x - 2)(x + 2)(2x + 3)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q096 | MCQ

C03-K03-G-MCQ-055

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ Select the correct answer:

A $(x - 1)(x + 2)(x^2 - 5x + 3)$

B $(x - 1)(x + 2)(x^2 - 5x + 4)$

C $(x - 1)(x + 2)(x^2 - 5x + 3) - 1$

D $(x - 1)(x + 2)(x^2 - 5x + 3)$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q097 | MCQ

C03-K03-G-MCQ-056

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:A $(x - 3)(x + 1)(x^2 - 2x + 2)$ (sign error)B $(x - 3)(x + 1)(x^2 - 2x + 2)$ C $(x - 3)(x + 1)(x^2 - 2x3)$ D $(x - 3)(x + 1)(x^2 - 2x + 2) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V02

Q098 | MCQ

C03-K03-G-MCQ-057

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:

A $(x - 1)(x + 1)(x^2 - x + 2) - 1$

B $(x - 1)(x + 1)(x^2 - x + 2)$ (sign error)

C $(x - 1)(x + 1)(x^2 - x + 2)$

D $(x - 1)(x + 1)(x^2 - x3)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q099 | WRITTEN

C03-T03-316-TR1d

Factorise $2x^3 - 3x^2 - x - 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q100 | WRITTEN

C03-T03-316-EX1j

Factorise $2x^3 - 7x^2 + 3x + 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q101 | WRITTEN

C03-T03-316-EX1k

Factorise $x^3 + 4x^2 + 2x - 4$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q102 | WRITTEN

C03-T03-316-EX11

Factorise $x^3 - 3x^2 + 2$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q103 | MCQ

C03-K03-G-MCQ-037

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:

A $(x - 2)(2x^2 + x + 1) - 1$

B $(x - 2)(2x^2 + x + 1)$ (sign error)

C $(x - 2)(2x^2 + x + 1)$

D $(x - 2)(2x^2 + x2)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q104 | MCQ

C03-K03-G-MCQ-049

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:A $(x - 2)(2x^2 - 3x - 3)$ (sign error)B $(x - 2)(2x^2 - 3x - 3)$ C $(x - 2)(2x^2 - 3x - 2)$ D $(x - 2)(2x^2 - 3x - 3) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q105 | MCQ

C03-K03-G-MCQ-050

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:A $(x + 2)(x^2 + 2x - 2)$ (sign error)B $(x + 2)(x^2 + 2x - 2)$ C $(x + 2)(x^2 + 2x - 1)$ D $(x + 2)(x^2 + 2x - 2) - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V03

Q106 | MCQ

C03-K03-G-MCQ-051

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:

A $(x - 1)(x^2 - 2x - 1)$

B $(x - 1)(x^2 - 2x - 2) - 1$

C $(x - 1)(x^2 - 2x - 2)$ (sign error)

D $(x - 1)(x^2 - 2x - 2)$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V04

Q107 | CHALLENGE WRITTEN

C03-T03-314-CH

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V03

Q108 | CHALLENGE WRITTEN

C03-T03-312-CH

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V02

Q109 | CHALLENGE WRITTEN

C03-T03-313-CH

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V04

Q110 | CHALLENGE WRITTEN

C03-T03-315-CH

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V04

Q111 | CHALLENGE WRITTEN

C03-T03-316-CH1

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V04

Q112 | CHALLENGE WRITTEN

C03-T03-316-CH2

For the polynomial above with $a = -12$, $b = 2$, determine all roots of $P(x) = 0$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | V04

Q113 | CHALLENGE MCQ

C03-K03-G-MCQ-023

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

A 56 (sign error)

B 56

C 57

D $56 - 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | V03

Q114 | CHALLENGE MCQ

C03-K03-G-MCQ-012

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

A $a = 10$; for $x + 2$, $a = -10$; linear factors correspond to roots.

B $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (minus 1).

C $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots (sign error).

D $a = 10$; for $x + 2$, $a = -11$; linear factors correspond to roots.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | V02

Q115 | CHALLENGE MCQ

C03-K03-G-MCQ-017

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

A $x - 1 - 1$ B $x - 1$ (sign error)C $x - 1$ D $x0$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | V04

Q116 | CHALLENGE MCQ

C03-K03-G-MCQ-031

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts. Select the correct answer:

A $x = \sqrt{3}, -\sqrt{3}$

B $x = \sqrt{3}, -\sqrt{4}$

C $x = \sqrt{3}, -\sqrt{3} - 1$

D $x = \sqrt{3}, -\sqrt{3}$ (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V04

Q117 | CHALLENGE MCQ

C03-K03-G-MCQ-058

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

A $a = -12, b = 2 - 1$

B $a = -12, b = 2$ (sign error)

C $a = -12, b = 2$

D $a = -12, b = 3$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | V04

Q118 | CHALLENGE MCQ

C03-K03-G-MCQ-059

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

A $x = 1, -2, 2 \pm \sqrt{10}$

B $x = 1, -2, 2\sqrt{10}$

C $x = 1, -2, 2 - \sqrt{10}$

D $x = 1, -2, 2 + \sqrt{10}$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | C03-K03-F01-VQ

Quiz M1 | QUIZ MCQ

C03-K03-QUIZ-MCQ-1

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

A $9 - 1$

B 9 (sign error)

C 9

D 10

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F02 | C03-K03-F02-VQ

Quiz M2 | QUIZ MCQ

C03-K03-QUIZ-MCQ-2

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

A $P(-1) = 0 - 1$

B $P(-1) = 0$ (sign error)

C $P(-1) = 0$

D $P(-1) = 1$

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | C03-K03-F03-VQ

Quiz M3 | QUIZ MCQ

C03-K03-QUIZ-MCQ-3

The remainder on division by a quadratic divisor has the form Select the correct answer:

A 1

B $ax + b$ C $ax + b$ (sign error)

D 0

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F05 | C03-K03-F05-VQ

Quiz M4 | QUIZ MCQ

C03-K03-QUIZ-MCQ-4

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

A 0

B 1

C *Syntheticdivision*D *Syntheticdivision* (sign error)

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F01 | C03-K03-F01-VQ

Quiz W1 | QUIZ WRITTEN

C03-K03-QUIZ-WRITTEN-1

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F03 | C03-K03-F03-VQ

Quiz W2 | QUIZ WRITTEN

C03-K03-QUIZ-WRITTEN-2

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | C03-K03-F06-VQ

Quiz W3 | QUIZ WRITTEN

C03-K03-QUIZ-WRITTEN-3

Factorise $x^3 - 6x^2 + 11x - 6$.

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 3 | Polynomial Theorems

F06 | C03-K03-F06-VQ

Quiz W4 | QUIZ WRITTEN

C03-K03-QUIZ-WRITTEN-4

If $x - 1$ and $x + 2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b .

WORKING AREA

FORMULA BANK

Remainder theorem

$$P(x) \div (x - a) \Rightarrow \text{remainder} = P(a)$$

Factor theorem

$$(x - a) \mid P(x) \Leftrightarrow P(a) = 0$$

Quadratic remainder

$$P(x) \div Q(x), \deg Q = 2 \Rightarrow R = ax + b$$

Two-value remainder

$$R(r_1) = P(r_1), \quad R(r_2) = P(r_2)$$

Synthetic division

$$P(x) = (x - a)Q(x) + P(a)$$

Factorisation route

$$P(a) = 0 \Rightarrow P(x) = (x - a)Q(x)$$

Concept 4 | Higher-Degree Equations

OFFICIAL LESSONS 3-17, 3-18, 3-19, 3-20

Higher-Degree Equations

Equation-first recall | use the same rail beside every question in this Concept

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Higher-Degree Equations

F01 | V01

Q001 | WRITTEN

C03-T04-17-WU1a

$$x^3 - 8 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q002 | WRITTEN

C03-T04-17-TR1a

$$x^3 = -1$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q003 | WRITTEN

C03-T04-17-EX1a

$$x^3 = 27$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q004 | WRITTEN

C03-T04-17-EX1b

$$x^3 = -125$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q005 | WRITTEN

C03-T04-17-EX1c

$$8x^3 = 1$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q006 | MCQ

C03-K04-G-MCQ-001

 $x^3 - 8 = 0$ Select the correct answer:

A $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$

B $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$

C $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$

D $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q007 | MCQ

C03-K04-G-MCQ-005

 $x^3 = -1$ Select the correct answer:

A $x = -1, (1 - \sqrt{3})/2, (1 + \sqrt{3})/2$

B $x = -1, (1 - \sqrt{3} - i)/2, (1 + \sqrt{3} i)/2$

C $x = -1, (1 - \sqrt{3} i)/2, (1 + \sqrt{3} i)/3$

D $x = -1, (1 - \sqrt{3} i)/2, (1 + \sqrt{3} i)/2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q008 | MCQ

C03-K04-G-MCQ-011

 $x^3 = 27$ Select the correct answer:

A $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/3$

B $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$

C $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$

D $x = 3, (-3 - 3\sqrt{3} - i)/2, (-3 + 3\sqrt{3}i)/2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q009 | MCQ

C03-K04-G-MCQ-012

 $x^3 = -125$ Select the correct answer:

A $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$

B $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$

C $x = -5, (5 - 5\sqrt{3} - i)/2, (5 + 5\sqrt{3}i)/2$

D $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/3$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V01

Q010 | MCQ

C03-K04-G-MCQ-013

 $8x^3 = 1$ Select the correct answer:

A $x = 1/2, (-1 - \sqrt{3} - i)/4, (-1 + \sqrt{3}i)/4$

B $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/5$

C $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$

D $x = 1/2, (-1 - \sqrt{3})/4, (-1 + \sqrt{3}i)/4$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q011 | WRITTEN

C03-T04-17-WU1b

$$x^3 - 2x^2 - 5x + 6 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q012 | WRITTEN

C03-T04-17-TR1b

$$3x^3 + x^2 - 8x + 4 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q013 | WRITTEN

C03-T04-17-TR1c

$$2x^3 - 3x^2 - x - 2 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q014 | WRITTEN

C03-T04-17-TR1d

$$x^3 - x^2 - 16x - 20 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q015 | WRITTEN

C03-T04-17-EX1d

$$x^3 + 6x^2 + 11x + 6 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q016 | WRITTEN

C03-T04-17-EX1e

$$x^3 + 3x^2 - 4x - 12 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q017 | WRITTEN

C03-T04-17-EX1f

$$3x^3 - 2x^2 - 3x + 2 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q018 | WRITTEN

C03-T04-17-EX1g

$$x^3 - x^2 - 4 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q019 | WRITTEN

C03-T04-17-EX1h

$$x^3 + 4x^2 + 2x - 4 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q020 | WRITTEN

C03-T04-17-EX1i

$$x^3 - 4x^2 + 9x - 10 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q021 | WRITTEN

C03-T04-17-EX1j

$$x^3 + x^2 - 8x - 12 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q022 | WRITTEN

C03-T04-17-EX1k

$$2x^3 + 7x^2 + 8x + 3 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q023 | WRITTEN

C03-T04-17-EX11

$$x^3 + 6x^2 + 12x + 8 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q024 | MCQ

C03-K04-G-MCQ-002

 $x^3 - 2x^2 - 5x + 6 = 0$ Select the correct answer:

A $x = -2, 1, 3 - 1$

B $x = -2, 1, 3$ (sign error)

C $x = -2, 1, 3$

D $x = -2, 1, 4$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q025 | MCQ

C03-K04-G-MCQ-006

$3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:

A $x = -2, 2/3, 2$

B $x = -2, 2/3, 1 - 1$

C $x = -2, 2/3, 1$ (sign error)

D $x = -2, 2/3, 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q026 | MCQ

C03-K04-G-MCQ-007

 $2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:

A $x = 2, (-1 - \sqrt{7})/4, (-1 + \sqrt{7})/4$

B $x = 2, (-1 - \sqrt{7} - i)/4, (-1 + \sqrt{7} i)/4$

C $x = 2, (-1 - \sqrt{7} i)/4, (-1 + \sqrt{7} i)/5$

D $x = 2, (-1 - \sqrt{7} i)/4, (-1 + \sqrt{7} i)/4$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q027 | MCQ

C03-K04-G-MCQ-008

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:
A $x = -2$ (repeated), 5 (sign error)B $x = -2$ (repeated), 5C $x = -2$ (repeated), 6D $x = -2$ (repeated), 5 - 1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q028 | MCQ

C03-K04-G-MCQ-014

$x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:

A $x = -3, -2, -1 - 1$

B $x = -3, -2, -1$ (sign error)

C $x = -3, -2, -1$

D $x = -3, -2, 0$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q029 | MCQ

C03-K04-G-MCQ-015

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:

A $x = -3, -2, 3$

B $x = -3, -2, 2 - 1$

C $x = -3, -2, 2$ (sign error)

D $x = -3, -2, 2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q030 | MCQ

C03-K04-G-MCQ-016

 $3x^3 - 2x^2 - 3x + 2 = 0$ Select the correct answer:

A $x = -1, 2/3, 2$

B $x = -1, 2/3, 1 - 1$

C $x = -1, 2/3, 1$ (sign error)

D $x = -1, 2/3, 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q031 | MCQ

C03-K04-G-MCQ-017

 $x^3 - x^2 - 4 = 0$ Select the correct answer:

A $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$

B $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$

C $x = 2, (-1 - \sqrt{7} - i)/2, (-1 + \sqrt{7}i)/2$

D $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/3$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q032 | MCQ

C03-K04-G-MCQ-018

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:

A $x = -2, -1 - \sqrt{3}, -1 + \sqrt{4}$

B $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3} - 1$

C $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$ (sign error)

D $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q033 | MCQ

C03-K04-G-MCQ-019

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:

A $x = 2, 1 - 2, 1 + 2i$

B $x = 2, 1 - 2 - i, 1 + 2i$

C $x = 2, 1 - 2i, 1 + 3i$

D $x = 2, 1 - 2i, 1 + 2i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q034 | MCQ

C03-K04-G-MCQ-020

$x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:

A $x = -2$ (repeated), 4B $x = -2$ (repeated), 3 - 1C $x = -2$ (repeated), 3 (sign error)D $x = -2$ (repeated), 3

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q035 | MCQ

C03-K04-G-MCQ-021

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:
A $x = -1$ (repeated), $-3/2$ (sign error)B $x = -1$ (repeated), $-3/2$ C $x = -1$ (repeated), $-3/2 + 1$ D $x = -1$ (repeated), $-3/2 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V02

Q036 | MCQ

C03-K04-G-MCQ-022

 $x^3 + 6x^2 + 12x + 8 = 0$ Select the correct answer:
A $x = -1$ (triple)B $x = -2$ (triple) $- 1$ C $x = -2$ (triple) (sign error)D $x = -2$ (triple)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q037 | WRITTEN

C03-T04-17-WU1c

$$x^4 - 7x^2 - 18 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q038 | WRITTEN

C03-T04-17-TR1e

$$x^4 - 5x^2 + 4 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q039 | WRITTEN

C03-T04-17-EX1m

$$x^4 - 5x^2 - 36 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q040 | WRITTEN

C03-T04-17-EX1n

$$x^4 - 6x^2 - 27 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q041 | WRITTEN

C03-T04-17-EX1o

$$x^4 - x^2 - 2 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q042 | MCQ

C03-K04-G-MCQ-003

 $x^4 - 7x^2 - 18 = 0$ Select the correct answer:

A $x = -3, 3, -\sqrt{2}, \sqrt{2}i$

B $x = -3, 3, -\sqrt{2} - i, \sqrt{2}i$

C $x = -3, 3, -\sqrt{2}i, \sqrt{3}i$

D $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q043 | MCQ

C03-K04-G-MCQ-009

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:

A $x = -2, -1, 1, 3$

B $x = -2, -1, 1, 2 - 1$

C $x = -2, -1, 1, 2$ (sign error)

D $x = -2, -1, 1, 2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q044 | MCQ

C03-K04-G-MCQ-023

 $x^4 - 5x^2 - 36 = 0$ Select the correct answer:

A $x = -3, 3, -2 - i, 2i$

B $x = -3, 3, -2i, 3i$

C $x = -3, 3, -2i, 2i$

D $x = -3, 3, -2, 2i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q045 | MCQ

C03-K04-G-MCQ-024

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:

A $x = -3, 3, -\sqrt{3} - i, \sqrt{3}i$

B $x = -3, 3, -\sqrt{3}i, \sqrt{4}i$

C $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$

D $x = -3, 3, -\sqrt{3}, \sqrt{3}i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V03

Q046 | MCQ

C03-K04-G-MCQ-025

 $x^4 - x^2 - 2 = 0$ Select the correct answer:

A $x = -\sqrt{2}, \sqrt{3}, -i, i$

B $x = -\sqrt{2}, \sqrt{2}, -i, i$

C $x = -\sqrt{2}, \sqrt{2}, -, i$

D $x = -\sqrt{2}, \sqrt{2}, +i, i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q047 | WRITTEN

C03-T04-17-WU1d

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q048 | WRITTEN

C03-T04-17-TR1f

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q049 | WRITTEN

C03-T04-17-EX1p

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q050 | WRITTEN

C03-T04-17-EX1q

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q051 | MCQ

C03-K04-G-MCQ-004

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:

A $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$ (sign error)

B $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

C $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{6}$

D $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5} - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q052 | MCQ

C03-K04-G-MCQ-010

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:
A $x = 1, 2, 3, 4$ B $x = 1, 2, 3, 5$ C $x = 1, 2, 3, 4 - 1$ D $x = 1, 2, 3, 4$ (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q053 | MCQ

C03-K04-G-MCQ-026

 $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ Select the correct answer:
A $x = 1, 2, 3(\text{repeated}) - 1$ B $x = 1, 2, 3(\text{repeated})$ (sign error)C $x = 1, 2, 3(\text{repeated})$ D $x = 1, 2, 4(\text{repeated})$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | V04

Q054 | MCQ

C03-K04-G-MCQ-027

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

A $x = -1, 3, 1 - i, 1 + i$

B $x = -1, 3, 1 - i, 2 + i$

C $x = -1, 3, 1 - i, 1 + i$

D $x = -1, 3, 1 - i, 1 + i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q055 | WRITTEN

C03-T04-18-TR1

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q056 | WRITTEN

C03-T04-18-EX1

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q057 | WRITTEN

C03-T04-18-EX2

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q058 | MCQ

C03-K04-G-MCQ-032

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

A $a = 1, b = -6$; other roots: $-2 - 1$

B $a = 1, b = -6$; other roots: -2 (sign error)

C $a = 1, b = -6$; other roots: -2

D $a = 1, b = -6$; other roots: -1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q059 | MCQ

C03-K04-G-MCQ-034

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a,b and the other root Select the correct answer:

A $a = 2, b = 2$; other roots: 1 (sign error)

B $a = 2, b = 2$; other roots: 1

C $a = 2, b = 2$; other roots: 2

D $a = 2, b = 2$; other roots: 1 - 1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V01

Q060 | MCQ

C03-K04-G-MCQ-035

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a,b and the other root Select the correct answer:

A $a = -7, b = 14$; other roots: $4 - 1$

B $a = -7, b = 14$; other roots: 4 (sign error)

C $a = -7, b = 14$; other roots: 4

D $a = -7, b = 14$; other roots: 5

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q061 | WRITTEN

C03-T04-18-WU1

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q062 | WRITTEN

C03-T04-18-TR2

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q063 | WRITTEN

C03-T04-18-EX4

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q064 | WRITTEN

C03-T04-18-EX5

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q065 | MCQ

C03-K04-G-MCQ-031

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

A $a = -3, b = 1$; other roots: $-1, 2 - i$

B $a = -3, b = 1$; other roots: $-1, 2 - i$

C $a = -3, b = 1$; other roots: $-1, 2 + i$

D $a = -3, b = 1$; other roots: $-1, 3 - i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q066 | MCQ

C03-K04-G-MCQ-033

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

A $a = -4, b = 6$; other roots: $1 - i, 3$

B $a = -4, b = 6$; other roots: $1 - i, 2$

C $a = -4, b = 6$; other roots: $1 - i, 2$

D $a = -4, b = 6$; other roots: $1 + i, 2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q067 | MCQ

C03-K04-G-MCQ-037

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

A $a = -4, b = 14$; other roots: $1 + 3, 2$

B $a = -4, b = 14$; other roots: $1 + 3 - i, 2$

C $a = -4, b = 14$; other roots: $1 + 3i, 3$

D $a = -4, b = 14$; other roots: $1 + 3i, 2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q068 | MCQ

C03-K04-G-MCQ-038

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

A $a = 0, b = 1$; other roots: $1 - 2i, -1$

B $a = 0, b = 1$; other roots: $1 - 2i, -2$

C $a = 0, b = 1$; other roots: $1 - 2, -2$

D $a = 0, b = 1$; other roots: $1 - 2 - i, -2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V03

Q069 | WRITTEN

C03-T04-18-EX3

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V03

Q070 | MCQ

C03-K04-G-MCQ-036

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

A $k = 9$; other roots: 3, 4 (sign error)

B $k = 9$; other roots: 3, 4

C $k = 9$; other roots: 3, 5

D $k = 9$; other roots: 3, 4 - 1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q071 | WRITTEN

C03-T04-19-WU1a

For an imaginary cube root of unity w , evaluate w^9 .

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q072 | WRITTEN

C03-T04-19-TR1a

For an imaginary cube root of unity w , evaluate w^{2010} .

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q073 | WRITTEN

C03-T04-19-EX1a

For an imaginary cube root of unity w , evaluate w^3 .

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q074 | MCQ

C03-K04-G-MCQ-040

For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:

A 1

B 2

C $1 - 1$

D 1 (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q075 | MCQ

C03-K04-G-MCQ-043

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

A 1 (sign error)

B 1

C 2

D $1 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V01

Q076 | MCQ

C03-K04-G-MCQ-046

For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:

A 2

B $1 - 1$

C 1 (sign error)

D 1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q077 | WRITTEN

C03-T04-19-WU1b

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q078 | WRITTEN

C03-T04-19-TR1b

For an imaginary cube root of unity w , evaluate $w^2 + w$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q079 | WRITTEN

C03-T04-19-EX1b

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q080 | MCQ

C03-K04-G-MCQ-041

For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:

A -1 B 0 C $-1 - 1$ D -1 (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q081 | MCQ

C03-K04-G-MCQ-044

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

A -1 (sign error)B -1 C 0 D $-1 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q082 | MCQ

C03-K04-G-MCQ-047

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

A 1

B $0 - 1$

C 0 (sign error)

D 0

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q083 | WRITTEN

C03-T04-19-WU1c

For an imaginary cube root of unity w , evaluate $w + 1/w$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q084 | WRITTEN

C03-T04-19-TR1c

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q085 | WRITTEN

C03-T04-19-EX1c

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q086 | MCQ

C03-K04-G-MCQ-042

For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:

A 0

B $-1 - 1$ C -1 (sign error)D -1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q087 | MCQ

C03-K04-G-MCQ-045

For an imaginary cube root of unity w , evaluate $(1+w)(1+w^2)$ Select the correct answer:

A $1 - 1$ B 1 (sign error)C 1 D 2

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V03

Q088 | MCQ

C03-K04-G-MCQ-048

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

A 1

B $0 - 1$

C 0 (sign error)

D 0

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q089 | WRITTEN

C03-T04-20-WU1

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q090 | WRITTEN

C03-T04-20-TR1

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q091 | WRITTEN

C03-T04-20-EX1

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q092 | WRITTEN

C03-T04-20-EX2

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q093 | WRITTEN

C03-T04-20-EX3

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q094 | MCQ

C03-K04-G-MCQ-050

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root
Select the correct answer:

A $a = -4$ or $9/4 + 1$

B $a = -4$ or $9/4 - 1$

C $a = -4$ or $9/4$ (sign error)

D $a = -4$ or $9/4$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q095 | MCQ

C03-K04-G-MCQ-051

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root
Select the correct answer:

A $a = -3$ or 2 B $a = -3$ or $1 - 1$ C $a = -3$ or 1 (sign error)D $a = -3$ or 1

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q096 | MCQ

C03-K04-G-MCQ-052

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root
Select the correct answer:

A $a = 5$ or 9 (sign error)B $a = 5$ or 9 C $a = 5$ or 10 D $a = 5$ or $9 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q097 | MCQ

C03-K04-G-MCQ-053

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root
Select the correct answer:

A $a = -6, -2$, or 10 B $a = -6, -2$, or 11 C $a = -6, -2$, or $10 - 1$ D $a = -6, -2$, or 10 (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q098 | MCQ

C03-K04-G-MCQ-054

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root

Select the correct answer:

A $a = -7, -3$, or 2

B $a = -7, -3$, or 3

C $a = -7, -3$, or $2 - 1$

D $a = -7, -3$, or 2 (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q099 | CHALLENGE WRITTEN

C03-T04-18-CH1

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q100 | CHALLENGE WRITTEN

C03-T04-19-CH1

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q101 | CHALLENGE WRITTEN

C03-T04-20-CH1

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q102 | CHALLENGE WRITTEN

C03-T04-17-CH1

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q103 | CHALLENGE WRITTEN

C03-T04-17-CH2

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q104 | CHALLENGE WRITTEN

C03-T04-17-CH3

Classify the stable pressure solutions as real or complex.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | V02

Q105 | CHALLENGE MCQ

C03-K04-G-MCQ-039

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

A $a = -1, b = 6$; other roots: $1 - \sqrt{2} - i, -2$

B $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -1$

C $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

D $a = -1, b = 6$; other roots: $1 - \sqrt{2}, -2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | V02

Q106 | CHALLENGE MCQ

C03-K04-G-MCQ-049

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

A 5

B $4 - 1$

C 4 (sign error)

D 4

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | V03

Q107 | CHALLENGE MCQ

C03-K04-G-MCQ-055

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root
Select the correct answer:

A $a = 2/3, 2$, or 7 B $a = 2/3, 2$, or $6 - 1$ C $a = 2/3, 2$, or 6 (sign error)D $a = 2/3, 2$, or 6

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q108 | CHALLENGE MCQ

C03-K04-G-MCQ-028

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

A $x^4 - x^3 + x^2 + x - 2 = 0$

B $x^4 - x^3 + x^2 + x - 2 = 1$

C $x^4 - x^3 + x^2 + x - 2 = 0 - 1$

D $x^4 - x^3 + x^2 + x - 2 = 0$ (sign error)

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q109 | CHALLENGE MCQ

C03-K04-G-MCQ-029

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels. Select the correct answer:

A $x = -1, 1, (1 - \sqrt{7})/2, (1 + \sqrt{7}i)/2$

B $x = -1, 1, (1 - \sqrt{7} - i)/2, (1 + \sqrt{7}i)/2$

C $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/3$

D $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F99 | V01

Q110 | CHALLENGE MCQ

C03-K04-G-MCQ-030

Classify the stable pressure solutions as real or complex Select the correct answer:

A Both types occur : $realx =$ $(1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x = 0$, there are 2 real physical levels.

C Both types occur : $realx =$ $(1 \pm \sqrt{7})/2$. If pressure is restricted to $x = 0$, there are 1 real physical level.

B Both types occur : $realx =$ $(1 \pm \sqrt{7}i)/2$. If pressure is restricted to $x = 0$, there is 1 real physical level.

D Both types occur : $realx =$ $(1 \pm \sqrt{7} - i)/2$. If pressure is restricted to $x = 0$, there are 1 real physical level.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | C03-K04-F01-VQ

Quiz M1 | QUIZ MCQ

C03-K04-QUIZ-MCQ-1

Solve $x^3 - 8 = 0$. Select the correct answer:

A $x = 2, -1 - \sqrt{3}, -1 + \sqrt{3}i$

B $x = 2, -1 - \sqrt{3} - i, -1 + \sqrt{3}i$

C $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{4}i$

D $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | C03-K04-F02-VQ

Quiz M2 | QUIZ MCQ

C03-K04-QUIZ-MCQ-2

A real cubic has root $1 + 2i$. Which root must also occur? Select the correct answer:

A $1 - 2$ B $1 - 2 - i$ C $1 - 1i$ D $1 - 2i$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | C03-K04-F03-VQ

Quiz M3 | QUIZ MCQ

C03-K04-QUIZ-MCQ-3

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

A -1 (sign error)B -1 C 0 D $-1 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | C03-K04-F04-VQ

Quiz M4 | QUIZ MCQ

C03-K04-QUIZ-MCQ-4

For the remaining quadratic factor, a repeated root requires Select the correct answer:

A $D = 0$ (sign error)B $D = 0$ C $D = 1$ D $D = 0 - 1$

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F01 | C03-K04-F01-VQ

Quiz W1 | QUIZ WRITTEN

C03-K04-QUIZ-WRITTEN-1

Solve $x^4 - 5x^2 + 4 = 0$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F02 | C03-K04-F02-VQ

Quiz W2 | QUIZ WRITTEN

C03-K04-QUIZ-WRITTEN-2

For $x^3 + ax^2 + bx + 10 = 0$ with root $1 + 2i$, find a , b , and the other roots.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F03 | C03-K04-F03-VQ

Quiz W3 | QUIZ WRITTEN

C03-K04-QUIZ-WRITTEN-3

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

Concept 4 | Higher-Degree Equations

F04 | C03-K04-F04-VQ

Quiz W4 | QUIZ WRITTEN

C03-K04-QUIZ-WRITTEN-4

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

WORKING AREA

FORMULA BANK

Difference / sum of cubes

$$x^3 - a^3 = (x - a)(x^2 + ax + a^2) \quad x^3 + a^3 = (x + a)(x^2 - ax + a^2)$$

Factor theorem

$$P(a) = 0 \iff (x - a) \mid P(x)$$

Biquadratic substitution

$$x^4 + bx^2 + c = 0 \quad u = x^2$$

Real-coefficient conjugates

$$a + bi \text{ root} \Rightarrow a - bi \text{ root}$$

Cube roots of unity

$$w^3 = 1, w \neq 1 \Rightarrow w^2 + w + 1 = 0$$

Repeated cubic root

$$D = b^2 - 4ac = 0 \quad \text{for the remaining quadratic}$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 04

Geometry and Equations

3 Concepts | 734 synchronized questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 04 | Geometry and Equations

CLASSROOM ROUTE

Concept 1 | Coordinate Geometry Basics**224 questions**

Official lessons: 4-1, 4-2, 4-3, 4-4, 4-5, 4-6

Concept 2 | Triangles and Lines**260 questions**

Official lessons: 4-7, 4-8, 4-9, 4-10, 4-11, 4-12, 4-13, 4-14, 4-15

Concept 3 | Circles**250 questions**

Official lessons: 4-16, 4-17, 4-18, 4-19, 4-20, 4-21, 4-22, 4-23, 4-24, 4-25, 4-26

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 1 | Coordinate Geometry Basics

OFFICIAL LESSONS 4-1, 4-2, 4-3, 4-4, 4-5, 4-6

Coordinate Geometry Basics

Equation-first recall | use the same rail beside every question in this Concept

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q001 | WRITTEN

C04-K01-41-WA-001

A (-4), B (8): distance AB

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q002 | WRITTEN

C04-K01-41-TR-005

A (-5), B (7): distance AB

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q003 | WRITTEN

C04-K01-41-EX-009

For two points A (−2) and B (6), find the distance AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q004 | WRITTEN

C04-K01-41-EX-013

For two points A (4) and B (10), find the distance AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q005 | MCQ

C04-K01-41-WA-001-MCQ

Find a (-4) , B (8) : distance AB:

A 11

B -12

C 12

D 13

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q006 | MCQ

C04-K01-41-TR-005-MCQ

Find a (−5), B (7): distance AB:

A 11

B −12

C 12

D 13

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q007 | MCQ

C04-K01-41-EX-009-MCQ

For two points A (−2) and B (6), find the distance AB:

A 7

B −8

C 8

D 9

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V01

Q008 | MCQ

C04-K01-41-EX-013-MCQ

For two points A (4) and B (10), find the distance AB:

A 5

B -6

C 6

D 7

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q009 | WRITTEN

C04-K01-41-WA-002

A (-4), B (8): internal point C in ratio 2:1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q010 | WRITTEN

C04-K01-41-TR-006

A (-5), B (7): internal point C in ratio 2:1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q011 | WRITTEN

C04-K01-41-EX-010

For two points A (−2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q012 | WRITTEN

C04-K01-41-EX-014

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q013 | MCQ

C04-K01-41-WA-002-MCQ

Find a (−4), B (8): internal point C in ratio 2:1:

A −4

B 4

C 5

D 3

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q014 | MCQ

C04-K01-41-TR-006-MCQ

Find a (−5), B (7): internal point C in ratio 2:1:

A 2

B −3

C 3

D 4

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q015 | MCQ

C04-K01-41-EX-010-MCQ

For two points A (−2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1:

A −4

B 4

C 5

D 3

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V02

Q016 | MCQ

C04-K01-41-EX-014-MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

A -8

B 8

C 9

D 7

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q017 | WRITTEN

C04-K01-41-WA-003

A (-4), B (8): external point D in ratio 2:3

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q018 | WRITTEN

C04-K01-41-TR-007

A (-5), B (7): external point D in ratio 2:5

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q019 | WRITTEN

C04-K01-41-EX-011

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q020 | WRITTEN

C04-K01-41-EX-015

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q021 | MCQ

C04-K01-41-WA-003-MCQ

Find a (−4), B (8): external point D in ratio 2:3:

A −28

B −27

C −29

D 28

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q022 | MCQ

C04-K01-41-TR-007-MCQ

Find a (−5), B (7): external point D in ratio 2:5:

A 13

B −13

C −12

D −14

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q023 | MCQ

C04-K01-41-EX-011-MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

A 18

B 19

C 17

D -18

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V03

Q024 | MCQ

C04-K01-41-EX-015-MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

A -1

B 1

C 2

D 0

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q025 | WRITTEN

C04-K01-41-WA-004

A (-4), B (8): midpoint M

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q026 | WRITTEN

C04-K01-41-TR-008

A (-5), B (7): midpoint M

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q027 | WRITTEN

C04-K01-41-EX-012

For two points A (−2) and B (6), find the coordinate of midpoint M of segment AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q028 | WRITTEN

C04-K01-41-EX-016

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q029 | MCQ

C04-K01-41-WA-004-MCQ

Find a (−4), B (8): midpoint M:

A 1

B −2

C 2

D 3

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q030 | MCQ

C04-K01-41-TR-008-MCQ

Find a (−5), B (7): midpoint M:

A 1

B 2

C 0

D −1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q031 | MCQ

C04-K01-41-EX-012-MCQ

For two points A (−2) and B (6), find the coordinate of midpoint M of segment AB:

A 3

B 1

C −2

D 2

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | V04

Q032 | MCQ

C04-K01-41-EX-016-MCQ

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:

A 7

B 8

C 6

D -7

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q033 | CHALLENGE WRITTEN

C04-K01-41-CH-017

Pods A (−8) and B (−3): fibre-optic tether length

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q034 | CHALLENGE WRITTEN

C04-K01-41-CH-018

Pods A (−8), B (−3): internal sensor point in ratio 1:4

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q035 | CHALLENGE WRITTEN

C04-K01-41-CH-019

Pods A (-8) , B (-3) : external outpost in ratio 3:4

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q036 | CHALLENGE WRITTEN

C04-K01-41-CH-020

Pods A (-8) , B (-3) : midpoint supply hub

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q037 | CHALLENGE MCQ

C04-K01-41-CH-017-MCQ

Find pods A (−8) and B (−3): fibre-optic tether length:

A 5

B 6

C 4

D −5

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q038 | CHALLENGE MCQ

C04-K01-41-CH-018-MCQ

Find pods A (−8), B (−3): internal sensor point in ratio 1:4:

A −8

B 7

C −7

D −6

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q039 | CHALLENGE MCQ

C04-K01-41-CH-019-MCQ

Find pods A (−8), B (−3): external outpost in ratio 3:4:

A 23

B −23

C −22

D −24

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q040 | CHALLENGE MCQ

C04-K01-41-CH-020-MCQ

Find pods A (−8), B (−3): midpoint supply hub:

A $-13/2$ B $11/2$ C $-11/2$ D $-9/2$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q041 | WRITTEN

C04-K01-42-WA-021

Point a has coordinates (2,3): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q042 | WRITTEN

C04-K01-42-WA-022

Point b has coordinates (2,-3): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q043 | WRITTEN

C04-K01-42-WA-023

Point c has coordinates (-2,3): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q044 | WRITTEN

C04-K01-42-WA-024

Point d has coordinates (-2,-3): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q045 | WRITTEN

C04-K01-42-TR-028

Point a has coordinates (-4,2): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q046 | WRITTEN

C04-K01-42-TR-029

Point b has coordinates (1,2): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q047 | WRITTEN

C04-K01-42-TR-030

Point c has coordinates (3,-2): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q048 | WRITTEN

C04-K01-42-TR-031

Point d has coordinates (-3,-1): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q049 | WRITTEN

C04-K01-42-EX-035

Point a has coordinates (-3,1): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q050 | WRITTEN

C04-K01-42-EX-036

Point b has coordinates (2,-8): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q051 | WRITTEN

C04-K01-42-EX-037

Point c has coordinates (4,4): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q052 | WRITTEN

C04-K01-42-EX-038

Point d has coordinates (-6,-4): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q053 | WRITTEN

C04-K01-42-EX-039

Point e has coordinates (-2,6): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q054 | WRITTEN

C04-K01-42-EX-040

Point f has coordinates (3,-3): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q055 | WRITTEN

C04-K01-42-EX-041

Point g has coordinates (-5,-7): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q056 | WRITTEN

C04-K01-42-EX-042

Point h has coordinates (1,4): determine its quadrant

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q057 | MCQ

C04-K01-42-WA-021-MCQ

Point a has coordinates (2,3): determine its quadrant:

A III

B IV

C I

D II

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q058 | MCQ

C04-K01-42-WA-022-MCQ

Point b has coordinates (2,-3): determine its quadrant:

A III

B IV

C I

D II

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q059 | MCQ

C04-K01-42-WA-023-MCQ

Point c has coordinates (-2,3): determine its quadrant:

A II

B I

C III

D IV

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q060 | MCQ

C04-K01-42-WA-024-MCQ

Point d has coordinates (-2,-3): determine its quadrant:

A II

B IV

C III

D I

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q061 | MCQ

C04-K01-42-TR-028-MCQ

Point a has coordinates (-4,2): determine its quadrant:

A II

B I

C III

D IV

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q062 | MCQ

C04-K01-42-TR-029-MCQ

Point b has coordinates (1,2): determine its quadrant:

A *III*B *IV*C *I*D *II*

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q063 | MCQ

C04-K01-42-TR-030-MCQ

Point c has coordinates (3,-2): determine its quadrant:

A III

B IV

C I

D II

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q064 | MCQ

C04-K01-42-TR-031-MCQ

Point d has coordinates (-3,-1): determine its quadrant:

A III

B I

C II

D IV

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q065 | MCQ

C04-K01-42-EX-035-MCQ

Point a has coordinates (-3,1): determine its quadrant:

A *IV*B *II*C *I*D *III*

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q066 | MCQ

C04-K01-42-EX-036-MCQ

Point b has coordinates (2,-8): determine its quadrant:

A *IV*B *I*C *II*D *III*

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q067 | MCQ

C04-K01-42-EX-037-MCQ

Point c has coordinates (4,4): determine its quadrant:

A I

B II

C III

D IV

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q068 | MCQ

C04-K01-42-EX-038-MCQ

Point d has coordinates (-6,-4): determine its quadrant:

A II

B IV

C III

D I

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q069 | MCQ

C04-K01-42-EX-039-MCQ

Point e has coordinates (-2,6): determine its quadrant:

A *IV*B *II*C *I*D *III*

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q070 | MCQ

C04-K01-42-EX-040-MCQ

Point f has coordinates (3,-3): determine its quadrant:

A II

B III

C IV

D I

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q071 | MCQ

C04-K01-42-EX-041-MCQ

Point g has coordinates (-5,-7): determine its quadrant:

A II

B IV

C III

D I

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V01

Q072 | MCQ

C04-K01-42-EX-042-MCQ

Point h has coordinates (1,4): determine its quadrant:

A II

B III

C IV

D I

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q073 | WRITTEN

C04-K01-42-WA-025

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q074 | WRITTEN

C04-K01-42-TR-032

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q075 | WRITTEN

C04-K01-42-EX-043

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q076 | WRITTEN

C04-K01-42-EX-046

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q077 | WRITTEN

C04-K01-42-EX-049

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q078 | MCQ

C04-K01-42-WA-025-MCQ

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

A $(-3, 2)$ B $(3, -2)$ C $(-3, -2)$ D $(-2, -3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q079 | MCQ

C04-K01-42-TR-032-MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

A $(-2, -1)$ B $(-1, 2)$ C $(1, -2)$ D $(-1, -2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q080 | MCQ

C04-K01-42-EX-043-MCQ

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:

A $(-4, -2)$ B $(4, 2)$ C $(-4, 2)$ D $(2, -4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q081 | MCQ

C04-K01-42-EX-046-MCQ

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis:

A (2, 1)

B (1, -2)

C (-1, 2)

D (1, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V02

Q082 | MCQ

C04-K01-42-EX-049-MCQ

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis:

A $(-3, 4)$ B $(3, -4)$ C $(-3, -4)$ D $(-4, -3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q083 | WRITTEN

C04-K01-42-WA-026

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q084 | WRITTEN

C04-K01-42-TR-033

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q085 | WRITTEN

C04-K01-42-EX-044

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q086 | WRITTEN

C04-K01-42-EX-047

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q087 | WRITTEN

C04-K01-42-EX-050

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q088 | MCQ

C04-K01-42-WA-026-MCQ

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q089 | MCQ

C04-K01-42-TR-033-MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

A $(1, -2)$ B $(-1, 2)$ C $(1, 2)$ D $(2, 1)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q090 | MCQ

C04-K01-42-EX-044-MCQ

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

A $(4, -2)$ B $(-2, 4)$ C $(4, 2)$ D $(-4, -2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q091 | MCQ

C04-K01-42-EX-047-MCQ

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis:

A (-1, -2)

B (-2, -1)

C (-1, 2)

D (1, -2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V03

Q092 | MCQ

C04-K01-42-EX-050-MCQ

For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis:

A $(3, 4)$ B $(4, 3)$ C $(3, -4)$ D $(-3, 4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q093 | WRITTEN

C04-K01-42-WA-027

For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q094 | WRITTEN

C04-K01-42-TR-034

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q095 | WRITTEN

C04-K01-42-EX-045

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q096 | WRITTEN

C04-K01-42-EX-048

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q097 | WRITTEN

C04-K01-42-EX-051

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q098 | MCQ

C04-K01-42-WA-027-MCQ

For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:

A $(-3, -2)$ B $(3, -2)$ C $(-2, 3)$ D $(3, 2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q099 | MCQ

C04-K01-42-TR-034-MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

A $(-1, -2)$ B $(1, -2)$ C $(-2, 1)$ D $(1, 2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q100 | MCQ

C04-K01-42-EX-045-MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

A $(-4, 2)$ B $(4, 2)$ C $(2, 4)$ D $(4, -2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q101 | MCQ

C04-K01-42-EX-048-MCQ

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin:

A (-1, 2)

B (2, -1)

C (-1, -2)

D (1, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | V04

Q102 | MCQ

C04-K01-42-EX-051-MCQ

For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin:

A $(-4, 3)$ B $(3, 4)$ C $(-3, -4)$ D $(3, -4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q103 | CHALLENGE WRITTEN

C04-K01-42-CH-052

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q104 | CHALLENGE WRITTEN

C04-K01-42-CH-053

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q105 | CHALLENGE WRITTEN

C04-K01-42-CH-054

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q106 | CHALLENGE MCQ

C04-K01-42-CH-052-MCQ

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

A $(-3, -4)$ B $(-4, -3)$ C $(-3, 4)$ D $(3, -4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q107 | CHALLENGE MCQ

C04-K01-42-CH-053-MCQ

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

A $(-3, 4)$ B $(3, 4)$ C $(4, 3)$ D $(3, -4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V01

Q108 | CHALLENGE MCQ

C04-K01-42-CH-054-MCQ

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

A $(3, 4)$ B $(-3, -4)$ C $(3, -4)$ D $(-4, 3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q109 | WRITTEN

C04-K01-43-WA-055

Distance between A (2, 6) and B (5, 10)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q110 | WRITTEN

C04-K01-43-TR-057

Distance between A (4, 3) and B (1, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q111 | WRITTEN

C04-K01-43-TR-058

Distance between A($\sqrt{3}$, -1) and B (0, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q112 | WRITTEN

C04-K01-43-EX-061

Distance between A (1, 2) and B (11, 7)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q113 | WRITTEN

C04-K01-43-EX-062

Distance between A (3, -1) and B (-1, 3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q114 | MCQ

C04-K01-43-WA-055-MCQ

Find distance between A (2, 6) and B (5, 10):

A 4

B -5

C 5

D 6

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q115 | MCQ

C04-K01-43-TR-057-MCQ

Find distance between A (4, 3) and B (1, 2):

A $\sqrt{10}$

B $2\sqrt{10}$

C (10)

D 1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q116 | MCQ

C04-K01-43-TR-058-MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):

A (7)

B 1

C $\sqrt{7}$ D $2\sqrt{7}$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q117 | MCQ

C04-K01-43-EX-061-MCQ

Find distance between A (1, 2) and B (11, 7):

A 1

B $5\sqrt{5}$ C $52\sqrt{5}$

D 5(5)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V01

Q118 | MCQ

C04-K01-43-EX-062-MCQ

Find distance between A (3, -1) and B (-1, 3):

A $4\sqrt{2}$

B $42\sqrt{2}$

C $4(2)$

D 1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q119 | WRITTEN

C04-K01-43-WA-056

A (3, -2), B (x, 1), distance 5: find x

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q120 | WRITTEN

C04-K01-43-TR-060

A $(-1, 3)$, B $(x, 1)$, distance $\sqrt{5}$: find x

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q121 | WRITTEN

C04-K01-43-EX-064

A (3, -1), B (x, 2), distance 5: find x

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q122 | WRITTEN

C04-K01-43-EX-065

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q123 | WRITTEN

C04-K01-43-EX-066

A $(-4, -2)$, B $(x, 3)$, distance $5\sqrt{5}$: find x

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q124 | MCQ

C04-K01-43-WA-056-MCQ

A (3, -2), B (x, 1), distance 5: find x:

A $x = -1, 7$ (one value only)B $x = -1, 7$ C $x = +1, 7$ D $x = -17$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q125 | MCQ

C04-K01-43-TR-060-MCQ

A $(-1, 3)$, B $(x, 1)$, distance $\sqrt{5}$: find x:

A $x = -20$

B $x = -2, 0$ (one value only)

C $x = -2, 0$

D $x = +2, 0$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q126 | MCQ

C04-K01-43-EX-064-MCQ

A (3, -1), B (x, 2), distance 5: find x:

A $x = -1, 7$ (one value only)B $x = -1, 7$ C $x = +1, 7$ D $x = -17$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q127 | MCQ

C04-K01-43-EX-065-MCQ

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y:

A $y = -5, -1$

B $y = +5, -1$

C $y = -5 - 1$

D $y = -5, -1$ (one value only)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V02

Q128 | MCQ

C04-K01-43-EX-066-MCQ

A $(-4, -2)$, B $(x, 3)$, distance $5\sqrt{5}$: find x:

A $x = +14, 6$

B $x = -146$

C $x = -14, 6$ (one value only)

D $x = -14, 6$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V03

Q129 | WRITTEN

C04-K01-43-TR-059

Distance from origin to A (2, -4)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V03

Q130 | WRITTEN

C04-K01-43-EX-063

Distance from origin to A (−3, 4)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V03

Q131 | MCQ

C04-K01-43-TR-059-MCQ

Find distance from origin to A (2, -4):

A 2(5)

B 1

C $2\sqrt{5}$ D $22\sqrt{5}$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | V03

Q132 | MCQ

C04-K01-43-EX-063-MCQ

Find distance from origin to A (−3, 4):

A 4

B −5

C 5

D 6

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q133 | CHALLENGE WRITTEN

C04-K01-43-CH-067

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q134 | CHALLENGE MCQ

C04-K01-43-CH-067-MCQ

Distance between (x,7) and (3x-1,-5) is 13: find x:

A $x = -23$

B $x = -2, 3$ (one value only)

C $x = -2, 3$

D $x = +2, 3$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q135 | WRITTEN

C04-K01-44-WA-068

Point on the x-axis equidistant from A (−2, 1) and B (3, 4)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q136 | WRITTEN

C04-K01-44-TR-070

Point on the y-axis equidistant from A (4, 2) and B (1, -1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q137 | WRITTEN

C04-K01-44-EX-072

Point on the x-axis equidistant from A (−1, 2) and B (4, 3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q138 | WRITTEN

C04-K01-44-EX-073

Point on the y-axis equidistant from A (−5, 2) and B (3, 5)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q139 | WRITTEN

C04-K01-44-EX-074

Point on the x-axis equidistant from A (−1, 2) and B (3, 8)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q140 | MCQ

C04-K01-44-WA-068-MCQ

Find point on the x-axis equidistant from A (−2, 1) and B (3, 4):

A (−2, 0)

B (−3, 0)

C (2, 0)

D (0, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q141 | MCQ

C04-K01-44-TR-070-MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q142 | MCQ

C04-K01-44-EX-072-MCQ

Find point on the x-axis equidistant from A (−1, 2) and B (4, 3):

A (−2, 0)

B (−3, 0)

C (2, 0)

D (0, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q143 | MCQ

C04-K01-44-EX-073-MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:A $(0, -5/6)$ B $(-1, 5/6)$ C $(0, 5/6)$ D $(5/6, 0)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V01

Q144 | MCQ

C04-K01-44-EX-074-MCQ

Find point on the x-axis equidistant from A (−1, 2) and B (3, 8):

A (−19/2, 0)

B (17/2, 0)

C (0, 17/2)

D (−17/2, 0)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q145 | WRITTEN

C04-K01-44-WA-069

Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q146 | WRITTEN

C04-K01-44-TR-071

Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q147 | WRITTEN

C04-K01-44-EX-075

Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q148 | WRITTEN

C04-K01-44-EX-076

Point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q149 | WRITTEN

C04-K01-44-EX-077

Point on $y = -x + 1$ equidistant from A (1, 2) and B (-3, 6)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q150 | MCQ

C04-K01-44-WA-069-MCQ

Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):A $(-13/8, 9/4)$ B $(13/8, 9/4)$ C $(9/4, 13/8)$ D $(13/8, -9/4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q151 | MCQ

C04-K01-44-TR-071-MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):

A (-4, 11)

B (4, -11)

C (-4, -11)

D (-11, -4)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q152 | MCQ

C04-K01-44-EX-075-MCQ

Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):

A (2, 1)

B (1, 2)

C (2, -1)

D (-2, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q153 | MCQ

C04-K01-44-EX-076-MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:A $(3, -2)$ B $(-3, 2)$ C $(3, 2)$ D $(2, 3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | V02

Q154 | MCQ

C04-K01-44-EX-077-MCQ

Find point on $y = -x + 1$ equidistant from A (1, 2) and B (-3, 6):

A (2, 3)

B (-2, 3)

C (3, -2)

D (-2, -3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q155 | CHALLENGE WRITTEN

C04-K01-44-CH-078

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q156 | CHALLENGE MCQ

C04-K01-44-CH-078-MCQ

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD:

A $3\sqrt{5}$

B $32\sqrt{5}$

C 3(5)

D 1

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q157 | WRITTEN

C04-K01-45-WA-079

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q158 | WRITTEN

C04-K01-45-TR-083

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q159 | WRITTEN

C04-K01-45-EX-087

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q160 | WRITTEN

C04-K01-45-EX-091

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of point P dividing segment BC internally in the ratio 3:2.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q161 | WRITTEN

C04-K01-45-EX-095

Given A (−4, 5), B (1, −5), and C (3, 0), find the coordinates of point P dividing segment BA internally in the ratio 1:2.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q162 | MCQ

C04-K01-45-WA-079-MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

A $(4/3, -11/3)$ B $(-4/3, 11/3)$ C $(4/3, 11/3)$ D $(11/3, 4/3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q163 | MCQ

C04-K01-45-TR-083-MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

A $(-15/4, -13/4)$ B $(15/4, 13/4)$ C $(-15/4, 13/4)$ D $(13/4, -15/4)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q164 | MCQ

C04-K01-45-EX-087-MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

A (2, 9/5)

B (-2, -9/5)

C (2, -9/5)

D (-9/5, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q165 | MCQ

C04-K01-45-EX-091-MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

A $(-3, 3)$ B $(-3, -3)$ C $(3, 3)$ D $(3, -3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V01

Q166 | MCQ

C04-K01-45-EX-095-MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

A $(-2/3, -5/3)$ B $(-5/3, -2/3)$ C $(-2/3, 5/3)$ D $(2/3, -5/3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q167 | WRITTEN

C04-K01-45-WA-080

Given A (−2, −3), B (3, 7), and C (5, 2), find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q168 | WRITTEN

C04-K01-45-TR-084

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q169 | WRITTEN

C04-K01-45-EX-088

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q170 | WRITTEN

C04-K01-45-EX-092

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q171 | WRITTEN

C04-K01-45-EX-096

Given A (−4, 5), B (1, −5), and C (3, 0), find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q172 | MCQ

C04-K01-45-WA-080-MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

A $(-2, 19/2)$ B $(2, 19/2)$ C $(19/2, 2)$ D $(2, -19/2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q173 | MCQ

C04-K01-45-TR-084-MCQ

Given A (−3, 4), B (1, −4), and C (−4, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

A (11, 18)

B (−11, −18)

C (11, −18)

D (−18, 11)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q174 | MCQ

C04-K01-45-EX-088-MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

A (-10, -15)

B (10, -15)

C (-15, 10)

D (10, 15)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q175 | MCQ

C04-K01-45-EX-092-MCQ

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

A (8, −11)

B (−8, −11)

C (−11, −8)

D (−8, 11)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V02

Q176 | MCQ

C04-K01-45-EX-096-MCQ

Given A (−4, 5), B (1, −5), and C (3, 0), find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

A (0, 15/2)

B (−1, −15/2)

C (0, −15/2)

D (−15/2, 0)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q177 | WRITTEN

C04-K01-45-WA-081

Given A (−2, −3), B (3, 7), and C (5, 2), find the coordinates of midpoint M of segment CA.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q178 | WRITTEN

C04-K01-45-WA-082

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q179 | WRITTEN

C04-K01-45-TR-085

Given A (−3, 4), B (1, −4), and C (−4, 3), find the coordinates of midpoint M of segment AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q180 | WRITTEN

C04-K01-45-TR-086

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q181 | WRITTEN

C04-K01-45-EX-089

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q182 | WRITTEN

C04-K01-45-EX-090

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q183 | WRITTEN

C04-K01-45-EX-093

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of midpoint M of segment AB.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q184 | WRITTEN

C04-K01-45-EX-094

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q185 | WRITTEN

C04-K01-45-EX-097

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q186 | WRITTEN

C04-K01-45-EX-098

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q187 | MCQ

C04-K01-45-WA-081-MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

A $(3/2, -1/2)$ B $(-1/2, 3/2)$ C $(3/2, 1/2)$ D $(-3/2, -1/2)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q188 | MCQ

C04-K01-45-WA-082-MCQ

Given A (−2, −3), B (3, 7), and C (5, 2), find the coordinates of centroid G of triangle ABC:

A (−2, 2)

B (−2, −2)

C (2, 2)

D (2, −2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q189 | MCQ

C04-K01-45-TR-085-MCQ

Given A (−3, 4), B (1, −4), and C (−4, 3), find the coordinates of midpoint M of segment AB:

A (0, 0)

B (−1, 0)

C (0, −1)

D (1, 0)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q190 | MCQ

C04-K01-45-TR-086-MCQ

Given A (−3, 4), B (1, −4), and C (−4, 3), find the coordinates of centroid G of triangle ABC:

A (−2, 1)

B (1, −2)

C (−2, −1)

D (2, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q191 | MCQ

C04-K01-45-EX-089-MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

A (0, 3)

B (3, 0)

C (0, -3)

D (-1, 3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q192 | MCQ

C04-K01-45-EX-090-MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

A (1, 2/3)

B (2/3, -1)

C (-2/3, 1)

D (2/3, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q193 | MCQ

C04-K01-45-EX-093-MCQ

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of midpoint M of segment AB:

A (−2, −1)

B (2, −1)

C (−1, 2)

D (2, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q194 | MCQ

C04-K01-45-EX-094-MCQ

Given A (−2, 1), B (6, −3), and C (1, 7), find the coordinates of centroid G of triangle ABC:

A (5/3, 5/3)

B (5/3, −5/3)

C (−5/3, 5/3)

D (−5/3, −5/3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q195 | MCQ

C04-K01-45-EX-097-MCQ

Given A (−4, 5), B (1, −5), and C (3, 0), find the coordinates of midpoint M of segment BC:

A (−2, −5/2)

B (2, −5/2)

C (−5/2, 2)

D (2, 5/2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | V03

Q196 | MCQ

C04-K01-45-EX-098-MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

A 1

B 2

C $(0, 0)$ D $(-1, 0)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q197 | CHALLENGE WRITTEN

C04-K01-45-CH-099

A (-2, 3), B (6, 1), C (0, y), CA=CB: find the centroid G

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q198 | CHALLENGE MCQ

C04-K01-45-CH-099-MCQ

A (-2, 3), B (6, 1), C (0, y), CA=CB: find the centroid G:

A (4/3, 2/3)

B (-4/3, -2/3)

C (4/3, -2/3)

D (-2/3, 4/3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q199 | WRITTEN

C04-K01-46-WA-100

Point with A $(-3, 2)$ symmetric about centre $(2, 6)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q200 | WRITTEN

C04-K01-46-TR-101

Point with A (1, -2) symmetric about centre (2, 3)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q201 | WRITTEN

C04-K01-46-TR-102

Point with A (3, 2) symmetric about centre (-1, 5)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q202 | WRITTEN

C04-K01-46-EX-103

Point with A $(-3, 1)$ symmetric about centre $(2, 0)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q203 | WRITTEN

C04-K01-46-EX-104

Point with A (7, 4) symmetric about centre (-3, 5)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q204 | WRITTEN

C04-K01-46-EX-105

Point with A (4, -8) symmetric about centre (-3, 2)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q205 | WRITTEN

C04-K01-46-EX-106

Point with A (0, -4) symmetric about centre (-4, 1)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q206 | WRITTEN

C04-K01-46-EX-107

Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q207 | MCQ

C04-K01-46-WA-100-MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:A $(7, -10)$ B $(-7, 10)$ C $(7, 10)$ D $(10, 7)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q208 | MCQ

C04-K01-46-TR-101-MCQ

Find point with A (1, -2) symmetric about centre (2, 3):

A (-3, 8)

B (3, 8)

C (8, 3)

D (3, -8)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q209 | MCQ

C04-K01-46-TR-102-MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):

A (-5, 8)

B (8, -5)

C (-5, -8)

D (5, 8)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q210 | MCQ

C04-K01-46-EX-103-MCQ

Find point with A (−3, 1) symmetric about centre (2, 0):

A (7, 1)

B (−7, −1)

C (7, −1)

D (−1, 7)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q211 | MCQ

C04-K01-46-EX-104-MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):

A (-13, -6)

B (13, 6)

C (-13, 6)

D (6, -13)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q212 | MCQ

C04-K01-46-EX-105-MCQ

Find point with A (4, -8) symmetric about centre (-3, 2):

A (-10, -12)

B (10, 12)

C (-10, 12)

D (12, -10)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q213 | MCQ

C04-K01-46-EX-106-MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):

A (8, 6)

B (-8, 6)

C (6, -8)

D (-8, -6)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | V01

Q214 | MCQ

C04-K01-46-EX-107-MCQ

Find point with A $(-4, 2)$ symmetric about centre $(-1, -3)$:A $(2, -8)$ B $(-8, 2)$ C $(2, 8)$ D $(-2, -8)$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q215 | CHALLENGE WRITTEN

C04-K01-46-CH-108

B (m , 5), C (4, n) symmetric about A (1, -1): find $m+n$

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F99 | V02

Q216 | CHALLENGE MCQ

C04-K01-46-CH-108-MCQ

B ($m, 5$), C ($4, n$) symmetric about A ($1, -1$): find $m+n$:A -10 B 9 C -9 D -8

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F02 | C04-K01-F02-VQ

Quiz M1 | QUIZ MCQ

C04-K01-QUIZ-MCQ-1

Point A $(-3, 4)$ lies in which quadrant?:A *II*B *I*C *III*D *IV*

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | C04-K01-F03-VQ

Quiz M2 | QUIZ MCQ

C04-K01-QUIZ-MCQ-2

Find the distance between A (1, 2) and B (4, 6):

A -5

B 5

C 6

D 4

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | C04-K01-F05-VQ

Quiz M3 | QUIZ MCQ

C04-K01-QUIZ-MCQ-3

Point P divides the segment from A (−2, 3) to B (6, 1) internally in ratio 1:3. Find P:

A (0, −5/2)

B (−1, 5/2)

C (0, 5/2)

D (5/2, 0)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F06 | C04-K01-F06-VQ

Quiz M4 | QUIZ MCQ

C04-K01-QUIZ-MCQ-4

Point B is symmetric to A (2, -1) about P (-3, 4). Find B:

A (-8, -9)

B (8, 9)

C (-8, 9)

D (9, -8)

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F04 | C04-K01-F04-VQ

Quiz W1 | QUIZ WRITTEN

C04-K01-QUIZ-WRITTEN-1

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F01 | C04-K01-F01-VQ

Quiz W2 | QUIZ WRITTEN

C04-K01-QUIZ-WRITTEN-2

For points A (−5) and B (7), find the point dividing AB externally in ratio 2:5.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F03 | C04-K01-F03-VQ

Quiz W3 | QUIZ WRITTEN

C04-K01-QUIZ-WRITTEN-3

If the distance between A (3, -1) and B (x, 2) is 5, find x.

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 1 | Coordinate Geometry Basics

F05 | C04-K01-F05-VQ

Quiz W4 | QUIZ WRITTEN

C04-K01-QUIZ-WRITTEN-4

Find the centroid of A (−2, 1), B (6, −3), C (1, 7).

WORKING AREA

FORMULA BANK

Number line

$$AB = |b - a| \quad M = \frac{a + b}{2}$$

Plane division

$$P = \left(\frac{nx_1 + mx_2}{m + n}, \frac{ny_1 + my_2}{m + n} \right)$$

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Centroid

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Symmetry

$$B = 2A - P \quad (\text{centre is the midpoint})$$

Concept 2 | Triangles and Lines

OFFICIAL LESSONS 4-7, 4-8, 4-9, 4-10, 4-11, 4-12, 4-13, 4-14, 4-15

Triangles and Lines

Equation-first recall | use the same rail beside every question in this Concept

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Triangles and Lines

F01 | V01

Q001 | WRITTEN

C04-K02-47-TR-003

Determine the shape of triangle ABC with A (3, 2), B (-1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V01

Q002 | WRITTEN

C04-K02-47-EX-005

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V01

Q003 | MCQ

C04-K02-47-TR-003-MCQ

Determine the shape of triangle ABC with A (3, 2), B (-1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

A *isosceles (not right – angled)*

B *scalene (not right – angled)*

C *scalene right – angled*

D *equilateral*

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V01

Q004 | MCQ

C04-K02-47-EX-005-MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

A *scalene (not right – angled)*

B *scalene right – angled*

C *equilateral*

D *isosceles (not right – angled)*

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V02

Q005 | WRITTEN

C04-K02-47-TR-002

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V02

Q006 | WRITTEN

C04-K02-47-EX-004

Determine the shape of triangle ABC with A (0, 2), B (-1, -2), C (4, 1).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V02

Q007 | MCQ

C04-K02-47-TR-002-MCQ

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4):

A scalene right – angled

B isosceles right – angled; right angle at A

C equilateral

D isosceles (not right – angled)

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V02

Q008 | MCQ

C04-K02-47-EX-004-MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):

A equilateral

B isosceles (not right – angled)

C scalene right – angled

D isosceles right – angled; right angle at A

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V03

Q009 | WRITTEN

C04-K02-47-EX-007

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V03

Q010 | MCQ

C04-K02-47-EX-007-MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):

A *equilateral*

B *isosceles (not right – angled)*

C *scalene (not right – angled)*

D *scalene right – angled; right angle at A*

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q011 | WRITTEN

C04-K02-47-WA-001

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q012 | WRITTEN

C04-K02-47-EX-006

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q013 | WRITTEN

C04-K02-47-EX-008

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q014 | MCQ

C04-K02-47-WA-001-MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):

A scalene right – angled

B isosceles right – angled; right angle at B

C equilateral

D isosceles (not right – angled)

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q015 | MCQ

C04-K02-47-EX-006-MCQ

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):

A *isosceles (not right – angled)*

B *scalene right – angled*

C *isosceles right – angled; right angle at B*

D *equilateral*

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | V04

Q016 | MCQ

C04-K02-47-EX-008-MCQ

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4):

A scalene right – angled

B isosceles right – angled; right angle at C

C equilateral

D isosceles (not right – angled)

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V01

Q017 | CHALLENGE WRITTEN

C04-K02-47-CH-009

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V01

Q018 | CHALLENGE MCQ

C04-K02-47-CH-009-MCQ

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q019 | WRITTEN

C04-K02-48-WA-010

Find the equation of the line through (3,-4) parallel to the x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q020 | WRITTEN

C04-K02-48-WA-011

Find the equation of the line through $(-3, 8)$ parallel to the y-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q021 | WRITTEN

C04-K02-48-TR-016

Find the equation of the line through (2,6) parallel to the x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q022 | WRITTEN

C04-K02-48-TR-017

Find the equation of the line through (3,5) perpendicular to the x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q023 | WRITTEN

C04-K02-48-EX-022

Find the equation of the line through $(-4,5)$ parallel to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q024 | WRITTEN

C04-K02-48-EX-023

Find the equation of the line through (2,1) parallel to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q025 | WRITTEN

C04-K02-48-EX-024

Find the equation of the line through $(-4, -7)$ parallel to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q026 | WRITTEN

C04-K02-48-EX-025

Find the equation of the line through (1,-2) perpendicular to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q027 | WRITTEN

C04-K02-48-EX-026

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q028 | WRITTEN

C04-K02-48-EX-027

Find the equation of the line through (3,-5) perpendicular to x-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q029 | WRITTEN

C04-K02-48-EX-028

Find the equation of the line through (2,-7) parallel to y-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q030 | WRITTEN

C04-K02-48-EX-029

Find the equation of the line through (4,3) parallel to y-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q031 | WRITTEN

C04-K02-48-EX-030

Find the equation of the line through $(-6, -2)$ parallel to y-axis.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q032 | MCQ

C04-K02-48-WA-010-MCQ

Find the equation of the line through (3,-4) parallel to the x-axis:

A $y = 4$

B $y = -4$

C $y = -3$

D $y = -5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q033 | MCQ

C04-K02-48-WA-011-MCQ

Find the equation of the line through $(-3, 8)$ parallel to the y-axis:

A $x = -4$

B $x = 3$

C $x = -3$

D $x = -2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q034 | MCQ

C04-K02-48-TR-016-MCQ

Find the equation of the line through (2,6) parallel to the x-axis:

A $y = 7$

B $y = 5$

C $y = -6$

D $y = 6$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q035 | MCQ

C04-K02-48-TR-017-MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q036 | MCQ

C04-K02-48-EX-022-MCQ

Find the equation of the line through $(-4,5)$ parallel to x-axis:

A $y = -5$

B $y = 5$

C $y = 6$

D $y = 4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q037 | MCQ

C04-K02-48-EX-023-MCQ

Find the equation of the line through (2,1) parallel to x-axis:

A $y = 2$

B $y = 0$

C $y = -1$

D $y = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q038 | MCQ

C04-K02-48-EX-024-MCQ

Find the equation of the line through $(-4, -7)$ parallel to x-axis:

A $y = 7$

B $y = -7$

C $y = -6$

D $y = -8$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q039 | MCQ

C04-K02-48-EX-025-MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:

A $x = 0$

B $x = -1$

C $x = 1$

D $x = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q040 | MCQ

C04-K02-48-EX-026-MCQ

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis:

A $x = -2$

B $x = -4$

C $x = 3$

D $x = -3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q041 | MCQ

C04-K02-48-EX-027-MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:

A $x = 4$

B $x = 2$

C $x = -3$

D $x = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q042 | MCQ

C04-K02-48-EX-028-MCQ

Find the equation of the line through (2,-7) parallel to y-axis:

A $x = -2$

B $x = 2$

C $x = 3$

D $x = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q043 | MCQ

C04-K02-48-EX-029-MCQ

Find the equation of the line through (4,3) parallel to y-axis:

A $x = 5$

B $x = 3$

C $x = -4$

D $x = 4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V01

Q044 | MCQ

C04-K02-48-EX-030-MCQ

Find the equation of the line through $(-6, -2)$ parallel to y-axis:

A $x = -6$

B $x = -5$

C $x = -7$

D $x = 6$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q045 | WRITTEN

C04-K02-48-WA-012

Find the equation of the line through $(-3,1)$ and $(-3,2)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q046 | WRITTEN

C04-K02-48-WA-013

Find the equation of the line through $(-2,3)$ and $(-5,3)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q047 | WRITTEN

C04-K02-48-WA-015

Find the equation of the line through (1,-1) and (4,0).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q048 | WRITTEN

C04-K02-48-TR-018

Find the equation of the line through $(-2,-2)$ and $(2,-2)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q049 | WRITTEN

C04-K02-48-TR-019

Find the equation of the line through $(-4,2)$ and $(-4,8)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q050 | WRITTEN

C04-K02-48-TR-021

Find the equation of the line through $(-2,-6)$ and $(-4,1)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q051 | WRITTEN

C04-K02-48-EX-031

Find the equation of the line through (4,0) and (4,3).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q052 | WRITTEN

C04-K02-48-EX-032

Find the equation of the line through $(-1,2)$ and $(-1,6)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q053 | WRITTEN

C04-K02-48-EX-033

Find the equation of the line through $(-5, -3)$ and $(-5, 2)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q054 | WRITTEN

C04-K02-48-EX-034

Find the equation of the line through $(-2,3)$ and $(-3,3)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q055 | WRITTEN

C04-K02-48-EX-035

Find the equation of the line through (4,-2) and (1,-2).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q056 | WRITTEN

C04-K02-48-EX-036

Find the equation of the line through (-3,6) and (2,6).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q057 | WRITTEN

C04-K02-48-EX-040

Find the equation of the line through (1,2) and (3,10).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q058 | WRITTEN

C04-K02-48-EX-041

Find the equation of the line through $(-1, -8)$ and $(4, 2)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q059 | MCQ

C04-K02-48-WA-012-MCQ

Find the equation of the line through $(-3,1)$ and $(-3,2)$:

A $x = 3$

B $x = -3$

C $x = -2$

D $x = -4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q060 | MCQ

C04-K02-48-WA-013-MCQ

Find the equation of the line through $(-2,3)$ and $(-5,3)$:

A $y = 2$

B $y = -3$

C $y = 3$

D $y = 4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q061 | MCQ

C04-K02-48-WA-015-MCQ

Find the equation of the line through (1,-1) and (4,0):

A $y = -x/3 - 4/3$

B $y = 4/3 - x/3$

C $y = x/3 - 4/3 + 1$

D $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q062 | MCQ

C04-K02-48-TR-018-MCQ

Find the equation of the line through $(-2, -2)$ and $(2, -2)$:

A $y = -2$

B $y = -1$

C $y = -3$

D $y = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q063 | MCQ

C04-K02-48-TR-019-MCQ

Find the equation of the line through $(-4,2)$ and $(-4,8)$:

A $x = 4$

B $x = -4$

C $x = -3$

D $x = -5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q064 | MCQ

C04-K02-48-TR-021-MCQ

Find the equation of the line through $(-2, -6)$ and $(-4, 1)$:

A $7x + 2y + 26 + 1 = 0$

B $7x - 2y + 26 = 0$

C $2(7x + 2y + 26) + 1 = 0$

D $7x + 2y + 26 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q065 | MCQ

C04-K02-48-EX-031-MCQ

Find the equation of the line through (4,0) and (4,3):

A $x = 3$

B $x = -4$

C $x = 4$

D $x = 5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q066 | MCQ

C04-K02-48-EX-032-MCQ

Find the equation of the line through (-1,2) and (-1,6):

A $x = -1$

B $x = 0$

C $x = -2$

D $x = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q067 | MCQ

C04-K02-48-EX-033-MCQ

Find the equation of the line through $(-5, -3)$ and $(-5, 2)$:

A $x = -6$

B $x = 5$

C $x = -5$

D $x = -4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q068 | MCQ

C04-K02-48-EX-034-MCQ

Find the equation of the line through (-2,3) and (-3,3):

A $y = 4$

B $y = 2$

C $y = -3$

D $y = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q069 | MCQ

C04-K02-48-EX-035-MCQ

Find the equation of the line through (4,-2) and (1,-2):

A $y = -1$

B $y = -3$

C $y = 2$

D $y = -2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q070 | MCQ

C04-K02-48-EX-036-MCQ

Find the equation of the line through (-3,6) and (2,6):

A $y = 5$

B $y = -6$

C $y = 6$

D $y = 7$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q071 | MCQ

C04-K02-48-EX-040-MCQ

Find the equation of the line through (1,2) and (3,10):

A $y = -4x - 2$

B $y = 4x - 2 + 1$

C 0

D $y = 4x - 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V02

Q072 | MCQ

C04-K02-48-EX-041-MCQ

Find the equation of the line through $(-1, -8)$ and $(4, 2)$:

A $y = -2x - 6$

B $y = 2x - 6 + 1$

C 0

D $y = 2x - 6$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q073 | WRITTEN

C04-K02-48-WA-014

Find the equation of the line through (2,-1) with gradient 3.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q074 | WRITTEN

C04-K02-48-TR-020

Find the equation of the line through $(-3,4)$ with gradient 2.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q075 | WRITTEN

C04-K02-48-EX-037

Find the equation of the line through (-1,5) with gradient -1.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q076 | WRITTEN

C04-K02-48-EX-038

Find the equation of the line through $(-3, -8)$ with gradient 1.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q077 | WRITTEN

C04-K02-48-EX-039

Find the equation of the line through (2,-1) with gradient -2.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q078 | MCQ

C04-K02-48-WA-014-MCQ

Find the equation of the line through (2,-1) with gradient 3:

A $y = -3x - 7$

B $y = 3x - 7 + 1$

C 0

D $y = 3x - 7$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q079 | MCQ

C04-K02-48-TR-020-MCQ

Find the equation of the line through $(-3,4)$ with gradient 2:

A $y = 2x + 10 + 1$

B 0

C $y = 2x + 10$

D $y = -2x + 10$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q080 | MCQ

C04-K02-48-EX-037-MCQ

Find the equation of the line through $(-1, 5)$ with gradient -1 :

A $y = x + 4$

B $y = x - 4$

C $y = -x + 4 + 1$

D $y = -x + 4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q081 | MCQ

C04-K02-48-EX-038-MCQ

Find the equation of the line through $(-3, -8)$ with gradient 1:

A $y = x - 5 + 1$

B $y = x - 5$

C $y = -x - 5$

D $y = 5 - x$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | V03

Q082 | MCQ

C04-K02-48-EX-039-MCQ

Find the equation of the line through (2,-1) with gradient -2:

A $y = 2x + 3$

B $y = -2x + 3 + 1$

C 0

D $y = -2x + 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V02

Q083 | CHALLENGE WRITTEN

C04-K02-48-CH-042

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V02

Q084 | CHALLENGE MCQ

C04-K02-48-CH-042-MCQ

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

A $y = 3$

B $y = 1$

C $y = -2$

D $y = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q085 | WRITTEN

C04-K02-49-WA-043

A (2, 3), B (4, -1), and C ($a - 1$, $2 - a$) are collinear. Find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q086 | WRITTEN

C04-K02-49-TR-045

A (2, 5), B (4, 9), and C (-1, a) are collinear. Find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q087 | WRITTEN

C04-K02-49-EX-047

A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q088 | WRITTEN

C04-K02-49-EX-048

A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q089 | WRITTEN

C04-K02-49-EX-049

A $(-3, -9)$, B $(5, 7)$, and C $(a + 3, 3a - 2)$ are collinear. Find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q090 | MCQ

C04-K02-49-WA-043-MCQ

A (2, 3), B (4, -1), and C ($a - 1$, $2 - a$) are collinear. Find a:

A $a = 8$

B $a = 6$

C $a = -7$

D $a = 7$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q091 | MCQ

C04-K02-49-TR-045-MCQ

A (2, 5), B (4, 9), and C (-1, a) are collinear. Find a:

A $a = 0$ B $a = -2$ C $a = 1$ D $a = -1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q092 | MCQ

C04-K02-49-EX-047-MCQ

A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:

A $a = 0$ B $a = -1$ C $a = 1$ D $a = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q093 | MCQ

C04-K02-49-EX-048-MCQ

A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a:

A $a = 8$ B $a = 6$ C $a = -7$ D $a = 7$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V01

Q094 | MCQ

C04-K02-49-EX-049-MCQ

A $(-3, -9)$, B $(5, 7)$, and C $(a + 3, 3a - 2)$ are collinear. Find a:

A $a = 6$ B $a = 4$ C $a = -5$ D $a = 5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q095 | WRITTEN

C04-K02-49-WA-044

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q096 | WRITTEN

C04-K02-49-TR-046

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through $(5,6)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q097 | WRITTEN

C04-K02-49-EX-050

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q098 | WRITTEN

C04-K02-49-EX-051

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q099 | MCQ

C04-K02-49-WA-044-MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1, 3)$:

A $2(x + y - 2) + 1 = 0$

B $x + y - 2 = 0$

C $x + y - 2 + 1 = 0$

D $x - y - 2 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q100 | MCQ

C04-K02-49-TR-046-MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

A $x + 3y - 23 = 0$

B $x + 3y - 23 + 1 = 0$

C $x - 3y - 23 = 0$

D $2(x + 3y - 23) + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q101 | MCQ

C04-K02-49-EX-050-MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$:

A $2(x + y - 1) + 1 = 0$

B $x + y - 1 = 0$

C $x + y - 1 + 1 = 0$

D $x - y - 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F03 | V02

Q102 | MCQ

C04-K02-49-EX-051-MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$:

A $2x - 5y + 1 + 1 = 0$

B $2x - 5y - 1 = 0$

C $2(2x - 5y + 1) + 1 = 0$

D $2x - 5y + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V02

Q103 | CHALLENGE WRITTEN

C04-K02-49-CH-052

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V02

Q104 | CHALLENGE MCQ

C04-K02-49-CH-052-MCQ

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

A $3x - 13y = 0$

B $2(3x + 13y) + 1 = 0$

C $3x + 13y = 0$

D $3x + 13y + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q105 | WRITTEN

C04-K02-410-WA-053

Find the gradient of line $y = 2x + 3$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q106 | WRITTEN

C04-K02-410-WA-054

Find the gradient of line $y = -3x + 2$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q107 | WRITTEN

C04-K02-410-WA-055

Find the gradient of line $x - 3y + 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q108 | WRITTEN

C04-K02-410-WA-056

Find the gradient of line $2x - y - 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q109 | WRITTEN

C04-K02-410-WA-057

Find the gradient of line $x + 3y + 3 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q110 | WRITTEN

C04-K02-410-TR-060

Find the gradient of line $y = 3x + 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q111 | WRITTEN

C04-K02-410-TR-061

Find the gradient of line $y = 4x - 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q112 | WRITTEN

C04-K02-410-TR-062

Find the gradient of line $y = -4x - 7$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q113 | WRITTEN

C04-K02-410-TR-063

Find the gradient of line $6x - 2y = 5$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q114 | WRITTEN

C04-K02-410-TR-064

Find the gradient of line $8x + 2y - 3 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q115 | WRITTEN

C04-K02-410-TR-065

Find the gradient of line $x + 4y - 4 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q116 | WRITTEN

C04-K02-410-EX-068

Find the gradient of line $y = 3x - 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q117 | WRITTEN

C04-K02-410-EX-069

Find the gradient of line $y = -3x + 4$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q118 | WRITTEN

C04-K02-410-EX-070

Find the gradient of line $y = x - 3$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q119 | WRITTEN

C04-K02-410-EX-071

Find the gradient of line $6x + 2y + 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q120 | WRITTEN

C04-K02-410-EX-072

Find the gradient of line $2x - 2y - 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q121 | WRITTEN

C04-K02-410-EX-073

Find the gradient of line $x + 3y - 2 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q122 | WRITTEN

C04-K02-410-EX-074

Find the gradient of line $y = 2x - 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q123 | WRITTEN

C04-K02-410-EX-075

Find the gradient of line $y = -x + 4$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q124 | WRITTEN

C04-K02-410-EX-076

Find the gradient of line $y = 3x/2 + 6$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q125 | WRITTEN

C04-K02-410-EX-077

Find the gradient of line $x + y + 3 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q126 | WRITTEN

C04-K02-410-EX-078

Find the gradient of line $2x + 3y + 8 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q127 | WRITTEN

C04-K02-410-EX-079

Find the gradient of line $3x + 6y - 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q128 | MCQ

C04-K02-410-WA-053-MCQ

Find the gradient of line $y = 2x + 3$:

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q129 | MCQ

C04-K02-410-WA-054-MCQ

Find the gradient of line $y = -3x + 2$:

A $m = -4$

B $m = 3$

C $m = -3$

D $m = -2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q130 | MCQ

C04-K02-410-WA-055-MCQ

Find the gradient of line $x - 3y + 1 = 0$:

A $m = 1/3$

B $m = 4/3$

C $m = -2/3$

D $m = -1/3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q131 | MCQ

C04-K02-410-WA-056-MCQ

Find the gradient of line $2x - y - 1 = 0$:

A $m = -2$

B $m = 2$

C $m = 3$

D $m = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q132 | MCQ

C04-K02-410-WA-057-MCQ

Find the gradient of line $x + 3y + 3 = 0$:

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q133 | MCQ

C04-K02-410-TR-060-MCQ

Find the gradient of line $y = 3x + 1$:

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q134 | MCQ

C04-K02-410-TR-061-MCQ

Find the gradient of line $y = 4x - 1$:

A $m = -4$

B $m = 4$

C $m = 5$

D $m = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q135 | MCQ

C04-K02-410-TR-062-MCQ

Find the gradient of line $y = -4x - 7$:

A $m = -3$

B $m = -5$

C $m = 4$

D $m = -4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q136 | MCQ

C04-K02-410-TR-063-MCQ

Find the gradient of line $6x - 2y = 5$:

A $m = 3$

B $m = 4$

C $m = 2$

D $m = -3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q137 | MCQ

C04-K02-410-TR-064-MCQ

Find the gradient of line $8x + 2y - 3 = 0$:

A $m = -5$

B $m = 4$

C $m = -4$

D $m = -3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q138 | MCQ

C04-K02-410-TR-065-MCQ

Find the gradient of line $x + 4y - 4 = 0$:

A $m = -5/4$

B $m = 1/4$

C $m = -1/4$

D $m = 3/4$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q139 | MCQ

C04-K02-410-EX-068-MCQ

Find the gradient of line $y = 3x - 1$:

A $m = 4$

B $m = 2$

C $m = -3$

D $m = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q140 | MCQ

C04-K02-410-EX-069-MCQ

Find the gradient of line $y = -3x + 4$:

A $m = -3$

B $m = -2$

C $m = -4$

D $m = 3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q141 | MCQ

C04-K02-410-EX-070-MCQ

Find the gradient of line $y = x - 3$:

A $m = 2$

B $m = 0$

C $m = -1$

D $m = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q142 | MCQ

C04-K02-410-EX-071-MCQ

Find the gradient of line $6x + 2y + 5 = 0$:

A $m = -4$

B $m = 3$

C $m = -3$

D $m = -2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q143 | MCQ

C04-K02-410-EX-072-MCQ

Find the gradient of line $2x - 2y - 1 = 0$:

A $m = 0$

B $m = -1$

C $m = 1$

D $m = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q144 | MCQ

C04-K02-410-EX-073-MCQ

Find the gradient of line $x + 3y - 2 = 0$:

A $m = -1/3$

B $m = 2/3$

C $m = -4/3$

D $m = 1/3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q145 | MCQ

C04-K02-410-EX-074-MCQ

Find the gradient of line $y = 2x - 1$:

A $m = 3$

B $m = 1$

C $m = -2$

D $m = 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q146 | MCQ

C04-K02-410-EX-075-MCQ

Find the gradient of line $y = -x + 4$:

A $m = -1$

B $m = 0$

C $m = -2$

D $m = 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q147 | MCQ

C04-K02-410-EX-076-MCQ

Find the gradient of line $y = 3x/2 + 6$:

A $m = -3/2$

B $m = 3/2$

C $m = 5/2$

D $m = 1/2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q148 | MCQ

C04-K02-410-EX-077-MCQ

Find the gradient of line $x + y + 3 = 0$:

A $m = 0$

B $m = -2$

C $m = 1$

D $m = -1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q149 | MCQ

C04-K02-410-EX-078-MCQ

Find the gradient of line $2x + 3y + 8 = 0$:

A $m = 2/3$

B $m = -2/3$

C $m = 1/3$

D $m = -5/3$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V01

Q150 | MCQ

C04-K02-410-EX-079-MCQ

Find the gradient of line $3x + 6y - 1 = 0$:

A 0

B $m = -1/2$ C $m = 1/2$ D $m = -3/2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q151 | WRITTEN

C04-K02-410-WA-058

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q152 | WRITTEN

C04-K02-410-TR-066

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q153 | WRITTEN

C04-K02-410-EX-080

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q154 | WRITTEN

C04-K02-410-EX-081

Find the equation of the line through $(-1, 2)$ parallel to $2x + 3y = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q155 | WRITTEN

C04-K02-410-EX-082

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q156 | MCQ

C04-K02-410-WA-058-MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:

A $3x + 2y - 14 = 0$

B $2(3x - 2y - 14) + 1 = 0$

C $3x - 2y - 14 = 0$

D $3x - 2y - 14 + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q157 | MCQ

C04-K02-410-TR-066-MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:

A $2x + y + 1 + 1 = 0$

B $2x - y + 1 = 0$

C $2(2x + y + 1) + 1 = 0$

D $2x + y + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q158 | MCQ

C04-K02-410-EX-080-MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:

A $2x + y - 4 = 0$

B $2(2x - y - 4) + 1 = 0$

C $2x - y - 4 = 0$

D $2x - y - 4 + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q159 | MCQ

C04-K02-410-EX-081-MCQ

Find the equation of the line through $(-1, 2)$ parallel to $2x + 3y = 0$:

A $2x + 3y - 4 + 1 = 0$

B $2x - 3y - 4 = 0$

C $2(2x + 3y - 4) + 1 = 0$

D $2x + 3y - 4 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V02

Q160 | MCQ

C04-K02-410-EX-082-MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

A $3x + 2y - 7 + 1 = 0$

B $3x - 2y - 7 = 0$

C $2(3x + 2y - 7) + 1 = 0$

D $3x + 2y - 7 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q161 | WRITTEN

C04-K02-410-WA-059

Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q162 | WRITTEN

C04-K02-410-TR-067

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q163 | WRITTEN

C04-K02-410-EX-083

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q164 | WRITTEN

C04-K02-410-EX-084

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q165 | WRITTEN

C04-K02-410-EX-085

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q166 | MCQ

C04-K02-410-WA-059-MCQ

Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$:

A $5x - 3y + 25 + 1 = 0$

B $5x - 3y - 25 = 0$

C $2(5x - 3y + 25) + 1 = 0$

D $5x - 3y + 25 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q167 | MCQ

C04-K02-410-TR-067-MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:

A $3x - 2y - 4 + 1 = 0$

B $3x + 2y - 4 = 0$

C $2(3x - 2y - 4) + 1 = 0$

D $3x - 2y - 4 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q168 | MCQ

C04-K02-410-EX-083-MCQ

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:

A $2(x - 2y - 12) + 1 = 0$

B $x - 2y - 12 = 0$

C $x - 2y - 12 + 1 = 0$

D $x + 2y - 12 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q169 | MCQ

C04-K02-410-EX-084-MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:

A $2x + 5y - 13 = 0$

B $2x + 5y - 13 + 1 = 0$

C $2x - 5y - 13 = 0$

D $2(2x + 5y - 13) + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | V03

Q170 | MCQ

C04-K02-410-EX-085-MCQ

Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$:

A $2x + 3y + 19 + 1 = 0$

B $2x - 3y + 19 = 0$

C $2(2x + 3y + 19) + 1 = 0$

D $2x + 3y + 19 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V03

Q171 | CHALLENGE WRITTEN

C04-K02-410-CH-086

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V03

Q172 | CHALLENGE MCQ

C04-K02-410-CH-086-MCQ

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

A $m/k = 1 (k = 2, m = 2)$

B $m/k = 3 (k = 2, m = 6)$

C $m/k = -2 (k = 1, m = -2)$

D $m/k = 2 (k = 3, m = 6)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q173 | WRITTEN

C04-K02-411-WA-087

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q174 | WRITTEN

C04-K02-411-TR-089

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q175 | WRITTEN

C04-K02-411-EX-091

Find the perpendicular bisector of segment A (−2, 3), B (2, 1).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q176 | WRITTEN

C04-K02-411-EX-092

Find the perpendicular bisector of segment A (−1, 2), B (5, −2).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q177 | WRITTEN

C04-K02-411-EX-093

Find the perpendicular bisector of segment A (0, 6), B (4, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q178 | MCQ

C04-K02-411-WA-087-MCQ

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:

A $x + 3y - 18 = 0$

B $x + 3y - 18 + 1 = 0$

C $x - 3y - 18 = 0$

D $2(x + 3y - 18) + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q179 | MCQ

C04-K02-411-TR-089-MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:

A $2x + 3y - 5 = 0$

B $2(2x - 3y - 5) + 1 = 0$

C $2x - 3y - 5 = 0$

D $2x - 3y - 5 + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q180 | MCQ

C04-K02-411-EX-091-MCQ

Find the perpendicular bisector of segment A (−2, 3), B (2, 1):

A $2x - y - 2 = 0$

B $2(2x - y + 2) + 1 = 0$

C $2x - y + 2 = 0$

D $2x - y + 2 + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q181 | MCQ

C04-K02-411-EX-092-MCQ

Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$:

A $3x - 2y - 6 + 1 = 0$

B $3x + 2y - 6 = 0$

C $2(3x - 2y - 6) + 1 = 0$

D $3x - 2y - 6 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V01

Q182 | MCQ

C04-K02-411-EX-093-MCQ

Find the perpendicular bisector of segment A (0, 6), B (4, 4):

A $2x - y + 1 = 0$

B $2x - y + 1 + 1 = 0$

C $2x - y - 1 = 0$

D $2(2x - y + 1) + 1 = 0$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q183 | WRITTEN

C04-K02-411-WA-088

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q184 | WRITTEN

C04-K02-411-TR-090

Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q185 | WRITTEN

C04-K02-411-EX-094

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q186 | WRITTEN

C04-K02-411-EX-095

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q187 | MCQ

C04-K02-411-WA-088-MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:

A $B = (2, 1)$

B $B = (1, 2)$

C $B = (-2, 1)$

D $B = (2, -1)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q188 | MCQ

C04-K02-411-TR-090-MCQ

Find the point B symmetric to A (0, 4) with respect to $2x - y - 1 = 0$:

A $B = (4, 2)$

B $B = (2, 4)$

C $B = (-4, 2)$

D $B = (4, -2)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q189 | MCQ

C04-K02-411-EX-094-MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:

A $B = (-3, 6)$

B $B = (3, -6)$

C $B = (3, 6)$

D $B = (6, 3)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | V02

Q190 | MCQ

C04-K02-411-EX-095-MCQ

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:

A $B = (2, 2)$

B $B = (-2, -2)$

C $B = (-2, 2)$

D $B = (2, -2)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V04

Q191 | CHALLENGE WRITTEN

C04-K02-411-CH-096

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V04

Q192 | CHALLENGE MCQ

C04-K02-411-CH-096-MCQ

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

A $B = (3, 4)$

B $B = (-3, -4)$

C $B = (-3, 4)$

D $B = (4, -3)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q193 | WRITTEN

C04-K02-412-WA-097

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q194 | WRITTEN

C04-K02-412-TR-099

Find the distance from (3,-2) to $x + 2y - 4 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q195 | WRITTEN

C04-K02-412-EX-101

Find the perpendicular distance from $(-3, 2)$ to $2x - 3y + 6 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q196 | WRITTEN

C04-K02-412-EX-102

Find the perpendicular distance from $(-3, 7)$ to $12x - 5y = 7$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q197 | WRITTEN

C04-K02-412-EX-103

Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q198 | WRITTEN

C04-K02-412-EX-104

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q199 | WRITTEN

C04-K02-412-EX-105

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q200 | WRITTEN

C04-K02-412-EX-106

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q201 | WRITTEN

C04-K02-412-EX-107

Find the perpendicular distance from (1,2) to $y = 3x + 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q202 | MCQ

C04-K02-412-WA-097-MCQ

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$:

A $3/\sqrt{13} = 3\sqrt{13}/13$

B $3 \cdot \sqrt{13}/13 + 1$

C $-1 + 3 \cdot \sqrt{13}/13$

D $-3 \cdot \sqrt{13}/13$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q203 | MCQ

C04-K02-412-TR-099-MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:

A $1 + \sqrt{5}$

B $-1 + \sqrt{5}$

C $-\sqrt{5}$

D $\sqrt{5}$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q204 | MCQ

C04-K02-412-EX-101-MCQ

Find the perpendicular distance from $(-3, 2)$ to $2x - 3y + 6 = 0$:

A $1 + 6 \cdot \sqrt{13}/13$

B $-1 + 6 \cdot \sqrt{13}/13$

C $-6 \cdot \sqrt{13}/13$

D $6/\sqrt{13} = 6\sqrt{13}/13$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q205 | MCQ

C04-K02-412-EX-102-MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:

A -6

B 6

C 7

D 5

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q206 | MCQ

C04-K02-412-EX-103-MCQ

Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:

A $4\sqrt{5} + 1$

B $4\sqrt{5} - 1$

C 0

D $4\sqrt{5}$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q207 | MCQ

C04-K02-412-EX-104-MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:

A 2

B 0

C -1

D 1

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q208 | MCQ

C04-K02-412-EX-105-MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:

A $\sqrt{13}$

B $1 + \sqrt{13}$

C $-1 + \sqrt{13}$

D $-\sqrt{13}$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q209 | MCQ

C04-K02-412-EX-106-MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:

A -10

B 10

C 11

D 9

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V01

Q210 | MCQ

C04-K02-412-EX-107-MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:

A $\sqrt{10}/5 + 1$

B $-1 + \sqrt{10}/5$

C $-\sqrt{10}/5$

D $\sqrt{10}/5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V02

Q211 | WRITTEN

C04-K02-412-WA-098

Find the perpendicular length from the origin to $3x - y = 5$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V02

Q212 | WRITTEN

C04-K02-412-TR-100

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V02

Q213 | MCQ

C04-K02-412-WA-098-MCQ

Find the perpendicular length from the origin to $3x - y = 5$:

A $1 + \sqrt{10}/2$

B $-1 + \sqrt{10}/2$

C $-\sqrt{10}/2$

D $\sqrt{10}/2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | V02

Q214 | MCQ

C04-K02-412-TR-100-MCQ

Find the perpendicular length from $(-1, 5)$ to $y = 3x - 2$:

A $-\sqrt{10}$

B $\sqrt{10}$

C $1 + \sqrt{10}$

D $-1 + \sqrt{10}$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V05

Q215 | CHALLENGE WRITTEN

C04-K02-412-CH-108

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V05

Q216 | CHALLENGE MCQ

C04-K02-412-CH-108-MCQ

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

A $\sqrt{10}$

B $1 + \sqrt{10}$

C $-1 + \sqrt{10}$

D $-\sqrt{10}$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q217 | WRITTEN

C04-K02-413-WA-109

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q218 | WRITTEN

C04-K02-413-TR-110

Find the area for A (1, 5), B (-3, -3), C (3, -6).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q219 | WRITTEN

C04-K02-413-TR-111

Find the area for A (−2, 1), B (2, −1), C (0, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q220 | WRITTEN

C04-K02-413-EX-112

Find the area for A (0, -6), B (2, -2), C (1, 5).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q221 | WRITTEN

C04-K02-413-EX-113

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q222 | WRITTEN

C04-K02-413-EX-114

Find the area for A (2, 5), B (0, 1), C (4, 2).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q223 | MCQ

C04-K02-413-WA-109-MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (-3, -1):

A 11

B 9

C -10

D 10

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q224 | MCQ

C04-K02-413-TR-110-MCQ

Find the area for A (1, 5), B (-3, -3), C (3, -6):

A 29

B -30

C 30

D 31

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q225 | MCQ

C04-K02-413-TR-111-MCQ

Find the area for A (−2, 1), B (2, −1), C (0, 4):

A 8

B 9

C 7

D −8

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q226 | MCQ

C04-K02-413-EX-112-MCQ

Find the area for A (0, -6), B (2, -2), C (1, 5):

A -9

B 9

C 10

D 8

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q227 | MCQ

C04-K02-413-EX-113-MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:A -34 B 34 C 35 D 33

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | V01

Q228 | MCQ

C04-K02-413-EX-114-MCQ

Find the area for A (2, 5), B (0, 1), C (4, 2):

A 8

B 6

C -7

D 7

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V06

Q229 | CHALLENGE WRITTEN

C04-K02-413-CH-115

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V06

Q230 | CHALLENGE MCQ

C04-K02-413-CH-115-MCQ

Find the area of the wound triangle A (−2, 1), B (1, 5), C (−21, −21):

A 6

B 4

C −5

D 5

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q231 | WRITTEN

C04-K02-414-WA-116

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q232 | WRITTEN

C04-K02-414-TR-117

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q233 | WRITTEN

C04-K02-414-EX-118

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q234 | WRITTEN

C04-K02-414-EX-119

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q235 | WRITTEN

C04-K02-414-EX-120

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q236 | MCQ

C04-K02-414-WA-116-MCQ

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

A $(-3/2, 1/2)$ B $(-1/2, -3/2)$ C $(1/2, 3/2)$ D $(1/2, -3/2)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q237 | MCQ

C04-K02-414-TR-117-MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

A $(-2, -3)$ B $(-2, 3)$ C $(3, -2)$ D $(2, 3)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q238 | MCQ

C04-K02-414-EX-118-MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

A $(-3, 3)$ B $(-3, -3)$ C $(3, 3)$ D $(3, -3)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q239 | MCQ

C04-K02-414-EX-119-MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

A $(-1, 1)$ B $(1, -1)$ C 0 D $(1, 1)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | V01

Q240 | MCQ

C04-K02-414-EX-120-MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

A $(-3, 1)$ B $(1, -3)$ C $(3, 1)$ D $(-3, -1)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V07

Q241 | CHALLENGE WRITTEN

C04-K02-414-CH-121

The line $-(k-2)x + (k-3)y = -3k + 5$ passes through a fixed point for every k . Find it.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V07

Q242 | CHALLENGE MCQ

C04-K02-414-CH-121-MCQ

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

A (4, 1)

B (1, 4)

C (-4, 1)

D (4, -1)

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q243 | WRITTEN

C04-K02-415-WA-122

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q244 | WRITTEN

C04-K02-415-TR-123

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q245 | WRITTEN

C04-K02-415-EX-124

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q246 | WRITTEN

C04-K02-415-EX-125

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q247 | MCQ

C04-K02-415-WA-122-MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

A $a = -4, -3, \text{ or } 6$ B $a = -3, -3, \text{ or } 6$ C $a = -4, -2, \text{ or } 6$ D $a = -4, -3, \text{ or } 7$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q248 | MCQ

C04-K02-415-TR-123-MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

A $a = -1, -1/2, \text{ or } 1$

B $a = 0, -1/2, \text{ or } 1$

C $a = -1, 1/2, \text{ or } 1$

D $a = -1, -1/2, \text{ or } 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q249 | MCQ

C04-K02-415-EX-124-MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a :

A $a = 11/10, 5/2, \text{ or } -9/2$

B $a = 11/10, 3/2, \text{ or } -7/2$

C $a = 11/10, 3/2, \text{ or } -9/2$

D $a = 21/10, 3/2, \text{ or } -9/2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | V01

Q250 | MCQ

C04-K02-415-EX-125-MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a :

A $a = 2, -3/2, \text{ or } 2$ B $a = 1, -1/2, \text{ or } 2$ C $a = 1, -3/2, \text{ or } 3$ D $a = 1, -3/2, \text{ or } 2$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V08

Q251 | CHALLENGE WRITTEN

C04-K02-415-CH-126

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F99 | V08

Q252 | CHALLENGE MCQ

C04-K02-415-CH-126-MCQ

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

A 15

B 16

C 14

D -15

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F01 | C04-K02-F01-VQ

Quiz M1 | QUIZ MCQ

C04-K02-QUIZ-MCQ-1

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

A isosceles (not right – angled)

B scalene (not right – angled)

C scalene right – angled

D equilateral

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F02 | C04-K02-F02-VQ

Quiz M2 | QUIZ MCQ

C04-K02-QUIZ-MCQ-2

Find the equation of the line through (2,-1) with gradient 3:

A 0

B $y = 3x - 7$ C $y = -3x - 7$ D $y = 3x - 7 + 1$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F06 | C04-K02-F06-VQ

Quiz M3 | QUIZ MCQ

C04-K02-QUIZ-MCQ-3

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:A $6/5$ B $-4/5$ C $-1/5$ D $1/5$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F08 | C04-K02-F08-VQ

Quiz M4 | QUIZ MCQ

C04-K02-QUIZ-MCQ-4

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

A $(-3, -3)$ B $(3, 3)$ C $(3, -3)$ D $(-3, 3)$

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F07 | C04-K02-F07-VQ

Quiz W1 | QUIZ WRITTEN

C04-K02-QUIZ-WRITTEN-1

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F05 | C04-K02-F05-VQ

Quiz W2 | QUIZ WRITTEN

C04-K02-QUIZ-WRITTEN-2

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F04 | C04-K02-F04-VQ

Quiz W3 | QUIZ WRITTEN

C04-K02-QUIZ-WRITTEN-3

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1 k + A_0)x + (B_1 k + B_0)y + C_1 k + C_0 = 0$$

Concept 2 | Triangles and Lines

F09 | C04-K02-F09-VQ

Quiz W4 | QUIZ WRITTEN

C04-K02-QUIZ-WRITTEN-4

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

WORKING AREA

FORMULA BANK

Triangle type

$$a^2 + b^2 = c^2 \quad (c \text{ longest})$$

Line through two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

Parallel / perpendicular

$$m_1 = m_2 \quad m_1 m_2 = -1$$

Point-to-line distance

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Triangle area

$$A = \frac{1}{2} \text{ base} \times \text{height}$$

Fixed point

$$(A_1k + A_0)x + (B_1k + B_0)y + C_1k + C_0 = 0$$

Concept 3 | Circles

OFFICIAL LESSONS 4-16, 4-17, 4-18, 4-19, 4-20, 4-21, 4-22, 4-23, 4-24, 4-25, 4-26

Circles

Equation-first recall | use the same rail beside every question in this Concept

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2 \quad x^2 + y^2 + 2gx + 2fy + c = 0$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2 \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Tangency and real circle

$$r = d(\text{centre, axis}) \quad r^2 > 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points} \quad \Delta = 0 : \text{tangent} \quad \Delta < 0 : \text{none}$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2} \quad x x_1 + y y_1 = r^2$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Radical axis

$$S_1 - S_2 = 0 \quad S_1 = 0, S_2 = 0$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Circles

F01 | V01

Q001 | WRITTEN

C04-K03-416-WA-001

Find the equation of the circle with centre (3,-2) and radius 5.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q002 | WRITTEN

C04-K03-416-TR-003

Find the equation of the circle with centre $(-3,1)$ and radius 2.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q003 | WRITTEN

C04-K03-416-TR-004

Find the equation of the circle with centre the origin and radius 4.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q004 | WRITTEN

C04-K03-416-EX-006

Find the equation of the circle with centre (3,4), radius 6.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q005 | WRITTEN

C04-K03-416-EX-007

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q006 | WRITTEN

C04-K03-416-EX-008

Find the equation of the circle with centre the origin, radius 5.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q007 | MCQ

C04-K03-416-WA-001-MCQ

Find the equation of the circle with centre (3,-2) and radius 5:

A $(x - 3)^2 + (y + 2)^2 = 25$

B $(x + 3)^2 + (y + 2)^2 = 25$

C $(x - 3)^2 + (y + 2)^2 = 25 + 1$

D 4

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q008 | MCQ

C04-K03-416-TR-003-MCQ

Find the equation of the circle with centre $(-3,1)$ and radius 2:

A $(x + 3)^2 + (y - 1)^2 = 4$

B $(x + 3)^2 + (y + 1)^2 = 4$

C $(x + 3)^2 + (y - 1)^2 = 4 + 1$

D 6

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q009 | MCQ

C04-K03-416-TR-004-MCQ

Find the equation of the circle with centre the origin and radius 4:

A $x^2 + y^2 = 16 + 1$

B 6

C 7

D $x^2 + y^2 = 16$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q010 | MCQ

C04-K03-416-EX-006-MCQ

Find the equation of the circle with centre (3,4), radius 6:

A 9

B $(x - 3)^2 + (y - 4)^2 = 36$

C $(x + 3)^2 + (y - 4)^2 = 36$

D $(x - 3)^2 + (y - 4)^2 = 36 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q011 | MCQ

C04-K03-416-EX-007-MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:

A 10

B $(x+2)^2 + (y-6)^2 = 12$

C $(x+2)^2 + (y+6)^2 = 12$

D $(x+2)^2 + (y-6)^2 = 12 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V01

Q012 | MCQ

C04-K03-416-EX-008-MCQ

Find the equation of the circle with centre the origin, radius 5:

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 10

D 11

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q013 | WRITTEN

C04-K03-416-WA-002

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q014 | WRITTEN

C04-K03-416-TR-005

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q015 | WRITTEN

C04-K03-416-EX-009

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q016 | WRITTEN

C04-K03-416-EX-010

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q017 | WRITTEN

C04-K03-416-EX-011

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q018 | WRITTEN

C04-K03-416-EX-012

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q019 | MCQ

C04-K03-416-WA-002-MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:A circle; centre $(3, -1)$, radius 2

B 3

C 4

D 5

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q020 | MCQ

C04-K03-416-TR-005-MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:

A 7

B 8

C circle; centre (2, 3), radius 3

D 6

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q021 | MCQ

C04-K03-416-EX-009-MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:

A 10

B 11

C 12

D circle; centre $(2, -4)$, radius 4

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q022 | MCQ

C04-K03-416-EX-010-MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:

A 12

B 13

C circle; centre (2, 0), radius 2

D 11

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q023 | MCQ

C04-K03-416-EX-011-MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:

A 13

B 14

C circle; centre $(-1, 2)$, radius 5D circle; centre $(-1, 2)$, radius 10

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | V02

Q024 | MCQ

C04-K03-416-EX-012-MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:

A 14

B 15

C circle; centre $(-2, 3)$, radius $\sqrt{13}$

D 13

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V01

Q025 | CHALLENGE WRITTEN

C04-K03-416-CH-013

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V01

Q026 | CHALLENGE MCQ

C04-K03-416-CH-013-MCQ

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

A 0

C centre (5, 3), $\text{radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13 + 1$

B centre (5, 3), $\text{radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13$

D centre (5, 3), $\text{radius}^2 = 13; (x-5)^2 + (y-3)^2 = 13 - 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q027 | WRITTEN

C04-K03-417-WA-014

Find the equation of the circle with centre (2,3) passing through (4,6).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q028 | WRITTEN

C04-K03-417-TR-017

Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q029 | WRITTEN

C04-K03-417-EX-020

Find the equation of the circle with centre (1,-3) through (-3,0).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q030 | WRITTEN

C04-K03-417-EX-021

Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q031 | WRITTEN

C04-K03-417-EX-022

Find the equation of the circle with centre (2,-1) through the origin.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q032 | WRITTEN

C04-K03-417-EX-023

Find the equation of the circle with centre the origin through (4,-3).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q033 | MCQ

C04-K03-417-WA-014-MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):

A $(x - 2)^2 + (y - 3)^2 = 13 + 1$

B 4

C $(x - 2)^2 + (y - 3)^2 = 13$

D $(x + 2)^2 + (y - 3)^2 = 13$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q034 | MCQ

C04-K03-417-TR-017-MCQ

Find the equation of the circle with centre $(-3,1)$ passing through $(4,-2)$:

A $(x + 3)^2 + (y - 1)^2 = 58$

B $(x + 3)^2 + (y + 1)^2 = 58$

C $(x + 3)^2 + (y - 1)^2 = 58 + 1$

D 7

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q035 | MCQ

C04-K03-417-EX-020-MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):

A $(x + 1)^2 + (y + 3)^2 = 25$

B $(x - 1)^2 + (y + 3)^2 = 25 + 1$

C 10

D $(x - 1)^2 + (y + 3)^2 = 25$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q036 | MCQ

C04-K03-417-EX-021-MCQ

Find the equation of the circle with centre $(-4,1)$ through $(3,-2)$:

A $(x + 4)^2 + (y - 1)^2 = 58$

B $(x + 4)^2 + (y + 1)^2 = 58$

C $(x + 4)^2 + (y - 1)^2 = 58 + 1$

D 11

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q037 | MCQ

C04-K03-417-EX-022-MCQ

Find the equation of the circle with centre (2,-1) through the origin:

A 12

B $(x - 2)^2 + (y + 1)^2 = 5$ C $(x + 2)^2 + (y + 1)^2 = 5$ D $(x - 2)^2 + (y + 1)^2 = 5 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V01

Q038 | MCQ

C04-K03-417-EX-023-MCQ

Find the equation of the circle with centre the origin through (4,-3):

A $x^2 + y^2 = 25$

B $x^2 + y^2 = 25 + 1$

C 12

D 13

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q039 | WRITTEN

C04-K03-417-WA-015

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q040 | WRITTEN

C04-K03-417-TR-018

Find the equation of the circle with diameter endpoints (3,4), (5,-2).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q041 | WRITTEN

C04-K03-417-EX-024

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q042 | WRITTEN

C04-K03-417-EX-025

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q043 | WRITTEN

C04-K03-417-EX-026

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q044 | WRITTEN

C04-K03-417-EX-027

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q045 | MCQ

C04-K03-417-WA-015-MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):

A $(x - 2)^2 + (y + 1)^2 = 17$

B $(x + 2)^2 + (y + 1)^2 = 17$

C $(x - 2)^2 + (y + 1)^2 = 17 + 1$

D 5

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q046 | MCQ

C04-K03-417-TR-018-MCQ

Find the equation of the circle with diameter endpoints (3,4), (5,-2):

A $(x - 4)^2 + (y - 1)^2 = 10 + 1$

B 8

C $(x - 4)^2 + (y - 1)^2 = 10$

D $(x + 4)^2 + (y - 1)^2 = 10$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q047 | MCQ

C04-K03-417-EX-024-MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):

A $(x - 1)^2 + (y - 3)^2 = 25 + 1$

B 14

C $(x - 1)^2 + (y - 3)^2 = 25$

D $(x + 1)^2 + (y - 3)^2 = 25$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q048 | MCQ

C04-K03-417-EX-025-MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):

A $(x + 1)^2 + (y + 2)^2 = 25$

B $(x + 1)^2 + (y + 2)^2 = 25 + 1$

C 14

D 15

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q049 | MCQ

C04-K03-417-EX-026-MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:

A 16

B $(x+2)^2 + (y-1)^2 = 2$ C $(x+2)^2 + (y+1)^2 = 2$ D $(x+2)^2 + (y-1)^2 = 2+1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V02

Q050 | MCQ

C04-K03-417-EX-027-MCQ

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:

A $x^2 + (y - 2)^2 = 25 + 1$

B 17

C $x^2 + (y - 2)^2 = 25$

D $x^2 + (y + 2)^2 = 25$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q051 | WRITTEN

C04-K03-417-WA-016

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q052 | WRITTEN

C04-K03-417-TR-019

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q053 | WRITTEN

C04-K03-417-EX-028

Find the equation of the circle with through $(0,0)$, $(-1,1)$, $(2,4)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q054 | WRITTEN

C04-K03-417-EX-029

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q055 | WRITTEN

C04-K03-417-EX-030

Find the equation of the circle with through (1,1), (5,-1), (-3,-7).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q056 | MCQ

C04-K03-417-WA-016-MCQ

Find the equation of the circle passing through (1,1), (2,-1), and (3,2):

A $x^2 + y^2 - 5x - y + 4 = 0$

B $x^2 + y^2 + 5x - y + 4 = 0$

C $x^2 + y^2 - 5x - y + 4 = 0 + 1$

D 6

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q057 | MCQ

C04-K03-417-TR-019-MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):

A 9

B $x^2 + y^2 - 3x - 7y + 6 = 0$ C $x^2 + y^2 + 3x - 7y + 6 = 0$ D $x^2 + y^2 - 3x - 7y + 6 = 0 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q058 | MCQ

C04-K03-417-EX-028-MCQ

Find the equation of the circle with through (0,0), (-1,1), (2,4):

A $x^2 + y^2 + 2x - 4y = 0$

B $x^2 + y^2 - 2x - 4y = 0 + 1$

C 18

D $x^2 + y^2 - 2x - 4y = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q059 | MCQ

C04-K03-417-EX-029-MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:

A $x^2 + y^2 + 6y - 16 = 0$

B $x^2 + y^2 - 6y - 16 = 0 + 1$

C 19

D $x^2 + y^2 - 6y - 16 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F02 | V03

Q060 | MCQ

C04-K03-417-EX-030-MCQ

Find the equation of the circle with through (1,1), (5,-1), (-3,-7):

A $x^2 + y^2 + 2x + 8y - 8 = 0$

B $x^2 + y^2 - 2x + 8y - 8 = 0 + 1$

C 20

D $x^2 + y^2 - 2x + 8y - 8 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V01

Q061 | CHALLENGE WRITTEN

C04-K03-417-CH-031

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V01

Q062 | CHALLENGE MCQ

C04-K03-417-CH-031-MCQ

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

A $x^2 + \frac{y^2}{2} - 6x - 6y + 8 = 0$; centre (3, 3),
radius $\sqrt{10}$

B $x^2 + \frac{y^2}{2} - 6x - 6y + 8 = 0$; centre (3, 3),
radius $\sqrt{10} + 1$

C $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3),
radius $\sqrt{10} - 1$

D 0

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q063 | WRITTEN

C04-K03-418-WA-032

Find the circle with centre (4,-2) tangent to the x-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$x x_1 + y y_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q064 | WRITTEN

C04-K03-418-TR-035

Find the circle with centre $(-2,3)$ tangent to the y-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q065 | WRITTEN

C04-K03-418-EX-038

Find the circle with centre $(-1,4)$ tangent to x-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q066 | WRITTEN

C04-K03-418-EX-039

Find the circle with centre (3,-4) tangent to x-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q067 | WRITTEN

C04-K03-418-EX-040

Find the circle with centre (5,-4) tangent to x-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q068 | WRITTEN

C04-K03-418-EX-041

Find the circle with centre $(-1,6)$ tangent to y-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q069 | WRITTEN

C04-K03-418-EX-042

Find the circle with centre (2,-5) tangent to y-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q070 | WRITTEN

C04-K03-418-EX-043

Find the circle with centre $(-3, -4)$ tangent to y-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q071 | MCQ

C04-K03-418-WA-032-MCQ

Find the circle with centre (4,-2) tangent to the x-axis:

A radius zero

B $(x - 4)^2 + (y + 2)^2 = 4$

C no real circle

D centre on the wrong locus

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q072 | MCQ

C04-K03-418-TR-035-MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:

A radius zero

B $(x + 2)^2 + (y - 3)^2 = 4$

C no real circle

D centre on the wrong locus

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q073 | MCQ

C04-K03-418-EX-038-MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:

A radius zero

B $(x + 1)^2 + (y - 4)^2 = 16$

C no real circle

D centre on the wrong locus

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q074 | MCQ

C04-K03-418-EX-039-MCQ

Find the circle with centre (3,-4) tangent to x-axis:

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 3)^2 + (y + 4)^2 = 16$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q075 | MCQ

C04-K03-418-EX-040-MCQ

Find the circle with centre (5,-4) tangent to x-axis:

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 5)^2 + (y + 4)^2 = 16$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q076 | MCQ

C04-K03-418-EX-041-MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:

A no real circle

B centre on the wrong locus

C radius zero

D $(x + 1)^2 + (y - 6)^2 = 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q077 | MCQ

C04-K03-418-EX-042-MCQ

Find the circle with centre (2,-5) tangent to y-axis:

A radius zero

B $(x - 2)^2 + (y + 5)^2 = 4$

C no real circle

D centre on the wrong locus

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V01

Q078 | MCQ

C04-K03-418-EX-043-MCQ

Find the circle with centre $(-3, -4)$ tangent to y-axis:

A centre on the wrong locus

B radius zero

C $(x + 3)^2 + (y + 4)^2 = 9$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q079 | WRITTEN

C04-K03-418-WA-033

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q080 | WRITTEN

C04-K03-418-TR-036

Find the circles through (1,2) tangent to both axes.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q081 | WRITTEN

C04-K03-418-EX-044

Find the circle with through $(-1, -2)$ tangent to both axes.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q082 | WRITTEN

C04-K03-418-EX-045

Find the circle with through $(-2, -1)$ tangent to both axes.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q083 | WRITTEN

C04-K03-418-EX-046

Find the circle with through (8,-1) tangent to both axes.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q084 | MCQ

C04-K03-418-WA-033-MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:

A centre on the wrong locus

B radius zero

C $(x+1)^2 + (y-1)^2 = 1$ or $(x+5)^2 + (y-5)^2 = 25$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q085 | MCQ

C04-K03-418-TR-036-MCQ

Find the circles through (1,2) tangent to both axes:

A no real circle

B centre on the wrong locus

C radius zero

D $(x-1)^2 + (y-1)^2 = 1$ or $(x-5)^2 + (y-5)^2 = 25$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q086 | MCQ

C04-K03-418-EX-044-MCQ

Find the circle with through $(-1, -2)$ tangent to both axes:

A centre on the wrong locus

B radius zero

C $(x+1)^2 + (y+1)^2 = 1$ or $(x+5)^2 + (y+5)^2 = 25$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q087 | MCQ

C04-K03-418-EX-045-MCQ

Find the circle with through $(-2, -1)$ tangent to both axes:

A centre on the wrong locus

B radius zero

C $(x+1)^2 + (y+1)^2 = 1$ or $(x+5)^2 + (y+5)^2 = 25$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V02

Q088 | MCQ

C04-K03-418-EX-046-MCQ

Find the circle with through (8,-1) tangent to both axes:

A radius zero

C no real circle

$$\text{B } (x-5)^2 + (y+5)^2 = 25 \text{ or } (x-13)^2 + (y+13)^2 = 169$$

D centre on the wrong locus

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q089 | WRITTEN

C04-K03-418-WA-034

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q090 | WRITTEN

C04-K03-418-TR-037

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and $(1,2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q091 | WRITTEN

C04-K03-418-EX-047

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q092 | WRITTEN

C04-K03-418-EX-048

Find the circle with centre on $y = 2x - 6$ through $(3,7)$ and $(-1,3)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q093 | MCQ

C04-K03-418-WA-034-MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 1)^2 + (y + 1)^2 = 25$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q094 | MCQ

C04-K03-418-TR-037-MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and $(1,2)$:

A centre on the wrong locus

B radius zero

C $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q095 | MCQ

C04-K03-418-EX-047-MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):

A centre on the wrong locus

B radius zero

C $(x - 2)^2 + (y - 3)^2 = 2$

D no real circle

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F03 | V03

Q096 | MCQ

C04-K03-418-EX-048-MCQ

Find the circle with centre on $y = 2x - 6$ through $(3,7)$ and $(-1,3)$:

A no real circle

B centre on the wrong locus

C radius zero

D $(x - 4)^2 + (y - 2)^2 = 26$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q097 | CHALLENGE WRITTEN

C04-K03-418-CH-049

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1).
Find its standard equation, centre and radius.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q098 | CHALLENGE MCQ

C04-K03-418-CH-049-MCQ

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

A $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} - 1$

B 0

C $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

D $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5} + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q099 | WRITTEN

C04-K03-419-WA-050

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q100 | WRITTEN

C04-K03-419-TR-051

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q101 | WRITTEN

C04-K03-419-EX-052

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q102 | WRITTEN

C04-K03-419-EX-053

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q103 | WRITTEN

C04-K03-419-EX-054

Find the range of k for $x^2 + y^2 + 2kx - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q104 | MCQ

C04-K03-419-WA-050-MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:

A all real parameter values

B radius squared is negative

C $k < -1$ or $k > 3$

D radius squared is zero

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q105 | MCQ

C04-K03-419-TR-051-MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:

A radius squared is negative

B $k < -\sqrt{6}$ or $k > \sqrt{6}$

C radius squared is zero

D all real parameter values

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

WORKING AREA

Concept 3 | Circles

F04 | V01

Q106 | MCQ

C04-K03-419-EX-052-MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:

A all real parameter values

B radius squared is negative

C $k < 5/4$

D radius squared is zero

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q107 | MCQ

C04-K03-419-EX-053-MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:

A radius squared is negative

B $k < -1/5$ or $k > 3$

C radius squared is zero

D all real parameter values

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F04 | V01

Q108 | MCQ

C04-K03-419-EX-054-MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle:

A radius squared is zero

B all real parameter values

C radius squared is negative

D $-1 < k < 4$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q109 | CHALLENGE WRITTEN

C04-K03-419-CH-055

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q110 | CHALLENGE MCQ

C04-K03-419-CH-055-MCQ

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

A 0

B $0 < k < 4$ C $0 < k < 4 + 1$ D $0 < k < 4 - 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q111 | WRITTEN

C04-K03-420-WA-056

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q112 | WRITTEN

C04-K03-420-TR-058

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q113 | WRITTEN

C04-K03-420-EX-060

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q114 | WRITTEN

C04-K03-420-EX-061

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q115 | WRITTEN

C04-K03-420-EX-062

Find the coordinates of the intersections of $(x-1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q116 | MCQ

C04-K03-420-WA-056-MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:

A no real intersection

B two intersection points

C (1, -1) and (-1, 3)

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q117 | MCQ

C04-K03-420-TR-058-MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:

A tangent

B no real intersection

C two intersection points

D (0, 2) and (2, 0)

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q118 | MCQ

C04-K03-420-EX-060-MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:

A no real intersection

B two intersection points

C (3, -1)

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q119 | MCQ

C04-K03-420-EX-061-MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

A tangent

B no real intersection

C two intersection points

D $(-2, 1)$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V01

Q120 | MCQ

C04-K03-420-EX-062-MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

A two intersection points

B $(-2, 1)$ and $(4, -1)$

C tangent

D no real intersection

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q121 | WRITTEN

C04-K03-420-WA-057

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q122 | WRITTEN

C04-K03-420-TR-059

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q123 | WRITTEN

C04-K03-420-EX-063

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q124 | WRITTEN

C04-K03-420-EX-064

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q125 | WRITTEN

C04-K03-420-EX-065

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q126 | MCQ

C04-K03-420-WA-057-MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:

A tangent

B no real intersection

C two intersection points

D 0 points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q127 | MCQ

C04-K03-420-TR-059-MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:

A 2 points : (1, -3) and (3, 1)

B tangent

C no real intersection

D two intersection points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q128 | MCQ

C04-K03-420-EX-063-MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:

A no real intersection

B two intersection points

C 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$
and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q129 | MCQ

C04-K03-420-EX-064-MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:

A no real intersection

B two intersection points

C 0 points

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V02

Q130 | MCQ

C04-K03-420-EX-065-MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:

A tangent

B no real intersection

C two intersection points

D 1 point : $(-2, 0)$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q131 | WRITTEN

C04-K03-421-WA-066

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q132 | WRITTEN

C04-K03-421-WA-067

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q133 | WRITTEN

C04-K03-421-TR-068

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q134 | WRITTEN

C04-K03-421-TR-069

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q135 | WRITTEN

C04-K03-421-EX-070

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q136 | WRITTEN

C04-K03-421-EX-071

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q137 | WRITTEN

C04-K03-421-EX-072

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q138 | WRITTEN

C04-K03-421-EX-073

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q139 | WRITTEN

C04-K03-421-EX-074

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q140 | MCQ

C04-K03-421-WA-066-MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

A tangent

B no real intersection

C two intersection points

D $-5 < k < 5$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q141 | MCQ

C04-K03-421-WA-067-MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

A two intersection points

B $r = \sqrt{5}$

C tangent

D no real intersection

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q142 | MCQ

C04-K03-421-TR-068-MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

A two intersection points

B $m = \pm\sqrt{3}$

C tangent

D no real intersection

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q143 | MCQ

C04-K03-421-TR-069-MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

A two intersection points

B $0 < r < \sqrt{10}$

C tangent

D no real intersection

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q144 | MCQ

C04-K03-421-EX-070-MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

A no real intersection

B two intersection points

C $m = \pm 4$

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q145 | MCQ

C04-K03-421-EX-071-MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

A $n < -10$ or $n > 10$

B tangent

C no real intersection

D two intersection points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q146 | MCQ

C04-K03-421-EX-072-MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

A no real intersection

B two intersection points

C $-5 \leq m \leq 5$

D tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q147 | MCQ

C04-K03-421-EX-073-MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

A two intersection points

B $r = \sqrt{2}$

C tangent

D no real intersection

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | V03

Q148 | MCQ

C04-K03-421-EX-074-MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

A tangent

B no real intersection

C two intersection points

D $r > \sqrt{17}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q149 | CHALLENGE WRITTEN

C04-K03-421-CH-075

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$.
Find the positive range of r .

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q150 | CHALLENGE MCQ

C04-K03-421-CH-075-MCQ

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$.
Find the positive range of r :

A $0 < r < 2 + 1$

B $0 < r < 2 - 1$

C 0

D $0 < r < 2$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q151 | WRITTEN

C04-K03-422-WA-076

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q152 | WRITTEN

C04-K03-422-TR-078

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q153 | WRITTEN

C04-K03-422-EX-080

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q154 | WRITTEN

C04-K03-422-EX-081

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q155 | WRITTEN

C04-K03-422-EX-082

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q156 | MCQ

C04-K03-422-WA-076-MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

A $\sqrt{14}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q157 | MCQ

C04-K03-422-TR-078-MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C 4

D $2\sqrt{r^2 + d^2}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q158 | MCQ

C04-K03-422-EX-080-MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $2\sqrt{5}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q159 | MCQ

C04-K03-422-EX-081-MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:

A $2\sqrt{r^2 - d^2} + 1$

B $3\sqrt{2}$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V01

Q160 | MCQ

C04-K03-422-EX-082-MCQ

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D 2

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q161 | WRITTEN

C04-K03-422-WA-077

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q162 | WRITTEN

C04-K03-422-TR-079

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q163 | WRITTEN

C04-K03-422-EX-083

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q164 | WRITTEN

C04-K03-422-EX-084

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q165 | MCQ

C04-K03-422-WA-077-MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

A $\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 - d^2} + 1$

C $k = \pm 2$

D $2\sqrt{r^2 + d^2}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q166 | MCQ

C04-K03-422-TR-079-MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

A $2\sqrt{r^2 + d^2}$

B $\sqrt{r^2 - d^2}$

C $2\sqrt{r^2 - d^2} + 1$

D $k = \pm\sqrt{31}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | V02

Q167 | MCQ

C04-K03-422-EX-083-MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

A $k = \pm 2$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$(x - h)^2 + (y - k)^2 = r^2$

$x^2 + y^2 + 2gx + 2fy + c = 0$

Chord and tangent

$L = 2\sqrt{r^2 - d^2}$

$xx_1 + yy_1 = r^2$

Through points / diameter

$r^2 = (x_1 - h)^2 + (y_1 - k)^2$

$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Two-circle position

D vs $r_1 + r_2, |r_1 - r_2|$

Tangency and real circle

$r = d(\text{centre, axis})$

$r^2 > 0$

Radical axis

$S_1 - S_2 = 0$

$S_1 = 0, S_2 = 0$

Line-circle position

$\Delta > 0 : 2 \text{ points}$

$\Delta = 0 : \text{tangent}$

$\Delta < 0 : \text{none}$

Pencil of circles

$S_1 + \lambda S_2 = 0$

Concept 3 | Circles

F06 | V02

Q168 | MCQ

C04-K03-422-EX-084-MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

A $2\sqrt{r^2 - d^2} + 1$

B $k = 0$ or $k = -2$

C $2\sqrt{r^2 + d^2}$

D $\sqrt{r^2 - d^2}$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q169 | CHALLENGE WRITTEN

C04-K03-422-CH-085

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q170 | CHALLENGE MCQ

C04-K03-422-CH-085-MCQ

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

A $k = -1$ or $k = 7$

B $k = -1$ or $k = 7 + 1$

C $k = -1$ or $k = 7 - 1$

D 0

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q171 | WRITTEN

C04-K03-423-WA-086

Find the tangent to $x^2 + y^2 = 25$ at (4,3).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q172 | WRITTEN

C04-K03-423-TR-088

Find the tangent to $x^2 + y^2 = 34$ at (5,3).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q173 | WRITTEN

C04-K03-423-EX-090

Find the tangents $x^2 + y^2 = 10$ at (3,-1).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q174 | WRITTEN

C04-K03-423-EX-091

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q175 | WRITTEN

C04-K03-423-EX-092

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q176 | MCQ

C04-K03-423-WA-086-MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):

A $4x + 3y = 25$

B $4x - 3y = 25$

C $4x + 3y = 25 + 1$

D the radius line

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q177 | MCQ

C04-K03-423-TR-088-MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):

A $5x + 3y = 34 + 1$

B the radius line

C $5x + 3y = 34$

D $5x - 3y = 34$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q178 | MCQ

C04-K03-423-EX-090-MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):

A $3x - y = 10$

B $3x - y = 10 + 1$

C the radius line

D 8

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q179 | MCQ

C04-K03-423-EX-091-MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:

A the radius line

B 9

C $-5x - 2y = 29$ D $-5x - 2y = 29 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V01

Q180 | MCQ

C04-K03-423-EX-092-MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:

A $-2x + y = 5$

B $-2x - y = 5$

C $-2x + y = 5 + 1$

D the radius line

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q181 | WRITTEN

C04-K03-423-WA-087

Find the tangents from (1,3) to $x^2 + y^2 = 5$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q182 | WRITTEN

C04-K03-423-TR-089

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q183 | WRITTEN

C04-K03-423-EX-093

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q184 | WRITTEN

C04-K03-423-EX-094

Find the tangents from (2,1) to $x^2 + y^2 = 1$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q185 | WRITTEN

C04-K03-423-EX-095

Find the tangents from (2,6) to $x^2 + y^2 = 20$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q186 | MCQ

C04-K03-423-WA-087-MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:

A $2x - y = 5$ and $-x + 2y = 5$

B $2x + y = 5$ and $-x + 2y = 5 + 1$

C the radius line

D $2x + y = 5$ and $-x + 2y = 5$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q187 | MCQ

C04-K03-423-TR-089-MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:

A $-3x + y = 10$ and $x + 3y = 10$

B $-3x - y = 10$ and $x + 3y = 10$

C $-3x + y = 10$ and $x + 3y = 10 + 1$

D the radius line

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q188 | MCQ

C04-K03-423-EX-093-MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:

A $-2x - y = 5$ and $x + 2y = 5$

B $-2x + y = 5$ and $x + 2y = 5 + 1$

C the radius line

D $-2x + y = 5$ and $x + 2y = 5$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q189 | MCQ

C04-K03-423-EX-094-MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:

A 12

B $y = 1$ and $4x - 3y = 5$ C $y = 1$ and $4x - 3y = 5 + 1$

D the radius line

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | V02

Q190 | MCQ

C04-K03-423-EX-095-MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:

A $-x + 2y = 10$ and $2x + y = 10 + 1$

B the radius line

C $-x + 2y = 10$ and $2x + y = 10$

D $-x - 2y = 10$ and $2x + y = 10$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V03

Q191 | CHALLENGE WRITTEN

C04-K03-423-CH-096

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V03

Q192 | CHALLENGE MCQ

C04-K03-423-CH-096-MCQ

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

A 0

B $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$ C $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) + 1$ D $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2) - 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q193 | WRITTEN

C04-K03-424-WA-097

Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q194 | WRITTEN

C04-K03-424-TR-099

Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q195 | WRITTEN

C04-K03-424-TR-100

Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q196 | WRITTEN

C04-K03-424-EX-102

Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q197 | WRITTEN

C04-K03-424-EX-103

Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q198 | WRITTEN

C04-K03-424-EX-104

Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q199 | WRITTEN

C04-K03-424-EX-105

Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q200 | MCQ

C04-K03-424-WA-097-MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$, the required value is:

A internally tangent

B intersecting at two points

C intersect at 2 points

D separate

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q201 | MCQ

C04-K03-424-TR-099-MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

A separate

B internally tangent

C intersecting at two points

D B : touch externally

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q202 | MCQ

C04-K03-424-TR-100-MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

A intersecting at two points

B E : intersect at 2 points

C separate

D internally tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q203 | MCQ

C04-K03-424-EX-102-MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

A C : touch internally

B separate

C internally tangent

D intersecting at two points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q204 | MCQ

C04-K03-424-EX-103-MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x+2)^2 + (y-2)^2 = 2$ from A-E., the required value is:

A separate

B internally tangent

C intersecting at two points

D C : touch internally

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q205 | MCQ

C04-K03-424-EX-104-MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

A intersecting at two points

B E : intersect at 2 points

C separate

D internally tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V01

Q206 | MCQ

C04-K03-424-EX-105-MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

A internally tangent

B intersecting at two points

C D : one is inside the other

D separate

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q207 | WRITTEN

C04-K03-424-WA-098

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q208 | WRITTEN

C04-K03-424-TR-101

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q209 | WRITTEN

C04-K03-424-EX-106

Find the circle with centre (3,-9) touching $(x-3)^2 + y^2 = 9$ internally.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q210 | MCQ

C04-K03-424-WA-098-MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:

A $(x - 3)^2 + (y - 1)^2 = 45$

B separate

C internally tangent

D intersecting at two points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q211 | MCQ

C04-K03-424-TR-101-MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:

A separate

B internally tangent

C intersecting at two points

D $(x + 3)^2 + (y + 9)^2 = 16$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | V02

Q212 | MCQ

C04-K03-424-EX-106-MCQ

Find the circle with centre $(3, -9)$ touching $(x - 3)^2 + y^2 = 9$ internally:

A $(x - 3)^2 + (y + 9)^2 = 144$

B separate

C internally tangent

D intersecting at two points

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q213 | CHALLENGE WRITTEN

C04-K03-424-CH-107

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V02

Q214 | CHALLENGE MCQ

C04-K03-424-CH-107-MCQ

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

A $(x + 1)^2 + (y + 2)^2 = 9 + 1$

B $(x + 1)^2 + (y + 2)^2 = 9 - 1$

C 0

D $(x + 1)^2 + (y + 2)^2 = 9$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q215 | WRITTEN

C04-K03-425-WA-108

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q216 | WRITTEN

C04-K03-425-TR-110

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q217 | WRITTEN

C04-K03-425-EX-112

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q218 | WRITTEN

C04-K03-425-EX-113

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q219 | MCQ

C04-K03-425-WA-108-MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

A $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

B $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$) + 1

C the radical axis only

D 4

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q220 | MCQ

C04-K03-425-TR-110-MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

A $x + y - 1 = 0 + 1$

B the radical axis only

C $x + y - 1 = 0$

D $x - y - 1 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q221 | MCQ

C04-K03-425-EX-112-MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

A 8

B $12x - 6y - 7 = 0$ C $12x - 6y - 7 = 0 + 1$

D the radical axis only

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V01

Q222 | MCQ

C04-K03-425-EX-113-MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:

A $x = 1/2 + 1$

B the radical axis only

C 9

D $x = 1/2$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q223 | WRITTEN

C04-K03-425-WA-109

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q224 | WRITTEN

C04-K03-425-TR-111

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q225 | WRITTEN

C04-K03-425-EX-114

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q226 | MCQ

C04-K03-425-WA-109-MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:

A 5

B $(-3, -1)$ and $(-1, 3)$ C $(-3, -1)$ and $(-1, 3) + 1$

D the radical axis only

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q227 | MCQ

C04-K03-425-TR-111-MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:

A the radical axis only

B 7

C (0, 2) and $(8/5, -6/5)$ D (0, 2) and $(8/5, -6/5) + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F09 | V02

Q228 | MCQ

C04-K03-425-EX-114-MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:

A 10

B (1, 2) and (2, 1)

C (1, 2) and (2, 1) + 1

D the radical axis only

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V04

Q229 | CHALLENGE WRITTEN

C04-K03-425-CH-115

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V04

Q230 | CHALLENGE MCQ

C04-K03-425-CH-115-MCQ

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

A $(-4, 3)$ and $(3, -4) - 1$

B 0

C $(-4, 3)$ and $(3, -4)$ D $(-4, 3)$ and $(3, -4) + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q231 | WRITTEN

C04-K03-426-WA-116

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q232 | WRITTEN

C04-K03-426-TR-117

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q233 | WRITTEN

C04-K03-426-EX-118

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q234 | WRITTEN

C04-K03-426-EX-119

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q235 | WRITTEN

C04-K03-426-EX-120

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q236 | MCQ

C04-K03-426-WA-116-MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

A $x^2 - y^2 + 2x + 6y - 15 = 0$

B $x^2 + y^2 + 2x + 6y - 15 = 0 + 1$

C the radical axis only

D $x^2 + y^2 + 2x + 6y - 15 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q237 | MCQ

C04-K03-426-TR-117-MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

A $x^2 + y^2 + 4x + 2y = 0$

B $x^2 - y^2 + 4x + 2y = 0$

C $x^2 + y^2 + 4x + 2y = 0 + 1$

D the radical axis only

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q238 | MCQ

C04-K03-426-EX-118-MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

A $x^2 - y^2 + 10x + 20y - 39 = 0$

B $x^2 + y^2 + 10x + 20y - 39 = 0 + 1$

C the radical axis only

D $x^2 + y^2 + 10x + 20y - 39 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q239 | MCQ

C04-K03-426-EX-119-MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3, 2)$:

A $x^2 + y^2 + 24x + 32y - 5 = 0 + 1$

B the radical axis only

C $x^2 + y^2 + 24x + 32y - 5 = 0$

D $x^2 - y^2 + 24x + 32y - 5 = 0$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | V01

Q240 | MCQ

C04-K03-426-EX-120-MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

A the radical axis only

B $x^2 + y^2 - 10x - 6y + 24 = 0$ C $x^2 - y^2 - 10x - 6y + 24 = 0$ D $x^2 + y^2 - 10x - 6y + 24 = 0 + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q241 | CHALLENGE WRITTEN

C04-K03-426-CH-121

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F99 | V05

Q242 | CHALLENGE MCQ

C04-K03-426-CH-121-MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

A $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

B $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 + 1$

C $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0 - 1$

D 0

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | C04-K03-F01-VQ

Quiz M1 | QUIZ MCQ

C04-K03-QUIZ-MCQ-1

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

A centre $(-2, -3)$, radius 5

B centre $(2, -3)$, radius 5

C centre $(2, 3)$, radius 5

D centre $(2, -3)$, radius 10

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | C04-K03-F05-VQ

Quiz M2 | QUIZ MCQ

C04-K03-QUIZ-MCQ-2

Determine the substitution discriminant when a line meets a circle at exactly one real point:

A tangent

B no real intersection

C two intersection points

D 0

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F06 | C04-K03-F06-VQ

Quiz M3 | QUIZ MCQ

C04-K03-QUIZ-MCQ-3

Determine the chord length for radius r and centre-to-chord distance d :

A $2\sqrt{r^2 - d^2}$

B $2\sqrt{r^2 + d^2}$

C $\sqrt{r^2 - d^2}$

D $2\sqrt{r^2 - d^2} + 1$

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F08 | C04-K03-F08-VQ

Quiz M4 | QUIZ MCQ

C04-K03-QUIZ-MCQ-4

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

A separate

B internally tangent

C intersecting at two points

D externally tangent

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F01 | C04-K03-F01-VQ

Quiz W1 | QUIZ WRITTEN

C04-K03-QUIZ-WRITTEN-1

Find the circle with centre (1,-2) passing through (4,2).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F05 | C04-K03-F05-VQ

Quiz W2 | QUIZ WRITTEN

C04-K03-QUIZ-WRITTEN-2

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F07 | C04-K03-F07-VQ

Quiz W3 | QUIZ WRITTEN

C04-K03-QUIZ-WRITTEN-3

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$

Concept 3 | Circles

F10 | C04-K03-F10-VQ

Quiz W4 | QUIZ WRITTEN

C04-K03-QUIZ-WRITTEN-4

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

WORKING AREA

FORMULA BANK

Standard / general circle

$$(x - h)^2 + (y - k)^2 = r^2$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Chord and tangent

$$L = 2\sqrt{r^2 - d^2}$$

$$xx_1 + yy_1 = r^2$$

Through points / diameter

$$r^2 = (x_1 - h)^2 + (y_1 - k)^2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Two-circle position

$$D \text{ vs } r_1 + r_2, |r_1 - r_2|$$

Tangency and real circle

$$r = d(\text{centre, axis})$$

$$r^2 > 0$$

Radical axis

$$S_1 - S_2 = 0$$

$$S_1 = 0, S_2 = 0$$

Line-circle position

$$\Delta > 0 : 2 \text{ points}$$

$$\Delta = 0 : \text{tangent}$$

$$\Delta < 0 : \text{none}$$

Pencil of circles

$$S_1 + \lambda S_2 = 0$$